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STRATEGY, STRUCTURE, AND ECONOMIC PERFORMANCE

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ABSTRACT

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This thesis examines the relationships among diversification strategy, organizational structure and economic performance in large U.S. industrial firms during the period 1949-1969. A sample of 246 companies that were among the largest 500 in 1949, 1959, or 1969 was selected and studied intensively.

Rather than focus on the amount of diversification that characterized a firm, a system of categories corresponding to the firm's strategy of diversification was developed. These categories were designed to capture the nature of the firm as viewed by top management and were defined in terms of the firm's overall commitment to diversity and the way in which it expanded by relating new businesses to old. In all, nine distinct categories of diversification strategy are defined. Structural categories were also defined corresponding to functional, functional with subsidiaries, product division, geographic division, and holding company forms of organization.

A census of the proportion of the largest 500 firms that fell in each strategic and structural category was taken for 1949, 1959, and 1969. The results showed that there had been a dramatic increase in the number of fully diversified firms using product division structures during this period. An examination of the relationship between strategic and structural change showed that diversification and divisionalization were closely related in the 1950s, but that in the 1960s firms adopted product division structures without first increasing their product-market diversities.

The financial performance of firms following various strategies of diversification and using different structures were found to differ significantly. While there was no direct relation between profitability or growth and diversity per se, performance did depend upon the way in which new businesses were related to old. In particular, it was found that vertically integrated firms and firms engaged in unrelated businesses (but not having policies of rapid growth by acquisition) were the poorest performers and firms that were somewhat diversified but related all of their activities to some central skill or strength were the highest performers.

The vertically integrated firms appeared to suffer from stagnant industries but their long term commitments to a single industry and heavy investment in fixed plant made it extremely difficult for them to diversify into more dynamic sectors of the economy. The poor performing unrelated business firms were found to be characterized by administrative systems more suitable to companies engaged in activities that are related to one another.

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Richard P. Rumelt
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TABLE OF CONTENTS

Chapter

I.	INTRODUCTION	1
	Background	1
	The Research Question	6
	Research Methodology	8
	Outline of the Study	9
II.	DEFINITIONS, CONCEPTS AND METHODOLOGY	14
	Concepts of Diversification	14
	Developing Categories of Diversification	18
	Using the Diversification Categories - Examples	43
	Summary of Diversification Categories	52
	The Data and Its Sources	72
	Summary and Comments	80
III.	THE CHANGING PATTERNS OF STRATEGY AND STRUCTURE	83
	Comparison With the Industry Count	84
	Strategies of Diversification	89
	Organizational Structure	112
	Strategy and Structure	121
	Summary	134
	APPENDIX 3A (ESTIMATION TECHNIQUE)	136
IV.	FINANCIAL PERFORMANCE CHARACTERISTICS OF THE CATEGORIES OF DIVERSIFICATION STRATEGY AND STRUCTURE	142
	Theory and Hypotheses	143
	The Analysis of Financial Data	161
	The Question of Industry	176
	Testing the Hypotheses	185
	A Question of Price	220
	Synthesis	225
	APPENDIX 4A (STATISTICAL PROCEDURES)	235

Chapter

V.	THE LOW PERFORMANCE STRATEGIES	249
	The Dominant Vertical Syndrome	249
	The Unrelated Passive Strategy	266
	Unrelated Passive Performance	271
VI.	SUMMARY OF FINDINGS AND IMPLICATIONS	
	Summary of Findings	274
	Implication of the Findings	283
	APPENDIX A	A-1
	BIBLIOGRAPHY	A-100

CHAPTER I

INTRODUCTION

Because its consequences may alter the fundamental nature of the firm and because it implies a substantial redeployment of resources and redirection of human energy, a decision to diversify is a strategic decision. However, diversification is neither a goal nor a plan by itself; each firm that diversifies must make specific choices as to the types of businesses it will enter, the degree to which it will build on past strengths and competences or require the development of new ones, and the total amount of diversity that is appropriate. In short, there are many strategies of diversification. The subject of this study is the interrelationships among diversification strategy, organizational structure and economic performance in large American industrial corporations. In addition to investigating the relative prevalence of various types of strategy and structure, the relationship between strategy and structure, and the association between these two variables and economic performance, we shall be concerned with the validation of a research method that combines the managerially meaningful but essentially descriptive concept of diversification strategy with the analytic power of statistical techniques applied to large samples.

Background

Although it is an increasingly common attribute of large industrial corporations, product line diversification does not fit neatly into any of the current theories of economic and business behavior. As Gort notes,

There has been so much discussion of diversification that the phenomenon seems perfectly natural and something to be taken for granted. But in fact it runs against a pervasive trend in our society toward greater specialization in most channels of economic life. The broader the market

for a given product and the more complex the production process, the more specialized one would normally expect producers to be. And yet, notwithstanding the growth in markets and the increasing complexity of technology, the trend for companies seems to be toward greater diversification rather than toward specialization.¹

Economists who have studied diversification have been chiefly concerned with developing objective measures of the diversity of a firm's product-market posture and in relating such measures to pricing behavior, research decisions, overall concentration, and in studying the public policy implications of an economy that is increasingly composed of diversified firms. Statistical methods have been used to show that diversification has been continually increasing since the 1920s,² that it has been greatest in industries that are based on the newer more sophisticated technologies,³ that it has been chiefly powered by internal growth rather than merger,⁴ and that its degree correlates with a number of other quantitative attributes of companies and industries.

While such studies have been valuable in describing the extent and locus of diversification activity, it was a historian, Alfred D. Chandler, using the traditional techniques of historical data gathering and analysis, who made one of the most important contributions to our understanding of the role of diversification in the development of the modern firm. By

¹ Michael Gort, "Diversification, Mergers, and Profits", in William W. Alberts and Joel E. Segall (editors) The Corporate Merger (Chicago, University of Chicago Press, 1966) p. 32.

² See Chandler's report of Livesay and Porter's data in Alfred D. Chandler, "The Structure of American Industry in the Twentieth Century: A Historical Overview", Business History Review, Autumn 1969, pp. 255-298.

³ Ibid., p. 274

⁴ Gort, "Diversification, Mergers, and Profits", p. 31.

tracing the patterns of development of 70 large corporations from 1909 to 1959, Chandler found that diversification had given rise to the need for a new administrative system that would ensure the efficient utilization of diverse resources and that the multidivisional structure came into use as an adaptive response to this need.⁵

In turn, the multidivisional structure has served to institutionalize diversification, has permitted the insulation of a set of business managers from the vicissitudes of the capital markets, and has created a type of managerial environment that encourages rapid redeployment of resources and places a premium on economic performance and the skills of the generalist. As a consequence of this new understanding, diversification can no longer be viewed as a simple quantitative increase in the number of products produced and sold by a firm. Instead, Chandler has demonstrated that diversification has been causally related to and eventually dependent upon the rise of new types of business enterprises that are qualitatively different from the more traditional form of corporation.

The insight that the diversified multidivisional firm is fundamentally different from the more primitive type of enterprise, posing a different set of problems to top management and behaving differently as an economic entity, has sparked a number of further insights and research efforts. Scott, taking a cue from Rostow, has developed a "stages" model of corporate growth and development that views the corporation as moving through several architypical forms as its product-market complexity increases.⁶ He sees the firm as developing from a small "one man show" to a functionally organized (e.g. sales, production,

⁵ Alfred D. Chandler, Strategy and Structure (Doubleday, 1942)

⁶ Bruce R. Scott, "Stages of Corporate Development" (unpublished paper, Harvard Business School, 1970)

research) single business company as management seeks to gain control over all the factors important to the production and sale of a particular product. Then, as the firm begins to diversify and develop multiple product lines, it adopts the multidivisional structure. Scott's central thesis is that these three stages are not only different in outward form, but that they impose different "ways of life" on management and require quite different styles and systems of reward, control, resource allocation, review, and new business development.

Bower⁷ has studied the process of capital investment decision making in a large diversified multidivisional company and reports that the managerial climate created for the division general managers by the headquarters office is of crucial importance and that traditional theories of how capital budgeting is carried out are neither descriptive of the actual process nor normatively useful.

Stopford,⁸ investigating the relationship between structure and behavior with regard to direct investment abroad, has stated that the relatively large number of generalist roles created within the multidivisional organization is an important factor in explaining the increased ease with which such firms formulate and implement multinational strategies. Vernon has generalized these findings with the conjecture that the multidivisional structure is the critical factor in permitting firms to grow through diversification and to make use of a large spectrum of skills without experiencing gross diseconomies of scale. "Once the basic change in structure is achieved", he says, "the

⁷ Joseph L. Bower, Managing The Resource Allocation Process (Boston, Division of Research, Harvard Business School, 1970)

⁸ John M. Stopford, "Growth and Change in the Multinational Firm" (unpublished doctoral dissertation, Harvard Business School, 1968)

diseconomies of scale which threaten any large organization are pushed off to a new, more distant level or, perhaps more accurately, . . . new economies appear sufficiently important to offset for an added period the diseconomies associated with large organizations."⁹

Wrigley's Study. Wrigley's¹⁰ recent survey of the 1967 Fortune 500 built on Chandler's pioneering study by recognizing that there are significant differences in the ways in which firms have elected to diversify. He stratified his sample population into four categories: firms that were not diversified (Single Product), firms that were primarily committed to a single business but had diversified to a small degree (Dominant Product), firms that expanded into new areas which bore either a technological or market relation to current activities (Related Product), and firms that diversified without regard to such relationships (Unrelated Product). In particular, the Dominant Product group, defined to consist of firms that obtained more than 70% of their sales from a single product area, represents a new and important distinction. His results are shown in Table 1-1.

Wrigley's important contribution is not just this simple census, but the fact that he discarded the time honored "product count" measure of diversification and employed instead a system of categories based on a qualitative evaluation of the firm's product-market scope and diversification rationale. Although other researchers have admitted that a measure of diversification based on the number of products the firm sells or the number of industries in which it participates gives no weight

⁹ Raymond Vernon, "Organization as a Scale Factor in the Growth of Firms", in J.W. Markham and G.F. Papanek (editors) Industrial Organization & Economic Development: In Honor of E.S. Mason (Boston, Houghton Mifflin, 1970) p.64.

¹⁰ Leonard Wrigley, "Divisional Autonomy and Diversification" (unpublished doctoral dissertation, Harvard Business School, 1970).

to the "differentness" of the various areas of activity or to their relative importance, most have been reluctant to abandon what seemed to be the only available "objective" measure.

Wrigley's categories, although not totally objective, at least permit one to differentiate among firms that have diversified in different ways (such as Litton Industries and General Electric) and among firms that are fully diversified and those that remain wedded to a single major product line (such as Dow Chemical and Ford Motor).

Table 1-1

WRIGLEY'S CENSUS OF THE 1967 FORTUNE 500

<u>Category</u>	<u>Percentage of Firms in Sample Falling in this Category</u>	<u>Percentage of Firms in Category Having Multidivisional Organizations</u>
Single Product	6 %	0 %
Dominant Product	14	64
Related Product	60	95
Unrelated Product	20	100

The Research Questions

The data in Table 1-1 shows the strategy of diversification and the multidivisional structure to be widely accepted in 1967 and Scott¹¹ has commented that this represents a dramatic shift from the immediate post-war era. In addition, Scott has observed that the Wrigley data calls into question theories of corporate growth and development that emphasize the importance of technology and integration rather than diversification and

¹¹ Scott, "Stages of Corporate Development", p. 42.

decentralization.¹² Galbraith,¹³ for example, is a proponent of the former view and sees the corporation as becoming evermore dedicated to a particular product-market area and increasingly immune to top management direction. Scott suggests that Chandler's model, linking technology to diversification and diversification to administrative structures that encourage innovation and preserve top management's control over strategy, more accurately describes the bulk of large modern corporations.

Considering the special properties that theorists and practitioners attribute to the diversified divisionalized firm and its increasing prevalence, there are a number of intriguing questions that deserve investigation: (1) Exactly how has the shift to diversification come about? Do firms generally move from the Single Product category to the Dominant, then to the Related, or do they jump past the Dominant (or perhaps the Related) stages? (2) How much of the change in the composition of the largest 500 has been due to changes by firms themselves and how much has been due to new firms entering the top 500 to replace those that have been acquired or dropped from the list due to insufficient growth? (3) Is the Dominant Product group stable, or does it consist of firms that are on their way to full diversification? If it is stable, do these firms face any special barriers to diversification or have they purposefully elected to limit their product-market scopes? (4) What is the relation between strategic change and structural change over time? Can changes in structure be directly correlated with changes in diversification strategy or are they parallel but independent trends?

One of the purposes of the present study was to seek answers to these and similar questions and, in the process, to develop an enriched model of the pattern of development of the large industrial corporation.

¹² Ibid., p. 39

¹³ John K. Galbraith, The New Industrial State (Boston, Houghton Mifflin, 1967)

In addition to extending the Wrigley census to cover a period of time and investigating the usefulness of alternative categories, this study is concerned with comparing the economic performance of firms belonging to different strategic and organizational categories. In a sense, this portion of the research repeats past efforts as several investigators have sought a relationship between diversification and profitability or growth. In no case, however, was a definite relationship found between economic performance and product count measures of diversification.¹⁴ Therefore, it was felt that if the strategic category method revealed significant performance differences among the various strategies of diversification, it would not only be a finding of high intrinsic interest, it would also lend support to our approach to the study of diversification. As will be seen, many significant differences were found, several of which were quite unexpected.

In looking at economic performance and diversification, we sought to compare groups of firms having different strategies of diversification in terms of their rates of growth in sales and earnings, their returns on investment, their price-earnings ratios, and their debt-to-capital ratios. In addition, marginal relationships between growth and profitability and, for the case of acquisitive conglomerates, between earnings growth and the price-earnings ratio are investigated. Since the specific hypotheses that were tested are many and complex, the discussion of them is deferred to Chapter IV.

Research Methodology

The population of interest in this study was taken to be the largest 500 U.S. industrials as listed annually by Fortune. The time span was chosen to be the twenty years between

¹³ For example, see Michael Gort, Diversification and Integration in American Industry (Princeton, Princeton University Press, 1962) p. 75-77.

1949 and 1969. While a longer time span might have been more revealing, data on the product lines, economic performance, and the organizational structure of firms prior to 1950 is sparse and difficult to locate. Given the choice between studying only those firms for which data was available and studying an unbiased random sample of large firms over a period for which more data was available, we preferred the latter.

The sample was constructed by selecting 100 firms randomly from the largest 500 in 1949, another 100 from the largest 500 in 1959, and another 100 from the largest 500 in 1969. Since a number of firms were selected in more than one of the three samples, the total sample size was 246 rather than 300. This sampling procedure allowed us to (1) investigate the composition of the largest 500 in 1949, 1959, and 1969, (2) study the changing composition of the largest 500, and (3) obtain groups of firms that were members of the 500 throughout a decade in order to have a sample representative of large firms during that time period.

Once the sample was obtained we placed each firm in one of the categories of diversification strategy and one of the categories of organizational structure in 1949, 1959, and 1969. The data used in making these classifications came from published sources and included company annual reports, prospectuses, books, articles, Moody's Industrials and less often from a variety of other sources. Financial data on most of the companies in the sample was obtained from a computer tape originally created by Standard and Poors and maintained by The Boston Company.

The information obtained from these sources was then placed on a computer file that contained for each firm in the sample a record of its strategic categorization, structural history, and a continuous record of its financial performance during the period 1950-1969. A number of computer programs were then written to manipulate this file so that analyses could be made of the composition of the largest 500, changes in this

composition, the relationship between changes in strategy and changes in structure, and the financial performance of firms falling in various categories of strategy and structure. As in any project of this type, the range of analyses performed greatly exceeded the scope of those presented in the final report.

In addition to the classificatory and financial data on each firm, a large amount of descriptive data on each firm in the sample was gathered from published sources. Not used in the quantitative analyses, this information played an invaluable role in the interpretation of statistical and census results.

Outline of the Study

Chapter II provides a description of the classification scheme that was employed and the research methodology. It begins by giving a detailed description of Wrigley's classification scheme and then turns to the development of an improved system of discrete categories of diversification strategy. Wrigley's four basic categories are maintained, though with slightly different definitions, and sub-categories of the Dominant, Related, and Unrelated groups are defined. In defining these sub-categories, we introduce the concept of "constrained" versus "linked" diversification in order to capture the difference between firms that built their diversification strategies on some central skill or strength and firms that continually branch out in new directions. In addition, we make a distinction between Unrelated class firms that are acquisitive conglomerates and those that operate unrelated businesses without engaging in rapid diversification through acquisition. Finally, a special sub-class of the Dominant category is defined for vertically integrated firms.

Following the definition of strategic categories, we provide a number of examples of how the classification system was applied. Brief descriptions of the development of firms over time are given and analyses of the reasons for their assigned classifications are discussed.

The system of structural classification is then presented and five categories of organizational structure are defined: functional, functional with subsidiaries, product division, geographic division, and holding company. Of these, the functional with subsidiaries category is new to the literature and corresponds to a functional organization that has created a few semi-autonomous product divisions in order to manage activities not directly connected to the major business.

Next, the research methodology is described. Source of data are given, the type of data processing used is discussed, and an evaluation of the replicability of the classification system is made by comparing the results obtained in this study with Wrigley's.

Chapter III is devoted to the presentation and analysis of the census data on the composition of the largest 500 firms in terms of the categories of diversification strategy and structure. It begins by comparing the breakdown of the top 500 by strategic category with a simpler measure of corporate diversity and we suggest that the simpler measure is not sufficiently rich to serve as a proxy for diversification strategy. Turning to the changing composition of the top 500, we investigate the degree to which internal change and the effects of entry to and exit from the top 500 are responsible for the overall observed change. We find that both factors have had similar effects and that they have been of about equal magnitude. Their net result has been a marked decrease in the number of Single business firms and a substantial increase in the number of Related business and Unrelated business firms.

A closer inspection of the patterns of strategic change shows that transition rates between major strategic categories have been fairly stable over the past twenty years with the only important exception being a recent increase in the transition rate into the Acquisitive Conglomerate class.

An investigation of the patterns of structural change

shows that during the period 1949-1969 the product division structure moved from a position of minor importance to the major organizational form employed by large corporations. Again, the sources of change are investigated and it is shown that while increasing diversification explains about one-half of the increase in divisionalization, the remainder of the increase must be attributed to a trend towards the adoption of the product division structure among almost all types of firms.

Chapter IV deals with the relative financial performance of firms following different strategies of diversification and having different organizational structures. After a brief overview of the major differences, the possibility of obtaining industry corrected results is discussed. We conclude that diversification strategy is so intimately mixed with industry that it is not possible to separate the two effects.

Next, a number of hypotheses regarding financial performance are explained and tested. The most important results are that (1) firms that have diversified to some extent but have restricted their range of activities to those that built on some central skill or competence that is common to all have shown substantially higher rates of profitability and growth than other types of firms, (2) Unrelated business firms that participate in science-based industries, while having rates of profit that are average in an overall sense, are less profitable than other types of science-based firms, (3) of the structure categories, the product-division is the best performing, (4) the poorest performing firms are the vertically integrated and Unrelated business (but not conglomerate) companies, and (5) firms with product division structures are faster growing and at least as profitable as other types of firms and also seem to rely less on high profits to power growth than other types of firms.

In Chapter V we look closely at the two strategic categories that showed the poorest financial performances: the vertically integrated group and the Unrelated Passive (unrelated

but not conglomerate) category. We suggest that the vertically integrated firms face special barriers to diversification so that, unlike many other less diversified companies, they find it very difficult to remedy low performance through diversification. In examining the properties of the Unrelated Passive group we find that most consist of a few unrelated clusters of businesses which are themselves diversified but internally related units. We suggest that this type of structure presents particularly difficult administrative problems and the histories of most of these firms indicate that they have not had either the opportunity or the willingness to solve them.

Chapter VI summarizes the findings of the study and points out the implications for managers and public policy makers. Following this chapter is an appendix which presents information in tabular form on each of the firms in the sample used in this research. Included in this table is data on each company's product line, rank in the 500, strategic classification, and structural classification for 1949, 1959, and 1969.

CHAPTER II

DEFINITIONS, CONCEPTS AND METHODOLOGY

Having noted the need for a large sample study of diversification and organizational change over time, we now turn to the development of a suitable conceptual framework. The goal is a set of concepts and measurement techniques that are managerially meaningful, capturing the essence of top management's view of diversification, while also being simple enough to permit the examination of a large number of firms. In developing this framework we shall make use of the concept of corporate strategy and introduce the idea of categories of diversification strategy, drawing heavily on the work of Wrigley. After reducing these categories to a set of operational definitions and measures, we shall look to Chandler and to Scott's "stages model" for guidance on the choice of organizational concepts. Then the sampling methodology and the sources of data used in the research will be discussed.

Concepts of Diversification

There exists no generally accepted definition or measure of diversification. An annual report to stockholders may call attention to a refrigerator manufacturer's diversification into home freezers while a statistical study on the extent of diversification in large firms may ignore any product addition that does not take a company into a new

industrial sector. Some describe an oil company's move into petrochemicals as diversification while others refer to it as vertical integration. Further complicating the issue is the recent unurge of interest in conglomerate corporations. Among the firms that have been called "highly diversified conglomerates" are Ford Motor, Crown Cork & Seal, General Electric and Pittsburgh Plate Glass, as well as Textron, LTV and Gulf & Western.

The lack of any clear-cut definition of diversification is both a problem and an opportunity for the researcher; although he must shoulder the burden of developing his own concept of diversification, he enjoys the freedom of being able to tailor that concept to suit his interests. For example, economists who have attempted to relate diversification activity to industry structure or to relate product additions to mergers have generally used a simple count of the number of products produced and sold by the firm as both the measure and the implicit definition of diversification. Berg,¹ on the other hand, has extensively studied the nature of the headquarters office of large diversified firms and has found that a classification system based on the method of growth--internal vs. by acquisition--best suits his needs.

¹Norman A. Berg, "What's Different About Conglomerate Management?", Harvard Business Review, May-June 1965, p.112.

Our interest in diversification stems from its interaction with the task of general management. In this context, diversification may be seen both as the result of general management decisions that are of great importance to the future of the firm and as the cause of problems in coordination, planning and control at the highest levels. In other words, our concern is with the strategic aspects of diversification, and rather than view the firm from the "bottom up", in terms of its detailed product portfolio, we shall look from the "top down".

In conceptualizing diversification as a strategy we are following Chandler's usage of this term. He defined strategy as "the determination of the basic long-term goals and objectives of an enterprise, and the adoption of courses of action and the allocation of resources necessary for carrying out these goals".² Then, by distinguishing among the strategies of "expansion of volume", "geographical dispersion", "vertical integration", and "product diversification", Chandler showed how each poses a different type of administrative difficulty and therefore tends to lead to a different form of organizational structure.

More recently, the Business Policy group at the Harvard Business School developed a more elaborate concept

² Alfred D. Chandler, Jr., Strategy and Structure, (Garden City, N.Y., Doubleday, 1966), p.16.

of corporate strategy.³ In the Harvard framework strategy is the result of a balanced consideration of a firm's skills and resources, the opportunities extant in the economic environment, and the personal desires of management tempered by its sense of social responsibility. Although this concept was not developed to deal specifically with diversification, it does suggest that the task of matching opportunity with corporate skills and strengths is the most important of top management's responsibilities and may be taken as the primary component of diversification strategy.

Using the range of skills possessed corporately by the firm as a point of departure, a diversification move is taken to be any entry into a new product-market activity that requires or implies an appreciable increase in the available managerial competence within the firm. Thus, the essence of diversification is taken to be a "reaching out" into new areas, requiring the development of new competences or the augmentation of existing ones. A new business activity may be related in several ways to a current activity but still require the understanding of a different production technology, of different marketing concepts and methods, or new approaches to investment decisions, planning, and control.

³ See for example Kenneth Andrews, The Concept of Corporate Strategy, (Homewood, Illinois, Dow Jones-Irwin, 1971).

A point that deserves emphasis is that this definition identifies diversification in terms of the degree to which a new product-market activity taxes the ability of the firm's management, particularly its general management, together with the administrative structure it has created. The fact that companies with operating skills in a given area can usually be purchased outright underlines the importance of management ability as one of the primary non-financial restrictions on entry into new businesses.

The concept of a diversification move applies only to a particular product-market addition. In order to compare the diversification postures of different firms, some way of expressing the cumulative totality of a firm's diversification moves is needed. Accordingly, we introduce the idea of a "diversification strategy". Combining Chandler's concept of diversification as a strategy and the Harvard view of strategy as the relation between competence and opportunity, we define a firm's diversification strategy as (1) the firm's commitment to diversity per se, together with (2) the strengths, skills or purposes that span this diversity, as evidenced by the way in which new activities are related to old activities.

Developing Categories of Diversification Strategy

The measurement technique that will be used to implement these definitions is a modified version of a classification system developed by Wrigley. Taking the term

specialization ratio (SR) as denoting the proportion of a firm's annual revenues attributable to its largest discrete product-market activity, Wrigley's original categories may be described as follows:⁴

Single Product: Firms with SR between 0.95 and 1.0. Such firms grow only through expansion in the scale of their operations.

Dominant Product: Firms with SR between 0.7 and 0.95. Such firms have diversified to a small degree but are still quite dependent upon and characterized by their major product-market activity.

Related Product: Firms with SR less than 0.7 which have diversified by adding new activities that are tangibly related to the collective skills and strengths possessed originally by the firm.

Unrelated Product: Firms that diversify (usually by acquisition) into areas that are not related to the original skills and strengths, other than financial, of the firm.

By identifying the specialization ratio as a measure of the "firm's commitment to diversity per se" it becomes clear that these categories represent one possible way of implementing our concept of strategic posture. In the early stages of this research the attempt was made to use these categories without modification so that the results would be directly comparable with Wrigley's. However, it soon became apparent that (1) the terms "Single Product" and "Dominant Product" tended to be interpreted too narrowly by those not intimately familiar with the methodology, (2) for the classifications to be replicable by other researchers more precise definitions were needed,

⁴ Leonard Wrigley, "Divisional Autonomy and Diversification", (unpublished doctoral dissertation, Harvard Business School, 1970), p.111-8.

(3) it would be useful to have some sub-categories to describe certain observable and persistent differences in kind among the firms falling in several of the categories, and (4) vertically integrated firms posed special problems in classification. In the light of this experience it was decided that using the categories as defined by Wrigley was less important than incorporating the knowledge gained from a larger sample taken over a longer time into an improved set of categories. The new categories that were developed were similar in spirit and broad outline to Wrigley's, but differed in a number of important respects. The structure of the new categories will be described by discussing each of these differences in turn.

Discrete Businesses. Wrigley's placement of firms such as Texaco and Deere in the Dominant Product class indicates that he employed a broader concept of what constitutes a "product" than is usually associated with this term. In order to minimize any confusion over this point, a revised set of category titles was adopted: Single business, Dominant business, Related business, and Unrelated business.

Underlying these category titles and central to the categorization scheme is the idea that a corporation's activities can be conceptually divided into a number of distinct businesses. By a distinct business we mean one that could be managed independently of the firm's other activities. Since interdependencies are difficult to observe or judge in a

static situation, we assess the separability of a particular business by evaluating the degree to which its basic nature and scope could be altered without meeting constraints imposed by other of the firm's businesses and without materially affecting the operation and strategic direction of other activities.

The difficulty of breaking a firm's operations into distinct businesses is greatly reduced by the stipulation that classification depends only upon the relative size of the company's largest distinct business, so that only one line of demarcation need be drawn among each firm's activities. However, even in making this one judgement it was found that standard lists of businesses, such as the Standard Industrial Classification categories, were not helpful. Each company had a unique history and had developed its own pattern of relationships among technologies, products, and markets. What was a discrete business for one firm was often an integral and non-separable part of a larger business in another firm.

Since the factors influencing interdependency among business activities varied from firm to firm, it was not feasible to develop an exhaustive description of the properties of a distinct business. Instead, the approach taken was to

- (1) tentatively identify a firm's largest discrete business,
- (2) assess the results of applying a number of different types of strategic change to this or other of the firm's businesses,

and (3) evaluate the degree to which any of these changes would involve strong interdependency effects among the firm's businesses. The three types of strategic change that were found to be most helpful in assessing strategic interdependencies are discussed below, along with examples of their application to specific situations.

1. A decision to drop the product-market activity entirely or, conversely, to greatly increase its relative size. Consideration of this very basic type of strategic change highlights the degree to which a business activity is functionally related to other activities. Ford, for example, could, at least theoretically, quit the truck business completely without material repercussions on its automobile business. On the other hand, if Ford were to drop one of its automobile model lines, there would be price, quality and volume implications for each of its other lines of automobiles. Thus, the automobile business, taken as a whole, is a discrete business separate from the truck business.

When this type of strategic change is applied to a vertically integrated firm, product flow interdependencies are almost invariably present. Alcoa, for example, could not drop any of its fabrication businesses without affecting its basic aluminium production activities, nor could a paper company change its product mix from one emphasizing boxboard to one emphasizing white paper without making compensating changes in

the types of trees used for pulp and in its pulp processing technology and capacity.

2. A decision to employ a different production technology or process or to use a different type of raw material. An evaluation of the effects of this type of strategic change aids in the assessment of the degree to which a business activity is bound to others by production technology. A company, for example, that produces a wide variety of die castings for the automobile, aircraft, and industrial machinery industries might find that a changeover to plastics in a few high volume items would not leave enough casting business to support the overhead costs of the facility. By contrast, an "auto parts" producer that made springs, batteries, valves, spark plugs, and valves necessarily employs a variety of production technologies and the decision to produce any of these products by a different method would not strongly affect the other activities. Thus the "diversified" casting company consists of a single discrete business while the auto parts company may be divided into several distinct businesses (barring inseparability on the marketing end).

3. A decision to significantly alter the price, quality, or services associated with the product. Consideration of this type of strategic change aids in the identification of products that are part of "product lines" rather than distinct businesses. For example, Xerox's paper business

cannot be strategically separated from its copying machine business, nor can Crown Cork and Seal alter the nature of its custom packaging design services without considering the effect on its container production business. Similarly, a firm that produces both large electronic computers and also designs special purpose systems cannot lower the price of its general purpose machines without producing problems for its special systems business.

These criteria should make it clear that while our method of identifying discrete businesses does require the exercise of judgement on the part of the evaluator, it is consistent with the "top down" approach to the study of diversification. While it necessarily introduces problems associated with the lack of totally objective standards, it is doubtful whether such standards are attainable in business research of this type. Many researchers have commented on the failings of the SIC categories as definitions of "businesses", but argue that such an approach is at least objective. We would argue in turn that even more "objective" measures can be found but that they bear even less of a relation to the real issues under investigation. In this research we have sought to strike a balance between dealing with aspects of a situation that are measurable and dealing with the aspects thought to be most relevant. When in doubt, we have tended to favor relevancy over exactness.

The Specialization Ratio. Following the definition of a discrete business given above, it was necessary to re-define the specialization ratio as follows:

Specialization Ratio (SR): The proportion of a firm's revenues that can be attributed to its largest single business in a given year.

Wrigley's choice of a specialization ratio of 0.7 as the dividing line between the Dominant, and the Related and Unrelated groups was based on the empirical evidence he gathered:

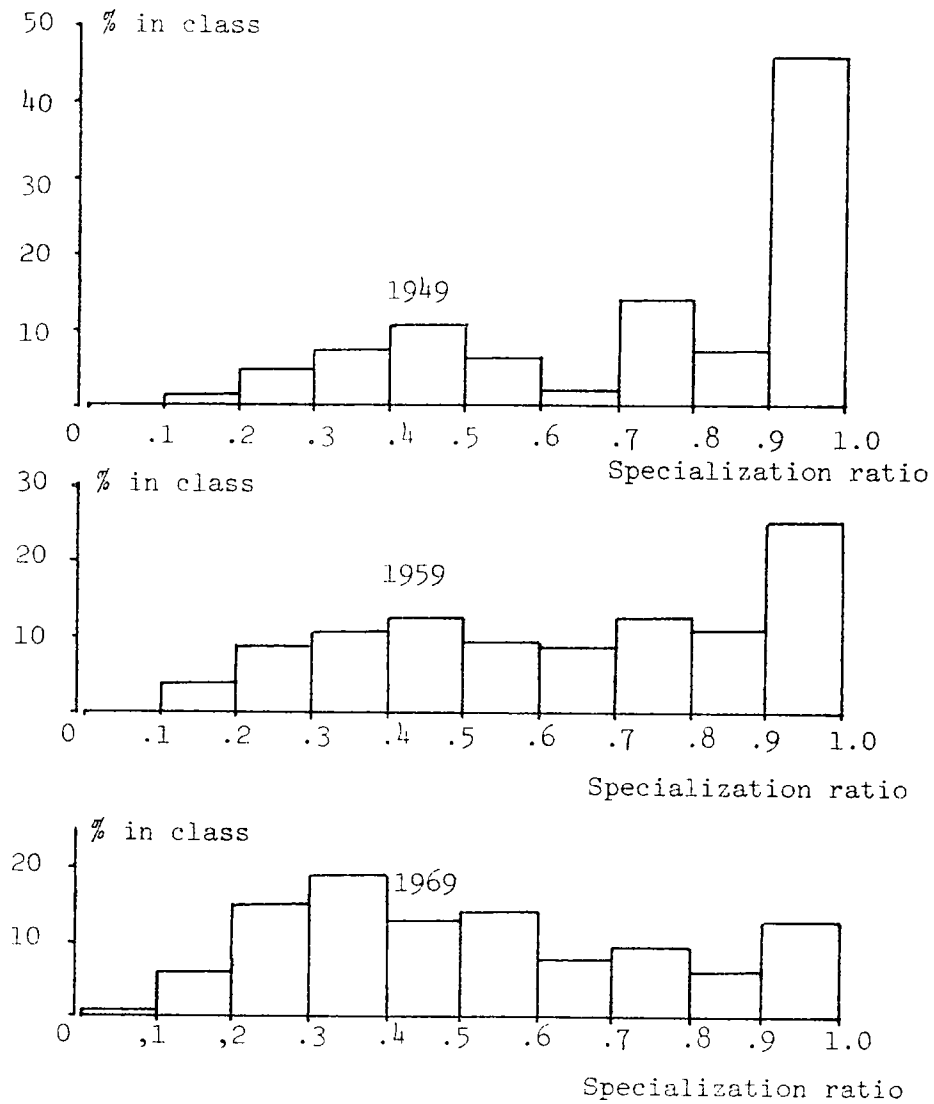
... at one stage in the research, a serious attempt was made to discover whether the basis of a diversification system could be the proportion of the "main" product in the total output. If this were successful it would have led to an ordered series of magnitudes of the kind (in per cent): 100; 90; 80; 70; 60; 50; 40; 30. However, the attempt met with a particular phenomenon among firms in the sample. While there were some 14 firms in the category 80-100 per cent, few firms were in the category 60-80 per cent. It seemed that in designing a strategy for expansion, businessmen did not adopt what might be called "half diversification"; they tended either to diversify by a small amount in relation to total output or to go the whole way.⁵

Figure 2-1 shows the distributions of specialization ratios for 1949, 1959, and 1969 that were obtained from the sample used in the present study. Due to many missing data points and the difficulty of identifying the specialization ratios of many highly diversified firms, these distributions must be considered suggestive rather than definitive. Still, their general shape tends to confirm Wrigley's findings.

⁵ Ibid., p.III-6.

Figure 2-1

FREQUENCY DISTRIBUTIONS OF SPECIALIZATION RATIOS
FOR 1949, 1959, AND 1969



Since no technical argument can be advanced for choosing any other figure in the range of 0.6 - 0.85, the 0.7 line of demarcation was retained.

The Related Ratio. A detailed inspection of the firms that Wrigley classed as Unrelated revealed some that this researcher judged belonged in the Related class, and vice versa. The judgements of several other individuals engaged in parallel research were obtained; the results revealed no consistent pattern of assessment. Although there was general agreement as to which businesses were unrelated to any other activity of the firm, there was no clear-cut agreement as to what constituted, in toto, an Unrelated firm. Consequently, a more precise definition of this class was sought.

The technique that was adopted was to introduce a new ratio, the related ratio, or RR, defined as the proportion of a firm's revenues attributable to its largest group of related businesses. For example, North American Rockwell's distribution of revenues by business area in 1969 was approximately as follows:⁶

⁶ This information and similar information on other companies presented elsewhere in this chapter were obtained from annual reports published by the firms in question.

<u>Business Area</u>	<u>Percentage of Total Revenues</u>	
Aircraft and missiles	14.3	) 61.2
Rocket engines	6.7	
Aerospace systems	19.5	
Aerospace electronics	20.7	
Auto parts	20.8	) 25.4
Industrial machine parts	4.6	
Textile machinery	3.8	
Graphic arts equipment	4.6	
Other	5.0	

The two major groups of related businesses within North American Rockwell were (1) the set of businesses related by aerospace technology and the NASA-government market and (2) the auto and industrial parts businesses. The former was the larger, in terms of revenues, so that North American's related ratio for 1969 is taken to be 0.612. As can be seen from this example, the related ratio is particularly helpful in dealing with firms which are active in several unrelated areas, each of which is a diversified business in its own right. It also minimizes the number of judgements that must be made. In this case, the questions of whether or not the textile machinery business and the graphic arts equipment business were related, or whether either or both was related to the parts manufacturing business, need not be answered as the related ratio would remain 0.612 in any case.

The dividing line between Related businesses and Unrelated business firms was taken to be a related ratio of 0.7. Thus North American would be classed as an Unrelated business firm in 1969. The seventy per cent cut-off was chosen because it seemed to match fairly well the judgements expressed by informed observers. Also, setting the critical RR equal to the critical SR ensures that a company cannot qualify for the Dominant category on the basis of its SR and, at the same time, qualify for the Unrelated category on the basis of its RR.

Constrained and Linked Relatedness. Before the work of actually classifying the firms selected for the study was begun, it was expected that it would be useful to sub-divide the Related business category into three sub-classes. These sub-classes would serve to differentiate among those firms that had diversified by relying chiefly on (1) relationships among markets served and distribution systems, (2) relationships based on similar production technologies, or (3) the exploitation of science-based research. It was found, however, that while individual diversification moves were easily placed in one of these sub-classes, few firms had used one type of relationship with enough consistency to warrant its use as characterizing the firm's strategic posture as a whole. Consequently, the attempt was abandoned.

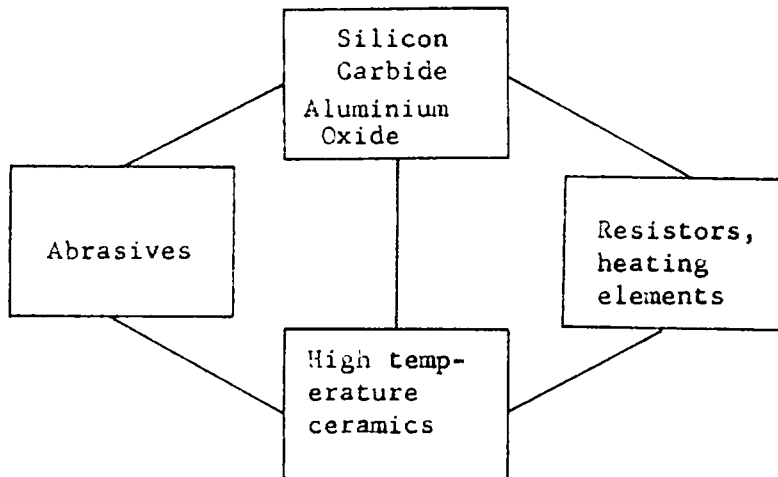
A search through the Related group for other types of patterns revealed that a distinction could be drawn between firms that had stayed relatively "close to home" and those which had related new activities to old in such a way that they were eventually active in businesses which, considered by themselves, were virtually unrelated. The former type of strategic posture was termed Related Constrained, and the latter Related Linked.

The change in strategic posture that occurred in Carborundum, Inc. between 1949 and 1969 is illustrative of this distinction. (See Figures 2-2 and 2-3). In 1949 all of Carborundum's businesses were closely related to the firm's basic strengths: the efficient production of high quality grains of silicon carbide and aluminium oxide, and the competence in the materials sciences necessary to engineer these materials to various uses. These materials were sold in bulk, fabricated into bricks and tiles for high temperature applications, bonded or coated to produce abrasives, or made into electrical resistors and heating elements. The firm was Related Constrained because, although it was diversified, each business was related to each other business and all could be seen as radiating from a common core.

During the 1950's, Carborundum began to produce a line of grinding, cutting, and polishing machines to complement its line of abrasives. Then, beginning in 1962 a Related

Figure 2-2

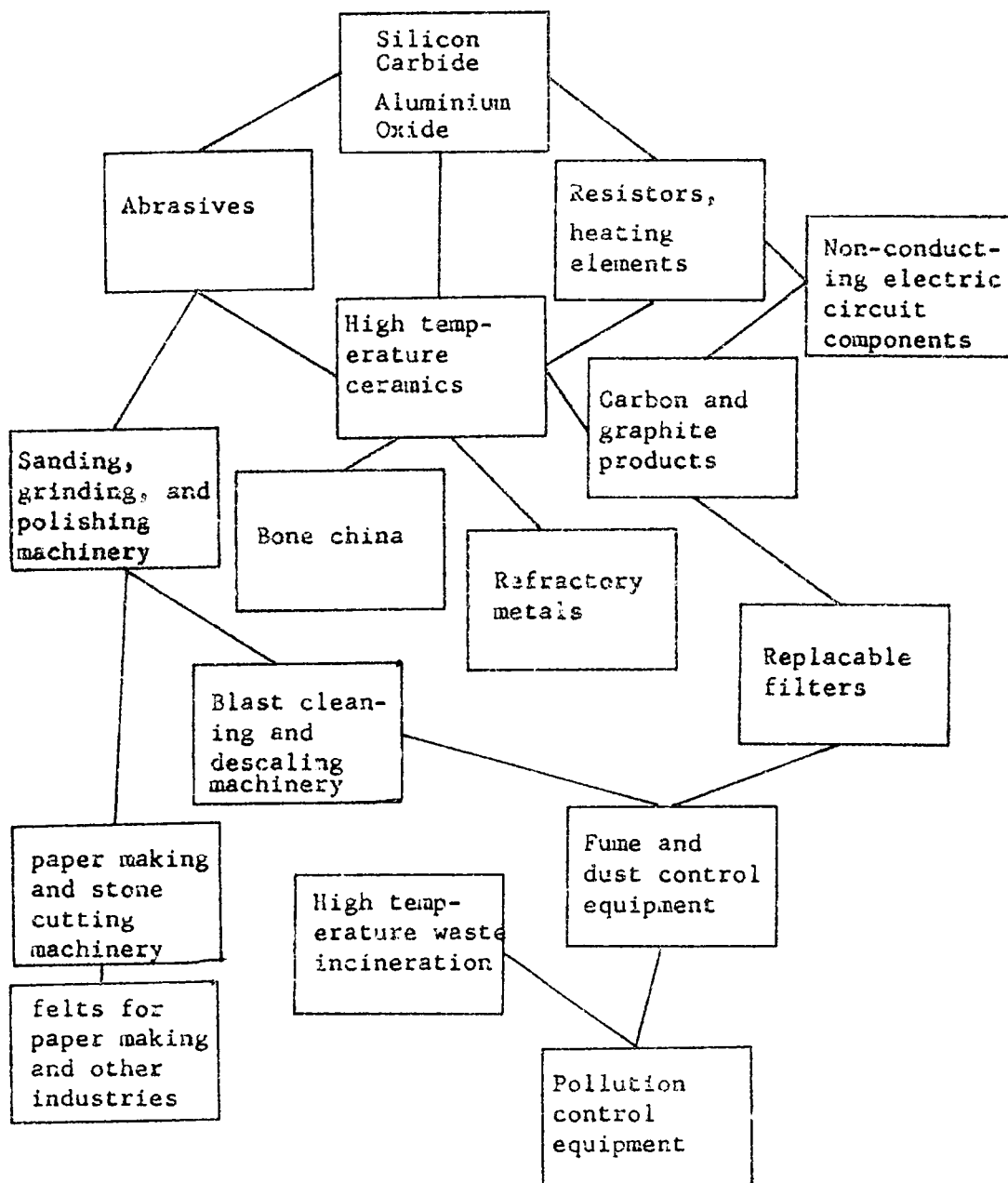
BUSINESSES OF CARBORUNDUM, INC. IN 1949



SOURCE: Company annual report.

Figure 2-3

BUSINESSES OF CARBORUNDUM, INC. IN 1969



SOURCE: Company annual reports.

Linked strategy was inaugurated. Skills acquired in manufacturing abrasive machines were applied to other types of industrial machinery, and non-abrasive cleaning and descaling equipment was added. Ceramics technology led to ceramic electrical components, and carbon and graphite materials, which in turn provided a base for expansion into filtering media. High temperature technology led to the refractory metals business. Recently, the company's activities in filtering media, cleaning machinery, general industrial machinery, and high temperature technology have been brought together through a position in the solid waste disposal and pollution control fields. Carborundum's 1969 posture is thus Related Linked; by adding new businesses in such a way that each was related to at least one - but often no more than one - of its current activities, the firm gradually became involved in a linked network of widely disparate businesses. Examples of other Related Linked firms in 1969 are General Electric, Borden, and Monsanto.

As aids to deciding if a firm was Related Linked or Related Constrained, these questions were found to be useful: (1) "Is each of the firm's businesses related to virtually all of its other businesses, taken one at a time?", and (2) "If not, did logical relationships exist between each business and some other business of the firm at the time it was added?". In cases involving a mix between the two postures, the firm was

classified according to the group of businesses (constrained or linked) that, in terms of revenues, was the larger.

The Dominant business group can also be sub-divided along these lines. The Dominant Constrained and Dominant Linked categories represent the same type of diversification pattern as the Related Constrained and Related Linked categories, except that total diversity is less, the largest single business contributing between 70 and 95 per cent of total revenues.

The Dominant Unrelated sub-class was created to deal with a Dominant business firm whose minor business or businesses were unrelated to its major business. An example is Ford Motor - while basically an automobile company, its electronics business (Philco) places it in the Dominant Unrelated category. The quantitative criteria for a company to be in the Dominant Unrelated class were $SR \geq 0.7$ and that revenues received from activities unrelated to the largest group of related businesses be larger than revenues due to businesses related, but not including, the largest single business. The latter requirement is equivalent to $RR < \frac{1}{2} (SR + 1)$.

Vertically Integrated Firms. Often having grown by absorbing their suppliers and industrial customers, vertically integrated firms possess what Thompson⁷ calls "long-linked

⁷ James D. Thompson, Organizations in Action, (New York, N.Y., McGraw-Hill, 1967), p.15.

technologies"; the company's major activities consist of stages in the sequential processing of a particular material from its raw form to a finished product, or a variety of products. These firms are mainly found in the oil, rubber, basic metals, and forest products industries.

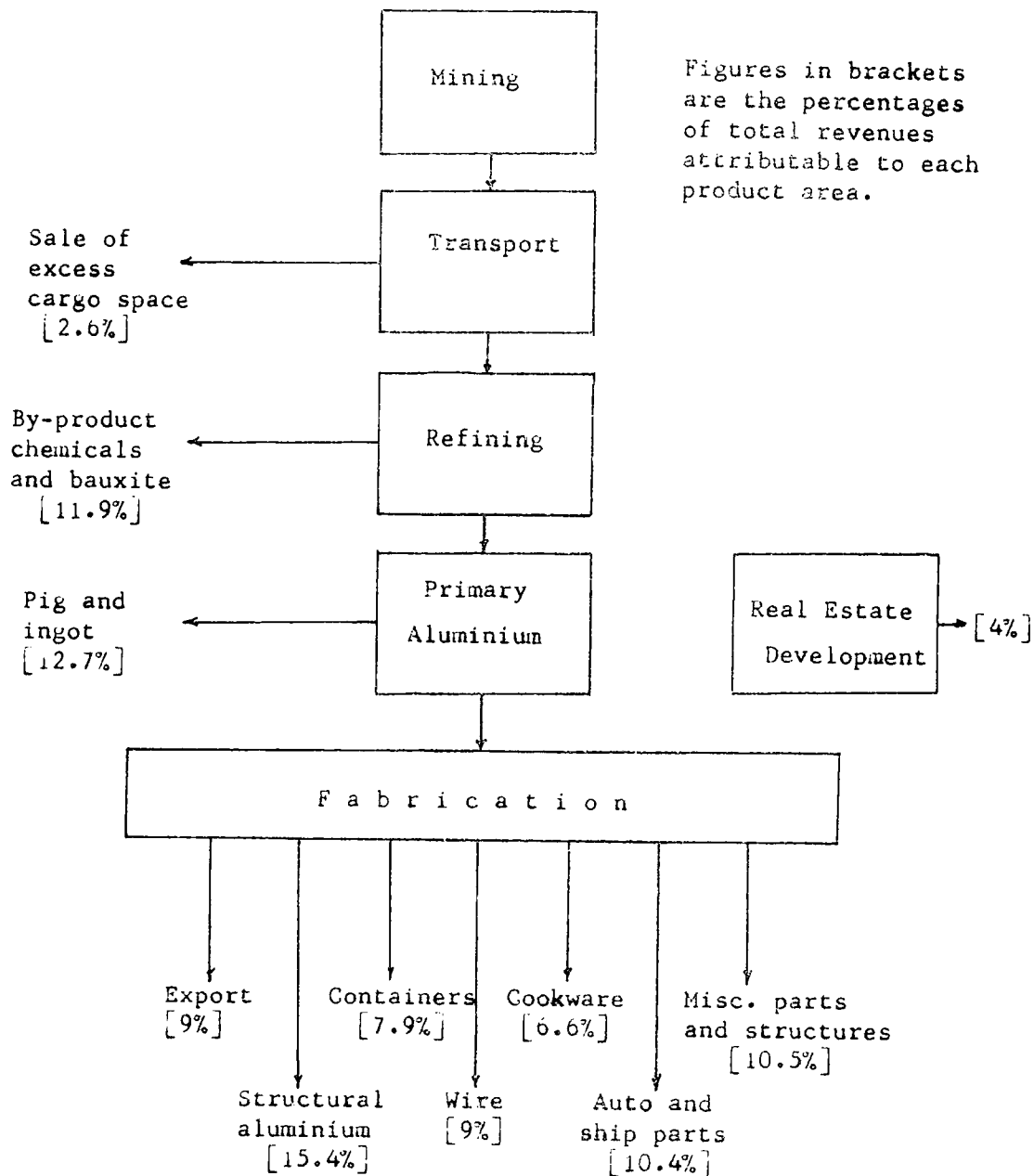
Before the research began, no special provisions were made for dealing with vertically integrated firms other than the stipulation that diversification was to be considered in terms of products sold. However, it was found that vertically integrated firms often did not seem to belong in any of the categories defined thus far. Attempts to classify these firms as either Single, Dominant, or Related businesses invariably required either too broad or too narrow a definition of "business". The two difficult questions that kept recurring were (1) Is a by-product a separate business or part of the business which produces it?, and (2) Under which conditions should a firm that has integrated forward into a wide variety of manufacturing activities be considered diversified?.

Examples of both these issues are provided by the Aluminum Company of America (Alcoa), whose materials flow and sales breakdown for 1969 are shown in Figure 2-4. Some descriptive statistics on Alcoa are:

Largest single product area (structural aluminum)	= 15.4% of sales
All aluminum fabrication	= 68.8% of sales
All aluminum and by-products	= 93.4% of sales

Figure 2-4

PRODUCT FLOW AND REVENUE BREAKDOWN FOR
THE ALUMINIUM COMPANY OF AMERICA IN 1969



SOURCE: Company annual report.

Simply stated, the problem is whether Alcoa is an aluminum company or a diversified producer of aluminum products. If Alcoa is the former, it should be classed as Dominant. However, applying this same broad view of a "business" to other firms results in Dow being a "chemical company", General Foods being a "food company", and so on. Such definitions would be absurd in a study of diversification, as they obscure the very differences with which we are concerned.

If, on the other hand, Alcoa is classed as a diversified producer of aluminum products, and therefore as a Related firm, another type of distortion has taken place: in terms of either assets or general management's tasks Alcoa is not really diversified. The preponderance of the company's assets are devoted to the extraction, transportation, and refining of ore, and the production of pig, sheet, bar, and tube aluminum, Alcoa's fabrication activities having historical roots in the need to increase primary demand by showing the variety of uses of aluminum. Much of the management task in this type of firm has to do with coordinating the elements in the processing chain; the emphasis is on achieving balance, efficient through-put, and ensuring that production capacity does not exceed or fall short of demand. The interdependencies among the businesses of a vertically integrated firm mean that, no matter how many different forms the end products take, general management must view the firm as a whole when considering

the effect of any change in operations or resource allocation.

Another example of the vertically integrated firm is United States Steel. It is a fully integrated steel producer which appears on most lists of "highly diversified firms" when the measure is product count. Yet this recent description of U.S. Steel's business, approved by its own management, highlights the inter-relatedness of the product-market activities of the vertically integrated company:

United States Steel Corporation is engaged in integrated steel operations, producing and selling iron, steel and related products in a variety of forms. The principal forms include: ingots, blooms, billets, slabs, tube rounds and skelp; structural shapes and piling; plates (and many other steel mill products) ... also, certain raw materials, pig iron, ferro-manganese, coal chemicals, and other items connected with, or closely related to, iron and steel making activities are produced and sold.

Integrated operations include the production of substantially all the iron ore and limestone, more than half the coal and part of the manganese ore used; the operation of coke ovens which supply the coke, coke oven gas and tar used; the generation of almost half the electric power required; and the operation of steamships, tugs, barges, and docks for the transportation of raw materials and steel.

The Corporation is engaged, through the American Bridge Division, in the fabrication and erection of bridges, buildings and other steel structures and in the manufacture of large diameter steel pipe. Various other divisions manufacture steel drums, oil field drilling and pumping equipment, and other items made of steel...

U.S. Steel produces and sells chemicals, a material amount of which is derived from the coal coking operations, through its divisions, USS Chemicals and USS Agri-Chemicals. These include agricultural chemicals, tar-based enamels for pipelines, other industrial chemicals and basic materials for the plastics fabrication operations.

Subsidiaries servicing the above operations include export distributors and a number of transportation companies including several common carrier railroads... Two transportation companies provide ocean transportation for the Corporation and others.

The products from the above operations are characterized by common raw material sources, interrelation of processed materials and substantial inter-plant, inter-division and inter-company transfers to such an extent that on a consolidated basis such products and the services related to them are considered to be a single line of business which accounted for approximately 95% of the Corporation's products and services sold, and in excess of 90% of its income before estimated United States and foreign taxes on income, in each of the years 1967 through 1970. (Emphasis added)⁸

In principle, vertically integrated firms like Alcoa and U.S. Steel are sufficiently different from other firms to warrant an entirely separate set of categories to describe their diversification postures. However, it did not seem wise to overly complicate an already expanded classification system to deal with this numerically small sub-set of all large firms. Therefore, a compromise approach was developed. Vertically integrated firms that had 95% or more of their sales in a single end-product business (such as sheet steel, petroleum refinery products, or white paper) were classed as Single businesses. Those in which the sales of all intermediate and end products associated with the vertical chain comprised less than 70% of total revenues were treated as either Related or Unrelated businesses, depending on the related ratio. The rest,

⁸ Prospectus for United States Steel Corp. \$150 million 7-3/4% Sinking Fund Debenture due 2001, (New York, N.Y., Feb. 23, 1971), p.6.

which were the majority, were placed in a new sub-class of Dominant business firms, the Dominant Vertical class.

The rationale for this procedure is that, regardless of the degree of variety of end product forms, the company that has the preponderance of its output in areas linked to a vertical chain of processing steps is, viewed from the top down, more akin to a Dominant business than to a truly diversified firm. When the output of the vertical chain is in only one product-market area (i.e., mill products or paper), there is, of course, no question about classifying the company as a Single business firm.

To make these ideas more precise, we define the Vertical ratio (VR) in any given year as the proportion of the firm's revenues that arise from all by-products, intermediate products and end products of a vertically integrated sequence of processing activities. The vertical ratio, specialization ratio, and related ratio are used in combination to classify the firm as follows:

Vertical ratio greater than or equal to 0.7:

Single business if SR is 0.95 or greater
Dominant vertical if SR is less than 0.95

Vertical ratio less than 0.7:

Related business if RR is 0.7 or greater
Unrelated business if RR is less than 0.7

Conglomerates: Generally thought to be a fairly recent phenomenon, the conglomerate is a company that has rapidly diversified into several unrelated areas via a relatively large number of mergers and acquisitions. While some writers have argued that firms like Westinghouse, selling products as unrelated as epoxy resin, television sets, and nuclear reactors, are conglomerates, we have classified such firms as related linked, because of the way in which they achieved diversification. Berg calls this type of firm a "diversified major" and has demonstrated that its approach to general management, particularly at the headquarters level, is quite different from that taken by conglomerates.⁹ He found that conglomerates tend to be quite "thin" at the top, concentrating few headquarters personnel in the finance, legal and planning functions. Their administrative structure is fashioned simply to watch over and allocate capital among a portfolio of businesses, there being no central R&D or central staff coordinating offices.

In a thorough study of the financial strategies and performances of acquisitive conglomerates, Lynch¹⁰ offers

⁹ Norman A. Berg, "Corporate Organization in Diversified Companies", (unpublished working paper, Harvard Business School, 1970), pp.32-36.

¹⁰ Harry H. Lynch, Financial Performance of Conglomerates, (Boston, Division of Research, Harvard Business School, 1971), pp.67-68.

evidence that their typically high rates of growth have been achieved by creating a positive feedback between their growth in earnings per share and the market-determined price-earnings ratio; a high price-earnings ratio permits the acquisition of firms to be accomplished with little dilution, which in turn insures a high growth in earnings per share, thus justifying and maintaining the high price-earnings ratio.

Although an acquisitive conglomerate is by definition an Unrelated business firm, not all Unrelated business firms are conglomerates. Companies such as Rockwell Manufacturing, Midland-Ross, and Pullman, while active in unrelated areas, have not experienced particularly high growth rates or price-earning ratios in the last twenty years, nor have they engaged in ambitious acquisition programs. As will be seen in Chapter III, this "Unrelated Passive" group is, numerically, at least as important as the group of true conglomerates.

The distinction between these two types of Unrelated business firms was incorporated in the classification scheme by defining two sub-classes to the Unrelated business category: the Acquisitive Conglomerate group and the Unrelated Passive group. The criteria used to delineate the Acquisitive Conglomerate class were essentially those developed by Lynch. To qualify as a member of the Acquisitive Conglomerate sub-class in any given year a firm had to fall in the Unrelated business category ($RR < 0.7$) and, over the preceeding five years, (1) had

to have experienced an average growth rate in earnings per share of at least 10% per year, (2) had to have made at least five acquisitions, three of which were diversification moves into new business areas unrelated to previous activities, and (3) had to have issued new equity shares whose cumulative market value (taken at the time of issue) was greater than the cumulative value of dividends paid during the same period. Taken together, these criteria identify firms which have been growing rapidly by acquisition of unrelated businesses, with the acquisition program chiefly financed through the issue of new equity shares.

Using the Diversification Categories - Examples

Careful definition of concepts and the use of a number of quantitative criteria increase the likelihood that a particular firm will be similarly classified by independent researchers with access to the same data. Nevertheless, that classification process is not algorithmic; it requires the exercise of a reasonable amount of judgement. In order to show how the classification process works, indicate the kind of company that each class represents, and communicate a feeling for the type of judgements that were made, a number of examples are presented.

GAF. Known as General Aniline and Film before 1968, GAF's most widely recognized brand name was Ansco, used to denote its line of photographic supplies and films. Upon the

The company was not vertically integrated and it was judged that all of its businesses were related in some way to some other of its businesses, so $RR=1.0$. It is clear from the sales breakdown that the largest single business contributed no more than 33% of total revenues. Whether or not the Ansco business is a discrete business is quite irrelevant for classification purposes - SR is obviously less than 0.7. Therefore GAF falls in the Related business class. The various chemical activities constitute a "constrained" type of diversification while the Ansco and Ozlid businesses are "linked" to the chemical business. Both groups provide 50 per cent of revenues so the decision as to whether GAF was Related Constrained or Related Linked in 1959 could go either way. It was classed as Related Linked on the basis of its direction of movement.

Between 1959 and 1969 GAF moved into office copiers, agricultural and pharmaceutical chemicals, slide projectors and prepared slides, acquired a manufacturer of a variety of business forms and in 1967 acquired Ruberoid. The latter firm was itself a diversified producer of asphalt roofing and siding, tile and sheet floor coverings, and asbestos belts and insulatory materials. GAF's 1969 sales breakdown was:

Chemicals	27
Ansco	24
Business forms, copiers reproduction, microfilm	13
Builders' roofing, siding flooring	29
Asbestos belts, filler, insulation	8

The Ruberoid acquisition was judged to be an unrelated move. Any relations between Ruberoid's business and GAF's were tenuous at best and there was no evidence that management had sought to build relationships. Hence, by 1969 RR had dropped to 0.64, placing GAF in the Unrelated business class. However, since GAF did not follow up its one unrelated move with others, in fact making only one other acquisition between 1967 and 1969 (which was horizontal), we judged it not to be a conglomerate. Hence, it was classed as Unrelated Passive in 1969.

Litton Industries. Incorporated in 1953, by 1959 Litton had rapidly diversified by acquisition into these major areas:

<u>Business</u>	<u>Percentage of Total Revenues</u>
Business calculators & accounting machines (Monroe)	38
Electronic communications, movie & sound recording	9
Military electronics and systems	38
Electronic components	14

In this case the classification rests on whether or not the business machines business is related to other of Litton's activities. We judged that it was, given Litton's experience in electronics and strategy of modernizing Monroe's line of calculators. In 1959, Litton was in the Related Linked category.

By 1969 Litton had made an astounding number of

acquisitions, many of them unrelated to past businesses.

The largest single business (internal guidance, navigation and control systems) contributed only 10 per cent of revenues.

The largest group of related businesses (calculators, computers, data systems, and electronic components) contributed 30 per cent of all revenues. With a ten year average growth in earnings per share of 23 per cent, a price earnings ratio of 36, and a growth in equity three times as large as retained earnings, Litton was clearly an Acquisitive Conglomerate.

Philip Morris. The manufacture and sale of cigarettes, regardless of the number of brands, is, by our definition, a single business. In 1949 Philip Morris derived 98% of its revenue from cigarettes and the remainder from pipe tobacco. Thus, in 1949 Philip Morris fell in the Single business category even if pipe tobacco were to be judged a different business.

By 1959, sales of the company's tobacco products had dropped to 86 per cent of total revenues. The remaining 14 per cent were generated by two acquired firms which specialized in glassine paper, packaging adhesives, textile processing chemicals and printing on difficult to print substances (i.e., slick transparent papers). While some of these activities had slight connections with Philip Morris' tobacco products packaging activity, the company's original competence in this area did not go beyond the minimum required and there was no evidence that potential relations were exploited. Hence, Philip Morris,

in 1959, had $SR=0.86$ and was classed as Dominant Unrelated.

By 1969 Philip Morris had acquired the Miller Brewing Company, and a number of other firms producing gum, razor blades, labels, toiletries, and surgical blades and hospital supplies. Tobacco products accounted for 75 per cent of revenues, and beer was the largest non-tobacco business, accounting for 15 per cent of total revenues. Thus, the company still fell in the Dominant Unrelated class. Note that categorization did not depend on whether gum, razor blades, medical products and packaging were related to one another or any other business. The company was Dominant because of tobacco and Dominant Unrelated because beer (unrelated to tobacco or any of the other activities) constituted more than one-half of non-tobacco activities.

AMAX. In 1949 AMAX (previously American Metal Climax) was an integrated producer of refined copper, lead, zinc, silver and gold. Ores were obtained both competitively and through mining subsidiaries and joint ventures. Fifty-four per cent of AMAX's sales were of copper, giving it an SR of 0.54. However, the company's mining, transportation and refining activities for copper, lead and zinc were not separate; the similarities in ores and refining processes among these metals having been exploited by vertically integrated operations encompassing all three. Sales of copper, lead and zinc were 74 per cent of total sales so $VR=0.74$. Thus, AMAX was in the

Dominant vertical category in 1949.

During the next decade AMAX moved into potash mining, oil and gas exploration and production, light metals, and merged with Climax Molybdenum. By 1959 the company's sales breakdown was:

<u>Business</u>	<u>Percentage of Total Revenues</u>
Copper	59
Lead, zinc	6
Tin, silver, and gold	14
Molybdenum	9
Uranium and vanadium	1
Potash	2
Oil and gas	9

Since $VR=0.65$, less than 0.7, AMAX no longer fell in the Dominant Vertical class. An $SR=0.59$ placed it out of the Dominant category and since the only candidates for unrelated activities (potash and oil and gas) had sales less than 30 per cent of total sales, AMAX was placed in the Related class. AMAX had obviously applied the constraints of "primary metals" and/or "mining and mineral extraction" to its diversification strategy so it was placed in the Related Constrained sub-category.

In the sixties AMAX moved into the aluminum business, first by acquiring smelting fabrication facilities and then by integrating back into mining. It also entered the concrete, general building products, and mobile homes businesses but sold off these operations and most of its oil and gas interests in 1966-7. In 1969 it merged with Ayrshire Collieries Corp., a

coal company. Its 1969 sales breakdown was:

<u>Business</u>	<u>Percentage of Total Revenues</u>
Coal, potash, oil	6
Copper, lead and zinc	36
Aluminum	37
Molybdenum and other refractory metals	21

The specialization ratio is difficult to estimate but is certainly less than 21 per cent. The constraints on the company's diversification did not seem to have changed so it was again placed in the Related Constrained group.

Cincinnati Milacron. In 1949 Milacron's (previously Cincinnati Milling Machine Corp.) basic business was the manufacture and sale of machine tools, machine tool parts and cutting fluid. The company also produced several chemicals used in cutting fluids and sold these and related chemicals to industrial customers for use in lubricating oils, food, soap, and other products. In addition, Milacron marketed a limited line of tools for home use and engaged in contract construction of machines designed and sold by others. While no sales breakdown was available, an examination of Milacron's sales over time, its sales breakdown in later years and total U.S. machine tool sales indicated that machine tools and cutting fluid accounted for substantially more than 70 per cent of total sales in 1949. Hence, Milacron was classed as Dominant Constrained in that year.

During the nineteen fifties Milacron introduced a line of grinding wheels, extended its chemical business into the area of "organo-metallic" compounds for use in paints and plastics and developed a line of heat treating and metal forming equipment. In 1959 sales of machine tools, cutting fluid and grinding wheels constituted 83 per cent of total sales, so that it still fell in the Dominant Constrained class.

Between 1959 and 1969 Milacron introduced a line of plastic processing machinery and a laminater for printed circuits. The machine tool business was radically altered by the new technology of numerical control and Milacron developed a competence in special purpose computers, software, and the design and development of automated systems for metal forming, machining and industrial process control. In 1969 Milacron began to market a special purpose digital computer and its sales of machine tools had dropped to 73 per cent of total sales. Thus it was categorized as Dominant Linked.

Summary of Diversification Categories

In summary four major categories of diversification strategy have been defined and all but the Single business category have been further divided into sub-categories, giving a total of nine different classifications. Figure 2-5 presents a flow diagram that describes the process of categorizing a firm and Figure 2-6 provides a graphical clarification of the relationships between the ratio criteria and the categories.

To facilitate further reference to the diversification categories that have been defined and discussed in this section the following condensed descriptions are provided.

A firm's diversification strategy is defined as its commitment to diversity per se together with the strengths, skills or purposes that span this diversity, as evidenced by the way in which business activities are related one to another.

The primary measure of diversity is taken to be the specialization ratio, defined as the proportion of a firm's revenues that is attributable to its largest discrete product-market activity. A discrete business (or product-market activity) is one which is strategically independent of the firm's other businesses in the sense that basic changes in its nature and scope can be made without meeting constraints imposed by other of the firm's businesses and without materially affecting the operation and strategic direction of other of the firm's businesses.

Businesses are related to one another when a common skill, resource, market, or purpose applies to each. A firm's related ratio is defined as the proportion of its revenues that are attributable to the largest group of businesses that are related on some way to one another. Each member of this group need only be related to one other business in the group (linked relatedness), though it may be related to all and all may be directly related to one another (constrained relatedness).

The vertical ratio is defined as the proportion of a firm's revenues attributable to all of the by-products, intermediate products and final products of a vertically integrated sequence of manufacturing operations.

In any given year a firm's diversification strategy may be described as corresponding to one of the following categories:

- I. Single business (abbreviated "S" or "SING") - firms that are basically committed to a single business. Among non-vertically integrated firms ($VR < 0.7$), Single business companies are those with specialization ratios of 0.95 or more. Among vertically integrated firms ($VR \geq 0.7$), those that have an end-product business that contributes 95% or more of total revenues are classified as Single business firms.
- II. Dominant business (abbreviated "D" or "DOM") - firms that have diversified to some extent but which still obtain the preponderance of their revenues from a single business. Among non-vertically integrated firms ($VR < 0.7$), those with specialization ratios greater than or equal to 0.7 but less than 0.95 are Dominant business firms. Among vertically integrated firms ($VR \geq 0.7$), those that do not qualify as Single business companies fall in the Dominant category.
 - A. Dominant Vertical (abbreviated "DV") - vertically integrated firms (having vertical ratios of 0.7 or more) which produce and sell a variety of end-products, no one of which contributes more than 95% of total revenues.
 - B. Dominant Constrained (abbreviated "DC") - non-vertical Dominant business firms that have diversified by building on some particular strength, skills, or resource associated with the original dominant activity. In such firms the preponderance of the diversified activities are all related one to another and to the dominant business.

- C. Dominant Linked (abbreviated "DL") - non-vertical Dominant business firms that have diversified by building on several different strengths, skills, or resources or by building on new strengths, skills or resources as they are acquired. In such firms the preponderance of the diversified activities are not directly related to the dominant business but each is somehow related to some other of the firm's activities.
- D. Dominant Unrelated (abbreviated "DU") - non-vertical Dominant business firms in which the preponderance of the diversified activities are unrelated to the dominant business.

III. Related business (abbreviated "R" or "REL") - non-vertically integrated firms which are diversified, having specialization ratios less than 0.7, and in which diversification has been primarily accomplished by relating new activities to old so that the related ratio is 0.7 or more.

- A. Related Constrained (abbreviated "RC") - Related business firms that have diversified chiefly by relating new businesses to a specific central skill or resource and in which, therefore, each business activity is related to almost all of the other business activities.
- B. Related Linked (abbreviated "RL") - Related business firms which have diversified by relating new businesses to some strength or skill already possessed, but not always the same strength or skill. By diversifying in several directions and exploiting new skills as they are acquired, such firms have become active in widely disparate businesses.

IV. Unrelated business (abbreviated "U" or "UNRL") - non-vertical firms which have diversified chiefly without regard to relationships between new businesses and current activities. Such firms are defined by a related ratio of less than 0.7.

- A. Unrelated Passive (abbreviated "UP") - Unrelated business firms that do not qualify as Acquisitive Conglomerates (see definition below).
- B. Acquisitive Conglomerates (abbreviated "AC") - Unrelated business firms which have aggressive programs for the acquisition of new unrelated businesses. More specifically, such firms are defined as having, over the past five years, (1) had an average growth rate in earnings per share of at least 10% per year, (2) made at least five acquisitions, at least three of which took the firm into businesses unrelated to past activities, and (3) issued new equity shares whose total value (using market prices at the time of issue) was at least as great as the total amount of common dividends paid during the same period.

Figure 2-5

ASSIGNING DIVERSIFICATION CATEGORIES

SR = Specialization Ratio

RR = Related Ratio

VR = Vertical Ratio

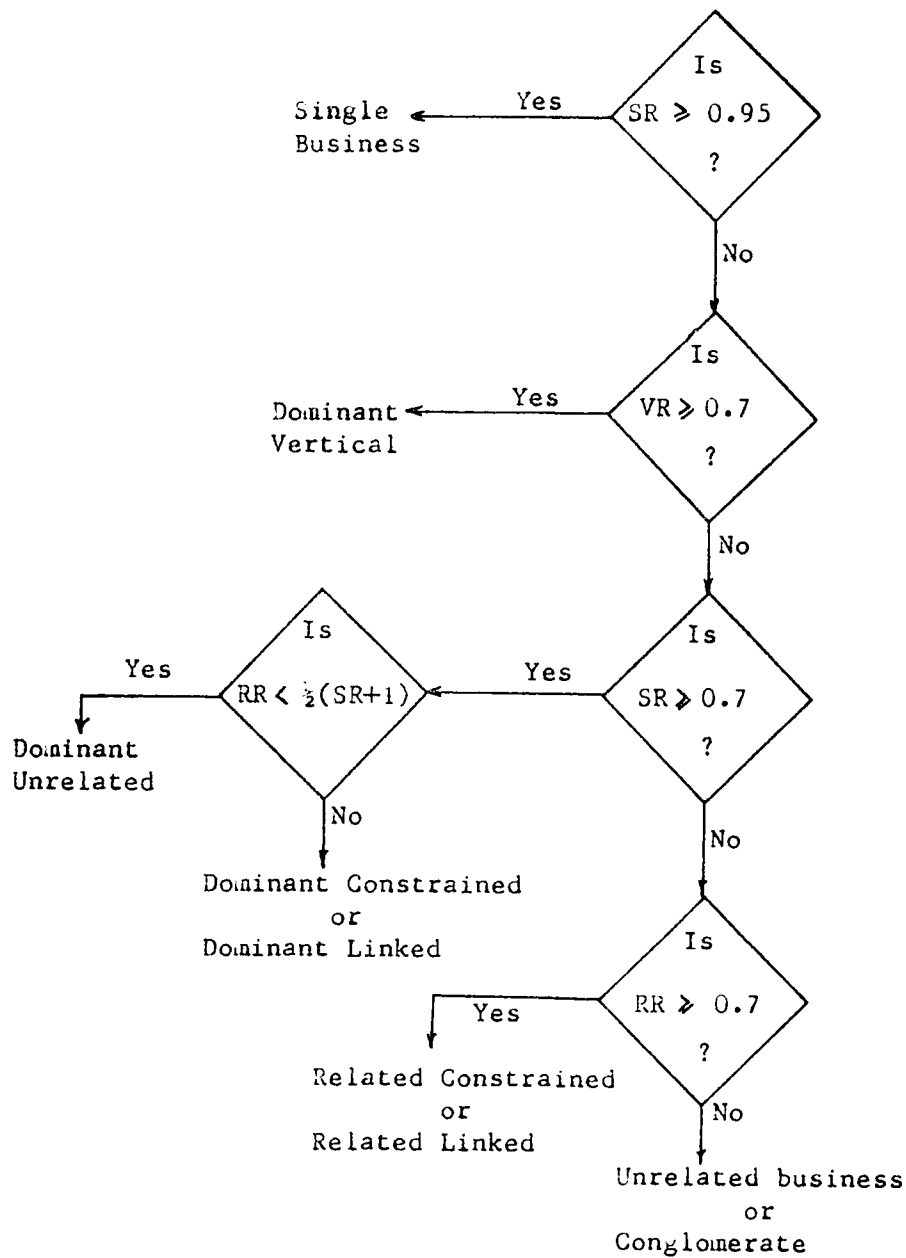
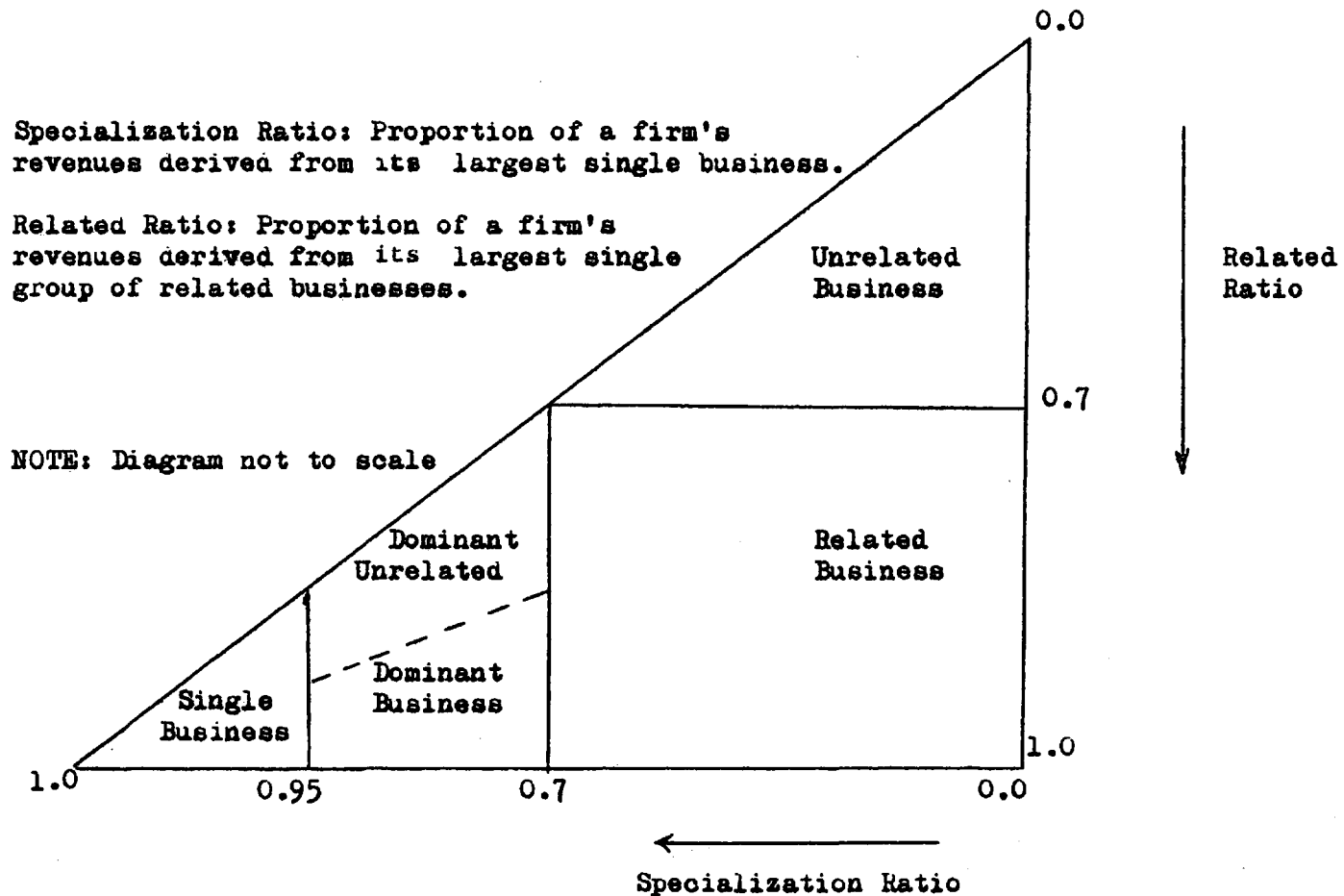


Figure 2-6

STRATEGIC CLASSES DEFINED IN TERMS OF
THE SPECIALIZATION RATIO AND THE RELATED RATIO



Categories of Organizational Structure

No two firms possess the same organizational structure and the number of possible structures is virtually unlimited. As was done in the case of diversification, the enormous complexity of reality will be simplified by looking at the firm from the "top down", focusing on the major organizational options open to top management and their potential impact on the general management task.

General management is the management of a complete economic unit: a firm or a sub-unit of a firm that has sufficient resources and autonomy to act on its own behalf in the market place. A manager whose responsibilities include the design, development, manufacture and sale of any particular product or set of products is a general manager as he commands a complete economic unit. On the other hand, a manager whose task is, for example, the efficient production of automobiles, trucks, and tractors, perhaps on a world-wide scale, is not a general manager; his responsibilities are functional and must be coordinated with the engineering, research, marketing, and finance functions on a routine basis.

Consideration of the commonly employed variations in the organizational locus of general management activity and the rationale for specialization of major organizational sub-units leads directly to three basic categories of organization. In the functional structure the major sub-units deal with business

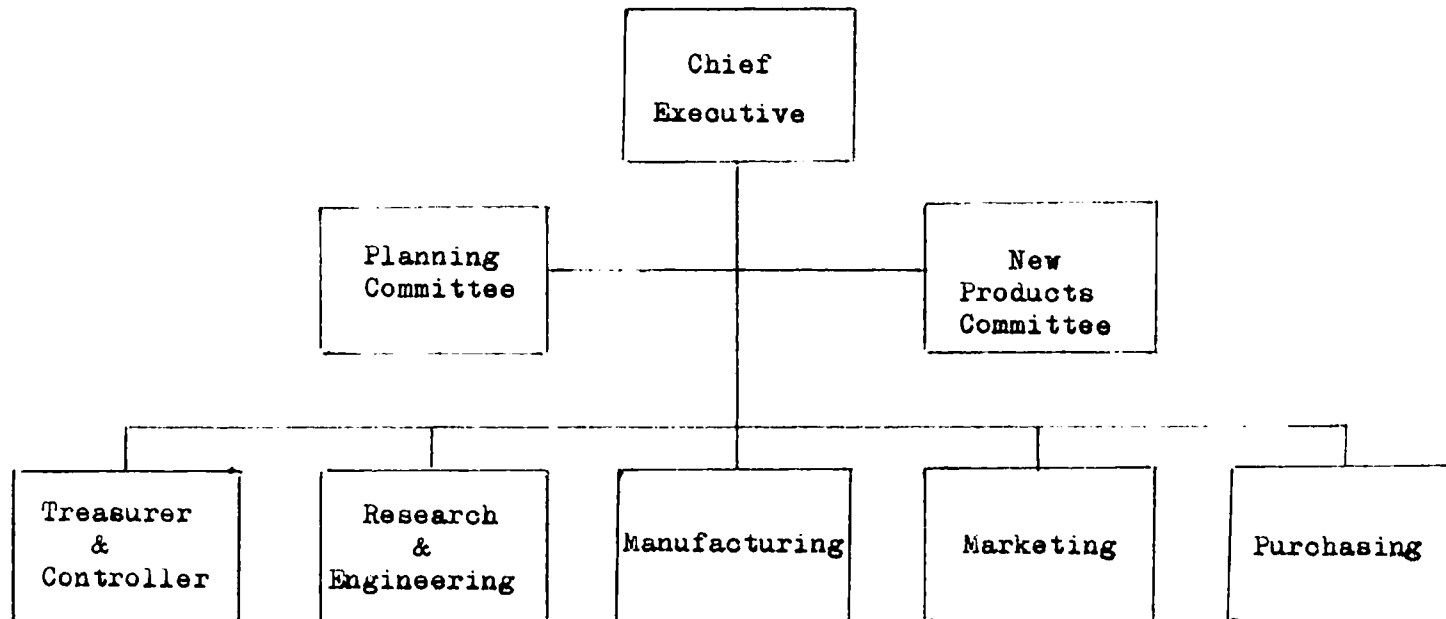
functions, such as sales, engineering and production, rather than complete businesses. General management occurs solely at the topmost level and the coordination of the functional sub-units is one of its important responsibilities. In product division organizations the organization is split into a number of quasi-autonomous divisions, each headed by a general manager and supplied with the resources necessary for it to operate as an independent economic entity. The product division structure implies the existence of at least two levels of general management and an increase in the number of general management roles over that of the functional structure.

Figure 2-7 shows the organization chart of a typical functional organization and Figure 2-8 that of a product division organization. Both charts are suggestive rather than definitive.

Organizational structure, however, consists of a great deal more than the differences that can be shown by charts. Systems of control, planning and information flow, methods of reward and punishment, the degree of delegation and techniques of coordination are among the important determinants of the way of life within the enterprise. The usefulness of the distinction between functional and multi-division organizations depends upon the degree to which these structures are associated with consistent differences among many other organizational characteristics. Recent research has shown that such uniformi-

Figure 2-7

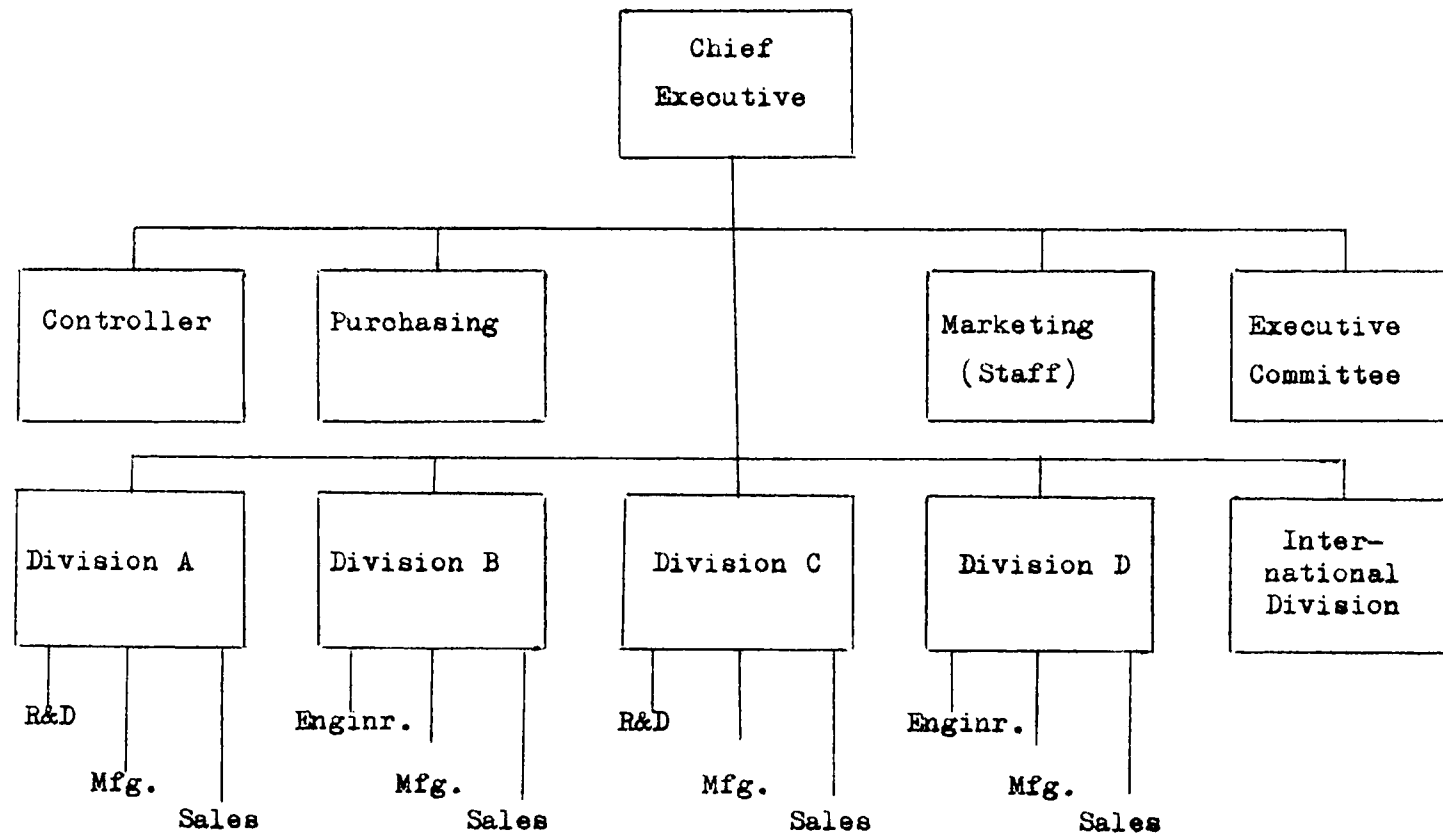
A SIMPLIFIED FUNCTIONAL ORGANIZATION



NOTE: There is great variety in the way different firms define functions and in the ways in which they are coordinated and related. This chart is simply illustrative of one possible pattern.

Figure 2-8

A SIMPLIFIED PRODUCT DIVISION ORGANIZATION



ties do exist.

Chandler's study of the development of seventy large U.S. corporations was, in part, devoted to an examination of how the strategy of diversification placed great strains on the functional type of organization and led to the adoption of the product division structure. Summarizing the impact of this change, Chandler concluded that the multi-division organization, by creating two levels of general management "removed the executives responsible for the destiny of the entire enterprise from the more routine operational activities and so gave them the time, information, and even psychological commitment for long-term planning and appraisal. Conversely, it placed the responsibility and the necessary authority for the operational administration in the hands of the general managers of the multifunction divisions".¹¹ He also noted that in addition to freeing a group of generalists from day to day problems, the headquarters office of a multi-division structure, because it is not associated with any particular business area, can develop staffs to supply objective information on the operations of each business, thus providing the framework for relatively unbiased evaluations of performance. These evaluations can then offer the division managers a more comprehensive view of their businesses than might be supplied by their own staff.

¹¹ Chandler, Strategy and Structure, p.382.

Following Chandler, Scott developed a paradigm of corporate development that views the firm as moving through successive stages as its product-market relationships become more elaborate. In this "stages model", the firm is seen as growing from a "one man show" to a functionally organized structure and then, as it develops multiple product lines, to a product division structure. Since the firms of interest in the present research have all grown past the point of being one man shows, that portion of the model will not be discussed here. The stages model predicts these differences between functional and multi-division organizations:

1. Research and development in functional organizations tends to focus on product and process improvement while in multi-division firms it also stresses the search for new products.

2. Performance criteria in functional firms tend to be based on technical and/or cost measures while product-division firms tend to employ market based measures such as return on investment and market share.

3. Rewards in multi-divisional organizations tend to be based on formal systems that relate rewards to market performance while functional firms tend to have less systematic reward systems and base them on functional efficiency and stability.

4. The major strategic choices considered by top management in functional firms tend to center on the desired degree of integration, market share and the breadth of the product line. In multi-division firms top level strategy is concerned with entry and exit from businesses, allocation of resources among businesses, the overall rate of growth, and the balance between internal expansion and growth by acquisition.

In addition, Scott has pointed out that multi-divisional organizations act as "a built-in 'school of management' training middle level general managers in the problems and opportunities associated with economic responsibility. As a result, this form of organization provides a pool of trained talent from which to draw, a pool from which a new group may be formed in a few days or weeks to take over and manage a new activity".¹³

In addition to the support provided by Chandler's data, Salter's¹⁴ research has tended to confirm the portions of the stages model that deal with reward and control systems and Ackerman¹⁵ has provided some additional support for the hypothesized resource allocation differences.

Salter has also suggested that a fourth type of structure, the geographic division organization, corresponds to a definite stage in corporate development. Somewhat similar to the product division structure, the geographic division structure is characterized by semi-autonomous divisions each of which operates in a different geographical area, though frequently all deal with the same range of products. Although we will show in Chapter III that there are few firms among the

¹³

Ibid., p.14.

¹⁴

Malcolm S. Salter, "Stages of Corporate Development", Journal of Business Policy, Vol. I, No.1, 1970, pp.40-51.

¹⁵

Robert W. Ackerman, "The Impact of Integration and Diversity on the Investment Process", (unpublished paper, Harvard Business School, 1969).

largest 500 with this type of structure, it is included in our classification scheme for the sake of completeness.

Many of the differences noted above seem to favor the multi-divisional structure, perhaps giving the impression that the functional structure is an outmoded holdover from the past. Such a conclusion is definitely unwarranted. While the multi-division structure is generally thought to be superior in situations in which the firm has many diverse products and the economic and technological environments are subject to rapid change, research by Lawrence and Lorsch¹⁶ and others has indicated that functional organizations tend to perform well or better than divisionalized organizations when environmental change is slow and the product line is not too broad.

All of these arguments offer assurance that these organizational categories do not just reflect superficial differences in how activities are grouped, but represent substantial differences in kind among a wide range of organizational properties. In addition to these three, two other categories were used in this research: the holding company and the functional with subsidiaries organizations. Although there is no empirical evidence on the subject, it is expected that these categories also represent organizational patterns that are significantly different in kind.

¹⁶ Paul R. Lawrence and Jay W. Lorsch, Organization and Environment, (Boston, Division of Research, Harvard Business School, 1967).

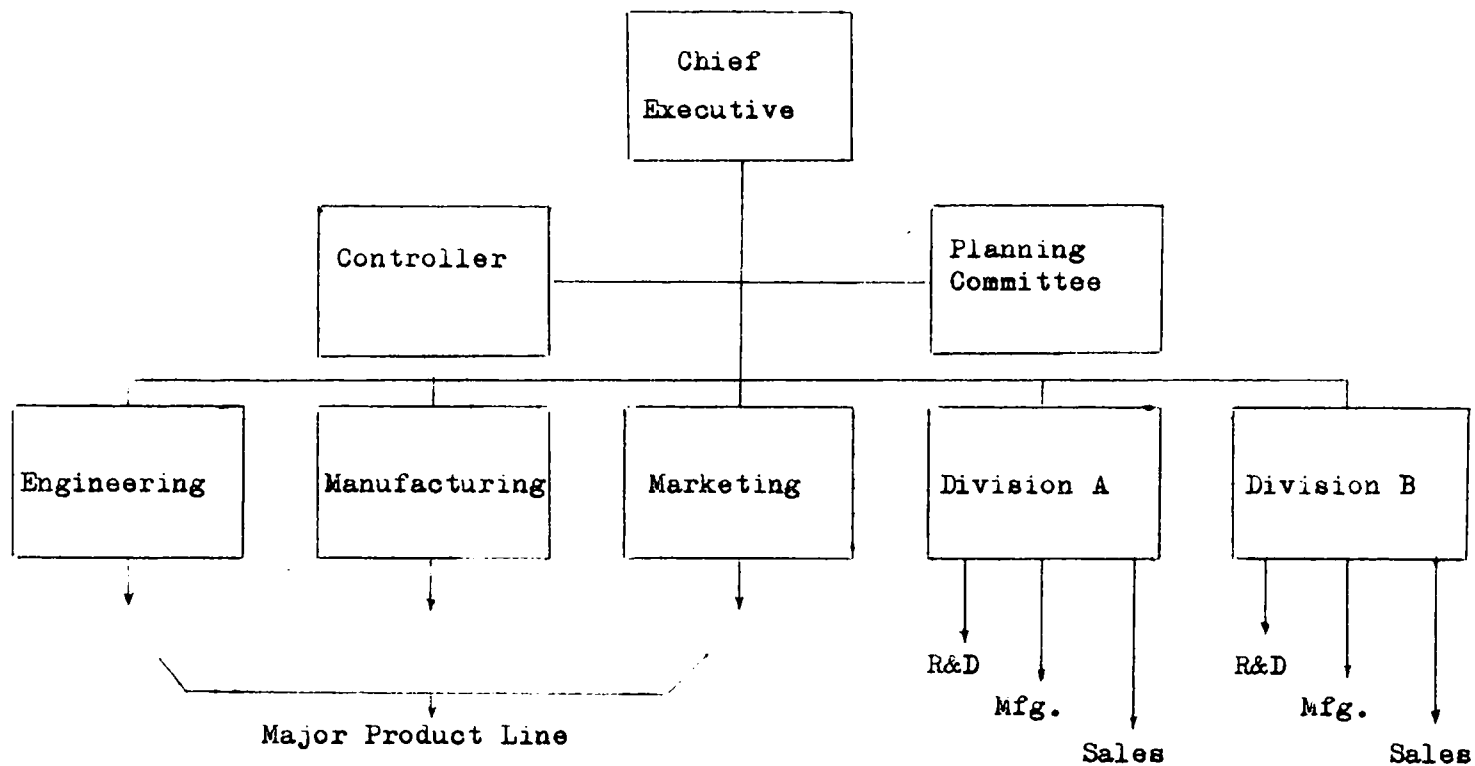
The holding company consists of a set of virtually independent firms that are owned and, at least legally, controlled by a parent corporation. Its distinguishing characteristic is the almost complete lack of management at the top. Each unit (or division) operates autonomously, with quarterly dividend payments to the parent often being the only visible connection between it and the rest of the company. The headquarters office is quite small, usually composed of only a few individuals who carry out financial, legal and other activities that must be performed by the parent.

It is important to note that "holding company" is also a legal term that refers to a particular type of corporation. While the legal "holding company" may have a structure similar to the one described here, there are holding companies that engage in much more thorough supervision of their constituent corporations. On the other hand, firms that are not legal "holding companies" may have organizations like the one described here. Thus, the holding company organization is not necessarily associated with firms that are legal "holding companies".

The functional with subsidiaries is a hybrid form of organization that does not seem to have received prior attention in the literature. It is often employed by vertically integrated firms with some side activities and by firms that are moving from functional to product division structures. Figure 2-9

Figure 2-9

A SIMPLIFIED "FUNCTIONAL/SUBS" ORGANIZATION



shows a typical organization of this type. As can be seen from the diagram, the functional with subsidiaries organization consists of a functional structure with one or more product divisions also reporting to top management (or in some instances to functional management). Usually the functional part of the organization controls most of the firm's activities and the divisions exist to carry out side ventures, new experimental businesses, or have been added by diversification through acquisition. The critical feature of this structure is the lack of a true multi-divisional type of headquarters office. General managers of the divisions are organizationally on a par with functional managers in the major area, and the central office must be active in day to day management of the major area and cannot gain a detached view of the firm as a whole. It is expected that the degree to which headquarters-division relationships in this type of firm approaches the product division pattern will be very sensitive to the detailed make-up of the headquarters office.

Summary In summary, the five organizational categorizational categories are as follows:

Functional: An organization in which the major sub-units are defined in terms of business functions of stages in the manufacturing process. Responsibility for coordination and product-market performance rests with the chief executive, his staff and committees. Vertically integrated firms fall in this class despite the common practice of referring to their process-linked sub-units as divisions.

Functional With Subsidiaries: An organization that is basically functional but which also has one or more separate product divisions (not necessarily true subsidiaries) which report to top management or, in some instances, to one of the functional managers. The distinguishing characteristic is that the general managers of the product divisions are organizationally on the same level or below the functional managers.

Product Division: An organization that consists of a central office and a group of operative divisions, each having the responsibility and resources needed to engineer, produce and market a product or set of products. In some instances the product divisions are clustered into groups and a third line of general management is placed between the division and headquarters. At some point below the division level, the organization reverts to a functional or geographic form.

Geographic Divisions: An organization that consists of a headquarters office and a group of operating divisions, each having the responsibility and resources needed to engineer, produce and market a product or set of products in a different geographic area. The areas may be portions of the United States or multi-national. At some point below the division level the form of organization reverts to a functional or product form.

Holding Company: An association of firms (or divisions) commonly owned by a parent corporation. Each firm is virtually autonomous and formal organization above the level of the individual firm is virtually non-existent.

Sampling Methodology and Data Sources

The population of interest in this research was taken to be the 500 largest U.S. industrial companies as listed annually by Fortune. To qualify as a member of the Fortune 500 a company must be (1) based in the United States, (2) derive at least 50% of its revenues from manufacturing or mining, (3) provide public reports on its financial status, and (4) be among the largest 500 such companies in terms of consolidated revenues. Excise taxes collected by the firm for the Federal Government are not included in revenues, a fact of some importance in the case of companies in the liquor and beer industry.

A study of the characteristics of the largest 500 firms cannot be based on a single sample as the set of corporations changes over time as firms grow with respect to one another and are merged or acquired. It was decided that the data requirements could be met with three independent random samples of the largest 500 firms in 1949, 1959, and 1969. The 1969 sample was constructed by taking the 100 firms that Wrigley had selected randomly from the 1967 Fortune 500, deleting those which were no longer among the largest 500 in 1969 and randomly selecting firms from the 1969 500 to take their places. The 1959 sample was constructed by randomly selecting 100 firms from the Fortune 500 in that year. The 1949 sample, however, could not be obtained in this way as Fortune first

began publishing its list of the 500 largest industrials in 1953. The only document that could be found that came close to a list of the largest 500 firms in 1949 was a Federal Trade Commission Report that gave (in alphabetical order) the 100, 200, 500, and 1000 largest industrial firms, in terms of assets, in 1950.

A sample was constructed from this list of the 500 largest firms by arranging them in a random order and then selecting the first 100 firms from this new list that met certain criteria. To be selected a firm had to be a U.S. industrial in the sense used by Fortune, be in existence in 1949, and not have had a sales to assets ratio that was substantially lower than that of other large firms. Unfortunately, this procedure could not avoid a bias against selecting companies with high sales and low assets in 1949, as they were not on the FTC list in the first place. This bias exists in the 1949 sample and its effect, though probably quite small, is difficult to estimate.

Due to the fact that some firms were selected in more than one of the samples, the three independent random samples of 100 firms each consisted in toto of 249 industrial firms. Three firms were subsequently dropped due to lack of sufficient data on their mix of products and organizations. This reduced the sample sizes for 1949, 1959 and 1969 to 100, 100, and 97 respectively and reduced the total number of individual firms

in cell samples to 246.

The Data and Its Sources

For each of the 246 firms a set of primary data was gathered and placed on a computer disc file. All subsequent analyses were carried out by manipulating the information in this file. Its contents included:

1. An abbreviated form of the corporate name.
2. The sample or samples (1949, 1959 or 1969) in which the firm appeared.
3. The firm's rank among the 500 largest industrials in 1959 and 1969.
4. If the firm was not in existence in 1949, the year it came into being and the way in which it was created: incorporated, resulting from a merger of two firms, with neither surviving, or previously part of a larger enterprise and divested.
5. If the firm was not in existence or no longer qualified as a U.S. industrial by the end of 1969, the last year in which it was a valid member of the sample and the reason for its exit: acquired, merged with another firm with neither surviving, no longer U.S. based, no longer an industrial firm (more than 50% revenue from other than manufacturing or mining).
6. The firm's specialization ratio in 1949, 1959, and 1969.
7. The firm's related ratio in 1949, 1959, and 1969.
8. The firm's vertical ratio in 1949, 1959, and 1969.
9. The firm's diversification posture category in 1949, 1959, and 1969.
10. The firm's organizational structure category in each year it was a valid member of the sample.
11. The four digit SIC code assigned to the firm by Standard & Poor's.

12. A list of the identifiable businesses in which the firm had been active during the time it was a valid member of the sample. For each business a short description, the appropriate SIC code, and the year of exit, if any, was recorded. In addition, if the firm had entered the business after it became a valid member of the sample, the year of entry, and the method (internal development or acquisition) was recorded.
13. Financial data for the years 1951-1970 for most of these firms still in existence in 1970. This data included annual consolidated figures for:
 - a. Net Sales
 - b. Common dividends
 - c. Earnings after tax and preferred dividends
 - d. Interest expense
 - e. Number of common shares outstanding, adjusted for splits and stock dividends
 - f. The average of the high and low stock prices for the year
 - g. Long term debt
 - h. Total equity
 - i. Total assets (or liabilities)

In addition to this formal data base, extensive files were kept which contained a variety of descriptive and interpretive information on each company. In all, the data gathering, cross-checking, classifying, and file construction portion of this research required approximately 1750 man-hours of work.

Information on the history of the firms, their entries into and exits from businesses and their product mix at particular points in time was gathered chiefly from Moody's Industrials, The Thomas Register of American Manufacturers, corporate annual reports to stockholders, and prospectuses accompanying the issue of corporate securities. When necessary, books and articles concerning a particular firm were also consulted. The

most constantly elusive information was the breakdown of the firms' revenues by product-area necessary for computing the specialization, related and vertical ratios. The richest source of this data proved to be prospectuses. The legal requirement for the inclusion of all relevant information in these documents often produced quite a different view of the firm from that presented in annual reports. Still, in many instances the product mix had to be estimated from data bracketing, but not including, the year in question. In a few cases, no estimates could be made.

Specialization ratios, related ratios, and, if applicable, vertical ratios for each firm for 1949, 1959 and 1969 were estimated from the available data on revenue breakdowns by product. Whenever a ratio could be estimated to within an accuracy of ± 0.05 it was recorded. In over one-third of the cases, this precision was unobtainable, either because the data was too sparse or because there was difficulty in identifying discrete businesses. However, since the ratios' major use was in determining diversification categories, the lack of accurate estimates was not a serious problem--for classification purposes it was only necessary to know if the specialization ratio lay in the range 0.7 to 0.95 or if it was above or below that range, and if the related and vertical ratios were above or below 0.7.

One of the nine strategic diversification categories was assigned to each firm for 1949, 1959 and 1969, excepting those years in which it was not a valid member of the sample. The major category was determined by the estimated value (or range) of the various ratios and the sub-category was assigned by examining the firm's rationale for diversification, its product mix, the methods of entry into each business and the degree to which potential relationships had been exploited. Examples of this process are given in an earlier section of this chapter. Most of this qualitative information was obtained by reading through twenty years of annual reports for each firm. In a few instances, the corporation's annual reports were so lacking in information or pitched at such an abstract level that books, newspapers and magazine articles and investors' surveys that dealt with the firm had to be located and consulted.

A point that should be mentioned is that in two cases (Avco and Canada Dry in 1949) strategic categories were assigned that did not coincide with the values of the specialization and related ratios for that year. In both instances, the ratios were close to the critical value 0.7 and their values in nearby years indicated that the 1949 values were atypical. Hence, classification was based on the more normal values of the ratios.

Information on organizational structures was gathered from a wide variety of sources. Chandler's Strategy and Structure briefly discussed the organizational histories of seventy large industrials and where his sample coincided with ours his observations on structure were used. A National Industrial Conference Board publication, Corporate Organization Structure by Stieglitz and Wilkenson, gave organizational charts for 1967 sixty-six firms, many of which were in the research samples. Other NICB books and pamphlets (see the bibliography) mentioned in passing or gave as examples the organizational structures of some of the firms in the sample. Two American Management Association publications were also useful; Understanding the Company Organization by White and Organizing the R & D Function by Stanley and White both provided charts and discussions of the organizational development that applied to several firms in the sample. Another useful source was Corporate Diagrams and Administrative Personnel of the Chemical Industry by Kern, which gave organization charts of chemical divisions and subsidiaries of oil companies and other diversified corporations.

However, over half of the information on structure was derived by careful reading of annual reports. Although any particular annual report, taken by itself, is a poor source of data on this subject, a twenty year sequence of annual reports often reveals a great deal, especially when the structure

at some point in time is known in advance. The titles assigned to key executives frequently provided important clues to structural change, as did discussions of how new acquisitions were integrated into the firm and descriptions of the responsibilities of newly promoted managers. Of course major reorganizations were almost always noted in annual reports and perusal of the preceding and following reports usually classified the natures of the previous and new structures.

Although the original research plan only called for assigning organization categories to the firms in the sample at three points in time, 1949, 1959 and 1969, because most structures were deduced by working forwards and backwards in time from evidence of a reorganization, data was available on structure on a year by year basis. Therefore, organization categories were assigned to each firm for each year it was a valid member of the sample.

Data on individual diversification moves came from annual reports and Moodys Industrials. The procedure followed was to first list every move that might be considered a diversification move for each firm in the sample. Then, given the benefit of this experience, moves were deleted that were judged to be basically product line extensions, follow-up moves to previous diversification moves, or simply insignificant.

Appendix A has been prepared to provide a succinct summary of the non-financial data used in this study. It contains a list of the 246 firms in the total sample and for each gives (for 1949, 1959, and 1969) its (1) important businesses, (2) rank in the 500, (3) specialization ratio, (4) related ratio, (5) assigned diversification strategy class, and (6) assigned structure class.

The financial information was obtained from a computer data bank maintained by the Boston Company. This data had been originally obtained from Standard and Poors, providing twenty years (1951-1970) of annual observations on sixty financial variables for about 1750 corporations. The financial reporting was on a consolidated basis and had been adjusted by Standard and Poors to provide reasonable inter-firm consistency in methods. One drawback associated with this data base is that it includes only firms which were in existence in 1970. Consequently, no financial data was available for the many companies in the 1949 and 1959 samples that were acquired in the late 1960s. As the accepted practice in maintaining such data banks is to eliminate the information on firms that become defunct, all other similar sources posed the same difficulty. The alternative of gathering the missing data from published sources was deemed too time consuming and better left to experts. Consequently, the financial data used in this research pertains only to firms still in existence in 1970, and is

therefore biased in favor of surviving rather than acquired companies.

Comparison With Wrigley. Since the 1969 sample used in the research was taken directly from Wrigley, a comparison can be made between the classification of these firms by Wrigley and the classifications assigned in this research. This comparison can serve to point out those portions of the system that are most sensitive to the researcher's subjective judgement. It should be noted that this researcher was careful not to let Wrigley's classifications influence those made for the current study; the comparison was not even attempted until the study was completed. The results of the comparison are shown below:

<u>Category</u>	Percentage in Category	
	<u>This Study 1969 Sample</u>	<u>Wrigley's 1967 Sample</u>
Single business	7.6%	6.0%
Dominant business	31.0	14.0
Related business	45.2	60.0
Unrelated business	16.2	20.0

Of the 87 firms that were common to both studies, 28 were classified differently. Investigation of the source of discrepancies revealed that 14% were due to changes in the status of the firms between 1967 and 1969 (Wrigley used 1967 data), 7% were due to firms Wrigley classed as Related falling in the Dominant Vertical category, and 25% were due to the stricter conditions for placement in the Unrelated category

used in this study. Fifty-four per cent of the discrepancies were due to differences in judgement or data, the former being more important. The pronounced disagreement on the proportion of firms in the Dominant and Related business categories was largely due to differences in judgement, the critical issue being the definition of a discrete business. The precise nature of the difference in the concept of a discrete business held in these two studies is difficult to determine, as the classification of firms was of secondary importance in Wrigley's study and he did not supply detailed operational definitions. It is hoped that the definitions and examples provided in this study will insure a greater degree of comparability with future research.

Summary and Comments

Starting from the conviction that there is usually an underlying logic behind the development of a corporation over time, a logic we call "strategy" which necessarily includes but need not be expressed in terms of the microeconomics surrounding the production and sale of each of the firm's products, we chose to study diversification and structure by viewing the firm as a whole rather than as a host of separate pieces. Thus, instead of concentrating on individual diversification moves or on the total number of product-market activities, we have attempted to take the viewpoint of the top level general manager and characterize the firm in terms of the total amount

diversify inherent in its strategic posture and the way in which business areas have been related to one another. This "top down" approach led to the definition of four major categories of diversification strategy which were further subdivided to give a total of nine distinct classes of strategy.

The procedure employed in classifying firms used a mixture of quantitative measure and qualitative assessments. Of these, the most crucial were the decision as to what constitutes the firm's largest discrete business and the evaluation of the nature of the interrelationships among the firm's businesses. While the use of this type of classification system permits the consideration of non-quantifiable data, the apparent preciseness of definite category boundaries must not obscure the fact that the researcher's judgement plays an important role in the technique. In the present context this type of semi-subjective technique has several strengths: (1) it permits the direct consideration of the most important, though non-quantifiable, aspects of the firm's strategy, (2) it is easy to understand and does not require the processing of complex multi-dimensional data, and (3) the constant need for judgement on the part of the researcher provides a continual check on the validity and usefulness of the conceptual framework. Of course, these advantages are only obtained by sacrificing a degree of objectivity and replicability. Since others have studied diversification by using "objective" measures

such as the product count and other indices, it was felt that creating another similar piece of research was not as important as exploring the possibility that a measurement scheme based on overall corporate strategy might yield different results. As will be seen in Chapter IV, this hope was borne out.

CHAPTER III

THE CHANGING PATTERNS OF STRATEGY AND STRUCTURE

The preceding chapters have served to develop a rationale and methodology for studying diversification strategy and organizational structure in large firms. In the process something of the spirit of our approach should also have been communicated. With this chapter we begin to report the research findings.

The general objective of this chapter is to present the results of applying the strategic and organizational classification schemes to each of the 246 firms in the sample and to begin the work of piecing together a composite picture of the changing patterns of strategy and structure among the largest 500 industrial corporations. As data are presented, the answers to a series of simple but important questions will be sought. How pervasive is the phenomenon of diversification among large industrial corporations? What proportions of the largest 500 firms fall in each of the diversification and structural categories and how have these proportions changed with time? To what extent are such changes due to actual strategic and organizational change by firms rather than the changing composition of the 500? Which organizational forms are associated with which strategies and is there any evidence of a

cause and effect relationship between strategic and structural change?

All of the estimates presented in this chapter were obtained by applying a weighting procedure to a stratification of the sample of 246 firms. The technique used to obtain these weighted estimates is described in Appendix 3A at the end of this chapter.

Comparison With the Industry Count.

The use of strategic categories based on a mixture of quantitative and qualitative criteria is a break with most of the traditional techniques for measuring diversification. Although one of Gort's¹ diversification indices and the measure used by Stopford² were essentially equivalent to the specialization ratio used in this research, using this ratio together with information on the degree and nature of the relationships among business areas to define discrete categories was an innovation of Wrigley's, whose work was the immediate predecessor to and stimulus for this study. Most prior research on diversification was based on some type of product count, usually employing the product classes of the Standard Industrial Classification (SIC) system. Because of the wealth of data

¹Michael Gort, Diversification and Integration in American Industry, (Princeton, Princeton University Press, 1962)

² John M. Stopford, "Growth and Change in the Multinational Firm", (Unpublished doctoral dissertation, Harvard Business School, 1968)

that has been accumulated on product count measures of diversification, it is of interest to determine the extent to which such measures correspond to our diversification categories and thus the extent to which they might serve as surrogates for the categories.

In order to carry out a comparison, the number of 2-digit and 3-digit industries in which each firm in the sample participated was determined for 1949, 1959 and 1969. Neither the 2-digit nor the 3-digit measure is actually a "product count" as the degree of aggregation is quite high in 3-digit categories and the 2-digit categories correspond to entire industrial sectors such as food processing (SIC 20), primary metals (SIC 33), etc. However, a true product count would require the use of at least the 4-digit level of detail and probably should be based upon 5-digit or 7-digit SIC categories. Nevertheless, a great many studies have used 3-digit and 2-digit categories as the measures of diversification both because of the difficulty of obtaining very detailed data and the observed high degree of correlation between measures at various levels of detail.

The data on 2-digit and 3-digit industry participation for each firm in the sample was obtained from annual reports, Moody's, and Standard and Poor's. In most cases, lists of the different SIC categories in which a firm was active were not used; it was found that they frequently conflicted with one

another and with the actual data on products. The philosophy that was followed was that if no mention of a product could be found in annual reports or standard references, it was either not produced or produced in inconsequential amounts.

Table 3-1 shows for each major strategic category the average number of 2-digit and 3-digit industries in which firms in that category were active and the standard deviations about these averages.

The overall pattern of average "industry count" is as expected; there was a trend in time towards increased diversification in all categories and the order of categories in terms of increasing numbers of industries participated in was Single, Dominant, Related, and Unrelated. However, the correspondence is far from precise. For example, several Single business firms were, in 1969, active in more than one industry sector. Central Soya, in 1969, was basically a soy bean processor, but had products in SIC 204 (milling), SIC 281 (chemical by-products), SIC 130 (poultry production) and SIC 287 (agricultural chemicals). However, the overwhelming preponderance of its revenues came from soy milling, so that it was classed as a Single business firm in that year.

The undue weight given to minor activities is one of the major arguments against the product or industry count measure of diversification. Using a 4-digit level of detail, Gort noted that

TABLE 3-1

PARTICIPATION IN 2 AND 3 DIGIT INDUSTRIES BY STRATEGIC CLASS

	1949		1959		1969	
	Mean	Standard Deviation	Mean	Standard Deviation	Mean	Standard Deviation
2 Digit Industries						
Single Business	1.33	0.88	1.23	0.50	1.39	0.77
Dominant Business	2.05	1.29	2.63	1.50	3.72	2.34
Related Business	2.50	1.72	3.37	2.08	4.63	2.52
Unrelated Business	4.29	2.87	5.14	2.57	7.90	3.03
3 Digit Industries						
Single Business	1.46	1.00	1.42	0.85	1.77	1.17
Dominant Business	2.87	1.90	3.87	2.31	5.75	3.59
Related Business	4.03	2.89	5.57	3.85	8.12	4.38
Unrelated Business	6.86	3.81	8.71	4.63	12.21	5.35

Note: The Smirnov test was used to test the significance of the differences in industry participation among classes, each class being tested against each other class. For participation in 2 digit industries, these differences were not significant at the 0.1 level: between Dominant and Related firms in all years, between Related and Unrelated firms in 1949 and 1959, and between Dominant and Unrelated firms in 1949. For participation in 3 digit industries, the only differences not significant at the 0.1 level were between Related and Unrelated firms in 1949 and 1959. All other differences were significant at the 0.02 level. It appears that an increase in the number of Unrelated firms in the sample would have altered these results.

It is rather surprising to find that for a majority of the 111 large enterprises, minor activities accounted for a substantial proportion of the total number of industrial activities in which the companies were engaged. In six of the thirteen groups (2-digit industries) at least one third of the manufacturing activities (4-digit product areas) individually contributed not more than 1 per cent to the manufacturing employment of their companies.³

The other major source of inconsistency is the arbitrary nature of the SIC classifications. A manufacturer of paper cups who adds a line of plastic cups is seen as entering a new industrial sector according to the SIC system. The strategic category technique, while not having the degree of "objectivity" generally accorded to the SIC system, permits the researcher to choose to ignore the difference between paper and plastic in this case and call the firm a Single business company.

A feeling for the degree of correspondence between the strategic categories and the industry count can be obtained by determining the industry count levels that provide the best possible discrimination among categories and measuring the resulting amount of overlap. For example, in 1969 the following scheme for assigning strategic classes using a 2-digit industry count would have produced the least number of misclassifications:

<u>Category</u>	<u>Trial criterion: Number of 2-digit industries engaged in</u>	<u>Percentage of firms in cate- gories misclassified</u>
Single business	1	23%
Dominant business	2-3	62%
Related business	4-6	59%
Unrelated business	7 or more	41%

³ Gort, Diversification and Integration, p.35

The outcome of such a scheme would have been the placement of over one half of the firms in categories other than were assigned to them on the basis of specialization ratios, related ratios, and qualitative information. The major conflicts are due to Dominant business firms that are active in several sectors and Unrelated firms that are active in few.

The inescapable conclusion is that the strategic categories reflect characteristics of firms that are significantly different from those measured by a product or industry count, and consequently, product or industry count measure cannot serve as surrogates for the strategic categories.

Strategies of Diversification

Weighted estimates of the percentages of the 500 largest industrial corporations that fell in the four major and nine minor categories of diversification strategy are shown in Table 3-2. The same information is portrayed graphically in Figure 3-1. It does not require sophisticated analyses to see the most dramatic pattern of change revealed by the data. Between 1949 and 1969 the number of truly diversified corporations more than doubled; the percentage of firms following Related or Unrelated Business strategies of expansion rising from 30% to 65% in this 20-year period. Clearly, there has been a basic change in the product-market scope of the largest U.S. industrial corporations.

The most striking change in any individual group is

TABLE 3-2

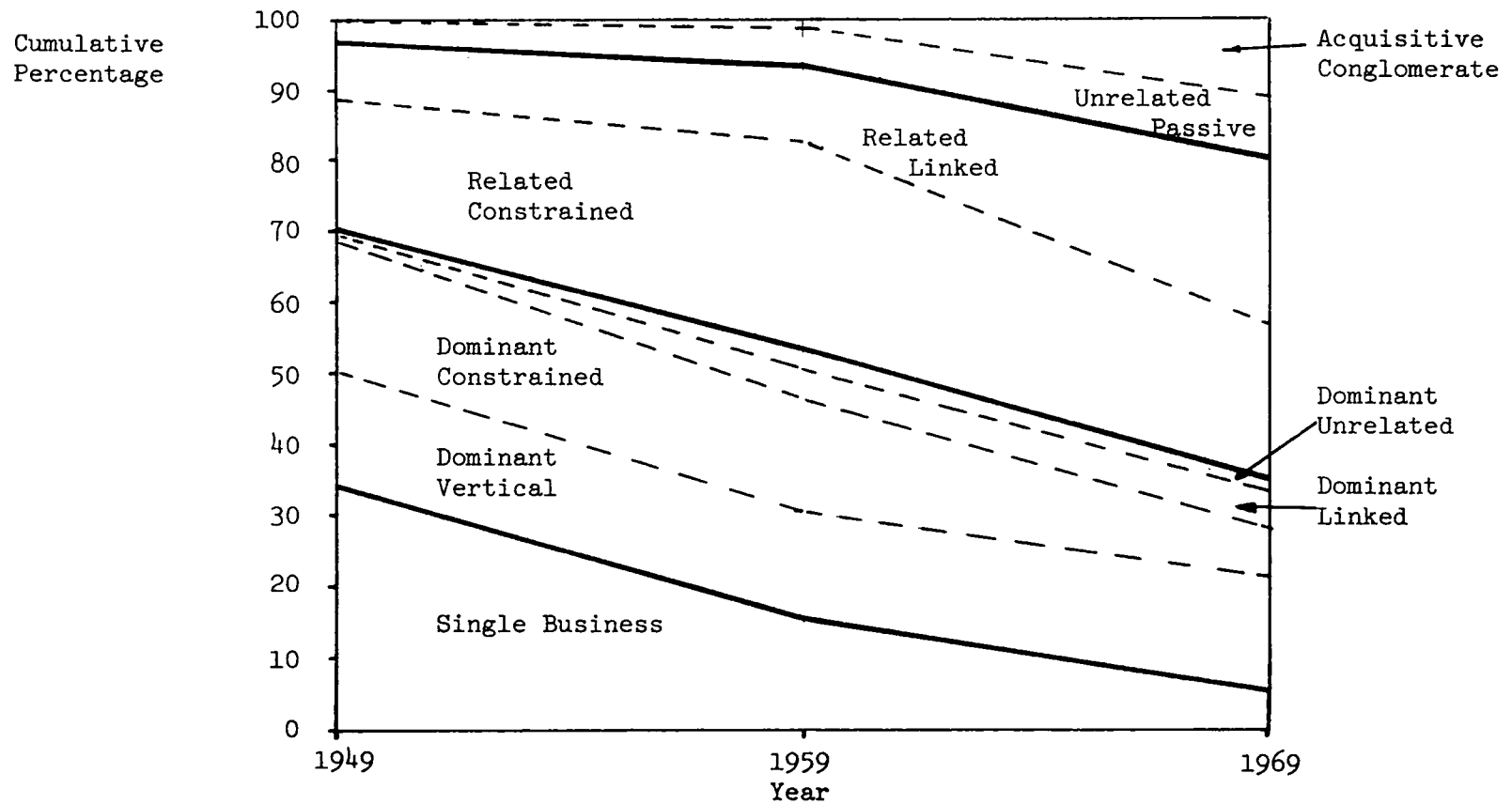
ESTIMATED PERCENTAGE OF FIRMS IN EACH STRATEGIC CATEGORY

Strategic Category	1949	1959	1969
Major Classes			
Single business	34.5	16.2	6.2
Dominant business	35.4	37.3	29.2
Related business	26.7	40.0	45.2
Unrelated business	3.4	6.5	19.4
Minor Classes			
Single business	34.5	16.2	6.2
Dominant Vertical	15.7	14.8	15.6
Dominant Constrained	18.0	16.0	7.1
Dominant Linked	0.9	3.8	5.6
Dominant Unrelated	0.9	2.6	0.9
Related Constrained	18.8	29.1	21.6
Related Linked	7.9	10.9	23.6
Unrelated Passive	3.4	5.3	8.5
Acquisitive Conglomerate	0.0	1.2	10.9
Number of firms used to derrive the estimates	189	207	183

Note: Estimates were made using the weighting procedure described in the text.

Figure 3-1

ESTIMATED PERCENTAGE OF FIRMS IN EACH
STRATEGIC CATEGORY



the decline in the number of Single business firms among the largest 500. Comprising more than one-third of the 500 in 1949, by 1969 firms deriving 95% or more of their revenues from one business had dropped to 6.2% of the total. Equally noteworthy was the increase in large firms that follow Unrelated business strategies. In 1949 this group accounted for only 3.4% of the 500 largest but by 1969 one out of every five firms fell in the Unrelated category. Somewhat surprising is the evidence that this increase was not all due to growing numbers of Acquisitive Conglomerates; in 1969 fully 44% of Unrelated business firms were Unrelated Passive and 60% of these had appeared since 1949.

The relative importance of the Dominant business category, as a whole, decreased slightly during this 20-year period, although it appears to have grown between 1949 and 1959 and then diminished by more than this amount in the second decade. The behavior of sub-classes of the Dominant category shed some light on how this happened. The Dominant Vertical group was extremely stable, accounting for about 75 of the top 500 firms throughout the period. As would be expected, most of these giant oil, steel, aluminum and forest products companies neither left the top 500 nor became truly diversified during these 20 years. The Dominant Constrained group, on the other hand, became much smaller between 1959 and 1969, dropping from 16% to 7.1% of the top 500; in fact, this decrease is the chief reason

for the overall drop in the size of the Dominant business category. The Dominant Linked and Dominant Unrelated sub-classes have both been small throughout the period, although displaying different patterns of change. The Dominant Linked group steadily increased in size, from 0.9% to 5.6% between 1949 and 1969 while the Dominant Unrelated group was at its largest in 1959, at 2.5%, and almost insignificant in size in 1949 and 1969, at 0.9% in both years.

The Related business category became increasingly important between 1949 and 1959, not quite doubling in size. It is particularly interesting that almost all of the increase in the Related category was in the Related Linked sub-class which tripled in size between 1949 and 1969, increasing from 7.9% to 23.6% of the top 500. The Related Constrained sub-class, by contrast, increased between 1949 and 1959, from 18.1% to 29.1%, but then dropped back to 21.6% by 1969.

Sources of Change. The basic pattern of change in the composition of the 500 between 1949 and 1969 was an increase in the Related and Unrelated categories at the expense of the Single business category. But how did this and other redistributions come about? Did most of the Single business firms of 1949 adopt strategies of diversification by 1969, or were they forced out of the 500, either by being acquired or displaced, and replaced by firms in the Related and Unrelated categories?

In order to separate the effect of strategic change

from that of entry and exit, it is useful to look at those firms that were among the largest 500 in both 1949 and 1969. An estimated 319 companies satisfied this condition, about 64% of the top 500 in either year. The distributions by category for these firms is shown in Table 3-3. Clearly, there was significant strategic change among these firms; 51.3% moved from one category to another between 1949 and 1969, and 95% of these moves were in the direction of increased diversification. The changes made by the firms that stayed in the 500 resulted in strategic class populations very much like those of the full 500. The only significant differences were a higher proportion of firms in the Dominant and a lower proportion in the Unrelated categories. These gaps were, of course, filled by the entry and exit effects.

Another way of looking at these phenomena is to determine for each category the amount of change due to shifting (i.e., firms moving out of or into that category but staying in the top 500), and the amount due to entry and exit. For example, between 1949 and 1969 the Single business category dropped from 34.5% to 6.2% of the top 500, a total change of -28.3 percentage points. This change can be allocated as follows: -15.7 due to exits, +1.8 due to entries, giving a net of -13.9 due to entry and exit; -14.8 due to firms moving from Single to other categories and +0.4 due to firms moving from others into the Single business category giving a net of -14.4 due to

TABLE 3-3
FIRMS IN 500 IN 1949 AND IN 1969

Strategic Category	Percentage in Each Category		
	1949	1959	1969
Major Classes			
Single business	29.2	16.1	7.3
Dominant business	38.7	40.1	44.5
Related business	29.2	38.0	44.5
Unrelated business	2.9	5.8	12.4
Minor Classes			
Single business	29.2	16.1	7.3
Dominant Vertical	19.7	18.2	21.9
Dominant Constrained	16.1	13.1	8.0
Dominant Linked	1.5	4.4	4.4
Dominant Unrelated	1.5	4.4	1.5
Related Constrained	20.4	28.5	24.1
Related Linked	8.8	9.5	20.4
Unrelated Passive	2.9	5.8	10.2
Acquisitive Conglomerate	0.0	0.0	2.2

shifting. Thus the total change of -28.3 can be viewed as due to -14.4 from shifting, and -13.9 from entry and exit. These results and those for the other three major categories are shown below in Table 3-4.

TABLE 3-4

Sources of Change 1949-1969

	<u>Category</u>			
	<u>Single</u>	<u>Dominant</u>	<u>Related</u>	<u>Unrelated</u>
% of 500 in 1949	34.5	35.4	26.7	3.4
% of 500 in 1969	<u>6.2</u>	<u>29.2</u>	<u>45.2</u>	<u>19.4</u>
Change	-28.3	-6.2	+18.5	+16.0
Portion due to shifting	-14.4	-2.9	+11.6	+ 5.7
Portion due to entry-exit	-13.9	-3.3	+ 6.9	+10.3
% Change due to shifting	50.9%	46.8%	62.7%	64.4%

As can be seen from this analysis, the effects of shifting and entry-exit were in the same directions in each category and were about equally important in explaining the reductions in the number of firms in the Single and Dominant categories. In the Related and Unrelated categories shifting tended to be more important than entry-exit. Thus, from the data in Tables 3-2, 3-3, and 3-4 we can conclude that in the period 1949 to 1969 (1) the number of Single business firms sharply decreased and the number of firms in the Related and Unrelated categories increased, (2) that this change was due to changes in strategy by about 164 firms and the displacement of about 181 firms and their replacement by an equal number, (3) that both strategic change and entry-exit had the effect

of decreasing the size of the Single business category and increasing the size of the Related and Unrelated business categories, and (4) that both forces were of comparable importance in producing the changes in the number of firms in each category.

Entry and Exit. Firms exited from the largest 500 either by being acquired or being merged with another large firm, or by having total revenues drop below the minimum required for inclusion. In the decade 1949-1959 acquisition and displacement were of comparable importance; of the estimated 120 firms that exited the 500 in this period, approximately 63 were acquired or merged and 57 displaced. In the period 1959-1969, acquisition and merger were of much greater relative importance, accounting for about 91 of the 127 firms that are estimated to have exited.

It was hypothesized that some strategic categories (namely the Single, Dominant Constrained and Related Constrained categories) would be more susceptible to acquisition and merger than the others. A statistical analysis of acquisition and merger rates, however, did not support this hypothesis. In fact the data does not support any hypotheses of unequal acquisition rates among the categories in either decade. In other words, there is no evidence to suggest that Single businesses, for example, were either more or less likely to be acquired than firms in any other category.

A statistical test did indicate that Single business and Dominant Constrained firms were more likely to be displaced than other categories of firms in the first decade.⁴ However, this pattern was not repeated in the second decade. Furthermore, it is clear that a firm's size is the major determinant of its likelihood of being displaced, so that any relationship between displacement rate and category is most likely simply a result of a more basic difference in the average size of firms in each category.

The entry rates in the first decade seem to favor the Unrelated category at the expense of the Single category, but again the differences are not statistically significant. This pattern was repeated during 1959-1969 but this time the differences were quite significant. Of all firms that were among the 500 in both 1959 and 1969, only 4.8% had adopted acquisitive conglomerate strategies by 1969. By contrast, 28.7% of all firms that entered the 500 in this period were acquisitive conglomerates. (This difference is statistically significant at the 0.1% level). All other entering firms were distributed among the categories in rough proportion to the number of firms already in them.

Strategic Change. Tables 3-5 and 3-6 show the detailed pattern of change during 1949-1959 and 1959-1969 respectively.

⁴ Binomial one-sided tests significant at the 10% level for Single business acquisitions and at the 5% level for Dominant Constrained acquisitions between 1949 and 1959.

TABLE 3-5

STRATEGIC CATEGORY TRANSITION RATES: 1949-1959

1949 Category	Rate of Acq. Disp.		Percentage of Non-exiting Firms Shifting to Each Category by 1959											
			SING	DOM	DV	DC	DL	DU	REL	RC	RL	UNRL	UP	AC
SING	9.9	19.9	57.0	33.9	6.0	15.1	7.3	5.5	9.1	7.3	1.8	0	0	0
DOM	9.6	13.0	1.6	75.2	44.2	23.0	3.2	4.8	20.0	20.0	0	3.2	3.2	0
DV	14.8	0	0	90.1	90.1	0	0	0	9.9	9.9	0	0	0	0
DC	6.3	25.7	3.6	55.2	0	51.6	3.6	0	34.0	34.0	0	7.2	7.2	0
DL	0	0	0	100.0	0	0	50.0	50.0	0	0	0	0	0	0
DU	0	0	0	100.0	0	0	0	100.0	0	0	0	0	0	0
REL	17.0	0	0	0	0	0	0	0	96.0	65.3	30.7	4.0	4.0	0
RC	18.2	0	0	0	0	0	0	0	94.3	94.3	0	5.7	5.7	0
RL	14.4	0	0	0	0	0	0	0	100.0	0	100.0	0	0	0
UNRL	33.4	0	0	0	0	0	0	0	0	0	0	100.0	100.0	0
UP	33.4	0	0	0	0	0	0	0	0	0	0	100.0	100.0	0
AC	--	--	--	--	--	--	--	--	--	--	--	--	--	--
NEW ENTRIES			8.1	35.5	5.3	25.4	4.8	0	45.9	30.6	15.3	10.5	5.7	4.8

TABLE 3-6

STRATEGIC CATEGORY TRANSITION RATES: 1959-1969

1959 Category	Rate of Acq. Disp.		Percentage of Non-exiting Firms Shifting to Each Category by 1969											
			SING	DOM	DV	DC	DL	DU	REL	RC	RL	UNRL	UP	AC
SING	26.1	10.5	44.2	34.3	21.4	8.6	4.3	0	17.2	12.9	4.3	4.3	4.3	0
DOM	17.7	1.8	1.5	70.3	36.7	17.6	13.1	2.9	20.0	9.3	10.7	8.2	4.9	3.4
DV	17.1	4.6	0	95.1	95.1	0	0	0	0	0	0	4.9	4.9	0
DC	25.5	0	0	57.2	0	43.9	9.6	3.7	35.4	23.2	12.2	7.4	7.4	0
DL	0	0	0	50.0	0	0	50.0	0	34.8	0	34.8	15.2	0	15.2
DU	0	0	16.7	50.0	0	0	33.3	16.7	16.7	0	16.7	16.7	0	16.7
REL	15.3	10.2	0	3.4	0	3.4	0	0	85.3	42.7	42.7	11.2	7.8	3.4
RC	15.8	9.3	0	4.7	0	4.7	0	0	86.6	58.5	28.1	8.7	8.7	0
RL	13.9	12.6	0	0	0	0	0	0	81.9	0	81.9	18.1	5.5	12.6
UNRL	18.2	10.5	0	0	0	0	0	0	0	0	0	100.0	66.0	34.0
UP	22.1	12.7	0	0	0	0	0	0	0	0	0	100.0	87.5	12.5
AC	0	0	0	0	0	0	0	0	0	0	0	100.0	0	100.0
NEW ENTRIES			4.8	14.2	9.4	0	4.8	0	47.6	19.0	28.6	33.4	4.8	28.6

These tables give (1) estimates of the percentage of firms in each category that were acquired or merged during the next ten years (the acquisition and displacement rates), (2) of firms that stayed in the 500, estimates of the percentage of companies in each category that had moved to or remained in each category ten years later (transition rates), and (3) estimates of the percentage of the firms that entered the 500 during the period that fell in each category and its end. For example, the first line of Table 3-4 indicates that of firms that were Single business in 1949, 9.9% had been acquired or merged and 19.9% displaced from the 500 by 1959. Furthermore, of those that remained (70.2% of the original 1949 group), 57% stayed in the Single business category, 33.9% moved to one of the Dominant categories, 6% to Dominant Vertical, 15.1% to Dominant Constrained, etc. Note that these "transition rates" refer only to firms that did not exit, and refer to the proportion of these that moved to another group, not to the absolute level of the flow.

These tables show a surprising degree of stability between decades in the pattern of transitions among the major categories. In the first decade 43% of Single business firms that stayed in the 500 moved to other categories, in the second decade 56% moved. In both decades roughly one-third of Single business firms moved into the Dominant class. Similarly, 25% of Dominant business firms moved to other categories in 1949-1959

and 30% in 1959-1969, and 20% of Dominant business firms moved into the Related class in both decades. In both decades very few firms moved from more diversified to less diversified postures. The only major differences between the decades were that (1) the percentage of Single business firms that moved into the Related category during 1959-1969 was twice as large as during 1949-1959, and (2) transition rates into the Unrelated category from all other categories were significantly larger in the second decade.

The detailed pattern of transitions among minor categories is more difficult to interpret directly. However, the effect of the changes in transition rates from the first to the second decade can be seen by determining what would have happened if the 1949-1959 transition rates had not changed, but had been repeated in 1959-1969. Keeping the 1959-1969 exit rates and entry pattern, but applying the 1949-1959 transition rates to the 1959 distribution of firms gives the results in Table 3-7.

TABLE 3-7

1949-1959 Transition Rates Projected Through 1969

<u>Category</u>	<u>Percentage of 1969 500 in Each Category</u>	
	<u>Actual</u>	<u>Projected Using 1949-59 Rates</u>
Single	6.2	7.5
Dominant	29.2	30.5
Related	45.2	46.7
Unrelated	19.4	15.3
Single	6.2	7.5
Dominant Vertical	15.6	13.4
Dominant Constrained	7.1	7.7
Dominant Linked	5.6	4.3
Dominant Unrelated	0.9	5.1
Related Constrained	21.6	31.3
Related Linked	23.6	15.4
Acquisitive Congl.	10.9	8.4
Unrelated Passive	8.5	6.9

The differences between these two distributions are purely due to the differences between the 1949-1959 and 1959-1969 patterns of strategic change. Clearly, the only net effects of this change that were of any significance were:

1. All of the firms that were Dominant Unrelated in 1959 dispersed to other categories by 1969 and the 1949-1959 pattern of movements into this class were not repeated.

2. Unlike the first decade, the second decade saw a great many Related Constrained firms move to the Related Linked category; 26% of the 1969 Related Linked firms had been Related Constrained in 1959. This change accounts for almost all of the discrepancy between the actual and projected populations of these categories.

3. In the second decade the rate of transition into the Passive Conglomerate category increased and some firms already among the 500 adopted Acquisitive Conglomerate strategies. In contrast, the only Acquisitive Conglomerates appearing between 1949 and 1959 had entered the 500 in that period.

This analysis also sheds some light on the behavior of the size of the Dominant category. In Table 3-2 the estimated percentage of firms in this category rose during 1949-1959 from 35.4% to 37.3% and then dropped to 29.2% by 1969. Even if the transition rates had not changed between decades the Dominant category would have become smaller, dropping to 30.5% rather than 29.2%. From Table 3-4 we know that entry-exit phenomena explain only 3.3 percentage points of this drop. Thus the reason the Dominant category became smaller during 1959-1969 was simply that there were fewer Single business firms in 1959 to move into the Dominant category, while the rate of transition out of the Dominant category remained virtually unchanged. This type of phenomenon is typical of systems characterized by largely one-way flows between categories with fairly stable transition rates. At the start the upstream category is full (the Single group) and the others nearly empty. As time passes, the down-stream categories increase in size but, since the up-stream categories are not being fully replenished, eventually the sizes of all but the last category drop to low levels.

This analogy suggests that the Related group may also decrease in size in the future, if transition rates into the Unrelated category remain the same. Shown in Table 3-8 are the projected sizes of the four major groups in 1979 and 1989 and 1999, under the assumptions that the 1959-1969 transition

rates and entry-exit patterns are repeated each decade.

TABLE 3-8

Projected Strategic Class Distributions

	<u>Percentage in Category</u>			
	<u>Single</u>	<u>Dominant</u>	<u>Related</u>	<u>Unrelated</u>
Projected 1979	3.3	22.6	46.0	28.1
Projected 1989	2.4	18.3	45.3	34.0
Projected 1999	2.1	15.7	44.3	37.9

We see the continuing decline in the Dominant group as it feeds the Related and Unrelated categories and also the fact that the Related group itself would begin to shrink as firms move to the Unrelated category and fewer enter it from the Single and Dominant categories. These projections should not be considered to be forecasts as the cetabus parabus assumptions are almost certainly not valid. Rather, the projections clarify the basic nature of the 1959-1969 transition pattern by showing what would happen if the identical pattern were repeated.

Paths to Diversification. The data presented so far indicates that the managements of a great many large firms saw the opportunity or felt the need to diversify during the past 20 years. Of all firms that were Single businesses or Dominant businesses in 1949 and were still among the largest 500 in 1969, 79% of the former and 45% of the latter had moved to categories of greater diversification by 1969. But, which firms were the diversifiers, and what strategies of diversification did they employ?

It is estimated that there were 52 Single business

firms in the 500 in 1949 that diversified by 1959, 41 moving to the Dominant and 11 to the Related category. The sample evidence on 1959-1969 is much more scanty, but the estimate is that 27 Single business firms diversified in this period, 17 becoming Dominant, 8 Related, and 2 Unrelated business companies.

Obviously, most of the firms that moved from Single business to Related or Unrelated business strategies passed through the Dominant category at some point of time. Therefore, it might be thought that the distinction between Single business firms that, 10 years later, obtained more than 30% of their revenues from new business areas and those that had not diversified to this degree is rather arbitrary. Were not, for instance, most of the firms that shifted from Single to Dominant postures between 1949 and 1959 simply "diversifying", having been caught in mid-stream in 1959? The evidence does not support such a conclusion. Leaving aside the special case of vertically integrated firms,⁵ there were 33 firms in the sample that were among the top 500 in both 1949 and 1959, and Dominant in 1959 (except Dominant Vertical). Of these, 15 had been Single

⁵ Of the firms in the 500 in 1949 and 1959, that appeared in the sample, 30 were Dominant Vertical in 1959, one having come from the 1949 Single business category. None of these moved out of the Dominant Vertical category by 1959.

businesses in 1949 and 18 had been Dominant. In the next decade, 1959-1969, these groups moved as follows:

<u>Group</u>	<u>Stayed Dominant</u>	<u>Became Related</u>	<u>Became Unrelated</u>	<u>Merged or Acquired</u>
1949-59: Sing.- Dom.	8	4	1	2
1949-59: Dom. - Dom.	7	7	2	2

While the numbers of firms involved in these movements are not sufficient to produce any statistically valid inference, it does appear that firms that went from the Single to the Dominant categories in the first decade were no more likely, and perhaps even less likely, to move on to the Related category in the next decade than firms that were Dominant in both 1949 and 1959. The hypothesized existence of this phenomena was, in fact, one of the major motives for defining the Dominant category in the first place. As the managers of many Dominant business companies seem either unwilling or unable to undertake further diversification, this category cannot be simply viewed as consisting of companies that are on their way to full diversification.

Most of the Single business diversifiers during both decades stayed close to home, entering only businesses that were closely related to ongoing activities. Of Single business diversifiers in the first decade, 14% moved into the Dominant Vertical, 35% into the Dominant Constrained, and 17% into the Related Constrained categories. In the second decade, 38% of Single business diversifiers were oil companies that moved

heavily into petrochemicals, plastics, fertilizers, and sometimes uranium production and coal mining, and another 38% consisted of firms entering the Dominant Constrained and Related Constrained categories. Table 3-9 gives a list of the Single business diversifiers in both decades.

Dominant business diversifiers also showed a very strong tendency to stay close to their major areas of competence in the first decade, but in the second decade the majority eschewed constrained diversification and moved into the Related linked and Unrelated categories. Shown below are the percentages of Dominant diversifiers that entered each Related and Unrelated sub-category during both decades.

<u>Decade</u>	<u>Estimated Number</u>	<u>Related Constrained</u>	<u>Related Linked</u>	<u>Passive Conglomerate</u>	<u>Acquisitive Conglomerate</u>
1949-59	32	86%	0%	14%	0%
1959-69	42	33%	38%	17%	12%

Most Dominant diversifiers came from the Dominant Constrained sub-class. During both decades about 44% of firms starting in this sub-class (and not exiting the 500) had diversified out of the Dominant category by the end of the decade. There were too few firms in the Dominant Linked and Unrelated sub-classes to warrant any conclusions about their propensity to diversify.

Very few vertically integrated firms were able to move into the Related or Unrelated categories in the period under study. In the first decade only 10% of Dominant Vertical

TABLE 3-9
SINGLE BUSINESS DIVERSIFIERS

Firm	Category Entered
Firms leaving the Single category between 1949 and 1959	
Calumet & Hecla Continental Oil Richfield Oil	Dominant Vertical
Container Corp. of America Dana Deere Ford Motor General Cable Pet Royal McBee White Motor	Dominant Constrained
American Enka Carrier Genesco Handy & Harmon	Dominant Linked
Chrysler General Host Phillip Morris	Dominant Unrelated
Eaton Yale & Towne Lockheed Aircraft Mead USM (United Shoe Machinery)	Related Constrained
Kelsey-Hayes	Related Linked
Firms leaving the Single category between 1959 and 1969	
Atlantic Richfield Gulf Oil Marathon Oil Mobil Oil Standard Oil of Ohio	Dominant Vertical
Brown Shoe Mohasco	Dominant Constrained
Hershey Foods	Dominant Linked
Ashland Oil Cluett, Peabody Coca-Cola	Related Constrained
Ward Foods	Related Linked
Champion Spark Plug	Unrelated Passive

Note: Corporate names are those in use in 1969 or at the time of acquisition or merger if the firm was no longer in existence by 1969.

firms moved out of the Dominant category. The three firms in the sample that accomplished this were General Mills and CPC International, both integrated grain millers that diversified into a broad spectrum of packaged foods, and AMAX, which was described in Chapter II. In the second decade Boise Cascade was the only firm observed to leave the Dominant Vertical group it entered the Unrelated category through an aggressive program of acquisition and internal development taking it away from timber and lumber into a wide range of construction related activities as well as education, office supplies, contract engineering, recreational products and computer software.

AMAX was the only one of the "heavy" vertically integrated firms in the sample that achieved full diversification. None of the other integrated primary metals, oil, or rubber manufacturers moved beyond the Dominant Vertical group. Significantly, the only major oil company that became Related was not integrated; Ashland, buying most of its crude and chemical feedstocks on the open market had, by 1969, expanded its petrochemical business and its activities in stone, gravel, and other road construction materials to 47% of revenues.

A list of the Dominant business diversifiers in both decades is provided in Table 3-10.

TABLE 3-10

DOMINANT BUSINESS DIVERSIFIERS

Firm	Category Entered
Firms leaving the Dominant category between 1949 and 1959	
ABEX AMAX American Optical Bell Borden CPC International General Mills Gillette National Cash Register North American Rockwell Owens-Illinois Pillsbury	Related Constrained
Curtiss-Wright Midland-Ross	Unrelated Passive
Firms leaving the Dominant category between 1959 and 1969	
Addressograph-Multigraph Pet Proctor & Gamble Stevens (J.P.) Stokely-Van Camp White Motor	Related Constrained
ACF Genesco Handy & Harmon Harsco Libby-Owens-Ford Times Mirror Time	Related Linked
Air Reduction Boise Cascade Martin	Unrelated Passive
Brunswick General Host	Acquisitive Conglomerate

Note: Corporate names are those in use in 1969 or at time firm was acquired or merged if it was no longer in existence by 1969.

Organizational Structure

It has been well documented that by the late 1960s most large corporations had adopted product division forms of organization. Wrigley estimated that 86%⁶ of the 500 largest industrials in 1967 had product division organizations and Fouraker and Stopford⁷ found 89.4% of 170 large U.S. firms with direct investments abroad were organized along divisional lines. In addition, Chandler's⁸ study, although not quantitative, showed that the earliest product division organizations appeared in the United States during the first decade of this century and that the long-term trend was towards the adoption of this structure by increasing numbers of large firms. Nevertheless, the data in Table 3-11 and Figure 3-2 represent what is believed to be the first quantitative study of this phenomenon over a significant period of time. The results were unexpectedly dramatic: Between 1949 and 1969 the estimated

⁶ As previously noted, the Wrigley sample, although a priori fair and unbiased, contained disproportionately few vertically integrated firms. Hence, his estimate of the percentage of product division structures is higher than the weighted estimates presented here.

⁷ Lawrence E. Fouraker and J.M. Stopford, "Organizational Structure and Multinational Strategy", Administrative Science Quarterly, June, 1968, p.59

⁸ Alfred D. Chandler, Strategy and Structure, (Garden City, N.Y., Doubleday, 1966)

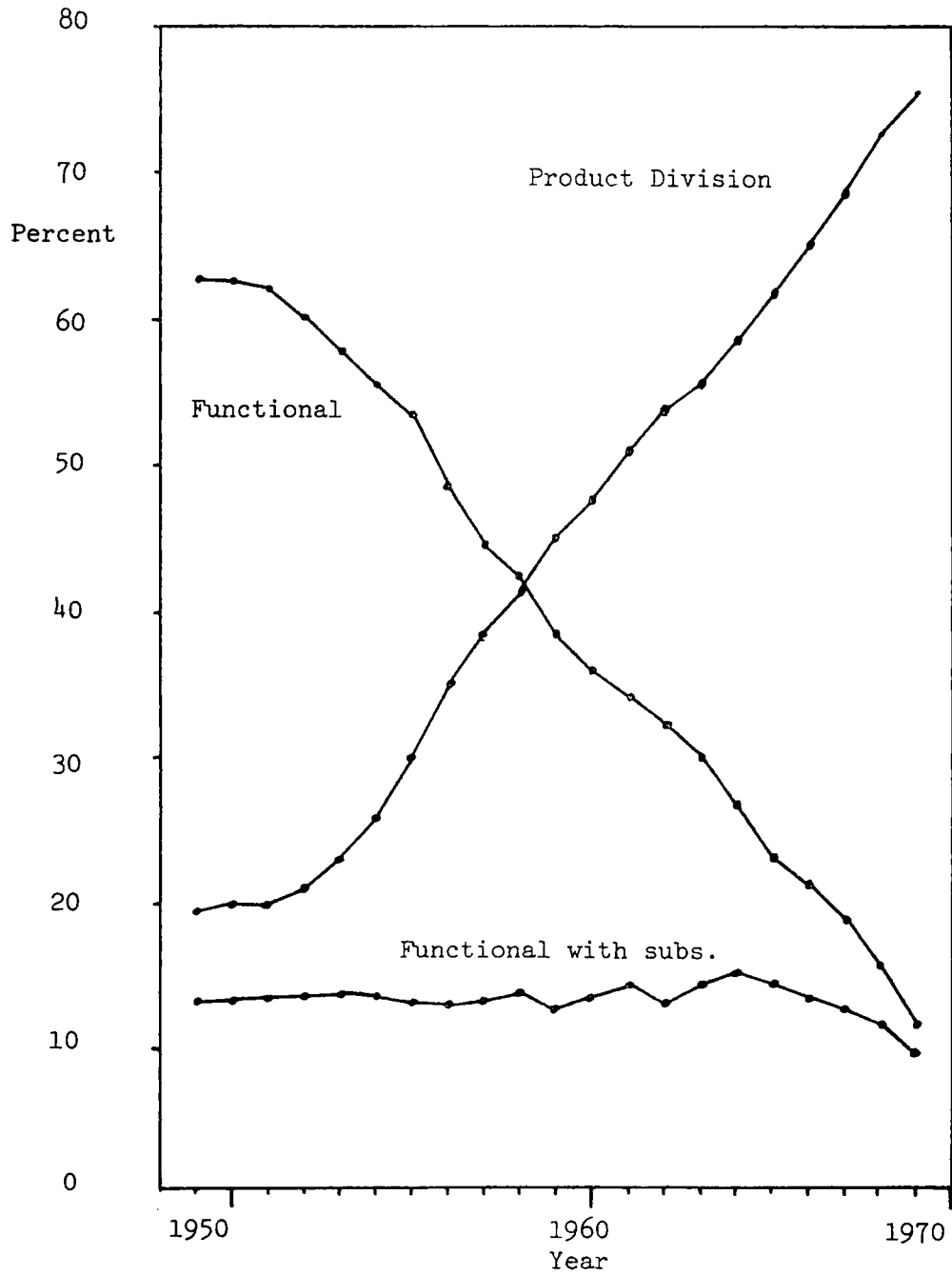
TABLE 3-11

ESTIMATED PERCENTAGE OF FIRMS IN EACH ORGANIZATIONAL CATEGORY

Organizational Category	1949	1959	1969
Functional	62.7	36.3	11.2
Functional with subs.	13.4	12.6	9.4
Product Division	19.8	47.6	75.5
Geographic Division	0.4	2.1	1.5
Holding Company	3.7	1.4	2.4
Number of firms used to derive the estimates	189	207	183

Figure 3-2

ESTIMATED PERCENTAGE OF FIRMS
IN EACH ORGANIZATIONAL CLASS: 1949-1969



percentage of firms among the largest 500 having product division organizations rose from 20.3 to 75.9 at a nearly constant rate of about 14 firms out of the 500 per year.

The necessary result of increasing numbers of product division firms among the 500 was, of course, decreasing numbers of functionally organized companies. Between 1949 and 1969, the percentage of functionally organized firms dropped from 62.2 to 11.2. As will be shown in the section on sources of change, most of this decrease was due to functionally organized firms adopting product division organizations rather than exiting from the 500.

The previously unstudied functional with subsidiaries category did not change sharply in size during the 20 years under study. It accounted for about 13% of the 500 in 1949 and 1959, then shrank somewhat to 9% in 1969. As expected, the geographic division category proved to be virtually unpopulated, accounting for only about 10 firms out of the 500 when at its maximum size. Also quite small, the holding company category decreased in importance in the first decade as managers moved to build true headquarters offices, then increased slightly in the 1960s as several acquisitive conglomerates with very "thin" top management groups entered the 500.

Sources of Change. As was done in the case of the strategic categories, changes in the percentage of firms in each organizational category were divided into two portions

corresponding to two major effects: entry-exit and organizational "shifting". This analysis demonstrated that the vast majority of the change in the distribution of organizational forms among the 500 was due to firms changing their form of organization rather than exiting and being replaced by firms with different structures. The calculations are shown in Table 3-12

TABLE 3-12

	<u>Functional</u>	<u>Functional/ Subsidiaries</u>	<u>Product Division</u>	<u>Geo. Division</u>	<u>Holding Company</u>
1949-59 Net Change	-26.0	-0.8	+27.3	+1.7	-2.3
Due to entry-exit	- 7.4	+0.1	+ 6.0	0	0
Due to shifting	-18.6	-0.9	+21.3	+1.7	-2.3
% Due to shifting	71.5%	111%	77.9%	100%	100.0%
1959-69 Net Change	-25.0	-3.20	+27.9	-0.7	+1.03
Due to entry-exit	- 7.9	0	+ 6.7	-0.7	+1.73
Due to shifting	-17.1	-3.20	+21.2	0	- .70
% Due to shifting	68.3%	100%	76.0%	0	-

Table 3-13 parallels Tables 3-5 and 3-6, providing estimates of the acquisition rates, entry rates, and inter-category transition rates for the organizational categories. As can be seen, the rate of flow into the product division category increased in the second decade, although it was substantial in both. The estimate that 1.2% of non-exiting product division firms reverted to functional structures between 1959 and 1969 is based on but a single observation: American Motors. There were no other cases in the sample of firms abandoning the product division structure.

TABLE 3-13

STRUCTURAL CATEGORY TRANSITION RATES

Starting Category	Percentage		Percentage Non-exiting Firms Ending in Each Category				
	Acq.	Displ.	F	F/S	PD	GD	H
1949-1959 Transition Pattern							
F	9.1	14.8	59.8	7.4	31.8	0.9	0
F/S	8.5	8.8	4.0	60.3	35.7	0	0
PD	28.0	5.6	0	0	100.0	0	0
GD	0	0	0	0	0	100.0	0
H	0	0	0	0	62.2	0	37.8
New Entries to the 500			31.1	10.1	53.6	5.3	0
1959-1969 Transition Pattern							
F	22.4	9.4	27.3	8.9	62.1	1.8	0
F/S	4.0	5.5	3.9	48.7	43.6	3.9	0
PD	17.4	6.5	1.2	0	98.8	0	0
GD	32.1	0	0	0	61.7	38.3	0
H	36.4	0	0	0	100.0	0	0
New Entries to the 500			14.2	4.8	71.4	0	9.6

It is also worth mentioning that the two firms in the sample that changed from functional to geographic division organizations in the first decade (Continental Oil and Foremost Dairies) created divisions based on areas of the United States. In the second decade, the two sample firms that entered this category formed world-wide geographic divisions. In 1967 Caterpillar Tractor split its functional organization into four major divisions, each responsible for a similar product line in a different region of the world. Dow Chemical had what is best described as a functional with subsidiaries structure until the mid 1960s, when it created a mixed structure with geographic, product and functional divisions which was classed as world-wide geographic divisions, as it was judged closer to this form than the other categories.

All of the firms in the sample whose organizations fell in the holding company category in 1949 developed true headquarters offices and strengthened administrative control over the divisions, moving to the product divisions category, by 1969. These companies were AVCO, ABEX, (American Brake Shoe), Borg-Warner, Genesco, Pullman, Questor (Dunhill), and Rockwell Manufacturing. Entering the 500 between 1959 and 1969, Bangor Punta had been a holding company when its sole business consisted of Cuban sugar plantations and its more recent strategy of being an Acquisitive Conglomerate did not alter the purely financial interest top management took in its

holdings. Also falling in the holding company category in 1969 was Ling-Temco-Vought. In 1959 LTV was classed as product division firm, but in 1967, at the zenith of its acquisition program, it was judged to have crossed the line and become a holding company, its acquisition of the Greatamerica Company, itself a holding company, being the deciding factor.

The functional with subsidiaries category did not change radically in size during the 20 years of observation; although many of its members shifted to product division organizations, the flow of firms moving into the category from the pure functional group kept the net change small. While it is tempting to think of the functional with subsidiaries structure as a transition structure adopted by firms in the process of moving from functional to product division structures, such a view is not borne out by the data. Of all firms that were in the top 500 in both 1949 and 1969 and dropped functional structures in favor of product division organizations in this period, it is estimated that only 3% used the functional with subsidiaries organization as an intermediate stage in the change (e.g. American Enka and Phillip Morris). The vast majority of firms observed moving from the functional with subsidiaries category to the product division category in this period had been in the functional with subsidiaries category in 1949. Of course, if the study had encompassed a time span longer than 20 years it is quite possible that the functional

with subsidiaries structure's role as an intermediate stage of organizational development would have been more evident. Nevertheless, if it is an intermediate and ultimately transient structure, it was little used as such during 1949 to 1969 and is remarkably stable, the median time to exit for firms in this category being at least 13 years.

The data on entrance rates in Table 3-13 indicates that in both decades the proportion of firms entering the 500 with product division structures was roughly similar to the proportion of firms in the 500 with product division structures. This tends to suggest that firms smaller than those in the 500, and in particular those that were growing at above average rates, were undergoing the same pattern of transition from functional to product division organizations as those in the 500.

The data on exit rates appears interesting because it seems that a disproportionate number of the firms that merged or were acquired in the first decade had product division structures. Out of the 11 firms in the sample that exited the 500 in 1949-1959 due to merger or acquisition, 5 had product division structures. If the likelihood of any firm being merged or acquired were independent of its structure, this result would have less than a 5% chance of occurring due to the random nature of the sample. The acquired firms with product division structure were Clinton Foods, Mengel, and

Pacific Mills, and the merging firms were Sperry and Remington Rand. Because the merging firms merged with each other, forming Sperry Rand, it can be argued that there were only 4 net exits from the product division class due to acquisition or merger. Such a result would not be statistically significant. Since in the second decade the acquisition rate for product division firms is not unusually high, there seems to be little to be concluded from the first decade acquisition rate anomaly other than the danger of reasoning from a very few data points.

Strategy and Structure

The purpose of this section is to describe the association between strategy and structure that was observed in the sample of 246 firms and to relate this data to existing theories and previous studies. A growing body of theory and empirical evidence support the idea that organizational structure is intimately related to the firm's strategy of expansion and the concepts and data developed by the major contributors to this field of study were reviewed in Chapters I and II. Thus, the issues addressed in this section are not new; the data to be presented does, however, provide the first quantitative portrayal of the strategy-structure interaction over a period of time.

It has already been shown that between 1949 and 1969 most of the largest 500 corporations became significantly diversified and adopted product division structures. It remains,

TABLE 3-14
STRATEGY AND STRUCTURE AMONG THE TOP 500

Strategic Class	Percentage in Each Structural Class				
	F	F/S	PD	GD	H
1949					
SINGLE	90.7	4.0	2.7	1.3	1.3
DOMINANT	56.8	23.2	17.1	0	2.9
DV	61.9	25.3	12.9	0	0
DC	53.1	18.8	22.4	0	5.6
DL+DU	50.0	50.0	0	0	0
RELATED	42.2	14.2	40.3	0	3.3
RC	49.0	17.8	30.9	0	2.3
RL	26.2	5.5	62.7	0	5.5
UNRELATED	0	0	61.2	0	38.8
UP	0	0	61.2	0	38.8
AC	—	—	—	—	—
1959					
SINGLE	90.6	5.8	0	3.6	0
DOMINANT	36.4	22.4	34.5	4.2	2.5
DV	41.7	20.8	34.5	3.0	0
DC	46.1	20.9	25.6	4.3	3.1
DL+DU	0	29.5	56.9	6.8	6.8
RELATED	20.1	8.3	71.6	0	0
RC	26.1	9.9	64.0	0	0
RL	4.0	4.0	92.0	0	0
UNRELATED	0	0	93.3	0	6.7
UP	0	0	91.8	0	8.2
AC	0	0	100.0	0	0
1969					
SINGLE	62.3	14.2	14.2	9.3	0
DOMINANT	20.7	17.5	60.3	1.5	0
DV	32.2	22.6	45.2	0	0
DC	14.2	6.2	73.5	6.2	0
DL+DU	0	17.8	82.2	0	0
RELATED	2.9	6.6	89.5	1.0	0
RC	4.1	9.7	86.3	0	0
RL	1.9	3.7	92.5	1.9	0
UNRELATED	0	2.3	85.3	0	12.4
UP	0	5.2	94.8	0	0
AC	0	0	77.8	0	22.2

however, to show that these changes were not separate parallel trends but reflect a cause and effect relationship between expansion strategy and organizational structure.

Table 3-14 shows the estimated percentage of the firms in each strategic class that fell in each organizational category in 1949, 1959, and 1969. The figures in this table are independent of the number of firms in the strategic classes and can be interpreted as indicating the likelihood of a firm having a particular type of structure, given its strategy. To simplify the discussion, Table 3-15 was constructed from this data.

TABLE 3-15

Incidence of Multi-divisional Structure

<u>Strategic Category</u>	Percentage of Firms in Strategic Category Having Product Division or Holding Company Structures		
	<u>1949</u>	<u>1959</u>	<u>1969</u>
Single business	5.3	0.0	14.2
Dominant Vertical	12.9	34.5	45.2
Dominant Constrained	28.1	28.7	73.5
Dominant Linked or Unrelated	0.0	63.7	82.2
Related Constrained	33.2	64.0	86.3
Related Linked	68.2	91.9	92.5
Passive Conglomerate	100.0	100.0	94.8
Acquisitive Conglomerate	-	100.0	100.0
All categories	23.9	49.0	78.4

In Table 3-15 the holding company and product division categories have been combined. Although having differing degrees of headquarters control over the divisions, both represent specialization on the basis of product-market activity rather than

function or geography and are similar in comparison to the functional and geographic categories. Also combined in Table 3-14 and 3-15 are the Dominant Linked and Dominant Unrelated categories. These classes were pooled due to the lack of enough data to arrive at meaningful separate organizational breakdowns.

Two distinct patterns are clearly visible in Table 3-15. First, in each year the greater the diversification of the firm the more likely it was to have a multi-divisional structure. With only one minor exception (Dominant Linked and Unrelated in 1949), the percentage of firms with multi-divisional structures constantly increases as one moves from Single to Dominant, Related, and Unrelated strategies and even as one moves toward increasingly diversified minor classes within each category. Second, with only two exceptions, the likelihood that a firm with a given strategy had a multi-divisional structure increased with time for all strategies. Thus, combined with a definite positive association between diversification and divisionalization was a secular trend towards divisionalization that effected firms in all strategic categories.

The total rise in the number of multi-divisional firms among the top 500 can therefore be ascribed to two different trends: Since more diversified firms tend to be more likely to divisionalize the trend towards diversification increased the number of multi-divisional firms among the 500, and the trend

towards divisionalization per se had a parallel effect. By projecting what would have happened if either trend had not occurred, it can be estimated that of the total increase in divisionalization among the 500 between 1949 and 1969, amounting to 256 firms, 53% was due to increased diversification and 47% to the secular trend towards divisionalization.

It should be noted that in the case of Single business firms there is some doubt as to the existence of the secular trend towards divisionalization. Between 1949 and 1959 the two Single business firms in the sample that had product division structures (Ford and Continental Container) moved out of the Single business category. In 1959 none of the Single business firms in the sample had product division structures. By 1969 the Single business group was quite small and the two firms observed to have product division structure (Donnelly and Interstate Brands) constituted 14.2% (weighted) of the entire class. Both had divisionalized in 1968-1969. Donnelly was particularly difficult to classify and its placement in the Single business group was somewhat arbitrary. Interstate Brands, a bread and cake producer, had announced a diversification program in 1967 and by 1969 had acquired a number of firms in the packaged and canned foods business. However, over 95% of its 1969 revenues still came from bread and cake. Instead of absorbing acquisitions into its functional structure or moving to a functional with subsidiaries structure, Inter-

state created separate divisions for bread, cake, and each of its acquisitions, apparently reacting more to the anticipation of further diversification than to the amount already achieved in 1969. Donnelly's 1969 divisionalization was not in response to increased diversification--management simply decided that the company's book printing and commercial printing activities would be carried out more effectively under separate general managements. Thus, great credence should not be placed in the correctness of the value 14.2%. Both cases, however, might be taken as indicative of an increasing propensity of firms to divisionalize even when product variety is low or when one product is still overwhelmingly dominant.

Among Dominant business firms the trend towards increased use of the product division structure was especially pronounced. In 1949 only 17% of Dominant business companies were divisionalized while by 1969 60% had adopted this form of organization. Table 3-14 shows that even many vertically integrated firms moved from functional to divisional structures, although the proportion of Dominant business having functional with subsidiaries structures remained fairly constant.

Most of the vertically integrated firms that adopted product division structure were in the rubber, forest products, and aluminum industries. Firms like Goodrich, Firestone, Kimberly-Clark, Scott, Kaiser Aluminum and Reynolds, despite their basic commitment to the processing of a particular material,

tended to emphasize the fabrication and sale of a variety of specially designed products. Having a scale sufficient to permit duplication of facilities, strategies of fabricating almost all output and aggressive marketing to a variety of customers led the managements of these firms to choose a product division structure. On the other hand, most of the vertically integrated oil and steel producers had functional or functional with subsidiaries structures.

The oil industry differs in so many essentials from other industries that often little is gained by comparing oil companies with other firms. In no other industry are such vast vertical processing chains almost entirely dedicated to producing a single virtually undifferentiated product that is sold directly to consumers. Most oil companies have employed functional or geographic division structures in 1969. The exceptions were Ashland (refinery products, petrochemicals, road construction and material), which was not integrated, and Continental Oil which had divided its petroleum operations into East Coast and West Coast divisions and had coal, chemicals, and plastics, and fertilizer divisions as well--actually a mixed geographic/product structure.

Steel firms have, if anything, tended to move to more centralized structures for their basic businesses, having subsidiaries carry out fabrication and marketing tasks that are not included in the primary activity of converting ore and coke

into mill products. Armco Steel and Keystone Steel, both of which sold relatively small amounts of standard mill products (52% and 20% of revenues in 1969 respectively), were the only steel companies observed to have product division structure in 1969.

Thus, those vertically integrated firms, which choose or belonged to industries which encouraged an emphasis on product rather than process and on marketing rather than production have been the ones that adopted product division organizations.

Structure and Diversity. We have seen that, in at least a statistical sense, diversification strategy explains a good deal of the variation in organizational forms observed among the top 500 firms. To what degree do firm to firm variations in diversity per se explain the differences in structure, and how much is gained by using strategy as the predictor of structure rather than simple diversity?

We take as a general measure of diversity the firm's specialization ratio. Table 3-16, shows, for the firms for which specialization ratios were obtained, the mean specialization ratios for firms in each of the major organizational categories. The geographic division and holding company categories are not considered here as not enough data on them was available.

TABLE 3-16

AVERAGE SPECIALIZATION RATIOS

<u>Structure</u>	<u>1949</u>	<u>1959</u>	<u>1969</u>
Functional	0.842 (114)	0.832 (62)	0.811 (17)
Functional/subs.	0.700 (27)	0.643 (22)	0.675 (14)
Product Division	0.562 (31)	0.492 (75)	0.469 (118)

Note: Figures in parentheses are the numbers of data points used in calculating each average.

This table again shows the general trend towards diversity over time and the tendency for more diverse firms to have product division structures. A two-way analysis of variance of this data reveals that the relationship between structure and diversity is statistically significant at the 1% level. Holding the effect of time constant, it was found that functional with subsidiaries firms had specialization ratios that averaged 0.152 less than those of functional firms and product division firms had specialization ratios that on average were 0.326 below those of functional firms.

To compare the explanatory power of strategic categories to that of a straight specialization ratio, the following question was asked: If all one knew was each firm's strategic class, how many mistakes would be made in guessing whether or not each firm had a product division structure, and how many mistakes would be made if one had to guess knowing only the

specialization ratio of each firm? In "guessing" structure from strategy, Table 3-14 was used. For a given strategy in a given year, if the table entry was greater than 0.5 it was predicted that a firm with that strategy had a product division structure. In 1949, this procedure would have resulted in 17.6% misclassifications and 16.2% in 1969. In "guessing" structure from specialization ratios, the detailed distributions of specialization ratios were examined and the value found which minimized the expected number of errors. For 1949 and 1969 these values of SR were 0.725 and 0.57 respectively giving percentage misclassifications of 30% and 27% in the same order. In other words, the best possible dividing line between product division and functional forms of structure was an SR of 0.725 in 1949 which, if used as such, would have resulted in 30% of all firms being erroneously classed as either functional or divisionalized.

In both 1949 and 1969 prediction of structure using only specialization ratios would have produced about 70% more errors than prediction using strategic categories. Consequently, we can conclude that in predicting structure the strategic categories are significantly superior to measures based only on the amount of diversity.

Strategic and Structural Change. If structure and strategy are casually related, not only will certain structures be associated with certain strategies of expansion, but strategic

change will tend to be matched by structural change. A simple test of this proposition is provided by Table 3-17.

TABLE 3-17

Percentage of Non-divisionalized Firms Shifting to
Multi-divisional Structures

Category	Non-diversifiers	Diversifiers	Significance of Difference
1949-1959			
Single business	0	15.1	.01
Dominant business	34.7	88.0	.10
Related business	55.7	-	-
1959-1969			
Single business	19.8	41.7	.05
Dominant business	52.0	61.2	Not significant
Related business	79.9	-	-

Note: Non-diversifiers are firms that did not move out of their major strategic category. Diversifiers are Single business firms that moved to the Dominant or Related categories and Dominant business firms that moved to the Related category. Tests of significance used were the Chi-square and exact binomial. The level of significance is the probability that a true condition of no difference could have produced the observed difference by chance.

The criterion used in this table is the percentage of all functional, functional with subsidiaries and geographic division firms that switched to a product division structure by the end of the decade. The evidence shows that in the first decade firms that moved to more diversified postures were more likely to divisionalize than similar firms that did not increase their degree of diversification. In the second decade, the pattern continued to hold for firms that started in the Single business category, but Dominant business firms seemed to divisionalize at the same rate whether or not they further diversified.

Since many of the 1959-1969 Dominant business diversifiers moved out of the Dominant category in the last few years of the decade, a time lag between strategic and structural change might be the explanation for the apparent lack of increase in divisionalization with diversification. As Chandler noted, after reviewing the organizational histories of General Motors, Du Pont, Standard Oil of New Jersey and Sears Roebuck,

Yet, structure often failed to follow strategy. In each of the four companies, there was a time lag between the appearance of the administrative needs and their satisfaction. A primary reason for delay was the very fact that responsible executives had become too enmeshed in operational activities.

On the other hand, as there were only 13 Dominant diversifiers during 1959-1969 in the sample, there is a good chance that even if a relationship between diversification and divisionalization did exist for this group, a larger sample would be required to reveal it.

In addition to the effect of strategy on structure, there is the possibility that structure affects strategy. If a Single or Dominant business firm adopts a product division structure, perhaps in response to relatively minor variations in products, it might be more likely to diversify further than if it had not reorganized. The general managers in the headquarters office, freed from responsibility over day-to-day operations, may gain the objectivity required to look for op-

⁹ Ibid., p.389

portunity beyond the current scope of their business. Furthermore, if diversification is attempted, it may proceed more rapidly within the administrative climate of the divisionalized firm. Table 3-18 was constructed to test this hypothesis. As in Table 3-17, the major strategic categories have been split into diversifiers and non-diversifiers, but here the criterion shown is the percentage of each group that had multi-divisional structures at the start of each decade.

Table 3-18

Percentage of Firms Having Multidivisional Structures
at the Start of Each Decade

Category	Non-diversifiers	Diversifiers	Significance of Difference
1949-1959			
Single business	0	13.2	0.005
Dominant business	6.7	33.1	0.025
1959-1969			
Single business	0	0	no difference
Dominant business	32.0	37.3	not significant

Note: Comments appended to Table 3-17 also apply to this table.

The evidence shows a strong connection between divisionalization and subsequent diversification in the first decade, but none in the second decade. Since no Single business firms in the sample had product division structure in 1959, there is no way to measure the effect of divisionalization on Single business diversification during 1959-1969. In the first decade divisionalized Dominant business firms were more likely to shift to the Related class, but, as in the

case of the strategy-structure relationship, in the second decade this association between structure and strategy was absent.

In conclusion, divisionalization and diversification were strongly associated in 1949-1959. In the next decade the evidence suggests that Single business firms again were likely to divisionalize upon diversifying. However, during 1959-1969 the pattern of Dominant business diversification changed and interdependence with structure and structural change, if in existence, was not evident within the sample. This suggests that up until the early 1960s the adoption of product division structures was strongly contingent upon the administrative pressures created by diversification but that in more recent years divisionalization has become accepted as the norm and managements have sought reorganization along product division lines in response to normative theory rather than actual administrative pressure.

Summary

During the period 1949-1969 basic changes took place in the diversification strategies and organizational structures of large industrial firms. Many companies that had previously had narrow product-market scopes elected to diversify into new businesses related to their past strengths and in the latter portion of this era the tendency for such firms to move into lightly related or unrelated businesses increased. The only

major exception to this pattern were the vertically integrated corporations. While many of these large industrials developed new business activities based upon by-products and the further processing of their main products, few chose to diversify into activities that did not make direct use of their raw material resources or primary production capacity.

The trend towards divisionalization paralleled the increase in diversification but was not wholly dependent upon it. While diversification and divisionalization were closely linked in the 1950, during the 1960s the link was less clear, although both trends continued unabated.

The effects of both types of change have been profound. In 1949 the typical large firm followed a Single business strategy and was functionally organized. By 1969, the modal strategy was Related Linked and more than three-quarters of large firms had product-division structures. This change has a dual nature: diversification serves to insulate the firm from the risks associated with each separate product-market activity but divisionalization places a particular unit of the firm in a close relationship with each business area. The result is a new type of industrial structure that is not adequately described by the traditional bureaucratic, oligopoly, industry structure, or competitive models of corporate behavior and influence.

APPENDIX 3A

ESTIMATION TECHNIQUE

Most of the analyses presented in this chapter were based on estimates of the proportions of the largest 500 industrial firms that fell in various categories at some particular time. It is necessary, therefore, to explain how these estimates were obtained. To simplify the exposition, the problem of estimating the percentage of firms in the Dominant Vertical category will be used as an example. By chance, it appears that the sampling errors associated with this category were the most serious, so that while similar procedures were applied to all categories, the Dominant Vertical category most clearly illustrates their need.

The random samples taken from the largest 500 firms in 1949, 1959, and 1969 can be used to provide estimates of the percentage of Dominant Vertical firms in each of these years. Of the 100 firms in the 1949 sample, for instance, 17 were classified as Dominant Vertical in 1949. Similarly, 18 of the 100 firms in the 1959 sample and 9 of the 97 firms in the 1969 sample were classified as Dominant Vertical in those years. Table A3-1 displays these results. The percentages shown in this table are, of course, only estimates of the true percentage of firms in the Dominant Vertical class in these years.¹⁰ It is also important to note that (1) some of the

¹⁰ The 95% confidence intervals for the three percentages are 10.1-23.9%, 11.1-24.9%, and 4.0-14.6% respectively.

firms in one sample also appear in other samples so that the total number of firms involved is 246 rather than 297, (2) although the strategic class of each firm was determined for all three years, the only information used in constructing Table A3-1 was the firms' classifications in the years corresponding to the samples to which they belonged.

Table A3-1

Dominant Vertical Firms in Each Sample

<u>Sample</u>	<u>Sample Size</u>	<u>Number Dominant Vertical</u>	<u>Percentage Dominant Vertical</u>
1949	100	17	17.0%
1959	100	18	18.0
1969	97	9	9.3

If the estimates are reasonably good, there is little incentive to develop more cumbersome procedures in order to use all of the data. However, in this case a rather serious sampling error seems to have occurred. The apparently striking drop in the proportion of Dominant Vertical firms between 1959 and 1969 is not consistent with other evidence. Investigation of the sources of this change revealed that of the 18 Dominant Vertical firms in the 1959 sample, 17 were still in this category and still among the top 500 in 1969. Furthermore, of all firms in all samples, 32 were Dominant Vertical in 1959 and 33 in 1969. Why then did the estimated percentage drop from 18.0 to 9.3? The only reasonable explanation is that, by chance, the 1969 sample contained an unusually small number of Dominant Vertical firms.

The very fact that we were able to detect this error suggests that a better estimate can be made, and such inconsistencies in the data can be resolved. One method that might be used to make fuller use of the available data would be to pool all the firms, counting the number that were Dominant Vertical among all firms in the largest 500 in a given year without regard to their sample membership. However, this procedure invites bias. Firms selected randomly from the 500 in 1949 and 1959, even though they may still be among the top 500 in 1969, are likely to be larger than the average firm of the 500 in 1969, and perhaps differ in other ways as well. To use all of the available data and avoid this type of bias we must take into account the fact that firms selected randomly from the top 500 in one year are not typical representatives of the top 500 in another year. This can be accomplished by defining seven mutually exclusive collectively exhaustive groups of firms, each group corresponding to a different pattern of membership in the top 500s of 1949, 1959, and 1969. One group consists of firms that were among the top 500 in 1959 but not in 1949 or 1969, another of those firms that were among the top 500 in 1959 and 1969 but not 1949, and so on. These seven groups are described in Table A3-2.

TABLE A3-2

Defining the Groups

Group	Symbol	Number of Sampled Firms in Group	Years in which group members were among the largest 500		
			1949	1959	1969
1	G ₄₉	18	X		
2	G ₅₉	14		X	
3	G _{49,59}	31	X	X	
4	G ₆₉	18			X
5	G _{49,69}	3	X		X
6	G _{59,69}	25		X	X
7	G _{49,59,69}	137	X	X	X
Total		246			

All of the firms among the top 500 in any of the years of interest fall among some subset of these groups. For example, the top 500 in 1969 is composed of $G_{69} + G_{49,69} + G_{59,69} + G_{49,59,69}$. For our purposes, the most important characteristic of these groups is that, for instance, the chance that a firm randomly selected from the top 500 in 1959 will fall in group $G_{59,69}$ is the same as the chance that a firm randomly selected from the top 500 in 1969 will fall in that group. Therefore, those firms of the 246 that fall in each group, regardless of their original sample memberships, constitute valid random samples from each group and may be used to estimate the properties of these groups. Then, if the true values of the group sizes are known, the properties of the appropriate groups can be combined to give an estimate pertaining to the top 500 in

any of the three years in question. Thus, all of the data can be employed without bias.

Clearly there is some true number of firms in each group which could be established by comparing lists of the largest 500. The problems of name changes, the lack of such a list for 1949, and the fact that we wish to use this procedure for the top 100 and top 200 as well, greatly complicate the task of counting. Hence, estimates were used. The size of each group was estimated by combining the independent estimates obtained from each sample in a weighted linear regression. The results were:

TABLE A3-3

<u>Group</u>	<u>Estimated Number of Firms</u>	<u>Estimated Proportion of the Largest 500 in the Relevant Years</u>
G49	102.3	0.2046
G59	48.1	0.0963
G49,59	78.3	0.1565
G69	108.7	0.2174
G49,69	17.7	0.0354
G59,69	71.9	0.1438
G49,59,69	301.7	0.6035

Returning to the problem of estimating the percentage of Dominant Vertical firms, we find the percentage of Dominant Vertical in each group for each year to which it applies, weight these estimates by the fraction of the top 500 represented by each group, and then sum over the appropriate years. For the case of 1969 calculations are shown below in Table A3-4.

TABLE A3-4

Sample Calculations

<u>Group</u>	(a) <u>Percentage</u> <u>Dom. Vertical</u>	(b) <u>Fraction of 1969</u> <u>500 in the Group</u>	<u>(a) x (b)</u>
G ₆₉	5.56	0.2174	1.21
G _{49,69}	33.33	0.0354	1.18
G _{59,69}	0.0	0.1438	0.0
G _{49,59,69}	21.90	0.6035	<u>13.22</u>
Weighted estimate =			<u><u>15.61%</u></u>

Similar calculations for 1949 and 1959 result in these estimates for the percentage of Dominant Vertical firms:

<u>Year</u>	<u>Percentage Dom. Vertical Among</u> <u>the Largest 500</u>
1949	15.7%
1959	14.8
1969	15.6

As can be seen, the sharp decrease of Table 3-1 is no longer evident. In addition to providing better estimates, the procedure just described also ensures that the estimates will be consistent with analyses of the sources of change. All of the estimates presented in this chapter were obtained in this manner.

CHAPTER IV

FINANCIAL PERFORMANCE CHARACTERISTICS OF THE CATEGORIES OF DIVERSIFICATION STRATEGY AND STRUCTURE

The purpose of this chapter is to present and interpret the results of an analysis of the average financial performance achieved by firms in each of the categories of diversification strategy and structure. The focus will be on the differences in performance among the categories, with membership in a category taken to be a relatively stable and continuing property of the firm.

The strategy of diversification, though creating complex problems of planning and administration, is generally thought to provide more than compensating benefits. Various writers have theorized that diversification should decrease the variability of a firm's earnings, permit the pursuit of both growth and profit without conflict, expedite adaptation to change and provide economies of scale in general management expertise and research and development. Accordingly, one would expect that the financial performance of diversified firms would be superior to that of non-diversified firms. In this chapter it will be shown that although this is true in a very general sense, there are several important exceptions and qualifications. In particular, the Unrelated Passive group has experienced

below average rates of profit and growth and the non-vertically integrated Single and Dominant business firms have, in some respects, out-performed Related business diversified corporations.

Additionally, it will be shown that although most of the differences in financial performance among categories can, in a sense, be attributed to a strong association between a firm's industry and its strategy, the association is so strong that it is virtually impossible to separate their relative effects. Consequently, diversification strategy cannot be viewed as simply the result of management action but must be seen as inextricably mixed with the technological, economic, and competitive characteristics of the industry or industries of which the firm is a part.

Before the empirical data is presented, we shall review the nature of the results that were expected. While several of these conjectures were not supported by the data, the hypotheses and the conceptual schemes that underlie them provide starting points for the analysis of the data and, more importantly, they provide a framework for distinguishing between empirical results that were expected and those that were surprising.

Theory and Hypotheses

Before the empirical portion of this study was begun a number of a priori expectations or hypotheses were

developed concerning the relative financial characteristics of the categories. Most were derived from five different, but not necessarily exclusive, views of diversification and the behavior of diversified firms. They were (1) the portfolio concept of diversification, (2) the theory that diversification is a response to declining rates of profit and growth in a firm's original business, (3) the theory that diversification and divisionalization are associated with high returns to scale in the general management function and the area of research and development, (4) the conjecture that diversified divisionalized corporations possess control, reward, and resource allocation systems that are basically different from those of undiversified firms, and (5) the observation that in certain circumstances the strategy of being an acquisitive conglomerate has proved startlingly successful, at least in terms of the growth achieved.

Portfolio Risk. One of the simplest and most often used arguments in favor of diversification is the reduction in total risk or variability in earnings that comes from spreading investment funds and effort among several businesses. Just as a mutual fund reduces the risk inherent in purchasing a single security by holding many, management can employ diversification to reduce the total risk borne by a corporation.

Until fairly recently it was commonly believed that the smaller the commitment made to any particular security

(or business) the more perfect the diversification.¹ However, the application of the modern Markowitz² diversification theory to the securities market has shown that the best possible diversification is achieved by portfolios consisting of a few carefully selected securities (often less than 10). The key requirement for efficient diversification is that negatively correlated returns be sought -- if one goes up the other goes down. This result suggests that "too much" diversification can increase the risk faced by a corporation and that optimal diversification, in the statistical sense, is not necessarily obtained by investing in completely unrelated areas. For example, the ultimate diversified company, participating in all sectors of the economy would, by definition, have an average return and still bear the risk associated with major economic cycles. A firm active in military, space, and consumer electronics might earn an equal or better return and obtain superior protection against economic cycles.

There are, however, variations in sales and earnings that are of greater interest to the business manager than those caused by economic cycles; it is the secular shift in demand

¹ Federal tax law, for example, gives tax benefits only to those mutual funds which have less than 5% of their assets in any particular security.

² Harry M. Markowitz, Portfolio Selection: Efficient Diversification of Investments (New York, John Wiley & Sons, 1959).

due to product obsolescence that is the major business risk of this type, as the "cycle" in this case promises no recovery from the downward slide. Thus American Car and Foundry's diversification away from railroad cars to escape the "cyclical nature of the business" and the major dairy companies' moves into chemicals and packaged foods can be seen as attempts to reduce the evident risk of remaining totally committed to a declining or stagnant product-market area. It is hypothesized that the risks associated with product life-cycles, rather than with variability per se, provide the primary motive for diversification and are those from which diversification can provide the most important relief.

The value of negatively correlated returns implies that the best defense against a declining market is not a new unrelated activity but a new product that is functionally related to the reasons for the declining sales or profitability of the old. Thus, the optimal response to a declining demand for, say, textbooks is not to emphasize fiction or move into electronics but to create a product that should grow for the very reason textbooks are declining. If, for example, college instructors are making greater use of original sources, one might offer a scholarly paper reprint service along with collected excerpts from primary sources. This type of strategy, one of response rather than escape, requires two things:

(1) a strategy of defining one's business in terms of a function rather than a product and (2) the possession of sufficient managerial, technological and marketing channel breadth and experience to permit rapid innovation and successful implementation of innovation.

The corporate characteristics required to best mitigate the risks associated with the product life-cycle are most closely represented by the Related business group, and in particular those Related business firms that have not diversified to such an extent that successes as well as failures are averaged out. Therefore, it was hypothesized that the Related Constrained category would experience the lowest risk in earnings and have an average rate of return at least as large as those of other categories. In addition, it was conjectured that the ranking of the categories, in order of increasing riskiness relative to return, would be: Related Constrained, Related Linked, Unrelated, Dominant and Single.

The Escape Paradigm. For a great many firms diversification is the means employed to escape from declining prospects in their original business areas. Poor absolute performance is often the result of participation in a highly competitive non-innovative slow growth industry or simply due to failure to cope with competition in a potentially promising area. Some examples are American Car and Foundry and American Brake Shoe, both of which moved away from the railroad equipment

industry in the 1950's, Armour's attempts to diversify away from meat packing, Borden, Pet and Fairmont's programs of decreasing their dependence on low margin dairy staples, General Foods' similar escape from flour milling, and Republic and U.S. Steel's ventures in housing.

If such firms can develop a strategy of diversification that builds on skills already possessed, namely a type of Dominant Constrained or Related Constrained strategy, it is expected that there would be a greater chance of success than if tenuously related or unrelated businesses are entered. The firms seeking escape through diversification that possess few exploitable skills are most likely to adopt Dominant Linked, Dominant Unrelated and Unrelated strategies, and will almost unavoidably use acquisition as the means of change. Since there is no a priori reason to expect acquisitions that do not build on current businesses to provide more than an average return and since, at least in the Dominant category, the original poor performing business will outweigh new activities, it is expected that Dominant Linked, Dominant Unrelated and Unrelated firms will generally have less than average returns on investment.

The above conjecture must be somewhat qualified for the case of science-based industries. In industries such as chemicals, electronics, and complex business and industrial machinery, Linked and Unrelated postures may well be the result,

perhaps the unexpected result, of science-based research. In such cases high performance would follow from the proprietary benefits that are typically associated with technological innovation. Consequently, the hypothesis offered is that firms that are not members of science-based industries that adopt Dominant Linked, Dominant Unrelated or Unrelated Passive strategies will experience returns on equity that are significantly lower than those achieved by firms in the same industry but following other strategies.

Organizational Scale. In a study of the organizational pre-conditions for direct investments abroad Stopford³ found a close association between research and development intensity, product innovation, the product division structure and the degree of direct foreign investment. In a subsequent paper, Fouraker and Stopford theorized that this relationship was due to the competitive advantage innovating firms have in foreign markets and the superior ability of product division organizations to train and use general managers and to bring a diversity of staff specialists, technologies and managerial experience to bear on opportunities in several areas. They concluded that their data demonstrated that:

. . .the growth of foreign markets and opportunities requires diversification, reorganization, and the training of many more general international managers. The

³ John M. Stopford, "Growth and Change in the Multinational Firm" (unpublished doctoral dissertation, Harvard Business School, 1968).

organizations that have been most successful in meeting this new challenge have been those (product division organizations) that had already developed the ability to produce general managers capable of controlling and guiding a heterogeneous, diverse enterprise.⁴

While foreign investment behavior is not investigated in our study, Fouraker and Stopford's conjecture that the product division structure provides special advantages in innovative activity and develops the general management skills necessary to exploit these advantages is of interest. A further elaboration on this theme was provided by Vernon who noted that such advantages could be seen as economies of scale. Arguing that large firms are better at the whole process of innovation -- not necessarily at the act of invention itself, but at the development and marketing activities which must be carried out to realize the benefits of innovation -- Vernon suggested that the product division structure is the key to gaining these advantages. The "constant process of search and adaptation, a process which continually pushes the firm into new products and new markets while permitting it to slough off the more senescent activities in which it has been engaged" he stated,⁵ was best carried out

⁴ Lawrence E. Fouraker and J.M. Stopford, "Organizational Structure and Multinational Strategy", Administrative Science Quarterly, June 1968, p.64.

⁵ Raymond Vernon, "Organization as a Scale Factor in the Growth of Firms", in J.W. Markham and G.F. Papanek (editors) Industrial Organization and Economic Development: In Honor of E.S. Mason (Boston, Houghton Mifflin, 1970), p.63.

in a product division framework due to its openness to change, explicit procedures for resource allocation and heavy emphasis on localized coordination across functions.

These observations suggest that the performance advantages of diversified divisionalized corporations, if such advantages exist, will be greater in environments calling for and producing relatively rapid technological change. More formally, it was hypothesized that the effects of the product division structure on growth and profitability will be more favorable in environments that require or encourage relatively rapid technological change. Note that this hypothesis does not imply that product division firms necessarily have growth and profit rates higher than those of functional firms. It merely states that the effects of the product division structure on performance will be relatively more favorable in environments characterized by change.

Systems Effects. The purpose of systems of reward, control, and resource allocation is to influence behavior and performance. If, as several writers have pointed out, diversified divisionalized firms employ systems that are significantly different from those used by low diversity functionally organized corporations, the effects of the difference might be observable in their respective records of financial performance.

As noted in Chapter II, Scott's stages model of corporate development calls attention to many of the ways in which diversified divisionalized firms differ from simpler forms of enterprise. Goals, he suggests, become increasingly quantitative and oriented towards return on investment as the firm becomes active in diverse markets. While objectives such as obtaining the largest market share, the most sales, or the highest margins may be useful in a oligopolistic environment, objective comparison between businesses that compete internally for corporate funds as well requires standards that are uniform and directly related to the efficient use of resources -- return on investment. The explicit and quantitative nature of the goals employed by divisionalized firms is matched by the creation of more formal systems of reward that tie promotion and bonuses to economic performance. The effects of these changes are compounded by the increase in the number of general management roles that comes with diversification and divisionalization. The ultimate general manager in a functional firm is the chief executive; for him promotion is irrelevant and only disastrous performance or age can usually unseat him. The division managers of a diversified firm, on the other hand, compete directly with one another as well as with outside firms, may receive substantial

rewards in terms of money and status for outstanding performance and are subject to transfer, dismissal or bonus reduction if their performance is below par. Scott states that:

... the important point is that ease of measurement by the numbers (profit numbers) gives high visibility for good performance, hence the potential for rapid promotions in competition with others. A manager thus has the opportunity to make his way by "getting results" as much as by getting on well with his immediate superiors. A divisional organisation with its open competition moves to another way of life, as removed from the functional organization and its bureaucratic traditions as the functional is from the family like organization of the entrepreneurial stage.⁶

This internal socio-economic system that creates pressure for the efficient use of capital also possesses the important property of being easily expandable. It is often simpler to create a new division to handle a new business than to merge its operations with those of a functional structure. Of course diversification by acquisition almost always requires a product division structure. The economic significance of these two structural attributes is that the conflict between the goals of high return and high rates of growth may be resolved by the divisionalized corporation.

Traditional microeconomic theory predicts that there will be a negative relation between the firm's rate of growth and its profitability. That is, in a one business equilibrium situation, high rates of growth are achieved by accepting lower rates of return on incremental investments. While the normative theory of growth is not sufficiently well developed to

⁶ Bruce R. Scott, "Stages of Corporate Development" (unpublished paper, Harvard Business School, 1970) p.37.

enable one to determine the optimal growth rate for a particular firm, many researchers have suggested that business executives often seem to be overly concerned with growth.

Baumol notes that:

... economists who have spent time observing the operations of business enterprises come away impressed with the extent of management's preoccupation with growth. Expansion is a theme that (with some variations) is dinned into the ears of stockholders and is constantly reported in the financial pages and the journals devoted to business affairs. Indeed, in talking to business executives one may easily come to believe that the growth of the firm is the main preoccupation of top management.⁷

Whether it stems from oligopolistic competition or simply the desire to preside over an ever larger enterprise, the pursuit of growth in a single business firm must be at the expense of profit. The divisionalized firm, by contrast, may be able to separate these goals to a much greater extent, pushing for maximum return in its current areas of operation and achieving growth through continuing diversification. The hypothesis that is offered to summarize this argument is that among firms with product division structures the relationship between growth and return on investment will not be as strong as among other types of firms.

Note that the above hypothesis does not imply that product division firms have, on average, either growth rates

⁷ William J. Baumol, Business Behavior, Value and Growth, (New York, Harcourt, Brace & World, 1967), p.87.

or rates of profit that are higher than those of other types of firms. While they may out-perform functional firms, there are special problems associated with diversification and divisionalization that could easily depress the overall performance of these firms. To many executives, Andrew Carnegie's advice "to put all your eggs in one basket -- and then watch the basket" is still sound. While the product division structure is based on the logical principle of having several "watchers" for several "baskets", the necessary emphasis on short term measures may result in short term performance. Furthermore, independent strategies to optimize the performance of each division do not necessarily produce optimum corporate performance. A large corporation has the capacity to bear risk and engage in long term development projects that exceeds that of each division considered separately. As Berg notes,

The divisions can no doubt judge what is in their own best interest better than the corporation can, but this ability is of little help to a corporate management interested in spending the limited resources available to the corporation in its interest. In addition, asking the divisions to take a broad corporate viewpoint when striking a balance between their own project expenditure and current profit is asking them to do a job which men at the corporate level themselves find exceedingly difficult to do (and which the divisions are ill-equipped to attempt).⁸

⁸ Norman A Berg, "Strategic Planning in Conglomerate Companies", Harvard Business Review, May-June 1965, p.86 .

The administrative difficulties associated with resource allocation in the diversified firm are compounded by the conceptual problem of defining suitable criteria for guiding the direction of diversification. Naive schemes based on investing in industries that have high rates of growth and profit ignore the facts that if these profits are based on proprietary skills there is little reason to expect that they will accrue to an inexperienced entrant or that existing firms can be purchased for less than their fair value, and that if an area's profitability and growth are due to a supply-demand imbalance, it probably will act as a "cash trap" as capital investment matches growth and prices drop to equilibrium levels. As many of the firms that diversified into the computer industry discovered, IBM's performance was not an "industry" phenomena and quite a few other diversifiers learned in the mid-1960s that the apparently lucrative chemical fertilizer and plastics chemicals businesses all too often absorbed large amounts of capital only to have margins drop drastically as total capacity exceeded demand. Clearly, a strategy of diversification cannot eliminate the need for careful consideration of the firm's special competences and undoubtedly greatly increases the complexity of the planning task.

In summary, while certain advantages connected with economies of scale in research, general management depth, openness to opportunity and performance oriented central

systems may be available to diversified divisionalized firms, the exploitation of these advantages is a major strategic and administrative challenge. Whether or not the average diversified firm has met and overcome this challenge to the degree necessary to produce superior performance is, at this point, a moot issue and more appropriately resolved by analysis of empirical data than further discussion. Thus the following hypothesis represents a question to be tested rather than an expectation: (1) Related business firms will, on average, have higher profitability, higher rates of growth and higher price-earnings ratios than other categories of firms, (2) Related Constrained firms will out-perform Related Linked firms in the above areas, and (3) Divisionalized Related business firms will out-perform those that are functionally organized.

Conglomerate Growth. Acquisitive conglomerates are a fairly recent phenomena; most of the firms adopting this strategy of expansion did so in the decade 1960-1969. Both their mechanics of growth and the question of their ultimate economic efficiency have been subjected to scrutiny by researchers, the investment community, and U.S. Senate subcommittees without definitive conclusions being reached on either issue.⁹

⁹ For a spectrum of views see Harry H. Lynch, Financial Performance of Conglomerates (Boston, Division of Research, Harvard Business School, 1971); Statement of Dr. Irwin Stelzer, Hearings Before the Senate Committee on the Judiciary, Subcommittee on Antitrust and Monopoly, Economic Concentration, Part 1, 88th Congress, 2nd session (Washington D.C., Government Printing Office, 1967), pp.181-202; and Thomas O'Hanlon, "The Odd News About Conglomerates", Fortune, June 15, 1967, pp.175-177.

One explanation of the conglomerate's ability to maintain a rapid pace of acquisition is a variation of the "bigger fool" theory of common stock valuation -- if dividends are zero the security is still worth what the next bigger fool will pay for it. More formally, if the investing public acts as if it determines the value of a share of common stock by multiplying earnings per share by a price earnings ratio and takes the price earnings ratio to be some increasing function of recent earnings per share growth, then the firm can establish positive feedback between price-earnings ratio, acquisition rate, and growth in earnings per share. No value has to be created and early stockholders benefit at the expense of later investors for, once the growth rate slows, equity prices must drop to levels warranted by current earnings. Lynch has shown that if investors do look only at earnings per share growth, such a strategy is possible and indeed seems to have been used by at least several conglomerates.

Another explanation is based on the idea that while the market feedback effect plays a role in conglomerate growth, value can be created. If (1) the return on capital of acquired firms can be improved, (2) if the capital structure of acquired firms can be rationalized, or (3) if there are any economic returns to scale through conglomeration, then the conglomerate's growth rate and price-earnings ratio may be warranted. As a

very simplified example, suppose that a firm currently earning \$1 million annually has a price-earnings ratio of 10 giving a market value of \$10 million. If this firm's management can convince the public that it has the know-how necessary to acquire firms earning a total of, say, \$20 million and to produce a 10% improvement in performance, the value of that extra potential \$2 million per year, capitalized at a price-earnings ratio of 10 is \$20 million. If this is added to the firm's current worth of \$10 million it produces a current price earnings ratio of 30 which, in turn, enables the company to carry out its acquisition program. Whether or not the acquired firms grow themselves is immaterial -- it is the improvement in return that creates value. Of course, the problem is that while such improvements may be possible the public is not well equipped to decide which potential conglomerate will be able to deliver on its promises.

The question of the ultimate economic efficiency of a conglomerate also involves the degree to which improvements in acquired firms can be affected but includes the larger issue of administrative control. Once growth slows, will the multi-industry Unrelated business firm be able to operate efficiently and set strategy effectively in all its constituent areas? The same pro and con arguments that were discussed for the case of Related business firms apply in even starker terms to the conglomerate. In addition, there is the very real problem of

holding on to and attracting superior general management talent when the atmosphere of explosive growth gives way to stability. Unfortunately, the very newness of the conglomerate phenomena prevents us from being able to bring much empirical evidence to bear on this question. While a few of the Unrelated Passive category firms could be described as Acquisitive Conglomerates that have moved past their rapid growth phase, most of the Unrelated Passive firms never were Acquisitive Conglomerates. Instead, some became unrelated quite slowly and others became unrelated through a single merger that was not followed by further important unrelated acquisitions.

The hypotheses that will be tested are designed to provide information on the degree to which conglomerate growth is related to abnormally high price-earnings ratios and the ability of acquisitive conglomerates to operate efficiently. First, we hypothesize that Acquisitive Conglomerates had price-earnings ratios that were significantly higher than those of other firms and that conglomerate price-earnings ratios were strongly related to the growth in earnings per share achieved. Secondly, we expect that both Acquisitive Conglomerates and Unrelated Passive firms had average returns on capital that were not significantly different from the overall average of other types of firms.

The Analysis of Financial Data

As was noted in Chapter II, the financial data used in the analysis of the performance characteristics of the categories of diversification strategy and organizational structure was obtained from a modified version of the Compustat data bank. This computer disk file contained financial histories for the period 1951-1970 for almost all of the firms in the sample that had survived through 1970 as well as data on about 1500 other corporations. The Compustat data, as well as data files containing information generated in this study, were accessed in a time-sharing mode through computer programs written expressly for this research. The programs contained routines for the manipulation of data, computation of simple averages, growth rates and standard deviations, multivariate regression, analysis of variance, and other statistical procedures.

The variables that were studied and the symbols used to identify them in tables and graphs are as follows:

1. GSALES: Annual rate of growth in net sales.
2. GERN: Annual rate of growth in earnings after taxes and preferred dividends.
3. GEPS: Annual rate of growth in earnings per common share.
4. SDEPS: Relative standard deviation of year-to-year variation of earnings per share about average long-term uniform growth trend. SDEPS is best thought of as the measure of the variability of annual growth rates.

5. PE: Price-earnings ratio of common stock.
6. ROC: After tax return on invested capital (net income plus interest divided by invested capital).
7. ROE: After tax return on equity (net income less preferred dividends divided by book equity).
8. E/C: Ratio of book equity to invested capital.
9. IFR: Internal financing ratio defined as the ratio of the market value of added equity to earnings after taxes (annual retained earnings plus market value of new shares issued divided by earnings after tax and preferred dividends).
10. RPR: Risk premium ratio for earnings per share defined as $(\text{GEPS} - 0.015) / \text{SDEPS}$ where 0.015 is taken as the available risk-free after tax rate of return. RPR is a measure of the degree to which growth above and beyond the risk-free rate is matched by increasing variability in earnings.

More detailed definitions of these variables together with descriptions of the computational procedures used to determine their values are given in Appendix 4A at the end of this chapter.

Although financial data was available on an annual basis from 1951-1970, knowledge of the firms' strategic categorization was limited to three points in time: 1949, 1959, and 1969. If a firm stayed in the same strategic category throughout a decade it is clear that its financial performance during that decade can be associated with that particular category. However, if a firm moved between categories, say from the Single business category in 1949 to the Related Constrained category in 1959, there is no unambiguous way of

associating its financial performance during that period with any particular category. Not only is the precise date of transition unknown, if such a concept is even meaningful, but the very fact that a firm was moving between categories suggests that its performance might not be representative of the long-term characteristics of firms in any category. Consequently, the analyses presented in this chapter are based only on data pertaining to decades in which a firm stayed in the same category throughout. For example, the average growth rate in earnings per share (GEPS) of Single business firms is estimated from 1959-1959 data on firms that were in the Single business category in both 1949 and 1959, and from 1960-1969 data on firms that were in the Single business category in both 1959 and 1969. Thus, the category averages presented in this chapter include only firms whose product-market scope's remained stable for at least ten years.

One exception to the requirement of at least a ten year residence in a category was made: the characteristics of Acquisitive Conglomerates were estimated by averaging data on all firms that fell in this category by the end of a decade. Because of the relative newness of this strategy, only two firms in the sample qualified as Acquisitive Conglomerates in both 1959 and 1969 (Colt and Grace), while ~~twelve~~ other firms adopted this strategy by 1969. Thus, it would have been im-

possible to obtain reasonable estimates of this category's average financial characteristics unless this modification in the rules for inclusion were made.

The elementary data points used in analyses were decade averages rather than annual observations. After the annual values of a financial variable for a firm had been found, two summary averages were obtained, one for 1951-1959 and one for 1960-1969. Category averages were then calculated by combining the appropriate decade averages and making corrections for time trends in average values (see Appendix 4A for details). While this procedure reduced the number of data points by a factor of ten, the existence of fairly long-lasting "firm effects" ¹⁰ prevents the direct use of annual data.

In calculating the decade averages, growth rates were obtained by making least-squares fits to a uniform growth model, simple arithmetic averages were used for the other variables, and both procedures included rules for omitting meaningless values and dealing with missing data points. Details are given in Appendix 4A.

Category Averages. Tables 4-1, 4-2, and 4-3 show the estimated mean values of the ten financial variables for

¹⁰ In other words, ten separate observations of Xerox's growth rate are not statistically equivalent to observing the growth rates of ten different office equipment companies.

TABLE 4-1

FINANCIAL CHARACTERISTICS OF THE MAJOR STRATEGIC CATEGORIES

	SING	DOM	REL	UNRL	AVG	F ₁	F ₂	F ₃	R ²	R _m ²
GSALES	7.17	8.03	9.14	14.24	9.01	8.82	3.57	14.00	0.147	0.735
(t-ratio)	-1.85	-1.88	.27	4.56		.001	.05	.001		
GERN	4.81	7.95	9.39	13.86	8.72	5.45	3.41	4.34	0.063	0.554
(t-ratio)	-2.72	-1.02	.96	3.11		.001	.05	.005		
GEPS	3.92	5.99	7.64	7.92	6.57	2.10	2.75	1.87	0.026	0.473
(t-ratio)	-2.01	-.84	1.67	.89		.1	.1	--		
SDEPS	16.13	22.40	16.43	31.19	20.11	5.11	2.77	2.30	0.061	0.690
(t-ratio)	-1.30	1.43	-2.47	3.15		.005	.1	.1		
PE	14.60	15.74	19.21	15.75	17.02	8.88	14.28	1.22	0.090	0.874
(t-ratio)	-2.70	-2.73	5.01	-1.23		.001	.001	--		
ROC	10.81	9.64	11.49	9.49	10.52	3.06	3.16	5.46	0.028	0.300
(t-ratio)	.35	-2.02	2.40	-1.08		.05	.05	.001		
ROE	13.20	11.64	13.55	11.92	12.64	2.32	2.60	6.05	0.027	0.280
(t-ratio)	.64	-2.18	2.13	-.72		.1	.1	.001		
E/C	.781	.752	.799	.696	.769	2.10	5.14	6.93	0.024	0.185
(t-ratio)	.51	-1.39	2.64	-2.72		.1	.01	.001		
IFR	.580	.640	.714	1.322	.732	19.00	0.94	22.00	0.169	0.465
(t-ratio)	-2.31	-1.14	-.24	3.32		.001	--	.001		
RPR	.378	.424	.669	.437	.516	5.79	7.94	0.30	0.070	0.946
(t-ratio)	-1.80	-2.30	4.10	-.89		.001	.001	--		

NOTE: Estimated mean values of E/C, IFR, PE, and RPR are expressed as ratios, all others are expressed as percentages. The t-ratios test the difference between an estimated category mean and the overall mean. F₁ tests for equality of all category means, F₂ tests just the equality of the means for the S,D, and R categories, and F₃ tests the equality of sub-category means (Table 4-2) with major category means. R² is the fraction of the variability explained by variations among major category means and R_m² is the fraction of the variability explained by all sub-category means that can be attributed to variations among major category means.

TABLE 4-2

FINANCIAL CHARACTERISTICS OF THE STRATEGIC CATEGORIES

	S	DV	DC	DLU	RC	RL	UP	AC	AVG	σ	F ₁	F ₂	R ²
GSALES	7.17	7.42	9.48	6.93	9.62	8.06	6.10	20.64	9.01	6.06	8.64	2.25	.200
t-ratio	-1.85	-2.14	.43	-.92	.95	-.90	-1.03	7.40			.001	.05	
GERN	4.81	7.34	9.08	8.10	10.39	7.15	7.78	18.64	8.72	8.73	4.19	1.91	.114
t-ratio	-2.72	-1.29	.23	-.19	1.81	-1.03	-.37	4.38			.001	.10	
GEPS	3.92	5.14	7.60	6.11	8.56	5.57	5.96	9.46	6.57	8.02	1.71	1.67	.055
t-ratio	-2.01	-1.46	.71	-.15	2.38	-.72	-.26	1.39			--	--	
SDEPS	16.13	22.79	23.15	16.63	15.91	17.58	20.90	39.28	20.11	18.63	3.17	1.28	.088
t-ratio	-1.30	1.17	.90	-.50	-2.13	-.78	.14	3.97			.005	--	
PE	14.60	15.68	15.92	15.41	19.19	19.27	13.77	17.43	17.02	5.45	4.31	5.27	.103
t-ratio	-2.70	-2.01	-1.11	-.79	3.77	2.37	-2.03	.29			.001	.001	
ROC	10.81	8.24	12.71	8.69	11.97	10.43	9.40	9.56	10.52	5.05	3.36	3.85	.094
t-ratio	.35	-3.68	2.34	-.97	2.72	-.10	-.75	-.73			.005	.001	
ROE	13.20	10.18	14.91	10.28	14.11	12.28	10.38	13.13	12.64	5.33	3.48	4.01	.096
t-ratio	.64	-3.76	2.35	-1.19	2.61	-.39	-1.44	.35			.001	.005	
E/C	.781	.724	.807	.758	.800	.798	.830	.591	.769	.142	4.86	2.56	.130
t-ratio	.51	-2.59	1.48	-.21	2.07	1.17	1.46	-4.83			.001	.025	
IFR	.580	.662	.609	.601	.729	.681	.616	1.970	.732	.400	17.82	0.90	.364
t-ratio	-2.31	-1.43	-1.70	-.88	-.07	-.73	-.99	11.94			.001	--	
RPR	.378	.329	.658	.221	.775	.431	.337	.516	.516	.466	2.61	3.04	.074
t-ratio	-1.80	-3.29	1.68	-1.45	5.26	-1.05	-1.30	0			.025	.01	

NOTE: Estimated mean values of E/C, IFR, PE, and IFR are expressed as ratios, all others are expressed as percentages. The t-ratios refer to the difference between the estimated category mean and the estimated overall mean. F₁ is the F-ratio for testing equality of all category means, F₂ tests equality of all category means except that of the Acquisitive conglomerate group (AC).² R² is the partial reduction in variance due to category effects (as distinct from time effects). Figures below F-ratios are levels at which they are significant.

TABLE 4-3
FINANCIAL CHARACTERISTICS OF THE STRUCTURE CATEGORIES

	F	F/S	PD	AVG	σ	F_1	R^2
GSALES (t-ratio)	8.55 -.85	6.49 -2.76	9.77 2.40	8.98	5.22	4.46 .025	0.082
GERN (t-ratio)	6.76 -3.03	9.57 .29	10.66 2.88	9.17	8.18	4.84 .01	0.045
GEPS (t-ratio)	5.08 -3.05	8.32 .71	8.63 2.05	7.37	7.71	3.46 .05	0.046
SDEPS (t-ratio)	20.13 -.18	23.90 1.00	19.82 -.52	20.47	19.88	0.50 --	0.005
PE (t-ratio)	14.86 -3.85	16.60 -.56	18.73 3.84	17.21	6.27	6.08 .05	0.045
ROC (t-ratio)	10.28 -.30	9.49 -1.07	10.75 .99	10.43	5.10	0.75 --	0.005
ROE (t-ratio)	12.28 -.35	11.09 -1.57	12.90 1.42	12.45	5.02	3.06 .05	0.041
E/C (t-ratio)	.774 1.29	.766 .40	.743 -1.43	.756	.144	1.49 --	0.014
IFR (t-ratio)	.634 -1.89	.671 -.57	.762 1.81	.713	.429	1.43 --	0.014
RPR (t-ratio)	.329 -2.97	.485 -.38	.667 3.07	.531	.700	5.31 .005	0.047

NOTE: Estimated mean values of E/C, IFR, PE, and RPR are expressed as ratios, all others are expressed as percentages. The t-ratios test the difference between an estimated category mean and the overall estimated mean. σ is the estimated standard deviation of the residual error. F_1 tests the hypothesis that all true category means are equal and the level at which it is significant is shown below each value of F . R^2 is the fraction of the total variability that was explained by variations among the estimated category means.

the major strategic categories, all the strategic categories, and the organizational categories respectively. These results were obtained by regression analysis with "zero-one" variables used to represent the effects of the different categories and time. The estimated mean values displayed are trend corrected, and their absolute magnitudes have been referred to the 1960-1969 period. It must be emphasized, however, that it is the differences between category means and not their absolute magnitudes that are important and to which tests of significance apply.

It was not necessary to display both the 1951-1959 and 1960-1969 means because interaction effects were not significant. That is, time trends in the financial variables seemed to affect all categories equally so that the 1951-1959 estimated means differ from those shown only by a constant amount for each variable, leaving the category differences unchanged.

Also shown in these tables are t-values for each category mean, the estimated residual standard deviation of each financial variable (σ), F-ratios for each set of category means and the level at which the R-ratios are significant, and the percentage of total variation explained by the categories (R^2). The standard deviation (σ) indicates the variability in the data that still remained after the effects of

time and the categories had been taken into account. The values of R^2 show the percentages of the original variability in the data that were eliminated by assigning firms to categories. As in most econometric studies, they are uniformly low.

Each t-value shows by its sign the direction of the deviation of an estimated category mean from the overall estimated mean and by its absolute value the significance of this deviation.¹¹ For example, in Table 4-2 the t-value corresponding to the estimated mean return on capital (ROC) of the Related Constrained category is 2.34. The fact that this t-value is positive shows that the estimated category means (12.71) is larger than the estimated overall mean (10.52) and the value indicates that this deviation is statistically significant at the 1% level (see note 11). It is important to remember, however, that two estimated category means may differ significantly from one another even if neither differs significantly from the overall estimated means.

The F-ratios show the degree to which the entire set of category means differ from one another. More precisely,

¹¹ For the numbers of data points involved here, the absolute values of "t" corresponding to various levels of significance are:
 (.1, $t = 1.28$), (.05, $t = 1.64$), (.025, $t = 1.96$), (.01, $t = 2.33$), (.005, $t = 2.58$), (.001, $t = 3.09$).

they provide a test of the hypothesis that all of the category means are actually equal. The level of significance of each F-ratio is shown immediately below it. Note that the F-ratios may not be significant even when certain pairs of estimated category means differ significantly. This is because the F-ratios take into account the idea that with, say, eight categories there are twenty-eight possible pairwise comparisons and random error will probably cause one of them to appear significant at the 5% level even if the true means are all equal.

In Table 4-2 the F_1 -ratios test the hypothesis "all true category means are equal" and the F_2 -ratios test the hypothesis "all true category means other than that of the Acquisitive Conglomerate category are equal". This distinction was necessary because Acquisitive Conglomerates were, in part, selected on the basis of their high rates of growth so that a significant F_1 for growth variables due to a high mean in the Acquisitive Conglomerate category is virtually meaningless. Hence F_2 , testing the differences among the other means, is the relevant statistic for the growth variables. When an F-ratio is significant the hypothesis of equality can be rejected and the conclusion that true category differences actually exist can be drawn.

Readers who are unfamiliar with some of these concepts and those who prefer a more detailed discussion of the

techniques employed are referred to Appendix 4A at the end of this chapter.

Preliminary Evaluation. Without, at this point, going too deeply into the nature and implications of the empirical data, an overall appraisal of the results leads to several general but important conclusions. Table 4-2 indicates that differences in diversification strategy accounted for significant amounts of the variation in all of the financial variables except growth in earnings per share. Furthermore, it is evident that while the Acquisitive Conglomerate category, which was known in advance to be a special case, was an important contributor to the total discriminatory power of the categories, financial differences among the other categories were significant for seven of the ten variables. Especially noteworthy is the fact that the variables providing the best measures of economic efficiency and overall investor appraisal of performance -- return on equity, return on capital and price-earnings ratio -- were strongly related to diversification strategy.

The performance differences revealed by Table 4-2 show a high degree of consistency; strategies that are associated with high profitability tend also to be associated with high rates of growth, lower amounts of variability, and higher price-earnings multiples. This pattern of consistency allows us to identify three clusters of strategies:

Performance

<u>High</u>	<u>Medium</u>	<u>Low</u>
Dominant Constrained	Related Linked	Dominant Vertical
Related Constrained	Single	Unrelated Passive

Acquisitive Conglomerate

What is particularly striking about these clusters is their heterogeneity. There is no evidence to suggest that performance varies smoothly with the degree of diversity. Rather, two "losing strategies" stand out: among undiversified and lightly diversified companies, the strategy of vertical integration has tended to be associated with poor performance, and among diversified firms the Unrelated Passive product-market posture has produced distinctly below average results. The dissimilarity of the two low performing strategies is intriguing and a close look at the reasons for the below average performance of these categories will be taken in Chapter V.

The fact that the categories are associated with significantly different levels of financial performance, regardless of the specific pattern of the differences, is in itself an important finding. In one of the most careful and comprehensive studies of U.S. corporate diversification, Gort tried to relate growth in assets (1939-1954) and return on equity (1947-1954) to a composite measure of diversification based on a 4-digit product count and an employment speciali-

zation ratio but found only very small and non-significant correlations.¹² In a more recent study, Eslick regressed return on equity for the period 1956-1965 against 4-digit product count for 450 of the firms in the Fortune 500. Again, no association was found.¹³ Since there appears to be no relation between profitability and overall measures of diversity, the connection between profitability and diversification strategy shown in Table 4-2 takes on extra importance. In addition to representing managerially meaningful distinctions among corporate diversification postures, the strategic categories and the "top down" approach to measurement reveal a consistent pattern of strategy associated performance differences that are not discernible when traditional measures of diversification are employed.

¹² Michael Gort, Diversification and Integration in American Industry (Princeton, Princeton University Press, 1962), pp.74-77. Gort did find, however, a moderate relation between the rate of asset growth and the rate of entry into new 4-digit industries, though the association is hardly surprising.

¹³ Statement of Prof. Donald F. Eslick, Economic Concentration, Part 8, pp. 4996-5026. Eslick's conclusions emphasized the existence of a moderate negative relation between return on equity and the 4-digit product count between 1959-1967, but since the association was not evident in the longer period 1956-1965, his argument is somewhat perplexing.

Why does the category technique reveal a relationship between diversification and performance when simple measures of diversity do not? The answer is twofold. First, the categories are actually a crude approximation to a multi-dimensional measure that takes into account several factors in addition to diversity per se. Secondly, the pattern of differences by the category technique indicates that it is not diversity itself but the central organizing principle which is used to manage diversity that is the critical factor in explaining performance differentials.

For example, neither return on equity nor return on capital seem to show any regular pattern of increase or decrease as one moves from the less to the more "diversified" categories. Table 4-1 indicates that profitability tends to be higher in the Single and Related major groups and lower in the Dominant and Unrelated groups, but that even this "down-up-down" pattern explains only 30% of the total variability explained by the sub-categories. The story told by the profitability pattern of the sub-categories (Table 4-1) seems to be: among Dominant businesses those that stayed close to home do well and those that are vertically integrated or have moved into linked and unrelated areas do relatively poorly. Again, among Related businesses those that have gone farther afield have done less well than those that have chosen or been

able to limit their product market scope. Unrelated businesses have experienced lower than average returns on capital but the Acquisitive Conglomerates have employed high leverage to boost their return on equity. A similar pattern can be seen for the growth variables, except that Acquisitive Conglomerate growth, without a per-share adjustment, is very much greater than that of any other category.

The degree to which the pattern of price-earnings ratios does not correspond to the profitability and growth variables is interesting. Although the Dominant Constrained group has experienced higher returns and growth than the Related Constrained group, its price-earnings ratio is not only below that of the Related Constrained group, but below the overall average. The Related Linked group, on the other hand, has consistently underperformed both the Dominant Constrained and Related Constrained groups but has the highest average price-earnings ratio of all. Why this is so cannot be determined from the financial data alone. However, we will return to this issue in a subsequent section and suggest that the price-earnings disparity results from a persistent bias among investors that may reflect very long-term expectations.

The structure categories also divided the companies into groups with significantly different levels of financial performance. The product division firms, in general outperformed functional and functional with subsidiaries type firms,

significant differences being found for six of the ten variables tested. The two types of functional firm, however, did not display similar patterns. The functional with subsidiaries firms had relatively low rates of sales growth but high growth rates in earnings while for true functional firms the opposite was true. In addition, while the profitability of true functional firms was close to that of product division firms, the average return experienced by functional with subsidiaries firms was lower than either. Although the reasons for these differences are not clear, it does seem that between them the two functional types can match either the earnings growth rate or profitability of the product division firms, but not both simultaneously.

The Question of Industry

Given that different strategies are associated with different financial consequences, it is natural to inquire as to whether such differences remain, disappear, or change in nature when each firm's major industry membership is taken into account. For example, while Related business firms outperformed Dominant business firms in aggregate, it might be that the opposite is true in some industries. Or, it might be that the aggregate difference is solely the result of unequal membership in industries with different average performance levels.

In the first attempt to develop industry corrected category differences, each firm was assigned a 3-digit SIC code that was the most representative of its major business activities. Firms that were so diversified that their primary industry was not determinable were placed in a special group. The value of the financial variables for each firm were then adjusted by subtracting from each decade average the overall decade average of the industry to which the firm belonged. Thus, the financial variables were converted into industry relatives. For example, two firms with returns on capital of 10% and 13% might belong to industries with average returns to capital of 8% and 14%, so that their industry relatives become 2% and -1% respectively. The industry averages were obtained from an analysis of all 1700 firms in the Compustat data bank.

The results of the analysis were negative; industry relatives did not differ significantly by category. However, several problems with the procedure cast doubt on the meaningfulness of the result. First, in some cases the average of all sampled firms in a given industry fell below the 1700 firm average and in other cases it was greater. This was essentially due to the small number of sampled firms in each 3-digit industry and meant that the average industry relatives of the categories were not strictly comparable. Secondly, the fewness

of the total number of firms in certain industries when combined with extreme values of particular variables produced spurious relatives. The business equipment group, for instance, had so few firms in it that Xerox's high growth rate gave all the other firms in the group, including IBM, below average relatives. While perhaps true in some strict technical sense, calling Hygrade's performance "relatively" better than IBM's seemed too much of an artifact of the sample to warrant serious attention.

In the second attempt to control for industry effects 2-digit industry groups were used. While the sample firms engaged in a total of twenty-one different 2-digit industries (including a special "conglomerate" group), the chemical group (28) was split into basic chemicals and pharmaceutical-toiletries groups and the transportation group (37) was split into land-based and aviation-aerospace groups to give a total of twenty-three distinct industries. Then, rather than use the industry relatives method, the industry effects and category effects were both represented by zero-one variables in regression analyses.

Theoretically, the regressions should have delivered measures of the relative effects of the categories on performance; measures that would be independent of industry associated differences. However, numerical problems arose

that were traced to multicollinearity¹⁴ -- the effects of industry and diversification strategy were not separable. At this point Table 4-4 was constructed and it became obvious that industry corrected results were not only elusive, but essentially unattainable and possibly meaningless.

Table 4-4 shows the number of sampled firms in each major industry group that fell in each strategic category in 1969, Table 4-5 displays the same information for 1949. The idea behind industry corrections is to compare firms with similar technological-market environments that have adopted different strategies. But Table 4-4 indicates that most firms have responded similarly to their industry environments. Paper (26), oil (29), rubber (30), and primary metals (33) firms have, for the most part, followed strategies of vertical integration. Textile (22), apparel (23), chemical (28), drug (283), electrical machinery (36) and instrument (38) companies are almost all in the Related categories. A closer look at the exceptions shows that they are chiefly due to the broadness of the industry categories. Of the three non-vertical forest products companies, two are really in building materials rather than paper. In the petroleum industry (29), one of the

¹⁴ Operationally a near singular regression matrix. For a discussion of this problem see John R. Meyer and Robert R. Glauber, Investment Decisions, Economic Forecasting, and Public Policy, (Boston, Division of Research, Harvard Business School, 1964), Chap.VIII.

TABLE 4-4

FREQUENCY OF DIVERSIFICATION STRATEGY BY INDUSTRY: 1969

SIC Industry	S	DV	DCLU	RC	RL	U
10 Mining, metals		1				
20 Food	6	4	3	8	4	2
21 Tobacco			1			
22 Textiles				2		
23 Apparel				1	1	
24 Lumber and wood products		1				1
25 Furniture and fixtures			1			
26 Paper and containers	1	6	1	1	1	
27 Printing and publishing	1		1		3	
28 Chemicals			1	4	7	5
283-4 Drugs, soap, toiletries			1	9	3	
29 Petroleum		7		2		
30 Rubber and plastics		4		1		
31 Leather and shoes			1			
32 Stone, clay and glass	1		1	1	4	
33 Primary metals	1	7			1	
34 Fabricated metals			2	1	1	1
35 Machinery, exc. electrical			7	8	4	4
36 Elect. mach. & electronics	1			2	8	3
37 Transportation equipment	1		4	2	5	3
372 Aircraft and aerospace			2	1		6
38 Instruments				1		2
39 Misc.					1	
99 Multi-industry					1	5

NOTE: Shown are the numbers of firms out of the entire sample that were in existence in 1969 and fell in each strategic class within each industry. The symbol DCLU stands for the combined Dominant Constrained, Dominant Linked and Dominant Unrelated categories.

fied firms and Unrelated business firms were few in number.

From one point of view the net result of this situation is that only the food processing and perhaps the aerospace industries have within them a sufficient number of firms in different categories to permit a controlled comparison. Such a comparison would, however, require a sample directed specifically at these industries and even then the results would hardly be of general interest. In addition, a more realistic 3-digit definition of industries would further strengthen the connection between industry and diversification strategy, making it close to impossible to control for industry. Taking a broader point of view, it seems evident that most of the strategy related performance differences are due to industry differences, but that the two effects are simply not separable. The higher performing industries tend to consist of mostly Related business firms and Related business firms tend to belong to higher performing industries. Which came first? The answer seems to be that they came together; the same conditions that produce above average performance -- science based proprietary strengths, growth in markets served along with rapid product innovation -- produce diversified firms. The strategies that were successful in the early part of the century -- mass production of basic durable goods and vertical integration to achieve control over primary industrial materials -- produced large firms committed to

single businesses that are now faced with mature, slowly growing, highly competitive non-proprietary product-market areas.

Implications. A definition of industry that is detailed enough to permit the comparison of close competitors is of no use when the effect of diversification is of primary interest. A diversified firm is not a close competitor with any one non-diversified firm. Broader definitions of industry reveal that diversification strategy and the general technological-market environment tend to be alternate aspects of the same phenomena. Thus, one cannot "eliminate" the industry effect on performance without also eliminating the effects of diversification strategy. The two are inextricably mixed.

This observation implies that a strict normative interpretation of performance differences is, in general, highly questionable. The empirical evidence that Related Constrained firms out-perform Dominant Vertical firms does not mean that Dominant Vertical companies should attempt to follow Related Constrained strategies. Both may be reasonable responses to different technological and economic environments. Since there are no diversified steel companies or vertically integrated electronics companies, a recommendation that the steel companies diversify because diversified firms are more

profitable would be based upon overly simplified logic. In a sense, it is as valid to think of the electronics company as being diversified because it is in a profitable science-based business as the reverse.

Testing the Hypotheses

To what extent do the empirical results on financial performance support the hypotheses advanced in the first section of this chapter? In answering this question we will deal with the hypotheses one at a time and then present a summary synthesis.

Portfolio Risk. Regarding the relative degrees of risk and return experienced by firms in various categories it was hypothesized that:

Related Constrained firms would experience the lowest variability in earnings and have an average rate of return at least as large as that of other categories. In addition, the rank order of the categories in terms of increasing variability relative to return was conjectured to be: Related Constrained, Related Linked, Unrelated, Dominant, and Single.

In developing a measure of return one must answer the questions "return to whom?" and "return on what investment?" While the stockholder can be used as a reference, returns to stockholders can depend as much on market swings and changing expectations as on performance. In addition, the market is essentially a device for equilibrating uncertainty, not passing on returns. If, for instance, it were

known with certainty that a firm would increase its earnings at a rate of 20% per year for the next five years, the current price of the company's stock would reflect that knowledge and returns over the next five years to stockholders would (theoretically) be quite low. Rapid price increases are the result of unexpected improvements in a firm's prospects.

Annual values of return on book capital or equity can be equally misleading. Fluctuations in these rates can reflect capital investment programs as much as changes in profitability and book value figures are sensitive to inflation and capital turnover rates. In addition, growing firms can have marginal rates of return substantially in excess of their average rates of return.

In the light of these considerations, and in view of the fact that the mechanisms that were hypothesized to lie behind risk-return variations with diversifications were based on flexibility in resource allocation, the firm's average rate of growth in earnings per share was taken as the measure of "return". Under uniform growth assumptions, GEPS is proportional to the firm's incremental return on investment and it also serves as a market-free proxy for stockholder benefit.

Ideally, "risk" should measure the potential variability in a firm's earnings stream. Unfortunately, the empiricist must make do with observed variability. We take as our

measure of risk the standard deviation of annual growth rates in earnings per share about the uniform growth trend (SDEPS). More formally, GEPS and SDEPS are obtained through a least-squares fit of the following expression to annual earnings per share data.

$$\text{Log} (\text{EPS}_t) = A + (\text{GEPS}) t + \tilde{\epsilon}_t$$

where t is an integer that measures time in years, and $\tilde{\epsilon}_t$ is the annual deviation from the uniform growth model. SDEPS is the estimated value of the standard deviation of $\tilde{\epsilon}_t$ and may be thought of as the variability in the rate of earnings per share growth. The estimated mean values of these measures for each category are shown in Table 4-2.

As can be seen from the F-ratios in Table 4-2, neither variable was found to be significantly affected by diversification strategy. Even though the Related Constrained category had an average growth rate in earnings per share that was 3.42 percentage points above that of the Dominant Vertical category ($p = 0.025$),¹⁵ the very high overall variability in GEPS kept the total results of classification from being significant. Similarly, though the value of SDEPS for the Related Constrained group was lower than the overall average ($p = 0.025$), the high total variability in SDEPS again kept the

¹⁵ This notation refers to the level of statistical significance of the statement just made.

overall fit from being significant (if the Acquisitive Conglomerate category is omitted). However, the fact that the direction of the deviation of the Related Constrained category was as hypothesized for both variables suggested that a composite measure of risk and return might prove more sensitive to category effects.

The risk premium ratio (RPR) that was used as a combined measure is adapted from a ratio suggested by Sharpe.¹⁶ While subject to the usual weaknesses of a single measure of a multi-factor situation, the risk premium ratio at least provides a consistent way of measuring return relative to risk. The ratio is defined for our purposes as

$$\text{RPR} = (\text{GEPS} - 0.015) / \text{SDEPS}$$

where 0.015 (1.5%) is taken as an approximation to the "risk-free" (government bonds) after tax return (and hence growth rate if all funds are retained) available to corporations. In essence, RPR measures the risk associated growth per unit of variability in growth. Leaving aside considerations of risk-return preference, the bigger RPR is, the better.

Table 4-2 shows that the average RPR for Related Constrained firms was 0.775, significantly above ($p=0.001$) the overall average of 0.516. Thus while Related Constrained

¹⁶ Sharpe, William F., "Mutual Fund Performance," Journal of Business, January 1966, pp. 119-138.

firms did not significantly surpass the others in growth in earnings per share or variability considered separately, their average growth rate relative to variability, RPR, was considerably superior, as predicted.

Contrary to what was hypothesized, the Dominant Constrained category also had a very high RPR (0.658) that did not differ significantly from that of the Related Constrained category. In fact, the picture that emerges from the risk premium ratio data is of two distinct and internally homogeneous groups: a high performing group consisting of the Related Constrained and Dominant Constrained firms and a low performing group that includes all other categories (see Table 4-6). The two groups differ at a high level of significance ($p = 0.001$) and none of the category mean RPRs within either group differ significantly from one another.

Diversification in itself has not served to reduce the relative variability of the firm's earnings stream. Dominant business firms have shown less relative variability than Unrelated business firms and Dominant Constrained firms have out-performed Related Linked firms in this measure. What has been important is the way in which the firm's businesses are related. It is the companies that have at least some product-market diversity but have a strategy of restricting these activities so that a central strength or skill spans all

of their activities — the Dominant Constrained and Related Constrained firms — that have shown the lowest variability in earnings relative to growth, and the absolute levels of GEPS for these firms provide assurance that the result is not due to stability associated with abnormally low growth.

While the superiority of Related Constrained firms was predicted, the hypothesized ordering of the other categories was not supported. Recalling the major theoretical argument that was advanced, it seems that rather than being wrong, the concept was not extended far enough. It was argued that the ability of a firm to "ride the product life-cycle", substituting new products for old was critical in reducing risk relative to return, and that this could be accomplished most effectively in firms in which variety was closely controlled; such firms are not so diversified that they cannot avoid the total systematic risk of the economy and are diverse enough to compensate for declining prospects in one product-market area with new products that are functional substitutes for the old product. This line of reasoning led to the hypothesis of Related Constrained superiority in return relative to risk. The fact that the Dominant Constrained (but not the Single) business category shared this superiority may mean that we overestimated the need for diversity and underestimated the value of controlled or channelled diversity in mitigating risk.

Table 4-6
RISK PREMIUM RATIOS

High Performing Group

Related Constrained	.775
Dominant Constrained	<u>.658</u>
Pooled Average	.741

Low Performing Group

Acquisitive Conglomerate	.516
Related Linked	.431
Single	.378
Unrelated Passive	.337
Dominant Vertical	.329
Dominant Linked & Unrelated	<u>.221</u>
Pooled Average	.374

A further interesting point is the poor risk return performance of the Dominant Vertical firms. Although these firms often appear "Related Constrained" in terms of their end products (e.g. Goodyear, Alcoa, etc.), it is obvious that their average risk premium ratio is drastically below that of true Related Constrained firms ($P = 0.001$). In addition to justifying the classification applied to these firms, this demonstrates that the commitment to a single material overrides the possibly beneficial effects of end product variety on risk-return performance.

The Escape Paradigm. Consideration of the motives and mechanism for escape through diversification from stagnant business areas led to this hypothesis:

Firms that are not members of science-based industries that adopt Dominant Linked, Dominant Unrelated and Unrelated strategies will experience returns on equity that are significantly lower than those achieved by firms in the same industry but following other strategies.

The reasoning was that low performance was the cause for adoption of these strategies, not the result.

As noted in the section on industry effects, industry and strategy are closely related and there are too few firms in the sample to permit a detailed definition of industry. Accordingly, the hypothesis was tested by dividing all firms into two large groups: those in science-based industries and all others. The science-based industries were chemicals (28),

business machines (357), electrical machinery and electronics (36), aerospace (372), and instruments (38). Two regressions were then computed, and in each a zero-one variable was used to indicate membership in a science-based industry. The variable, SCI, was assigned a value of one if the firm's major businesses were in any or all of the science industries and zero otherwise. The first regression simply served to remove the relative premium in return gained by science-based firms. The equation was

$$ROE = A + B_1 (SCI) + B_2 (DLU) + B_3 (U) + B_4 (TIME) + \tilde{\epsilon}$$

where B₁, B₂, B₃, and B₄ represent the effects of science-based industry, Dominant Linked or Unrelated strategy, Unrelated strategy and time on return on equity.

The results were:

<u>Coefficient</u>	<u>Value</u>	<u>t-ratio</u>	<u>P</u>
A (intercept)	11.90	-	-
B ₁ (SCI)	3.70	6.00	.001
B ₂ (DLU)	-1.17	-1.11	-
B ₃ (U)	-2.13	-2.04	.02
B ₄ (TIME)	0.08	.02	-

$$R^2 = 0.116$$

where "P" is the level at which the t-ratio is significant,¹⁷ and R² is the proportion of variance explained by the regression.

¹⁷ Here the t-ratios test the hypothesis "B_i = 0".

The science-based industries clearly receive a return on equity premium, estimated here as 3.7 percentage points. The regression shows that when the science industry premium is removed, Unrelated business firms have significantly lower returns on equity. This is in contrast to the results of Table 4-1 (where the science premium is not removed) in which the Unrelated category does not differ significantly from the average in return on equity. Thus, if the fact that Unrelated business firms generally are members of science-based industries is taken into account, their relative profitability is seen to be poor.

The Dominant Linked and Unrelated (hereafter DLU) firms, on the other hand, are not significantly below the average return on equity when science premiums are removed.

The second regression was similar in format to the first but split the Unrelated and DLU groups into those in science-based industries and those that were not. The resulting coefficients of interest are shown in Table 4-7.

Table 4-7
EFFECTS ON ROE

Strategy	Science-based Industries	Other Industries
Unrelated	-3.82	-1.20
(t-ratio)	(-1.60)	(-1.02)
P	.05	-
DLU	12.61	-1.66
(t-ratio)	(2.48)	(-1.55)
P	.005	.1

The startlingly high coefficient for DLU firms in science-based industries is based on only one company. Xerox. While significant in a strict statistical sense, it would be foolish to generalize from such a result. Unrelated business firms in science-based industries are seen to have returns on equity that are lower than those of other firms when the usual premium received by science-based firms is taken into account. Although there is a tendency for Unrelated and DLU firms in non-science industries to have lower returns on equity, neither coefficient is significant.

The original conjecture was that non-science based firms adopting Unrelated or DLU strategies did so because of low profitability and that this poor performance would not be entirely remedied by the diversification strategy. The evidence, however, indicates that the performance of these firms was somewhat lower, but not by enough to support the hypothesis. While it was not expected that science-based firms in the Unrelated and DLU categories would have unusual rates of return, the data in Table 4-7 shows that they do: Unrelated science-based firms have significantly lower returns on equity than other science-based firms. Which are the science-based Unrelated firms? Companies like AVCO, Air Reduction, Curtiss-Wright, FMC, GAF, Grace, Lear, Litton, LTV, Martin-Marietta, North American Rockwell, Olin, TRW, and Whittaker. These firms have an average return on equity of 11.07%, which is

near the average of all other firms. But the average science-based company has a return on equity of 15.68%. Evidently, the strategy of moving from a science-based business to an Unrelated business posture has, on the average, either been due to or resulted in relatively poor profitability.

Testing the Divisionalization Effect. Earlier it was argued that the flexibility in resource allocation, the economies of scale in the general management function, and the openness to change that theorists ascribe to multi-divisional organizations would be of the greatest relative benefit in environments subject to fairly rapid change, particularly technological change. Specifically, it was hypothesized that the effects of product division structure on growth and profitability would be relatively more favorable in environments that require or encourage rapid technological change.

To test this hypothesis, return on equity (ROE) and growth in earnings per share (GEPS) were chosen as the measures of growth and profitability. The science-based industries were taken as most representative of environments subject to technological change: chemicals (28), business machines (357), electrical machinery and electronics (36), aerospace (372), and instruments (38). Restated in these more specific terms, the hypothesis becomes:

The difference between the average return on equity (and growth in earnings per share) of product division firms and the average of non-divisionalized firms will be greater in science-based industries than in other industries.⁸

Table 4-3 can be used to find the average return on equity and growth rate in earnings per share for all product division and all non-divisionalized firms:

<u>Structure</u>	<u>Average ROE</u>	<u>Average GEPS</u>
Product division	12.90	8.63
Non-divisionalized	<u>11.93</u>	<u>6.02</u>
Difference	0.97	2.61
Significance	not sig.	0.025

In aggregate, product division firms have an average growth rate in earnings per share that is substantially higher than that of non-divisionalized firms; both types of firms, however, appear to have about the same level of overall profitability.

Turning now to the question of relative performance in science-based and other industries, the data for return on equity (Table 4-8) not only does not support the hypothesis, but tends to indicate that the converse may be true. The results show that science-based firms receive substantial profit premiums regardless of their type of organization. In addition, we find that product division and non-divisionalized firms have equal profitability in non-science based areas and that product division firms in science-based industries

⁸ Non-divisionalized firms consist of those with functional or functional with subsidiaries structures.

TABLE 4-8
ROE BY INDUSTRY TYPE AND STRUCTURE

	Science- based <u>industries</u>	Non-science based <u>industries</u>
Product division	14.56	11.60
Non-divisionalized	16.26*	11.55
Difference	-1.70	0.05
Significance	not sig.	not sig.

Estimated product division effect = -0.23 (not sig.)

Estimated science industry effect = +3.12 (p = 0.001)

Note: Critical level of significance is 5%. Tests on differences and estimated effects are for equality with zero. Estimated product division and science industry effects obtained by two-way analysis of variance.

Based only on 8 data points.

TABLE 4-9

GEPS BY STRUCTURE AND INDUSTRY TYPE

	<u>Science- based industries</u>	<u>Non-science based industries</u>
Product division	9.79 (60)	7.57 (56)
Non-divisionalized	8.42 (8)	5.81 (92)
Difference	1.37	1.76
Significance	not sig.	not sig.

Average Product division effect = 1.73 Composite fit significant
at the 0.025 level.

Average Science industry effect = 2.29

Note: Average effects obtained from a two-way analysis of
variance.

See note to Table 4-9 for other comments.

have, if anything, lower profitability than other firms in these areas, although the difference is not statistically significant. Unfortunately, so few of the sampled firms in science-based industries had non-divisionalized structures over a ten year period that statistically significant results on this hypothesis are virtually out of the question. It would require about 85 (rather than 8) observations in the non-divisionalized science-based category to make the observed difference significant.

The results for growth in earnings per share (Table 4-8) also suffer from too few non-divisionalized science-based firms, but in this case it is clear that the overall superiority of product-division firms in growth is not just due to membership in science-based industries. The averages shown in Table 4-9 can be closely replicated if a growth rate premium of 1.73% per year is associated with product division structure and an independent premium of 2.29% with membership in a science-based industry. However, again there is no evidence that product division structures are especially beneficial in a science-based environment.

In summary, it appears that:

1. Firms in science-based industries have average rate of return on equity and growth in earnings per share that are higher than the average rates of firms in other industries. A science base is associated with estimated return and growth per year premiums of 3.12% and 2.29% respectively.

2. The vast preponderance of science-based firms have product division structures. There are too few non-divisionalized science-based firms to permit a statistically meaningful comparison with product division firms in these industries.

3. When the effects on performance of structure are separated from the effects of industry membership, the product division structure appears to be associated with higher growth rates in earnings per share, but has no particular impact on return on equity.

Although the original hypothesis was not supported, there is not enough data on non-divisionalized science-based firms to consider it refuted. The real problem, however, is that the original hypothesis was based on a rather naive view of the relationship between environment, strategy, and structure. Clearly, the non-divisionalized firms in science-based industries, although few, are not suffering: they have an average return on equity of 16.26% and an average annual rate of growth in earnings per share of 8.42%, both of which are respectably above the overall averages. These are firms like Abbott Labs (49-49), Addressograph-Multigraph (60-69), Kodak (49-58), National Cash Register (49-69), Pfizer (49-59) and Rohr (49-59) that, while having various degrees of diversity, also have developed particularly successful major product lines: antibiotics, speciality business machines, cameras and film, accounting machines, and aviation and aerospace power equipment respectively. There were no examples in the sample

of surviving science-based firms that followed their major product line into the maturity and decline phase of its life-cycle without diversifying further and adopting a multi-divisional structure to manage the more complex enterprise that resulted. This suggests that in science-based industries diversification is not simply one of several alternative strategies for expansion, but an eventual necessity for long term survival. As long as a firm has a "winner" on its hands, it can make high returns and grow rapidly within the framework of a relatively simple product-market scope and organizational structure. However, if prospects in its major area decline, alternative sources of profit and growth must be sought. Similarly, if research and development efforts produce results, their application often lies in areas other than those with which the firm is familiar.

One of the implications of this view is that a comparison between product division and non-divisionalized firms in science-based industries can shed little light on the question of the value of product division structure -- what is actually being compared are firms in different stages of development with differing organizational requirements. Thus, as managers of science-based firms seem to act purposefully and avoid the "wrong" strategies of staying with a declining business until the bitter end or diversifying without divisionalizing, the relative merits of these options are not empi-

rically measurable.

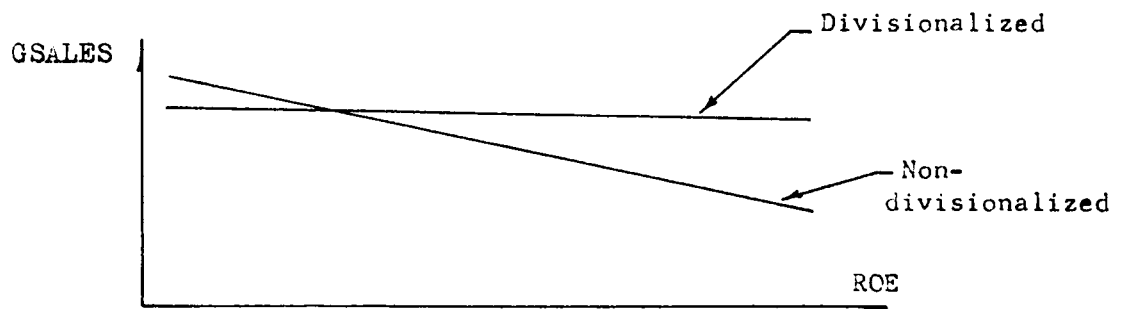
Tending to view strategy and structures as effects of performance rather than causes also brings into question the degree to which the observed "science premium" can be considered to be independent of the strategies and structures of firms in science-based industries. One could argue that technologically based products enjoy a sort of temporary miniature monopoly status and thereby produce superior performance and that this effect has little to do with the fact that most science-based firms are diversified and divisionalized. On the other hand, it could equally well be claimed that the continued health, growth and risk taking ability of firms in these industries have been dependent upon management's ability to institutionalize the process of innovation. Continuous innovation, in turn, may not require diversified divisionalized firms for its initiation, but they are its certain eventual result.

Testing for Systems Effects. One of the more subtle hypotheses advanced was that among product division firms the relationship between growth and profitability would not be as strong as among other types of firms. Economic theory predicts that growth and profits will tend to be negatively related because, in an economy in equilibrium, an increase in a firm's long-term growth rate can only be obtained by accepting a lower rate of return on marginal investments.

Taking growth in sales (GSALES) and return on equity (ROE) as the measures of growth and profit, Figure 4-1 shows the hypothesized situation. While divisionalized firms may have a negative relation between growth and profit, their special control, reward, and resource allocation systems should act to reduce the marginal amount of return they have to sacrifice to gain a unit of growth.

FIGURE 4-1

HYPOTHEZED RELATIONSHIP



Theoretically, the lines in Figure 4-1 represent the average marginal opportunity sets over growth and profit faced by the two groups of firms. Unfortunately, there is no way of observing an individual firm's growth-profit opportunity set, and less meaningful data must be used. Following the practice of past research in the area, it was assumed that some general growth-profit relationship exists for large groups of firms and that each company's growth-profit performance can be thought of as a particular point in the overall relationship. Thus, regression can be used to estimate the general

growth-profit relationship from single observations of growth and profit for individual firms.

In setting up the regression it is important to determine both lines simultaneously so that the difference between them can be subjected to direct tests of significance. If the two growth-profit relationships are assumed to be

$$\text{GSALES} = \alpha_0 + \beta_0 (\text{ROE})$$

for divisionalized firms, and

$$\text{GSALES} = \alpha_1 + \beta_1 (\text{ROE})$$

for other firms, they can be combined by introducing a zero-one variable, PD, where

$$\text{PD} = \begin{cases} 1 & \text{for product division firms} \\ 0 & \text{for other firms} \end{cases}$$

giving

$$\text{GSALES} = \alpha_0 + \alpha_1 (\text{PD}) + \beta_0 (\text{ROE}) + \beta_1 (\text{ROE}) \text{PD} + \beta_2 (\text{TIME}) + \tilde{\epsilon}$$

where the TIME variable has been included to represent changes in the overall level of growth rates and $\tilde{\epsilon}$ is the random error term. The individual data points on GSALES and ROE that were used were the 1951-1959 and 1960-1969 averages for each firm. Thus, TIME is also a zero-one variable.

Several regressions were computed, all based on this type of equation. It was found that (1) GSALES and ROE were positively rather than negatively related for all firms and for each sub-group, (2) that industry terms did not improve or change the regression and the coefficients of such terms

were not significantly different from zero, and (3) that when a strategy variable corresponding to "Related business" and "other" was added, the structure terms were the only ones that were significant. In other words, product division structure (or its absence) was a much more important contributor to the explanatory power of the regression than Related business strategy (or its absence), and once the product division variable was included the additional information supplied by the Related business variable was insignificant. The results of the best regression obtained are shown in Figure 4-2.

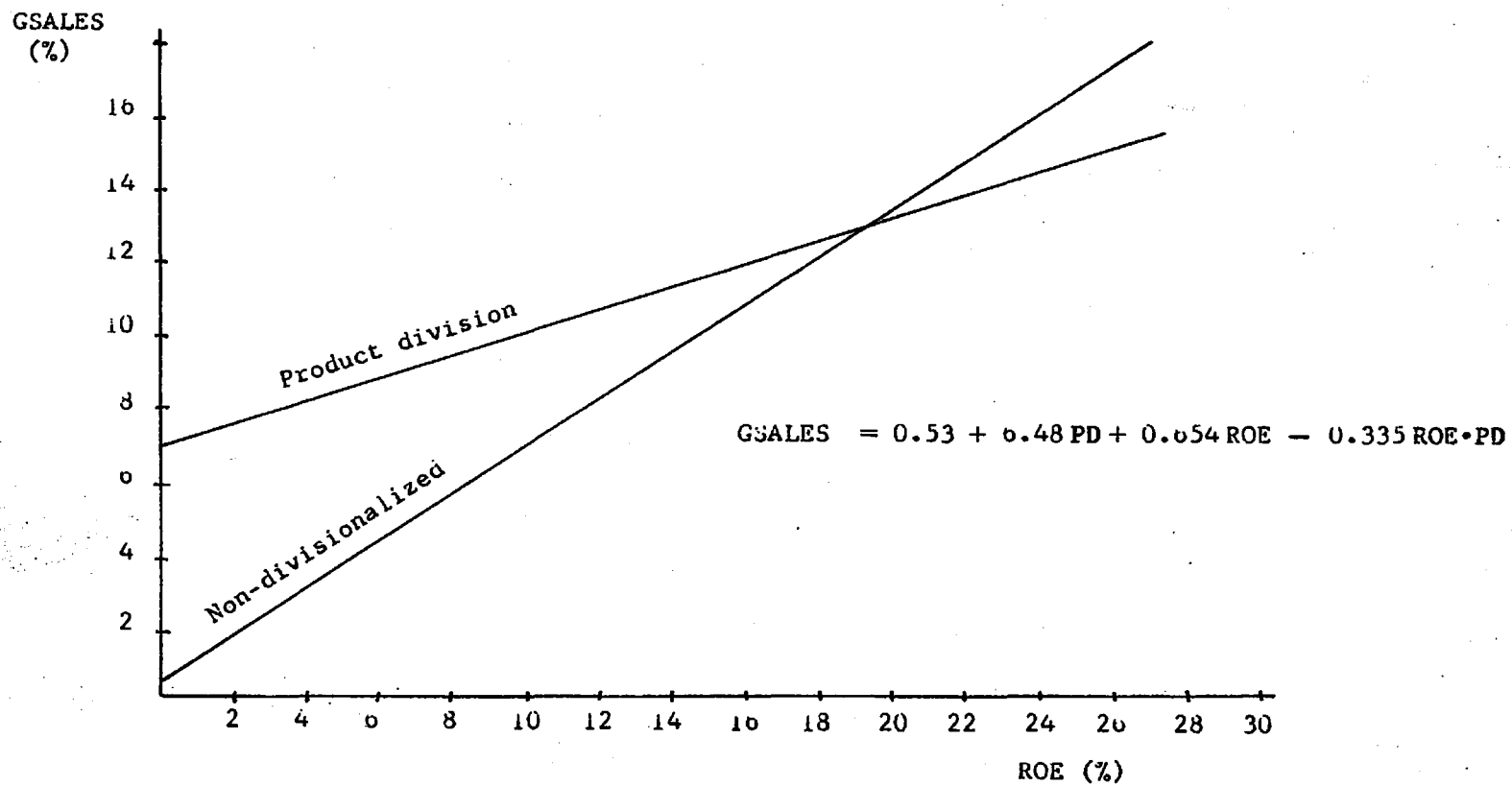
The overall relationship between growth and profit for all firms was estimated to be

$$\text{GSALES} = 4.80 + 0.455 \text{ ROE} \quad (\text{at } 1960\text{--}1969 \text{ levels})$$

with $R^2 = 19.1\%$. The t-ratio of the ROE coefficient was 6.87. Why was a positive relationship found rather than the negative one predicted by economic theory? Clearly the estimated relationship does not represent the opportunity set available to an individual firm: management cannot choose to increase both growth and profit at the same time or there would be no firms at the low end of the spectrum. Instead, the relationship seems to indicate that firms do not have homogeneous opportunity sets and that the same situations that produce above average profit also are associated with above average

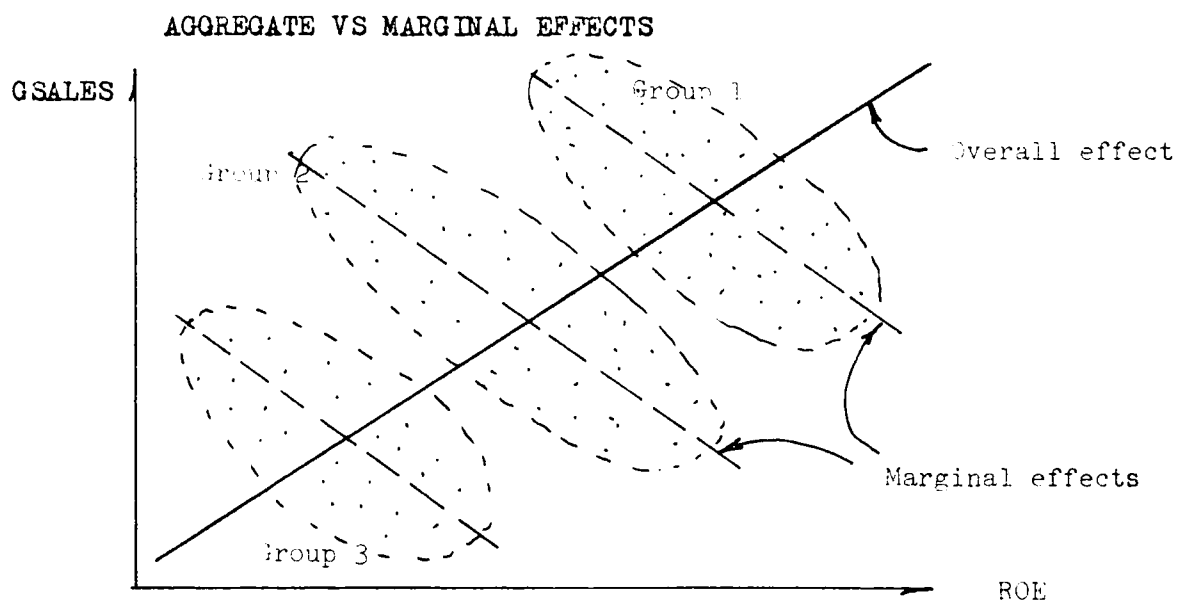
FIGURE 4-2

GSALES VS. ROE REGRESSION RESULTS



growth, and vice versa. Then, even if there is a marginally negative relationship between growth and profit within a number of special groups of firms considered separately, the aggregate effect is a positive relationship. This is illustrated in Figure 4-3.

FIGURE 4-3



Each group, of course, could consist of only a single firm, in which case what we are observing are the relative degrees to which firms have been able to insulate themselves, via proprietary skills, invention, or simple market power, from equilibrating economic forces. The implication is that the results shown in Figure 4-2 cannot be used to test the original hypothesis and must be evaluated in the context of a positive growth-profit relation among firms.

The regression shown in Figure 4-2 was quite

significant and indicates that (1) the average product division firm has both a higher rate of growth in sales and a higher rate of profit than the average firm with another type of structure, and (2) that growth and profitability are less closely related among product division firms than among other firms.

Among non-divisionalized firms, a difference in rate of return on equity of 1% is usually associated with a difference in sales growth rates of 0.53%, while among product-division firms the same difference in profitability is associated with a 0.32% difference in sales growth rates. The difference in slopes, estimated to be -0.21, was negative at the .006 level of significance.

Another way of looking at the data is that among firms with low levels of return on equity, product division firms have higher rates of sales growth and among firms with high rates of return on equity, non-divisionalized firms tend to have the highest rates of sales growth. The crossover point in return on equity is estimated to be 19%, but has a very large standard error.

Removing the Acquisitive Conglomerates from the regression does not change the signs of any coefficients or alter their significance, though the estimated crossover point drops to 12%. This is because the Acquisitive Conglomerates

have uniformly high rates of sales growth but the full spectrum of profit rates and consequently affect only the size of the intercept of the product division growth-profit line, not its slope.

In conclusion, the general concept behind the original hypothesis seems to be supported. Among product division firms, sales growth is less dependent upon profitability than among firms with functional or functional with subsidiaries structures.

Testing Overall Performance. In addition to the hypothesis of relative growth-profit independence, it was hypothesized that:

1. Related business firms will, on average, have higher profitability, higher rates of growth, and higher price-earnings ratios than other categories of firms;
2. Related Constrained firms will out-perform Related Linked firms in these measures;
3. Divisionalized Related business firms will out-perform those that are functionally organized.

As has already been noted in the preliminary evaluation of the financial data, the Related category does out-perform other firms in most measures and the Related Constrained sub-group has performance levels generally higher than that of the Related Linked sub-group. Table 4-10 displays results for all three hypotheses: average values of ROC, ROE, GSALES, GEPS, and PE for each group in question were computed

TABLE 4-10

RELATED BUSINESS PERFORMANCE

	ROC	ROE	GSALES	GEPS	PE
Difference between Related businesses and all others	+1.61 (.01)	+1.51 (.02)	+.20	+1.63 (.10)	+3.71 (.001)
Difference between Related Constrained and Related Linked businesses	+1.54 (.10)	+1.83 (.10)	+1.56	+2.99 (.05)	-.08
Difference between divisionalized Related businesses and Functionally organized Related businesses	+.30	+.80	+3.10 (.05)	-1.86	+1.08

NOTE: Figures shown are differences between estimated mean values of the variables. ROC, ROE, GSALES, and GEPS are expressed as percentages so that the differences are in terms of percentage points. Figures in parentheses indicate the level of significance of the difference. Levels of significance less than 0.10 are omitted.

(some were obtained directly from Tables 4-1 and 4-2), and the values shown in Table 4-10 are the differences between category averages and the significance of tests on these differences.

As can be seen, the Related group as a whole has higher profitability and higher price-earnings ratios than other firms. In addition, its average growth rate in earnings per share tends to be higher, though its sales growth is no different from the average of other firms. Looking inside the Related category, we find the Related Constrained group significantly out-performing the Related Linked group in growth in earnings per share, and tending to have a higher level of profitability. Further investigation shows that the difference in growth in earnings per share directly reflects a difference in earnings growth and is not due to differential dilution effects. Somewhat surprisingly, the relative under-performance of the Related Linked group is not reflected in its average price-earnings ratio which tends to be somewhat higher than that of the Related Constrained category.

Comparison between Related business firms with product division structures and Related business firms with functional or functional with subsidiaries structure reveals only one significant difference: the divisionalized group has a higher rate of sales growth. Other differences also favor

the divisionalized Related business firms but are not large enough to be statistically meaningful.

In summary, the first two hypotheses are strongly supported and the third, except for the case of sales growth, is not.

Conglomerate Performance. As a partial test of the theory of conglomerate growth it was hypothesized that:

1. Acquisitive Conglomerates will have price-earnings ratios that are significantly higher than those of other firms, and conglomerate price earnings ratios will be strongly related to the growth in earnings per share achieved.
2. Both Acquisitive Conglomerates and Unrelated Passive firms will have average returns on capital that are not significantly different from the overall average of other firms.

Table 4-11 shows the tests of these conjectures. Average values of ROC, ROE, E/C, GSALES, GERN, GEPS, IFR, and PE for the Acquisitive Conglomerate category, the Unrelated Passive category and the Unrelated category as a whole are compared to the average values for other firms and the differences are presented and tested to determine if they differ significantly from zero.

The data makes it quite obvious that the two subgroups of the Unrelated business category have radically different financial policies and levels of performance. The conglomerates achieve very high growth rates in sales and earnings, have average or slightly below average returns to

TABLE 4-11

UNRELATED BUSINESS PERFORMANCE

	ROC	ROE	E/C	GSALES	GERN	GEPS	PE	IFR
Difference between average of Unrelated category and that of other firms	-1.15	-.81	-.082 (.005)	+5.85 (.001)	+5.75 (.001)	+1.51	-1.42	+1.657 (.001)
Difference between average of Acquisitive conglomerate category and that of other firms	-1.02	+1.52	-.189 (.001)	+12.36 (.001)	+10.54 (.001)	+3.07 (.10)	+1.43	+1.310 (.001)
Difference between average of Unrelated passive category and that of other firms	-1.17	-2.31 (.10)	+1.064 (.05)	-3.05	-.99	-.64	-3.42 (.025)	-.122

NOTE: Figures shown are differences between estimated mean values of the variables. ROC, ROE, GSALES, GERN, and GEPS are expressed as percentages so that the differences for these variables are in terms of percentage points. Figures below differences are levels at which they are significant. Levels of significance less than 0.10 are omitted.

capital, but employ high leverage to boost their return on equity (and to finance their rapid growth). The Unrelated Passive firms, by contrast, have below average growth in sales, have slightly below average returns to capital, but have unusually low leverage, driving their average return on equity well below that of other firms.

The conglomerates tend to be well known, but which are the Unrelated Passive firms? A partial list includes American Standard, Bell Intercontinental (previously Bell Aircraft), Boise Cascade, Champion Spark Plug, Curtiss-Wright, DiGiorgio, GAF, Martin-Marietta, Midland-Ross, North American Rockwell, Olin, Pullman, Stewart-Warner, Sybron, TRW and USM (previously United Shoe Machinery). These firms derived more than 30% of their revenues from activities unrelated to their largest group of related business but failed to qualify as Acquisitive Conglomerates in either 1959 or 1969. Recall that to be classified as an Acquisitive Conglomerate an Unrelated business firm had to (1) have a five year average growth in earnings per share of at least 10% per year, (2) have made at least five acquisitions in the past five years, three or more of which took the firm into new unrelated businesses, and (3) have had a five year average IFR greater than one. If the firm met these restrictions anytime during the five years immediately prior to the classification date, it was

placed in the Acquisitive Conglomerate category. Many of the Unrelated Passive firms diversified so slowly or so long ago that they did not meet the requirements. Others became Unrelated business firms as the result of one or two major acquisitions but did not follow up these initial moves with further unrelated acquisitions.

Turning to the hypotheses, we find that Acquisitive Conglomerates have an average growth rate in earnings per share that is higher than that of other firms, though the difference is only marginally significant, and an average price-earnings ratio that is no different from the average of other firms. Related business firms have an average price-earnings ratio that is markedly higher than that of the conglomerates. This result is somewhat surprising because the usual theories of conglomerate growth are based on the use of a high price-earnings ratio to obtain favorable share exchange ratios in non-cash acquisitions.

That Acquisitive Conglomerate price-earnings ratios have been, on the average, quite ordinary was also found by Weston and Mansinghka who concluded that:

A dilemma confronted the conglomerate firms with depressed price/earnings ratios. If they merged with firms with higher price earnings (ratios), the conglomerates would suffer dilution in earnings per share which would at least initially depress growth rates in earnings per share. The disadvantage of a differentially lower price/earnings ratio in merger activity, can be offset by offering non-equity forms of securities in

exchange or utilizing delayed equity forms such as convertibles and warrants.¹⁸

The high leverage employed by conglomerates (average 40% debt or preferred) certainly suggests that this may be the case, particularly so in that Weston and Mansinghka also found that a high proportion of conglomerate debt consisted of convertible securities.

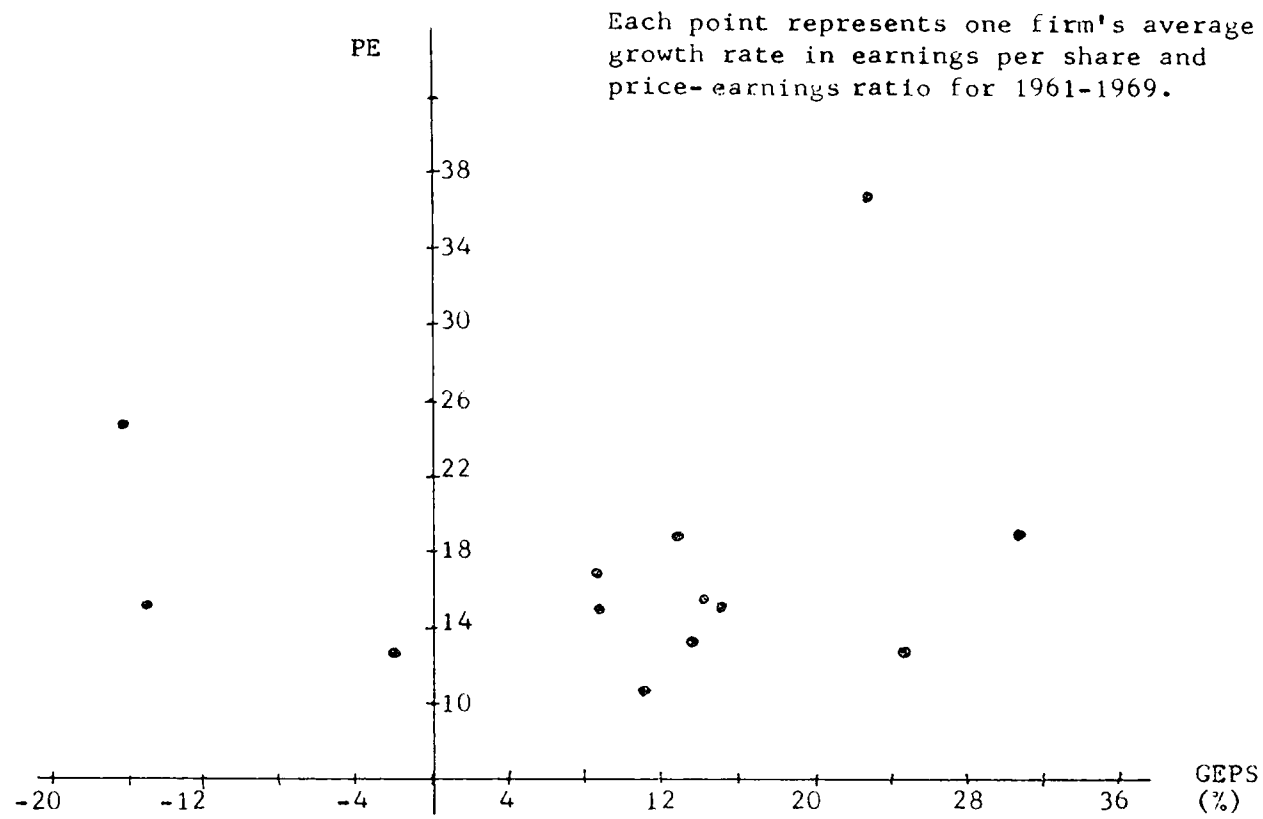
But, what about the "feedback" effect between growth in earnings per share, price-earnings ratio, and acquisition rate that is supposed to enable conglomerates to bootstrap their way to growth? Figure 4-4 shows a scatter plot of price-earnings ratio and growth in earnings per share, both averaged over 1960-1969, for the thirteen Acquisitive Conglomerates in the sample for which meaningful values could be calculated. The coefficient of correlation between price-earnings ratio and growth in earnings per share for these firms is 0.10 and is not statistically significant. If similar data for 1962-1967 is studied, the same lack of correspondence between earnings per share growth rate and price-earnings ratio is evident.

We can only conclude that not all Acquisitive Conglomerates relied on high price-earnings ratios to power their

¹⁸ J. Fred Weston and Surenda K. Mansinghka, "Tests of the Efficiency Performance of Conglomerate Firms", Journal of Finance, September, 1971, p.926.

FIGURE 4-4

PE vs. GEPS FOR ACQUISITIVE CONGLOMERATES



growth. Some, such as Litton, Whittaker, and Grace certainly correspond to the popular view. But many others, such as Bangor Punta, Colt, LTV, U.S. Industries and White Consolidated achieved quite rapid earnings per share growth with below average price-earnings ratios. LTV, for example, had an average PE during 1962-1967 of 10, and during the same period Colt achieved an average annual increase in earnings per share of 41.4% with a price earnings ratio of 13.4 and U.S. Industries experienced a 81% annual growth rate with a price-earnings ratio of 10.4. Apparently, this was accomplished by heavy use of convertible securities, concentrating on the acquisition of firms with very low price-earnings ratios or firms whose stock was not publicly traded, and perhaps the use of "incentive" purchase arrangements in which the seller's compensation is deferred and based on future growth.

In summary, the first hypothesis was not supported and the second, concerning profitability, was only partially supported. Acquisitive Conglomerate profitability does not differ from that of other firms, but Unrelated Passive companies had a decidedly lower return on equity. While it may be a means of avoiding exceptionally poor performance, the Unrelated Passive strategy of expansion has not, on average, proved to be a means of achieving above average growth or profitability. Indeed, as noted in a previous section, many

Unrelated Passive firms are primarily active in science-based industries and in comparison to other firms in these industries, their performance has been inferior.

A Question of Price

The price of a corporation's stock not only represents the investing public's evaluation of management quality and future prospects for earnings and dividends, but is, in itself, an important element of a firm's ability to diversify. Several writers have noted the strong statistical relationship between bullish stock markets and merger activity,¹⁹ and the mechanisms that allow firms with relatively high price-earnings ratios to obtain immediate improvements in earnings per share by acquiring firms at a price that implies a lower multiple are well known to both theorists and executives. Since the ease or difficulty with which a firm can manage the financial aspects of moving into a new business area through acquisition depends in part upon its market determined price-earnings ratio, the degree to which the price-earnings multiple varies with diversification strategy, either as cause or effect, deserves scrutiny.

¹⁹ See Markham, Jesse W., "Survey of the Evidence and Findings on Mergers", National Bureau of Economic Research, Inc., Business Concentration and Price Policy (Princeton, N.J., Princeton University Press, 1955), p.157, and Nelson Ralph L., Merger Movements in American Industry: 1895-1956 (Princeton, N.J., Princeton University Press, 1959), pp 64.

In Table 4-2, and elsewhere in the text, we have shown that firms in the Related business category had an average price earnings ratio that was much higher than that of other categories. In addition, the fact that the Related Linked category underperformed the Related Constrained category in terms of growth, stability, and profitability, yet had an equal or higher PE multiple has been noted. In order to test the validity of this impression and put the argument in more precise terms, a simple valuation model was developed and predicted versus actual results were examined on a category by category basis.

Regression equations employing a number of different combinations of variables as predictors of price-earnings ratios were tested, and the one that had the best overall predictive power is shown below.

$$PE = 10.97 + 4.8 (\text{TIME}) + 0.127(\text{GEPS}) - 0.075(\text{SDEPS}) + 3.72(\text{RTN})$$

$$(\text{t-ratios}): (7.5) \quad (3.22) \quad (-3.58) \quad (1.64)$$

$$R^2 = 0.48; \quad F\text{-ratio} = 24.9 (4,337).$$

Here, TIME is a zero-one variable that is set to zero for the period 1951-1959 and one for 1960-1969. PE, GEPS, and SDEPS are the previously defined decade averages of the price earnings ratio, growth rate in earnings per share, and relative standard deviation in earnings per share growth, and are expressed as percentages. RTN stands for the corporate

retention rate — the ratio of retained earnings to earnings after taxes and preferred dividends.²⁰

All coefficients were significant at the .05 level and the GEPS and SDEPS coefficients were significant at the .001 level. Since the purpose of the regression was to obtain a general valuation formula, rather than follow the rule of omitting firms that moved between categories (a rule used for other computations in this chapter), all sampled firms on which data was available were included. Decade averages were used because comparability with other data was desired. The fact that GEPS is a fitted rather than ratio measure of growth makes the use of ten year averages more acceptable than it otherwise would be.

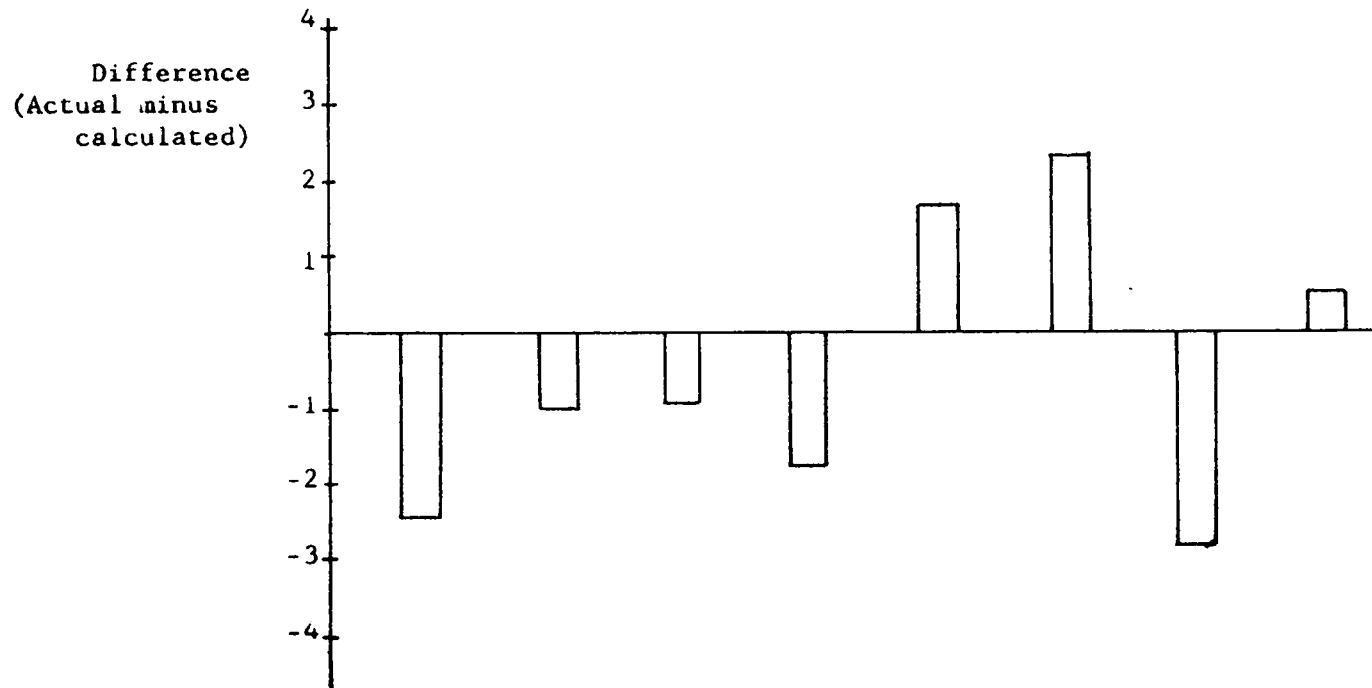
Since the regression formula is linear, it can be used to predict category means as well as individual data points. By substituting the category mean value for GEPS and SDEPS shown in Table 4-2 into the regression equation (along with mean values for RTN which are not shown), and setting TIME = 1, predicted mean PE's for each category were obtained. They are shown in Table 4-12 along with the actual average category PE's, the difference between these two, and the

²⁰ RTN had an overall mean value of 0.513 and the only category which had a mean RTN significantly different from this was the Acquisitive Conglomerate group (mean RTN = 0.754).

TABLE 4-12

THE PRICE-EARNINGS MULTIPLE GAP

	S	DV	DC	DLU	RC	RL	UP	AC	AVG
Actual PE	14.60	15.68	15.92	15.41	19.19	19.27	13.77	17.43	17.02
Calculated PE	17.02	16.64	16.84	17.19	17.50	16.94	16.55	16.89	17.02
Difference	-2.42	-.96	-.92	-1.78	+1.69	+2.33	-2.78	+.54	0.00
t-ratio	-2.44	-1.15	-.81	-.81	+2.34	+2.11	-1.59	+.33	---
Level of significance	.02	--	--	--	.01	.02	.06	--	



significance of the differences.

The results confirmed what we had suspected. Even when the effects of earnings per share growth rate, stability, and retention rate are taken into account, the Related category has an unusually high mean price-earnings ratio, and the "PE gap" is most prominent in the Related Linked category. Only the Single business category had a significant negative "PE gap". Dividing the regression residuals by strategy increased the total R^2 from .48 to .58 and the resulting decrease in variance was significant at the 0.5% level.

Why does the Dominant Constrained group, which out-performs the Related Linked group in every measure (though both are above average) have an average price-earnings ratio of 15.9 while the Related Linked category's average is 19.3? One could argue that the price-earnings ratio is a measure of expected rather than past performance, but our use of decade averages would remove all but the most persistent disparity between historical and expected performance. The only plausible explanation for the continuing existence of this difference over twenty years is that investors believe that the Related Linked firms are much more likely to be able to maintain their performance over the very long term. Whether

or not this belief is justified is not determinable but the idea that performance based chiefly upon a single major business is ultimately more transient than that growing from diversity does seem reasonable.

Another conclusion that can be drawn from the analysis of price-earnings ratio differences is that an unusually high multiple is not necessary in order to play the conglomerate game of trading shares for earnings. Single business firms can be a tempting target for many would-be conglomerates: with price-earnings multiples well below average, average returns on capital and conservative attitudes towards debt, even many Dominant Vertical firms could gain both profitability and higher earnings per share by acquiring companies in this group. That the executives of Single business firms realize this may be one of the reasons why the category is rapidly shrinking towards non-existence.

Synthesis

Our most important single finding was that the categories did separate firms into groups that displayed significant and consistent differences in financial performance. This result not only established the validity of the specific set of categories we used but also provided support for our approach to studying diversification and organizational structure. That an admittedly approximate system for capturing the

essence of top management's goals and concept of what a corporation's business is and shall become is superior to traditional measurement techniques in detecting the performance implications of diversification implies that the growing understanding of the general management task can be usefully brought to bear on issues best studied by non-clinical methods.

We found that although many of the performance differences among the categories could be thought of as reflections of industry differences, the close connection between strategy, structure and the firm's economic environment and heritage of industry-based strengths and weaknesses makes it difficult to separate these effects, either conceptually or mathematically. Is the steel industry a poorer performer than the chemical industry because of inexorable economic laws, or has the strategy of vertical integration itself helped to isolate steel firms from opportunities to apply their skills for new businesses based on other materials, markets or technologies? Have Related business firms performed well because they tend to be active in science-based industries, or do science-based industries owe their continued health to the strategic and structural attributes of the firms that constitute them? We cannot answer these questions but in their asking we are proposing a point of view which recognizes

that in the long run firms create their environments as well as react to them.

We discovered that although performance differences existed among the major categories of diversification strategy, breaking the major categories into sub-categories more than doubled the explanatory power of our model. In most cases, performance differences between Vertical and Constrained Dominant firms, between Constrained and Linked Related firms and between Passive and Acquisitive Unrelated firms were greater than those among the non-differentiated major categories. That we were surprised by this finding makes it all the more important as it implies that the firm's economic performance is more closely associated with the type rather than the extent of its product-market scope and with the way in which businesses are related to one another rather than their number.

The heterogeneity of the major categories and the pattern of several important sub-category differences are clearly visible in Figures 4-5 and 4-6. These graphs also serve to illustrate our finding that the Related Constrained and Dominant Constrained categories were the top performers in almost all measures. While the Acquisitive Conglomerates exceeded all other categories in terms of both absolute and per-share growth, their profitability was average, their

FIGURE 4-5

CATEGORY MEANS: RPR vs. ROC

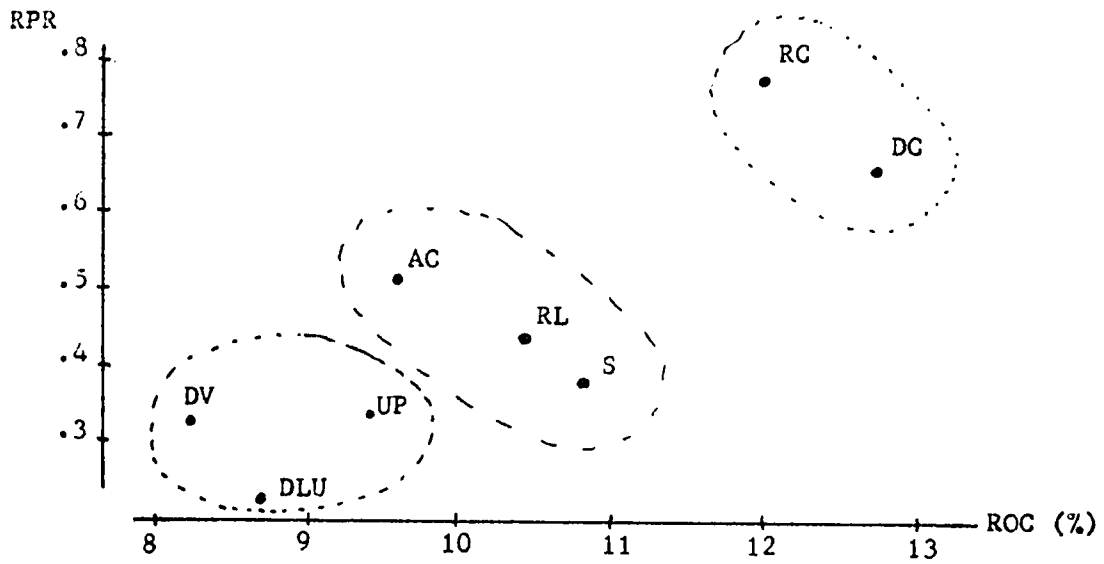
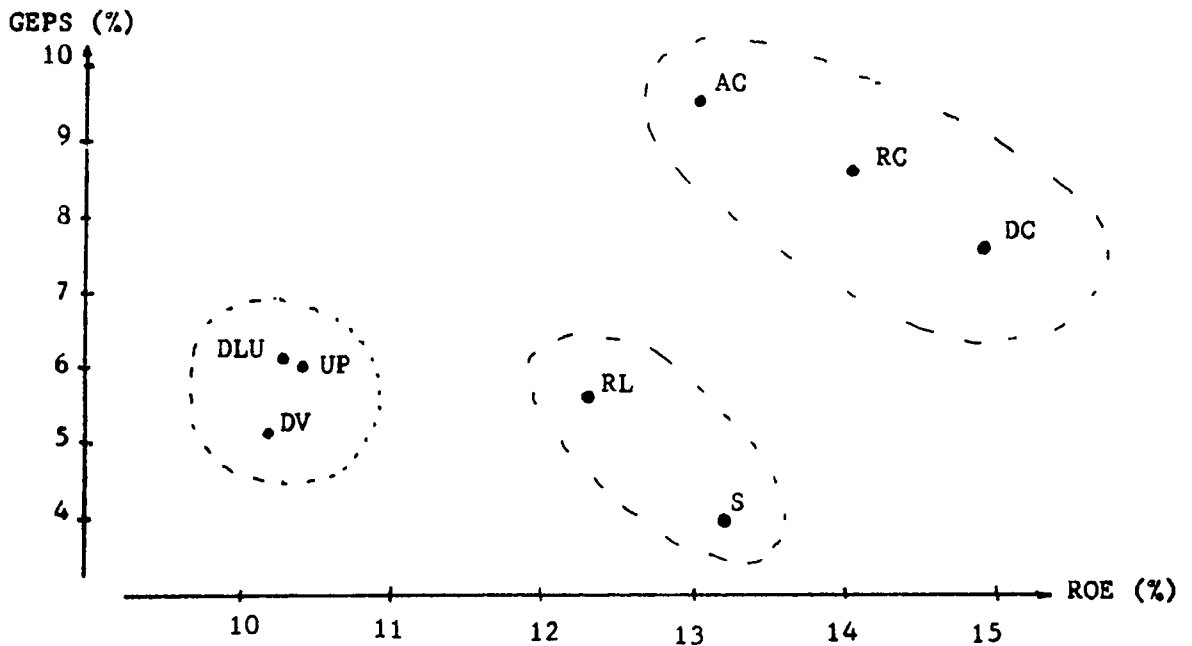


FIGURE 4-6

CATEGORY MEANS: GEPS vs. ROE



relative stability extremely poor, and the dismal performance record of the Unrelated Passive group casts some doubt on the conglomerate's ability to maintain its profitability, much less its growth, once it ceases being "acquisitive".

The Dominant Constrained and Related Constrained firms both are based on controlled diversity. Neither committed to a single business nor fully diversified multi-industry firms, their strategies reflect a managerial decision to undertake only those business activities that relate to and can draw strength from some product, skill, or market characteristic that is common to all; these are firms that need not resort to metaphysics to define their strategies yet are diverse enough to make the allocation of resources among competing uses a continuing challenge.

From the data presented in Chapter III we know that the Dominant Constrained and Related Constrained categories together account for about one-third of the largest 500 firms, that 74% of the former and 86% of the latter had product division structures by 1969, and that the Related Constrained group has been quite stable in size while the Dominant Constrained category has diminished in size as many of its members moved to more diversified postures, most of them having entered the Related Constrained category. Few of the firms that joined the diversification movement of the 1960s entered either

category, although a fair number of Related Constrained firms adopted Related Linked strategies during this period.

Why have these firms been consistently high performers? Andrews has suggested that "diversification is often an illusory diversion from the opportunities a company is best able to capitalize", ²¹ and their postures of controlled diversity certainly reflect a rejection of the mutual fund concept of general management. Still, it would be foolish to constrain one's product market scope to a dying industry and the apparent success of the Dominant Constrained and Related Constrained firms might depend to a large extent upon a process of self-selection: poorly performing companies in these categories may well elect to adopt Linked or Unrelated strategies of diversification, leaving behind those who remain successful. Therefore, controlled diversity is probably not the cause of high performance; it is rather that high performance eliminates the need for greater diversification.

Some support for this view comes from the performance of the Dominant Vertical firms. Like the Dominant Constrained firms, the Dominant Vertical companies have remained largely committed to a single business, in practice a single

²¹ Kenneth R. Andrews, The Concept of Corporate Strategy (Homewood, Illinois, Dow Jones-Irwin, 1971), p.40.

raw material. Unlike the Dominant Constrained firm, however, the strategy of vertical integration has created such a huge concentration of assets in raw material processing and produced such an internally interdependent enterprise, that escape from poor performance through diversification becomes exceedingly difficult. In Chapter V we will explore this "Dominant Vertical Syndrome" in greater depth.

One of our more surprising findings was the poor performance of the Unrelated Passive firms. Before the data gathering portion of the research was undertaken, it had been assumed that Unrelated business firms and "conglomerates" were pretty much one and the same. Even after qualitative evidence began to show that many Unrelated businesses were neither very highly diversified nor aggressive acquirers of other firms, it was felt that the distinction between Unrelated Passive and Acquisitive Conglomerates was a bit obscure. The financial data, however, indicated that these two types of strategy are associated with radically different financial policies and levels of economic performance. The Unrelated Passive category had the lowest rate of sales growth, very high variability in earnings per share, the next to lowest return on capital and, curiously, the most conservative capital structure.

The true conglomerates, on the other hand, lived up to most expectations. Despite heavy rates of dilution, their rapid absolute growth gave this category the highest average growth rate in earnings per share, although it also had the highest relative variability. Heavy use of debt levered a below average return on capital into an above average return on equity. The only unexpected results were that the Acquisitive Conglomerates' mean price-earnings ratio was quite average and that growth in earnings per share and price-earnings ratio were virtually un-correlated for these firms.

We also found that both types of Unrelated business firms tended to be heavily involved in science-based industries. A direct comparison between Unrelated business in science-based industries and other types of firms in these industries showed that they were significantly less profitable than most of the firms they compete with.

Turning to the issue of structure, we found that firms with product division structures significantly outperformed those with other forms of organization in all growth measures and in risk adjusted growth in earnings per share. Their returns on capital and equity were about average. Although few firms in science-based industries were not divisionalized, there was some indication that the same patterns held in both science and non-science based industries: product

division firms were equally profitable but faster growing.

The finding that product division firms not only grow faster overall but have been able to separate growth and profitability to a greater extent than other types of firms was by far the most intriguing result we obtained concerning structure. One could argue, as Reid ²² does, that growth associated with low profits is the result of management emphasizing "its interests" at the expense of those of stockholders. The evidence, however, is that growth and profitability are positively related and the special achievement of many product division firms has been to increase total profits more rapidly than functional firms that have the same level of economic efficiency (return on capital). This ability, we argued, is due to the special reward, control, planning, and resource allocation systems used by product division organizations. The control and reward system creates pressure for maximum return in current operations and it is the independent function of general management to allocate capital to areas that promise growth. Unlike many functionally organized firms, and particularly unlike a vertically integrated firm, the diversified divisionalized company does not have to reinvest in marginal activities just to "stay in the game"; its

²² Samuel Richardson Reid, Mergers, Managers and the Economy (New York, McGraw Hill, 1968), p.168.

strategy permits and its structure facilitates a range of policies tailored to individual businesses. Some areas can be used as cash generators or even dropped entirely; others may provide stable long term performance and still others consist of risky, but potentially lucrative new ventures. As a result, there is less conflict on the corporate level between the goals of profitability and growth.

Finally, we investigated the pattern of price-earnings ratios in detail and found that Related business firms, and especially Related Linked firms, seem to have been given earnings multiples that are higher than would be expected from their performance. Of course, the result could be spurious -- more sophisticated methods for computing expected price-earnings ratios might eliminate the apparent "PE gap". Still, the gap is considerable: Related Linked firms had average growth rates and rates of return that were slightly lower than those of other firms, yet had the highest average price-earnings ratio. Whether this gift from the market is only apparent, due to consistent irrationality, or long-term and as yet unrealized expectations cannot be determined. In any case, the Related Linked firms' average price-earnings ratio of 19.3 does not seem to have tempted their managements to rely heavily on mergers or acquisitions. Their sales and earnings growth rates are average, their earnings per share growth rate is somewhat below average but quite stable, and their rate of issuance of new shares is quite normal.

APPENDIX 4A

STATISTICAL PROCEDURES

This appendix contains descriptions of the algorithms that were used in computing the values of financial variables for firms, estimated category means and associated statistics. In addition precise financial definitions of the variables studied are provided.

Growth Rates. Given a sequence of annual observations of a variable, $X_1, X_2, X_3, \dots, X_n$, there are several interpretations of the "average growth rate" that best describes the sequence. The simplest growth rate measure is the annual compounded rate of growth which, applied to X_1 , would produce X_n after $n-1$ years. This summary value G_c , is defined as

$$G_c = \left[\frac{X_n}{X_1} \right]^{1/n-1} - 1$$

and providing X_1 and X_n are both positive, is unambiguous. The problem with this measure is that it depends only on values of X_1 and X_n . Thus, if the sequence of values of X_i is 5,6,7,8,9, 10,5, the compound annual growth rate is zero. While this cannot be disputed, the measure clearly does not reflect the "typical" or "average" annual growth rate. Changing the period in question slightly can greatly alter the value of G_c when growth is not actually smooth.

Frequently the attempt is made to avoid this dilemma by using an arithmetic average. Defining $g_i = (X_i/X_{i-1}) - 1$, the average arithmetic growth rate is

$$G_A = \frac{1}{n-1} \sum_{i=1}^{n-1} g_i$$

The problem with this measure is its consistent upward bias. Given the sequence: 1,2,1,2,1,2,..., the value of G_A is 0.25 even though no net growth occurs. This is simply because we are averaging ratios whose bases differ.

The average growth rates used in this research were computed by a third method: regression fit to a uniform growth model. The time spans involved were nine or ten years and observations were annual. We assumed that the "true" growth during this period was uniform and exponential but that each observation was subject to multiplicative error:

$$\tilde{X}_t = A e^{gt} \tilde{Y} \quad (1)$$

where $\tilde{X}_t$ is the observed value of the variable, g is the "true" uniform growth rate, $\tilde{Y}$ is a random error term with a mean of one and A is a constant. Taking the logarithm of (1) we have

$$\text{Log } \tilde{X}_t = \text{Log } A + gt + \tilde{\epsilon} \quad (2)$$

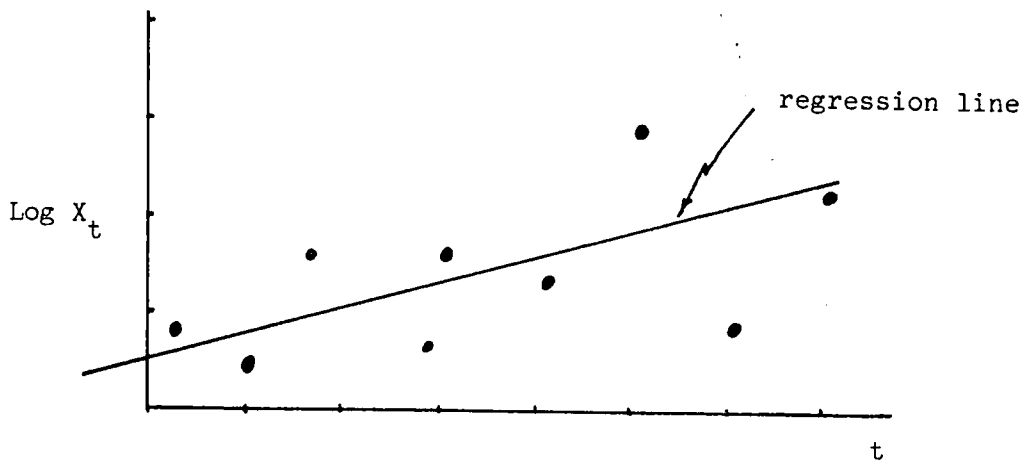
where $\tilde{\epsilon}$ is the now additive error with zero mean.

When values of X_t are zero or negative logarithms cannot be taken. This is to be expected as growth rates based on non-positive terms are not meaningful in a financial context (earnings going from -1 to 1 produce a growth rate of -100%). Our procedure was to drop observations that were non-positive, treating them like missing values.

Using equation (2) as the basis for a linear regression

gives estimates of the value of g and the standard deviation of the error, σ . A graphical view of the procedure is given in Figure 4A-1.

FIGURE 4A-1



Values of X_t close to zero can produce absurd estimates of g . If, for instance, earnings per share drops to \$0.01 for one year and then returns to the usual level of about \$3.00, a one-year growth rate of 30,000% is included in the regression. This is easy to detect when computations are done by hand, but processing close to one-thousand growth rates with a computer requires an automatic procedure for detecting such problems. What was done was to use the estimated value of the standard deviation of to screen out observations that were more than three standard deviations below regression estimates. The regression was then recomputed with the outliers omitted and the screen reapplied until all observations passed. If, after the final regression was obtained, more than four observations were missing or had to be dropped, the estimates of g and σ were discarded and special missing value codes returned to the main program.

Financial Variables. The financial variables used in this research were obtained from financial data in the Compustat data bank. Wherever possible, these figures represent the consolidated financial position of the firm with some adjustments of account definitions to ensure maximum comparability among firms. Data was available on an annual basis for 1951-1970 for most of the firms in the sample that remained in existence through 1970. The data consisted of reported values rather than pro forma accounts prepared after acquisitions, and adjustment factors were available to correct price and per-share statistics for stock splits and stock dividends. The definitions of the variables and the symbols used to represent them are given below.

GSALES: The annual mean uniform growth rate in net sales. Net sales is defined as gross revenues less returns and discounts. The mean annual growth rate was computed by the regression technique.

GERN: The annual mean uniform growth rate in earnings available for distribution to common stockholders. Earnings net for common is defined as net after-tax income less preferred dividends and after-tax extraordinary income or expenses. Since preferred dividends represent a fixed claim on income, they are more like interest than a return to owners from a managerial point of view, so earnings net for common is a better measure of profit than net income and is the only valid numerator for return on equity and earnings per share. The mean annual growth was computed by the regression technique.

GEPS: The annual mean uniform growth rate in earnings per share. Earnings per share is defined as the ratio of earnings net for common to outstanding common shares adjusted for splits and dividends. The mean annual growth rate was computed by the regression technique.

SDEPS: The relative standard deviation in earnings per share about the uniform growth trend. SDEPS is obtained from the regression used to compute GEPS. It is not the standard deviation of earnings per share but a ratio quantity that can be thought of as the standard deviation in annual growth rate.

PE: The price-earnings ratio. Annual average share price is approximated as the mean of the high and low for the year and adjusted for subsequent stock splits and dividends. Annual values of PE are the ratios of average share prices to earnings per share. After each annual value of PE was determined, non-meaningful values of annual PE were screened out. The rules used were (1) drop any negative values of annual PE, (2) drop values of PE for years in which corporate return on capital was less than 2%, and (3) drop values of PE for years in which earnings per share had dropped to less than 75% of the previous year's level. Since earnings per share can be zero or negative while price is always positive, negative earnings or unusually low levels of earnings do not create proportional changes in price. Hence, the above rules served to screen out those values of PE that bore little relation to the concept of a market capitalization rate applied to earnings. The rules were derived by trial and error with the researcher and several other individuals supplying judgments as to what constituted a meaningful price-earnings ratio. After screening, the arithmetic average of the remaining values was obtained. If more than four annual values had been dropped or were missing, the average was discarded and a missing value code returned.

ROC: The average return on invested capital. Invested capital was defined as total assets less current liabilities. The numerator of this annual ratio was after-tax net income plus interest on long-term debt (i.e., returns to all contributors of capital). ROC was computed as the arithmetic average of annual values.

ROE: The average return on equity. Annual return on equity was taken to be the ratio of earnings net for common to book equity. ROE was computed as the arithmetic average of annual values.

E/C: The average ratio of equity to capital. Annual values of this ratio were taken to be book equity divided by invested capital (defined above). E/C was computed as the arithmetic average of the annual ratios.

IFR: Internal financing ratio. This measure, suggested by Lynch,²³ is defined as the ratio of increases in equity valued at market rates to earnings. More precisely, IFR, for the period in question, was taken to be the market value of new shares issued during the period plus earnings retained during the period all divided by total earnings net for common during the period. It is helpful in understanding the meaning of IFR to know that under conditions of uniform growth,

$$\text{IFR} = \text{RTN} + \text{PE} \times \text{GSHR}$$

where RTN is the rate of earnings retention, PE is the price earnings ratio and GSHR is the rate of growth in common shares outstanding. Normally, high values of IFR (values larger than 1.0) are associated with heavy issuance of shares to finance mergers and acquisitions.

RPR: The average risk premium ratio. RPR is defined as

$$\text{RPR} = (\text{GEPS} - 0.015) / \text{SDEPS}$$

and is the risk associated growth per unit of variability in growth. Its interpretation is discussed in the text of Chapter IV.

Category Estimates. The estimated mean values of the financial variables for each category were obtained by analysis of variance. The procedure is theoretically and computationally equivalent to dummy variable regression but the development and meaning of the test statistics is easier to follow in an analysis of variance context. We will not present a rigorous symbolic description as many standard references are available.²⁴

23 Lynch, Conglomerate Performance, p. 69.

24 See, for example, S. R. Searle, Linear Models (New York, John Wiley & Sons, 1971).

Instead, we will demonstrate how the procedure applies in a particular case. The example we shall use is the derivation of category means of PE for the categories of organizational structure.

Table 4A-1 shows a summary tableau of the data pertaining to PE and organizational structure. Decade average PEs for each firm that did not move between categories during that decade have been averaged by categories within decades, across decades and over all data points.

The data in Table 4A-1 shows a pronounced time trend. In addition, the product division (PD) structure appears to be associated with larger PEs. The problem of separating the effects of structure and time is complicated by the unequal number of data points in different cells. To what degree is the higher 1960-1969 average PE due to a time trend versus the existence of relatively more PD structures in the second decade? To separate these two effects we construct a linear model:

$$PE = m + a_1 F + a_2 FS + a_3 PD + b_1 T_1 + b_2 T_2 + \tilde{\epsilon} \quad (1)$$

where m is some overall mean, a_1 , a_2 , and a_3 are the differential effects of functional, functional with subsidiaries and product division structure on PE; b_1 and b_2 are the differential effects of the first and second decades, and $\tilde{\epsilon}$ is a random error term. The important assumption implicit in this model is that time affects all categories equally and the category effects do not change over time. The validity of this assumption will be tested later.

TABLE 4A-1

INITIAL TABLEAU

	F	F/S	PD	Combined
1951-1959	10.57 (51)	11.78 (15)	13.65 (26)	11.64 (92)
1960-69	13.80 (15)	16.87 (13)	18.87 (87)	17.98 (115)
Combined	11.30 (66)	14.14 (28)	17.67 (113)	15.16 (207)

Total sum of squared observations = 58,092

Equation 1 is not usable as it stands because it is over-determined. For example, we could set m to any value at all and still find a 's and b 's that produce the same predicted category and decade PEs. It is not hard to show that two coefficients of equation 1 are redundant and we shall elect to omit m and b_2 giving

$$PE = a_1 F + a_2 FS + a_3 PD + b_1 T_1 + \tilde{\epsilon} \quad (2)$$

Here, a_1 , a_2 and a_3 will be the estimated 1960-1969 category means and b_1 is the amount that must be added to obtain 1951-59 estimated category means. We assume that $\tilde{\epsilon}$ is normally distributed.

Determining these coefficients is essentially a regression problem and involves the solution of a set of linear equations. However, for the simple case at hand, it can be shown that

$$b_1 = \frac{(92)(11.64) - (51)(11.3) - (15)(14.14) - (26)(17.67)}{92 - \frac{51^2}{66} - \frac{15^2}{28} - \frac{26^2}{113}}$$

and that a_1 , a_2 and a_3 can be obtained by subtracting b_1 from each 1951-1959 category average and then recomputing the total pooled category averages:

$$F: a_1 = 14.86$$

$$FS: a_2 = 16.60$$

$$PD: a_3 = 18.73$$

which are the estimated category means shown in Table 4-3. The estimated means for 1951-1959 are 4.6 less in each case.

To see how well this linear model has fit the averages of Table 4A-1, we can compute the predicted cell values and the deviations of actual from predicted. This is shown in Table 4A-2.

TABLE 4A-2

ACTUAL MINUS PREDICTED				
	F	FS	PD	AVG
1951-59	0.31	-0.23	-0.48	0.0
1960-69	-1.05	0.26	0.14	0.0
AVG	0.0	0.0	0.0	0.0

As can be seen from the table, the linear model provides a fairly good fit to the averaged data. The analysis of variance procedure provides a precise measure of the quality of the fit and also indicates which portions of the model are most important.

The sum of the squared values of all 207 observations, SST, is 58,092 in this example. If there were no variability in the data, so that each observation were equal to the observed average, the sum of squared observations, SSM, would be $207 (15.16)^2 = 47586$. The difference between SST and SSM represents the variability about the mean in the raw data. We call this SST_m :

$$SST_m = SST - SSM = 10,506.$$

The basic idea behind analysis of variance is to partition the variability into components that can be associated with different sources of variability. The completed analysis is shown in Table 4A-3.

Here SST_m has been partitioned into portions due to time trend effects, category effects and residual error. The final column of the table shows the proportions allocated to each source.

The residual error portion, when divided by the appropriate degrees of freedom,²⁵ gives an estimate of the variance of the error term, $\tilde{\epsilon}$, in the model. The square root of this variance, σ , is shown in Table 4-3.

The first line of Table 4A-3 tells us that the portion of sum-squares due to introducing the time trend term in the model is 2055, or 19.6% of the total. In addition, since there are only two time periods in question, the estimated variance due to the time trend is 2055. The F-ratio is the ratio of this variance to the residual variance. The F-ratio of 52.3 is quite large, indicating that the time trend effect is much larger than random variability can account for. More precisely, the F-ratio tests the hypothesis " $b_1 = 0$ ", or more generally, "all time coefficients are equal." Standard tables of the F distribution are widely available and show that an F-ratio of 52.3 with 1 degree of freedom in the numerator and 203 in the denominator is significant at at least the 0.001 level. This means that there is less than one chance in one thousand that random error effects could have produced this large an F-ratio if time effects were actually absent. We thus reject the hypothesis " $b_1 = 0$ " and proceed on the assumption that time correction is necessary. In many other cases time effects were not significant, particularly when growth rate variables were being analyzed. We took the critical level to be 0.1 and assumed time corrections were necessary if it was equaled or exceeded. When the time trend terms were not significant, the linear model was discarded and the simple pooled averages for each

²⁵ Two-hundred seven data points less 4 coefficients in the model leaves 203 degrees of freedom for estimating residual variance.

TABLE 4A-3

ANALYSIS OF VARIANCE

Source of variation	Sum of squares corrected for the mean	Degrees of freedom	Mean square	F-ratio	Level of significance	R ²
Time	2055	1	2055.0	52.30	0.001	0.196
Categories	478	2	238.5	6.08	0.05	0.045
Error	7973	203	39.3			0.759
Total	10506	206				1.000
Testing for Interaction						
Linear model	2533	3	844.0	21.40	0.001	0.241
Interaction	34	2	17.0	0.43	not sig.	0.003
Error	7939	201	39.5			0.756
Total	10506	206				1.000

category were taken as the estimated category means. Analysis of variance was then carried out on just the category effects.

The second line of Table 4A-3 indicates that category effects (differences among the categories) created 4.5% of the variability and that the variance among categories is estimated to be 238.5. The F-ratio is 6.08 which is significant at the 0.05 level. It is very important to note that these figures refer to the explanatory power of the categories after the effects of time have been taken into account. It is this F-ratio and R^2 that appear on Table 4-3.

The next step is to check the validity of the linear model. That is, we wish to check the original assumption that time trends affect all categories equally and category differences are stable over time. When these assumptions are not true, it is said that interaction effects are present.

We test for interaction effects by comparing the linear model to the actual category averages in each decade. More precisely, we assume a new model which has a term for each cell in Table 4A-1 and test the degree to which it is a significant improvement over the simple linear model. As can be seen from the lower portion of Table 4A-3, interaction effects had an F-ratio of only 0.43 and were not significant. We took the critical level of significance to be 0.05 on interaction effects and it was not equaled or exceeded for any of the variables analyzed.

If interaction effects had been significant, the concept of a single category mean could not have been used as

category means would have been changing relative to one another over time.

The t-ratios shown in Table 4-2 (and Tables 4-1 and 4-2) test the hypothesis "this category mean is equal to the overall mean". They also differ very little from t-ratios that test the hypothesis "this category mean is equal to the weighted average of the means of all other categories". For cases in which the time effects are not significant, the t-ratios are obtained from this relationship:

$$t = \left(\frac{\Delta}{\sigma} \right) \sqrt{N \cdot N_c / (N - N_c)}$$

where Δ is the difference between the estimated category mean and the overall mean, N_c is the number of data points in the category in question, N is the number of total data points, and σ is the residual error standard deviation. When time effects are present, more complex formulas are necessary.

For the number of data points in this study, the distribution of t under the null hypothesis is very close to normal with mean zero and standard deviation one. Thus the t-ratios can be viewed as "standard deviation" in order to gain an intuitive feel for their significance.

CHAPTER V

THE LOW PERFORMANCE STRATEGIES

As noted in Chapter IV, a simple way of summarizing the pattern of performance differences among the diversification categories is that there are two widely disparate low performance strategies: vertical integration and the Unrelated Passive product-market posture. The gap between the average level of performance of these firms and that of firms with other strategies was quite significant and can in no way be attributed to simple variation about the mean. What makes this finding particularly important is that these two low performing categories represent polar extremes of strategy -- a complete commitment, not only to a single industry, but to controlling an entire sequence of processing stages, versus the operation of a set of unrelated business enterprises. The purpose of this chapter is to explore the reasons for the low performance of these two groups. In particular, we will look closely at why Dominant Vertical firms have so infrequently sought to improve their prospects through diversification and why the Unrelated Passive strategy, which imposes few constraints on the pursuit of opportunity, has not even produced average levels of performance.

The Dominant Vertical Syndrome

Of the largest 500 U.S. industrials, about 15% have been vertically integrated Dominant business firms during the twenty years studied. The size of these firms, however, magnifies the economic impact of their behavior and performance -- 44% of the Dominant Vertical companies were among the largest 100 in 1969.

Considering the maturity of the industries in which they participate and the relatively undifferentiated nature of the products they sell, it is not too surprising that the average growth rate and return on capital of Dominant Vertical firms is well below the norm for large corporations. What is more difficult

to explain is the reluctance or inability of these firms to change their strategies in the face of continued poor performance. In the period under study, only five Dominant Vertical companies were observed to move on to Related or Unrelated business strategies: two were grain milling and converting firms that moved easily into packaged foods, AMAX diversified into a variety of mining activities, Boise Cascade had less than the usual amount of timber holdings for a forest products company and expanded rapidly into construction and other businesses, and Di Giorgio moved by divestiture and acquisition from being an integrated fresh fruit company to the production and sale of lumber, soft drinks, branded consumer products and other unrelated products. None of the integrated oil, paper, primary metals, meat packing, or rubber companies in the sample managed to move into the Related business (or Unrelated business) category; firms in heavy industries have acted as if diversification and integration were incompatible strategies.

Like the Dominant Vertical firm, the Dominant Constrained company derived the preponderance of its revenues from a single major business. Unlike the vertically integrated firms, companies in the Dominant Constrained category exhibited a strong tendency to abandon their one-business postures in response to deteriorating performance. A comparison of the 1951-1959 performance of Dominant Constrained firms that stayed Dominant Constrained during the next decade with that of Dominant Constrained firms that moved to strategies of greater diversification during the next decade showed that lower levels of growth and profitability were closely associated with a subsequent decision to diversify:

<u>Firms that were Dominant Constrained in 1949-1959</u>	<u>1951-1959 Average ROE</u>	<u>GEPS</u>
And stayed Dominant Constrained during the decade 1960-1969	17.4	11.7
And diversified out of the Dominant Constrained group in 1960-1969	13.4	6.7
Significance of the difference	.02	.10

Yet the "low" performing Dominant Constrained firms that sought a remedy in diversification were more profitable and faster growing than the average Dominant Vertical company.

In the sections that follow we will look at some of the reasons why Dominant Vertical firms have not, as a group, meaningfully diversified. The major barriers to diversification are: mature industries and near commodity products, problems in the transferability of skills, production oriented management, complex functional organizations, and low performance itself. All of these affect other types of firms as well, but in the vertically integrated enterprise they come together with a combined impact that greatly diminishes the ability and willingness of management to undertake the task of strategic change.

The Strategy of Integration. By the first decade of this century industrialization and a wave of combinations had culminated in the formation of many large-scale integrated businesses. In the manufacturing sector, integration came earliest and most completely in industries that were extractive or agriculturally based: oil, lumber and paper, iron and steel, copper and nickel, rubber, meat packing and fresh fruit. The creation of integrated firms was spurred by the economies of scale available from the creation of nation-wide sales and marketing organizations and the construction of ever larger manufacturing plants, for once a strategy of great size was adopted, the control of raw material sources became a major competitive

weapon.¹

The growth companies of the early decades of the twentieth century, by the mid-1950's many vertically integrated firms faced problems of slow growth, eroded margins, and industry over-capacity. Usually the suppliers of basic materials to other manufacturers, these firms found themselves tied to strategies of efficient operation of complex capital intensive processes in a time when consumer marketing, product innovation, engineering of special products rather than mass production, and science-based research were becoming the types of skills and production methods most highly rewarded by the marketplace. The steady growth in the demand for oil and refinery products is the only major exception to this pattern, though this has been an essentially fortuitous phenomenon; the oil industry neither foresaw nor developed the major markets for its products.

Requiring huge commitments of capital, the extractive industries can be among the most profitable in times of rapidly rising demand. However, products like steel sheet, paperboard, and petrochemical feed stocks are valued chiefly as raw materials in the manufacture of a vast variety of final products. Non-differentiated because it is their economic function to be standardized, the profitability of these products is largely contingent upon the balance between capacity and demand. In a mature slowly growing industry, the constant oligopolistic struggle to be a low cost producer and provide flexible and geographically dispersed capacity almost ensures industry over capacity (on the average) and poor profitability for all.

¹ For a detailed discussion of the development of the strategy of integration see Alfred D. Chandler, "The Beginnings of 'Big Business' in American Industry", Business History Review, Vol. XXXII, Spring 1959, pp. 1-31.

The steel industry's constant problem with capacity is well known and the aluminum industry, once considered a "growth industry", now shares this problem. Commenting on the state of the aluminum business in 1971, Forbes noted that

The villain here is overproduction. "Inventories are at record levels and prices are terrible," summarizes analyst David Healy of Burnham & Co. "I calculate that on a Free World basis that aluminum consumption would have to increase 31% next year to use up excess inventory and return the industry to a 100% operating rate. No such increase is likely." Some analysts think supply could exceed demand for three more years.²

That economic theory explains and predicts the existence of such phenomena indicates that these competitive patterns are not pathological -- they are the results of mature industries producing near commodity products by processes that require quantum investments of capital.

Of course, what is a problem for a firm's management is not necessarily a problem for society as a whole. Given the low level of technological change, there is no reason for the integrated extractive industries to earn more than a minimal return. What does cause concern and require explanation is the tendency for the managements of Dominant Vertical firms to constantly reinvest in their low return business. Bethlehem Steel, for example, recently completed the construction of its new Burns Harbor mill, representing an investment of more than one billion dollars. In money terms, this is a larger project than IBM's 360 venture. Yet, even if successful, analysts believe the new mill will do little more than arrest Bethlehem's decreasing earnings per share and push its return on capital back up to 6%.³

² Forbes, January 1, 1972, p. 102

³ Interview with industry analysts at Paine, Weber, Jackson and Curtis, Boston.

The lack of diversification by the steel industry makes it a special case even among vertically integrated firms. Some Dominant Vertical firms in other industries have embarked on diversification programs, but only a few companies have treated new businesses as strategically important rather than as side ventures. Among oil companies, Standard Oil of Ohio, Continental Oil and Phillips Petroleum derive substantial portions of their revenues from other than basic refinery products. In 1969, Continental obtained 10% of its revenues from coal production (Consolidated Oil, acquired in 1965) and another 13% from plastics chemicals and fertilizer. Ohio Standard almost slipped into the Related category in 1969 with 71% of its revenues coming from refinery products and crude oil and the balance from coal, chemicals, fabricated plastic products, and metals. But, in January 1970, British Petroleum obtained effective control of Ohio Standard and the combined company had more than 80% of its sales in crude oil and refinery products.

Phillips Petroleum has devoted more research effort to developing specialty chemicals and plastic products than most other oil companies and the effect on its sales mix has been significant. In 1969 Phillips derived 73% of its revenues from crude oil and refined products, 6% from natural gas, and 21% from synthetic fibers, synthetic rubber, fertilizer, plastic pipe and other molded products, vinyl sheet and a paperboard and plastic coated container business. If there is a pattern in the oil industry, it is that the larger world-wide integrated companies have been the least prone to diversify.

Among paper companies, neither the full line paper and paperboard firms nor the cellulose wadding consumer products companies (e.g., Kimberly-Clark and Scott Paper) have attempted to move beyond the confines of their industry. Boise Cascade started as a lumber company and, rather than try to become another Weyerhaeuser, moved into building materials of all types and containers. The critical strategic difference exhibited by

Boise was to then fully exploit the business implications of its product-market activities. It adopted a product division structure in about 1959 and moved into factory built homes, mobil homes, land development and recreation products. On the container side, Boise built up a business that included plastic, metal, foil, and paper containers. Finally, in a number of unrelated moves, the company moved into the engineering and construction of industrial plants, computer software, and the distribution of office supplies. Partly aided by its initial small size, Boise's rapid diversification also came from a willingness to look beyond lumber and other wood based products for opportunity.

The major rubber companies have complex businesses that are difficult to classify. Tires (and wheel rims) are the major product of each of the Big Four, and all have developed fairly large businesses devoted to selling the chemicals and yarns associated with the integrated manufacture of tires. Yet none have really devoted major resources to their two main opportunities to diversify: (1) to become full line producers of automobile, truck and farm equipment parts or (2) to engage in real science-based research in order to become chemical innovators rather than just producers of bulk chemicals.

It is somewhat ironic that while Goodyear and Goodrich have tried to develop aerospace materials and electronics businesses, requiring skills far removed from their basic competences, several aerospace companies have diversified into the auto parts field.

Non-extensible Technology. In his chronicle of the development of the first diversified firms in the early part of this century, Chandler placed heavy emphasis on the importance of the impetus supplied by new technologies and new markets:

In those industries most affected by the new markets and new technology, growth came . . . by going overseas and still more by diversification Diversification came when leading companies in these technologically advanced industries realized that their facilities and the scientific know-

how of their personnel could be easily transferred into the production and sale of new goods for new markets. For those enterprises whose energies were concentrated primarily in merchandising, diversification came because of the changing markets in the city and then in the suburbs.⁴

The technologically advanced industries (chemicals, electrical and later electronic equipment, and internal combustion engine based products) were not only science-based, but encompassed knowledge and techniques that were not specifically related to a single product, material or process. This special property of extensibility was and still is a key influence on the ease with which a firm can diversify. The production of steel and paper and more recently the manufacture of spacecraft and atomic power plants are all dependent upon some scientific knowledge and complex technologies, but little of the vast store of know-how associated with these activities has proven applicable to other products. Extensible technologies, by contrast, have found wide application in areas far removed from those to which they were first applied. Electrical science was first concentrated in the business of creating capital goods for the generation of power. Widespread electrification led in turn to a proliferation of consumer and industrial products that utilized electric motors as power sources. The development of radio, television, transistors, radar, electronic control systems, and computers led firms that had skills in the electrical sciences into a vast array of related businesses. Similarly, chemical science has proven applicable to product areas as diverse as textiles and antibiotics. The crucial point is not so much that these technologies are science-based as that it is in these areas that science-based research has provided the greatest diversity of fruitful results. This fact has been one of the primary determinants of diversifi-

⁴ Alfred D. Chandler, Strategy and Structure, (Garden City, N.Y., Doubleday, 1966), p. 51.

cation strategy. A textile manufacturer, no matter how finely honed his skills, simply has fewer opportunities for transferring those skills to other product-market areas than the average producer of electronic devices.

Most vertically integrated firms embody an industrial pattern that pre-dates the rise of the new technologies. The nature of the products and the strategy of integration both coincided with and supported the development of technologies that were based on process rather than product, and on improving the efficiency of production rather than changing the nature of the end product. While research efforts in the new technologies have continually blurred the boundaries between sciences, industries and products, the production oriented inward-looking research in steel, oil, paper, and non-ferrous metals companies has chiefly served to create less expensive or improved steel, gasoline, paper or other metals.

Still, the problem of relatively non-extensible technology has not proven to be an insuperable barrier to the diversification programs of many non-integrated firms. When neither a firm's technology nor its markets have been particularly conducive to a transfer of skills or strengths to new areas, careful sequences of moves and more than the usual amount of imagination have enabled quite a few to move to Related business product-market postures. For example, in 1949 American Chain and Cable had well over 70% of its revenues coming from the production and sale of chain, steel wire cable, and valves and other simple cast shapes. Almost as an incidental side-line, it also made a specialized type of hardness testing instrument that was well respected but had a limited market. Its strategy of diversification during the next twenty years was three-fold: (1) it integrated forward into the production of brakes, clutches, hoists and overhead cranes as these products employed both its cables and its casting and foundry capabilities, (2) it exploited

its growing position in hoists and cranes by moving firmly into stackers, palletizing equipment, conveyers and other materials handling equipment, (3) it built on its reputation in the hardness testing field by moving gradually into data recording instruments, gauges, and data loggers and eventually into industrial controls, telemetry, and other electronic instruments. Recently it has sought to draw on both branches of its diversification program by developing equipment and systems for automated warehouses.

Another example is Union Tank Car's (now Trans Union) diversification strategy. While its competitor, American Car and Foundry, made an unrelated and eventually abortive move into defense electronics, Union Tank exploited its experience in tank car leasing by moving into truck leasing, the operation of truck and rail bulk terminals, built on its growing financial skills by entering the credit and loan field and then combined its experience in transport and finance by taking a position in international trade services. At the same time, it utilized its tank car manufacturing skills by moving into the water and chemical tank business, entering the waste treatment and pollution control fields and introducing a line of home water conditioners. However, that the potential for expansion of corporate competence can easily fall short of management's ambitions is also illustrated by Union Tank's experience -- an attempt to build missile silos under government contract was resoundingly unsuccessful and eventually resulted in the resignation of the company's chief executive.

While these and other companies found it possible to diversify away from relatively non-extensible technologies, the complexity of the linking relationships employed made it anything but an easy task. For vertically integrated companies the problem is intensified by the process nature of their technologies and the illusion of diversification provided by forward integration into fabricated products.

For raw materials companies, the most obvious opportunities for transferring technological skills to other areas involves the production of products that are close substitutes for their original products. The strategy of vertical integration creates a large commitment of capital and human energy to a particular product, not to a product function. As a consequence, other products that perform the same function are not seen as logically related to the companies business but as threats to the firm's continued viability. For steel company executives, aluminum and plastics are not viewed as natural partners to steel in a structural materials business, but as competing substances which can undermine the economic performance of the ore-to-mill production system. For an oil company, petrochemicals are a way to increase the profitability of oil production and refining -- the distillation of coal and petrochemical feedstocks is a technology to be feared rather than welcomed. This attitude is not irrational; it is a natural consequence of large scale investment in capital intensive and inflexible processing systems.

The same principle affects the strategy of forward integration into products that use the integrated firm's major product as a raw material. Dominant Vertical companies frequently refer to this as "diversification", but in practice they look upon fabricated products as simply one way of ensuring utilization of the firm's basic investments in raw material sites and processing capacity. Even at Boise Cascade, where a strategy of rapid diversification later took the firm completely out of the Dominant Vertical category, the president, R.V. Hansberger, noted in 1963 that "the security of diversification came as a by-product of a plan to improve the use of our trees."⁵

Alcoa's strategy to increase primary demand for aluminum by demonstrating how it could be used to create a variety of products called for forward integration into cookware, wire,

⁵ Boise Cascade Corp., Annual Report to Stockholders, 1963

containers, branded foil and a number of other fabricated products. The goal of stimulating demand was achieved, but the company's so-called "diversification" posture has never served to insulate it from cycles in the demand for and supply of aluminum. As long as the final product businesses of the vertical chain have strategies of making aluminum pans and aluminum wire (or steel pipe or rubber gloves) rather than exploiting the full business potential of their product-market positions, the net result is merely an intensification of the firm's dependence on a single material. This occurs because each new fabrication activity creates a corresponding need for expensive high fixed cost basic processing capacity.

By-product diversification can offer the integrated firm a means of moving into new business areas, but the scope of action is limited by the growth rate of the primary business. Steel firms were suppliers of benzene long before the oil companies began to produce this valuable chemical building block in volume. But a steel company's sales of by-product materials (in this case from coke ovens) is limited by its rate of steel production. Meat packers face the same problem in fully exploiting their chemical by-products as do paper companies in the area of cellulose chemicals.

Management and Organization. Barriers to diversification can also arise when a one-industry strategy is institutionalized through organizational structure and the values and attitudes of management.

The functionally structured organization, departmentalized on the basis of production stages, may be an aid to efficient operation, but it also severely limits the number of generalist roles and thus the number of executives who perceive and are responsible for the corporation as a whole. In addition, the partitioning of an integrated firm into departments (e.g. extraction, processing, mill and converting), tends to promote a style of investment decision that focuses on capacity and

"balance" rather than rate of return and the consideration of alternate uses for capital. In a study of the investment decision process in both integrated paper companies and paper divisions of diversified divisionalized firms, Ackerman found that

Investment analyses in the integrated firm were carefully confined to one division and excluded the impact on profitability or the need for additional capacity in other divisions linked to it by product transfers In short, the corporate executives sought ways to isolate the effect of capacity changes to individual units A difficulty arose, however, if the desired balance between paper making and converting capability was upset by a major addition at the mill stage. The definition of converting plants was then encouraged by the corporation to restore an internal discrepancy . . . Such behavior, by itself, does not make economic sense. It may, in fact, provide one explanation for the recurrent tendency in this industry to extend capacity additions to paper to converted products at the expense of prices and, ultimately, profitability at the converting end of the business.⁶

In contrast, Ackerman found that the "paper division" managers of the diversified firms tended to include sub-unit interrelationships in their analyses of major investments.

Allocation systems that promote reinvestment are, in turn, part of a more general tendency for the attractiveness of investments to be judged in terms of historical and industry performance. A new plant may seem justified if it promises a 10% return if alternative products yield only 9%, but even more lucrative returns available from extra-industry investment may not even be considered. Thus, a steel firm may make the best possible use of its capital within the context of the steel business but in actuality be making a poor economic decision. This tendency is undoubtedly intensified by the heavy depreciation cash flows, often thought of as representing "costless capital", that are available to capital intensive producers.

⁶ Robert W. Ackerman, "The Impact of Integration and Diversity on the Investment Process" (unpublished paper, Harvard Business School, 1969) p. 14-15.

The one industry strategy, in addition to promoting questionable reinvestment rates, is usually associated with the development of values and attitudes that assign significant existential worth to the products or processes themselves. One cannot read the literature on the steel, auto, oil, rubber or paper industries without realizing that executives in these businesses look upon them as intrinsically exciting and worthwhile and take great pride in the process of production itself. The effects of this orientation on the propensity to diversify are obvious. In testimony before the Senate Subcommittee on Antitrust and Monopoly, Joel Dirlam, an economist, expressed the opinion that this value set was a major reason for the lack of diversification in the steel industry:

I would like to just mention an industry which has not diversified, which is vertically integrated, and yet has complained perhaps more vigorously than any other industry over a period of years before this committee and elsewhere about its low level of return and profits. I refer to the steel industry.

The absence of diversification of large steel firms is, perhaps, surprising. One of the motives for diversification is said to be that firms will endeavor to move from relatively low-return lines to more profitable businesses. Large firms in steel have been unwilling or unable to diversify out of steel, in spite of their sizable internal flows of funds, and in spite of the fact that the industry is characterized by excess and obsolescent capacity and by steadily increasing competition from substitutes. A partial explanation . . . may be that steel executives regard themselves not as having gone into a business but having adopted a way of life It is interesting that when one of the more profitable steel companies, National Steel, speaks of diversification, it means -- not that it is going to move into other industries but that it will attempt to reduce the reliance on the automobile industry for outlets.⁷

⁷ Statement of Dr. Joel Dirlam, Hearings Before the Senate Committee on the Judiciary, Subcommittee on Antitrust and Monopoly, Economic Concentration, Part 2 (Washington, Government Printing Office, 1965) p. 767.

Attitudes such as these are reminiscent of the situation that faced Alfred Sloan when he took over the management of then struggling General Motors corporation. In setting the future strategy of the concern he found it necessary to explicitly call into question the tendency for management to regard the company's purpose as selling a particular set of products:

"The primary objective of the corporation, therefore, we declared was to make money, not just to make motor cars."⁸

Of course, loyalty to a product or industry may not be dysfunctional and may even be required for the effective operation of complex enterprise. However, such attitude may also induce over investment and strongly inhibit strategic change.

Size and Performance. Even when the management of an integrated business decides to channel funds and energy into new areas, the absolute size of the original business can render invisible the impact of all but the most vigorous diversification programs. For example, several of the major oil companies that we classified as Dominant businesses have petrochemical, plastic and fertilizer operations that, if split apart from the parent firm, would be among the largest 500 industrials in their own right. For the very largest corporations this is an almost insoluble dilemma. Piecemeal additions to the businesses of companies like U.S. Steel, General Motors and Standard Oil of New Jersey can produce but insignificant changes in earnings and their rate of growth. In addition, the close scrutiny of the Federal Trade Commission virtually blocks the large merger route to diversification for these giant firms; even when unrelated businesses are entered, the public and the government may not permit the large increase in overall concentration of economic

⁸ Alfred P. Sloan, Jr., My Years With General Motors (New York, Macfadden, 1965) p. 64.

power implied by significant diversification of the industrial giants.

Size is also a problem for large but not giant enterprises. Not only may the financial effects of diversification be relatively small in comparison to the firm's major activity, but the organizational sub-units dealing in diversified areas must operate in the shadow of the company's large functional structure. The most frequent structural form adopted by integrated firms with diversified activities is the functional with subsidiaries organization. While this structure allows diversified businesses to operate apart from the firm's main business, the general managers of diversified sub-units are hierarchically on a par with the functional department managers of the major business. Not only must the division general manager compete for capital and top management attention with several large units that are really just parts of one business, but the general office remains wedded to the major business and is ill-equipped to objectively judge the merits of business projects in the diversified areas. Two distinct styles of appraisal and supervision are required of top management in a functional with subsidiaries structure, one for coordinating the functional departments and dealing with the strategic issues that affect the major business, often on a day-to-day basis, and another for dealing with general managers of diversified semi-autonomous divisions. It will be a rare top management team that can simultaneously serve both needs. More often, the diversified business must struggle to cope with resource allocation, appraisal, and control systems that were developed for the integrated business.

Coupled with large size, the relatively poor performance of the Dominant Vertical firms makes strategies of diversification all the more difficult to implement. Low price-earnings ratios mean that the acquisition of attractive

businesses will result in substantial dilution. This is, in turn, especially expensive to these firms as their stock values are almost entirely based on dividend rates so that dilution implies a substantial additional cash drain if current stockholders are to be satisfied.

On the other hand, paying cash for acquired companies or investing cash in new business development projects requires very large commitments if diversification is to have any visible influence on total earnings. Yet the oligopolistic nature of competition in the integrated industries makes constant infusions of cash necessary just to maintain position. In a business where expensive raw material sites and costly plants are the key competitive weapons it is difficult to disinvest -- like the poker player who has so far matched the bets of others, the integrated business keeps reinvesting because although winning is improbable, loss is certain if it does not.

Summary. Although the strategy of vertical integration is a reasonable and rational response to certain technologies and market conditions, these very conditions have become associated with mature and less profitable industries. Escape from these industries, however, is particularly difficult for the large vertically integrated firm. Its technologies tend to be based on process rather than science or product function and are not readily transferable. Large size implies that the scale of investment in new businesses must be large of noticeable changes in corporate performance are to be achieved, but low price-earnings ratios and high pay-out policies make such investments financially quite difficult. Finally, the integrated businesses train few generalists and their attitudes and the organizational structure they preside over tend to inhibit strategic change. Thus, unlike non-integrated firms, the Dominant Vertical companies have not, in general, significantly diversified in response to low performance.

The Unrelated Passive Strategy

In Chapter IV we found that, among diversified firms, those that followed the Unrelated Passive strategy of diversification had exhibited lower levels of profitability and growth and smaller risk premium ratios. While this category has not been a large one in numerical terms, it grew substantially in the ten years between 1959 and 1969. Since the Acquisitive Conglomerates have focused a great deal of attention on the concept of expansion into completely unrelated areas, the relatively poor performance of the Unrelated Passive firms, which have developed unrelated product-market postures without the conglomerate's tactic of rapid and continuous acquisition, takes on special interest.

In general, an Unrelated business firm was defined as one which derived less than 70% of its revenues from its largest group of somehow related businesses. This category was then further subdivided into Unrelated Passive and Acquisitive Conglomerate categories. The Acquisitive Conglomerates were selected on the basis of their rates of growth, rates of acquisition, and rates of acquisition of new unrelated businesses. While it might therefore be thought that the Unrelated Passive firms consist chiefly of conglomerates whose rapid growth phase is over, it was found that this simple screen split the overall Unrelated group into two quite distinct sub-groups.

While an Acquisitive Conglomerate has usually expanded by acquiring a large number of individual companies over a relatively short period of time, an Unrelated Passive firm tends to have developed in one of three other ways: (1) evolution from a pre-World War II holding company, (2) the merger of two or three already large and often diversified firms, or (3) a strategy of mixed related and unrelated business expansion, usually undertaken in order to diversify away from some major business area. In order to clarify these three types of development, several examples will be discussed.

Holding Company. AVCO was started in the late 1920's to serve as a holding company for American Airways (now American Airlines) and several aviation equipment firms. By 1945 AVCO controlled American Airlines, Pan American Airways, Consolidated Vultee, Lycoming, New York Shipbuilding, Crosley (electronics, broadcasting and refrigerators), New Idea (farm equipment) and several other firms. This portfolio of unrelated businesses had been built partly due to war contracts but chiefly as a defensive investment against government action which finally forced AVCO to sell its interests in both airlines and Consolidated Vultee in 1945. AVCO then bought Bendix (home washers) but suffered such losses in this area that it quit all of its home appliances businesses by 1956, leaving electronics and aircraft engines and parts as its largest business.

AVCO's transition from being a holding company to an operating company occurred in this period and management's decision to drop all consumer products was followed by heavy internal investment in laboratories oriented towards military and space research and development. By 1959, AVCO's largest business was contract research on advanced materials, electronics, and missile re-entry technology.

Between 1949 and 1964 AVCO had followed an Unrelated Passive strategy -- operation of Unrelated businesses with little or no growth by acquisition. In 1964, however, defense sales dropped sharply and management began to look outside the current product-market scope of the company for new opportunity. Delta Acceptance Co., a finance company, was acquired in 1964, followed by the acquisition of Bay State Abrasives in 1965. By 1969 AVCO had moved by acquisition into life insurance, the credit card business, land development, motion pictures, and steel wire. Its largest single business in 1969 was the production of helicopter engines (31% of sales) and it was classified as an Acquisitive Conglomerate.

Pullman's early development was roughly analogous to AVCO's. A holding company created in 1927 to control both Pullman-Standard, which manufactured railroad "sleeping cars" and freight cars, and the Pullman Co., a firm that operated the sleeping cars, providing maintenance and personnel, on a contract basis. In 1947, the government ordered the sale of the sleeping car operations portion of the company's business and the firm acquired the Kellogg (M.W.) Co., a company engaged in the design, fabrication and erection of plants and equipment for the oil and chemical industries, with the proceeds. Between 1950 and 1959 Pullman's moved into the truck trailer business through acquisition and into plastic chemicals production by plants built by its Kellogg division; still, the headquarters office remained small and Pullman was essentially a holding company. Between 1959 and 1963 Pullman became an operating company, creating a true product division structure. In the process it exited from the chemical business, sold off a small steel mill that had supplied steel to both divisions, and dropped several unprofitable Kellogg products: pressure vessels, heat exchangers, and rocket casings. Between 1963 and 1969, Pullman acquired a company that designed and built furnaces for the steel industry and developed, by internal investment, a number of new businesses related to plant construction and truck trailer and railroad car manufacture. Pullman was classified as Unrelated Passive throughout the period 1949-1969; "passive" only in the sense that it did not expand by rapid acquisition of new unrelated businesses, Pullman actually followed Related Constrained strategies of diversification in two separate unrelated areas.

A third example is Rockwell Manufacturing. Formed to implement a depression survival plan of joining together a number of manufacturing companies, Rockwell Manufacturing retained a holding company structure until about 1960. It produced gas and oil meters, cash registers, taxi meters, pumps, and power

tools for wood and metal working. A program of marketing rationalization by the combination of similar sales groups led to the combination of about 20 separate businesses into five divisions in 1960.

Large Merger. The 1961 merger of the Martin Co. and American Marietta, one of the largest mergers in history, created an Unrelated Passive firm of giant size. Martin was a manufacturer of airframes and missiles while Marietta was itself a diversified company that sold lime, concrete, and pulverizing machinery as well as paints, ink, coatings, dye, and other chemical products. The newly formed Martin-Marietta quickly sold off most of Marietta's paint and chemical businesses and also divested itself of the pulverizing machinery and some other activities. The company's only major diversification move since the merger was the acquisition (through a tender offer) of Harvey Aluminum, a smaller but integrated aluminum company.

The Martin-Marietta merger resulted in an Unrelated Passive company that showed little of the dynamism that management had hoped for. Both sales and earnings showed negative rates of growth during 1961-1969 and return on equity averaged about 11%.

Another example of an Unrelated Passive firm that was formed by merger is Olin (Olin Mathieson Chemical prior to 1969). Mathieson Chemical had been a producer of bulk heavy chemicals that sought diversification in the 1952 acquisition of E.R. Squibb & Sons, a manufacturer of antibiotics and other ethical drugs. Olin Industries was itself an Unrelated business company at the time of the merger, producing arms, ammunition, explosives, cellophane, lumber, flooring, copper and other non-ferrous metals, and dry cells. After the merger the company initiated a venture in primary aluminum. Between 1954 and 1969, Olin made no further large acquisitions, though in 1968 Squibb was spun-off and merged with Beech-Nut. Olin's return on capital has averaged about 7% since the merger and growth has been both erratic and very low. While the reasons for Olin's poor record are not entirely

clear, Fortune⁹ has suggested that top management had an exceptionally difficult time reaching agreement on corporate goals and in creating a suitable structure to control and coordinate the firm's activities.

Two other firms that became Unrelated chiefly through one or two mergers are North American Rockwell and Sybron. The 1967 merger of North American Aviation and Rockwell Standard brought together a firm in the aerospace and electronics business with one producing auto parts, machinery parts and light aircraft and textile machinery. Sybron was formed by the 1965 merger of Ritter and Pfaulder, producers of tanks and waste treatment equipment and dental and hospital equipment respectively.

At the risk of generalizing from too few examples, it seems that when a rapidly diversifying firm merges with another equally large firm, the degree of acquisition activity drops sharply and the managements turn to work on the problem of creating an administrative system to knit two large very different firms together.

Mixed Diversification. Up until 1963, Champion Spark Plug was a single product company. In that year, however, it began a diversification program with the acquisition of a producer of test equipment for quality control. In short order the firm acquired companies producing fire detection equipment, steel bar, paint sprayers and coating equipment, X-ray tubes and cartridge tape recorders. More cautious than an acquisitive conglomerate, Champion seems to buy firms with highly proprietary products that may benefit from its financial resources or multi-national skills.

American Standard began to diversify away from its plumbing, heating and heat exchanger business in 1955 with establishment of a division to explore the commercial feasibility of

⁹ Fortune, September 1958, p. 212.

products related to or based upon the use of atomic power. Management stated that fluctuations in residential housing construction were the primary reason for its decision to increase the "engineered products" portion of its business.

American Standard developed a line of military aircraft guidance system components in the late 1950s but sold off this activity in 1960 due to its "heavy capital requirements." During the 1960s the company relaxed its policy of internally generated diversification and acquired an instrument producer, a company making safes and office record-keeping materials, and Westinghouse Air Brake. The latter was by far the largest and contributed 14% of the firm's 1969 revenues.

Also following mixed strategies were Boise Cascade, Di Giorgio, Curtiss-Wright and USM (United Shoe Machinery). In some of the firms which became Unrelated businesses via the mixed route, there appeared to be no consistent rationale behind their diversification moves. Others, however, seem to have had relatively clear strategies of diversification but also were willing to enter Unrelated areas if the right situation presented itself.

Unrelated Passive Performance

While not far from the overall average, the profitability and rates of growth of Unrelated Passive firms were low in comparison to the averages of other diversified firms. In addition, as shown in Chapter IV, many Unrelated Passive firms had the majority of their activities in science-based industries and, in comparison to other firms in these industries, their performance was particularly poor. Relatively low profitability in science-based industries is also a characteristic of Acquisitive Conglomerates, and Unrelated Passive firms have the additional problem of low rates of growth.

The type of data on which this study is based does not lend itself to explanations of why the performance of the Unrelated Passive firms has been relatively poor. The ways in

which these firms developed and their special strategic and structural attributes do, however, suggest some hypotheses.

The attainment of higher than average profitability by an Unrelated business firm would seem to call for at least three things: (1) careful selection of constituent businesses, (2) willingness to divest poor performers, and (3) an administrative structure that creates pressure for profit performance and contains generalists capable of making resource allocation decisions and intervening in division activities if problems arise. When the Unrelated Passive firms are judged on these criteria, we find that many, though not all, fail to measure up.

The Unrelated Passive firms that developed from holding companies had usually operated for a length of time without any real central office or formal control and resource allocation systems. It is significant that when such firms made the transition to being operating companies with full product division structures, the divestiture of several product lines or businesses was almost always an immediate consequence. Having little or no strategic direction from the top and with inter-divisional transfers of cash made difficult by the holding company's legal status, each business entity of such a firm developed on its own. The managements of these holding companies and quasi-holding companies frequently cited the value of diversification as a defense against adversity, but apparently did not follow strategies of channeling funds from mature to growing businesses. Under such circumstances, it would be surprising if the overall corporate performance were anything but average.

In each case of the creation of Unrelated Passive firms through large mergers, one of the merger partners had already been diversifying fairly rapidly through acquisition. When such a firm merges with another large firm whose businesses are unrelated to its own, severe strategic and administrative problems can result. Neither a pure "unrelated" approach (e.g., Texton) nor a single concept of relatedness suffices to guide the development

of the resultant firm, and administrative systems built around different clusters of core skills are difficult to combine. Not only are synergistic benefits of such a merger difficult to obtain, but the approximately equal size of the merger partners can produce complex struggles for influence and resources, and unlike a merger of firms in the same business, these tensions cannot be resolved by combining like facilities.

Firms that became Unrelated Passive by following "mixed" diversification strategies also had strategic and administrative problems similar to those faced by the "large merger" group. In addition, for many of these firms the low performance that was the initial motive for diversification was not alleviated by the acquisition of unrelated enterprises.

Thus, our conjecture is that the Unrelated business strategy calls for an essentially administrative rather than operations oriented approach to general management. When the enterprise consists of a set of disparate businesses, the basic general management task is the "management of managers" and it requires special types of control, reward, and resource allocation systems. However, because of the way they developed, most firms in the Unrelated Passive group have either (1) weak administrative structures or (2) administrative systems designed for and bound to Related business strategies. Unlike the Acquisitive Conglomerates, which tend to view each business as an independent unit, many of the Unrelated Passive firms had tried to exploit tenuous synergies among unrelated businesses or attempted to knit together groups of unrelated businesses that shared some common aspect (such as being in the "amusement business", or being "consumer products" businesses).

CHAPTER VI

SUMMARY OF FINDINGS AND IMPLICATIONS

The aim of this study has been to add to the store of knowledge concerning the development, form, and economic performance of large industrial corporations. In the preceding chapters we have explained our approach to the study of diversification, presented data, and provided interpretations of the findings. Data by itself, however, is not a guide to action and the interpretation of isolated facts is but the first step towards an integrated overview of a topic. Now, therefore, we turn to the task of drawing together the threads of evidence into a coherent set of conclusions and discussing their implications.

Summary of Findings

Seventy percent of large firms fell in the Single or Dominant business categories of diversification strategy in 1949. By 1969, over one-half of these firms had either moved to Related or Unrelated business strategies or had been acquired and had their places in the top 500 taken by more diversified firms. At the same time, the product division structure, employed by 20% of large firms in 1949, moved to a position of predominance, being used by 75% of large firms in 1969. Thus, the revolution in the form of the large American corporation, which was identified and traced in its early stages by Chandler, progressed rapidly towards its culmination during this period.

The change in the composition of the largest 500 firms came about through two distinct processes: strategic change by firms that were already members and the entry of new firms into the top 500 to replace these that had been merged, acquired, or had dropped below the size limit for inclusion. Our data showed that, except for the case of Acquisitive Conglomerates, the effects of both of these processes were of about equal magnitude.

In addition, it is worth noting that role of major mergers (involving firms of approximately equal size) in the overall increase in diversification was negligible.

Few would dispute our very general conclusion that the average large firm became substantially more diversified and more than likely changed its structure from functional to product division during the period 1949-1969. Such a result could have been obtained with techniques much simpler than those that were employed. What insights, then, did the concepts of strategy and structure and the method of classification contribute that would not have been revealed by more primitive measures, such as the product count?

Diversification. First, we found that although there was a strong and continual trend towards increased diversification, it was far from uniform. Some firms were more likely than others to increase the diversity of their product-market postures and there was a degree of predictability as to which strategy of expansion a diversifying firm would adopt. The number of Single business firms among the top 500 decreased sharply between 1949 and 1959, dropping from slightly more than one-third of all large firms to less than one-fifteenth. Most of the Single business diversifiers stayed close to their original areas of business competence, entering the Dominant Constrained category. While the Dominant Constrained group was itself experiencing an exodus to more diversified postures, the flow of new Dominant Constrained firms from the Single business group kept it from decreasing in size until the 1960s, by which time the Single business category had become too small to provide a steady flow of newly diversifying firms.

The tendency for newly diversifying firms to stay close to home in their search for opportunity was also evident among Dominant Constrained firms in the 1950s. Dominant Constrained companies that diversified further between 1949 and 1959 strongly tended to enter the Related Constrained category; although they

decreased their dependence on their largest single business, these companies continued to restrict their scope of activity to businesses that built on some central skill or strength. However, in the 1960s, Dominant Constrained diversifiers entered the Related Constrained and Related Linked categories with about equal frequency. The increased willingness to look farther afield for opportunity that this implies was one of the most obvious differences between the two decades. Firms diversifying from within the top 500 and those that entered it by growth were much more likely to adopt Related Linked or Unrelated business strategies during the 1960s than in the 1950s.

The most stable of the categories was the Dominant Vertical group. Almost none of these large raw materials processing companies moved into the Related or Unrelated business categories, few were merged or acquired, and few of the firms that entered the top 500 were Dominant Vertical. The poor economic performance of most of these firms suggests that their slowness to diversify was due more to the existence of special barriers to diversification than to complacency. In particular, their historical development and economic purpose have left these firms with technologies that are process rather than product oriented, products that are commodities and, hence, attitudes and practices that favor selling rather than marketing, large investments in capital intensive primary processing plants that promote capacity utilization rather than return on investment analyses of new projects, departmentalized but functional structures that overshadow new business ventures, and managerial attitudes that tend to discourage any search for opportunity that might take the firm outside of current industry boundaries. Because of these and other similar factors, the vertically integrated firm tends to be a special case among large American corporations. If current trends continue, the years to come will increasingly show large firms as splitting into two camps: the majority of modern divisionalised firms following Related and Unrelated

business strategies and the Dominant Vertical holdovers of an earlier pattern of industrial structure.

Among Related business firms, in 1949 those that were Related Constrained outnumbered Related Linked companies by more than two-to-one. By 1969, the Related Linked sub-group was slightly larger than the Related Constrained. Though both sub-groups of the Related Category grew during these twenty years, most of the overall increase came in the Related Linked group and most of that came in the 1960s. During the 1950s, virtually none of the Related Linked growth was due to changes in strategy by already large firms -- its increase in this period was purely due to firms that broke into the top 500. During the 1960s, however, both factors were at work. Dominant and Related Constrained firms moved frequently to Related Linked strategies and 29% of all new entrants to the 500 were in this strategic category.

A similar but exaggerated pattern governed the growth of the Acquisitive Conglomerate category. There were no Acquisitive Conglomerates in the sample in 1949, but by 1969 they constituted 10.9% of the top 500 and most had entered the 500 in the 1960s. Of firms that were among the largest 500 in both 1949 and 1969, only 2.2% had become Acquisitive Conglomerates by 1969.

Thus, we see not only a trend towards increased diversification but an increasing tendency for experienced diversifiers to adopt Linked rather than Constrained strategies of diversification and a definite change in the composition of the top 500 due to fast growing Acquisitive Conglomerates. Both of these changes in the expansion paths used by large corporations became more pronounced in the 1960s.

Structure. Of the five categories of structure we defined, only the functional, functional with subsidiaries, and the product division groups contained significant numbers of firms. At its maximum extent during the period of study the geographic division

category only contained an estimated 10 of the largest 500 firms and the holding company category was only slightly larger with 19 firms.

The functional with subsidiaries category was fairly stable in size, chiefly because many of the companies with this structure were in the Dominant Vertical class, the most stable strategic class.

Our data gave strong support to Chandler's proposition that "structure follows strategy", but forced us to add the dictum "structure also follows fashion." Between 1949 and 1969 the relative numerical positions of the functional and product division categories more than reversed: starting at 63% of the top 500 in 1949, the functional structure was only used by 11% of large firms in 1969 while the product division organization increased in usage from 20% to 76% of the top 500 in the same period. Investigation of the association between strategy and structure showed that (1) firms that are members of categories of diversification strategy that correspond to greater amounts of product-market diversity are more likely to have product-division structures, (2) during the 1950s, firms that shifted to categories of greater diversification were more likely to also change their structures from functional to product division than functional firms that did not change strategy, and (3) during the 1960s this relationship did not hold -- while many firms changed from functional to product division structures, those that underwent strategic changes were no more likely to divisionalize than other firms. We can only conclude that the product division structure has become so well known and understood that modern firms tend to adopt it more readily than the companies of a generation ago. The firms Chandler studied often adopted the divisionalized structure only after the old system began to collapse under the burden of diversified operations. By contrast, many corporations of the 1960s, even those that were not highly diversified, split their operations into semi-autonomous divisions

without having suffered great administrative strains. Some divisionalized because they planned to diversify, others wanted to create greater accountability for product performance, even among closely related products, and still others simply believed that the product division structure was a better form of organization.

Performance. Our findings with regard to the population of the categories of strategy and structure are necessarily analytic -- they cannot serve to validate the systems of classification because no other types of information were included in the analyses. In looking at the relative financial performance of the categories, however, we are also looking at the degree to which the categories provide a meaningful map of a portion of reality. If the categories serve to divide firms into groups that are internally more homogenous than the sample as a whole, then we have evidence that the information provided by the classification system extends beyond that which was used to classify the companies.

The two most important findings regarding the economic performance of different categories of diversification strategy were that (1) the system of classification revealed differences in financial results and policies that were not discernable if simple measures of diversity were employed to discriminate among firms and that (2) performance differences among the strategic categories were more closely linked to the way in which the firm related new businesses to old than to overall diversity.

If measures of diversity, such as the product count, correlated with profitability or growth, one could easily argue that category associated performance differences were merely a reflection of a more basic correlation between profits or growth and diversity. However, no such correlation with simple measures of diversity has been found. We are thus led to the conclusion that our system for capturing the essence of top

management's goals and concept of the corporations purpose and scope is a better predictor of financial performance than simple measures of diversification. This result helps to validate our system of categories by providing assurance that it reflects at least a portion of reality.

We found that although performance differences existed among the major categories of diversification strategy (Single, Dominant, Related and Unrelated), the explanatory power of the system was more than doubled by breaking these categories into sub-categories. This showed that (1) among Dominant business firms, the vertically integrated group (Dominant Vertical) was low performing and the group that had limited the scope of its product-market commitment (Dominant Constrained) was among the highest performing, (2) among Related business firms, the Related Constrained group was high performing and the Related Linked group average or slightly below average, and (3) among Unrelated business firms, the Unrelated Passive group was among the poorest performing and the Acquisitive Conglomerates were average in profitability and substantially above average in growth. The heterogeneity of the major categories indicates that performance differences are not really related to diversity per se, but are more a function of the firm's strategy for dealing with growth and diversity.

The Dominant Constrained groups were unquestionably the best overall performers and both strategies are based upon the concept of controlled diversity. Neither totally dependent upon a single business or true multi-industry firms, these companies have strategies of entering only those businesses that build on, draw strength from, and enlarge some central strength or competence. While such firms frequently develop new products and enter new businesses, they are loath to invest in areas that are unfamiliar to management.

We found that although Dominant Constrained firms tended to move on to Related business strategies in response

to deteriorating performance, Dominant Vertical firms had, on the average, much poorer performance levels but were either unable or unwilling to employ diversification as a remedy. Scott calls these firms the "heavies" of American industry and our evidence supported his conjecture that they face special barriers to diversification and are among the poorest performers.¹

The distinction between Unrelated Passive and Acquisitive Conglomerate strategies was found to correspond to two radically different sets of financial policies and levels of economic performance. Unrelated Passive firms had the lowest rates of growth in sales, poor rates of return on capital, high variability in growth in earnings per share and also the most conservative capital structures. The Acquisitive Conglomerates, by contrast, had very high rates of growth in sales and earnings, experienced heavy dilution but still achieved above average rates of growth in earnings per share, and used leverage to boost a below average return on capital to an above average return on equity. Somewhat surprisingly, we found that Acquisitive Conglomerate price-earnings ratios were quite average and that there was no correlation between conglomerate price-earnings ratios and growth rates in earnings per share. It seems that the "bootstrap" model of conglomerate growth, whereby the firm uses a high price-earnings ratio to make favorable acquisitions and the resulting rapid growth in earnings per share to keep its price-earning ratio high, does not apply to all Acquisitive Conglomerates. Instead, many firms with low price-earnings ratios achieved rapid growth by acquisition by heavy use of convertible securities, deferred payments, and rationalization of the capital structures of acquired firms. By buying companies with their own excess debt capacities or with promises of future gains in earnings per share, these firms obtained rapid increases in size without abnormally

¹ Bruce R. Scott, "Stages of Corporate Development" (unpublished paper, Harvard Business School, 1970) p. 42.

high price-earnings ratios.

The low performance of the Unrelated Passive firms came as a surprise as many observers have grouped these companies with the Acquisitive Conglomerates. While they are active in several unrelated business areas, as are the true conglomerates, most of the Unrelated Passive companies had neither the administrative structure nor the aggressive acquisition programs of the conglomerates. Many of the Unrelated Passive firms were created by the merger of two already diversified, but unrelated, firms and others developed patterns of business activities that were similar to those of the firms formed by merger. We suggested that the way in which these firms developed has resulted in weak administrative systems. The marriage of two large Related business firms to produce an Unrelated business company results in control by managers who have spent their careers dealing with related businesses and are used to control, information, and reward systems that presuppose a degree of similarity among businesses and familiarity, on the part of top management, with the details of many activities. In the combined firm, however, resource allocation decisions may focus on which "side of the house" gets supported rather than on individual investment proposals and growth through future unrelated diversification moves is not easier for the Unrelated Passive firm than for the Related business firm.

We found that firms with product division structures had rates rates of growth that, on the average, were significantly higher than those of firms with other type of organizational structures and that their rates of profit were at least as high as those of other firms. More interesting was the finding that, among firms with product division structures, growth in sales was less directly related to profitability than among firms with other types of structures. Stated somewhat differently, differences in the growth rates of functionally organized firms can, to an extent, be explained by differences in their rates of profit; among product division firms, however, differences in rates of profit were asso-

ciated with much smaller differences in rates of growth. This seems to indicate that firms with product division structures do not rely as heavily on current profits to power their growth as functionally organized firms. We conjectured that this result was due to the special control, resource allocation, and reward systems used by product division firms that allow a high degree of separation between the goals of current profitability and growth. In fact, the two levels of general management that characterize the product division firm may be thought of as being concerned with current return on investment (division management) and growth through investment in new products and businesses (top management).

Implications of the Findings

This study has been chiefly devoted to facts, their discovery and interpretation. In discussing the implications of our findings we move to the larger realm of conjecture and belief. While reasonable men may draw different implications from our data, our closeness to this work and intimate familiarity with its assumptions and results creates a responsibility to record what we believe to be its important implications.

In the twenty years between 1949 and 1969 the structure of industry in the United States underwent a remarkable change, a change that was not foreseen by most observers. As in the earlier portions of the century, big business carried out the lion's share of industrial activity, but concentration in individual industries did not increase appreciably. Instead, the boundaries between various industries became more and more blurred as firm's expanded their product-market scopes to include several industrial sectors. While in 1949 the typical member of the Fortune 500 was a functionally organized company that was largely committed to a single major business, by 1969 the modal strategy was Related Linked diversification and over three-quarters of large firms had adopted product division structures.

Many present day observers of the industrial situation still fail to recognize either the extent or the impact of the revolution in industrial structure that has occurred. In Galbraith's important and widely read book, The New Industrial State,² for example, he claims that the technological requirements of large modern industrial enterprise have created corporations which are inflexible, uncontrolled from the top, and which have superseded the competitive markets in their need to plan over long time spans, make vast capital commitments, and maintain production flow through long-linked production systems. In particular, Galbraith claims that the imperatives of technology and size have acted to vest corporate power in the technostructure. In painting his picture of the modern corporation, Galbraith argues that:

1. Modern technology requires an ever increasing time span between the conception of a product or project and its completion and the size of markets and enterprises implies the advance commitment of vast amounts of capital.

2. Technology is based on specialization and specialization requires an increasingly complex administrative structure to coordinate the work of specialists. Furthermore, top level managers no longer have the expertise to directly influence or control the nature of the products or services that are produced by the corporation.

3. Because time spans are long, costs high, and because large scale organization is inflexible and difficult to redirect, uncertainty in the final outcome of a project cannot be tolerated. Hence the firm supersedes market uncertainty by planning, the manipulation of consumer tastes, inter-firm agreements, and vertical integration.

4. The imperatives of technology and the ability of the firm to supersede the market make it necessary and possible for power over critical decisions to pass from the hands of owners and

² John K. Galbraith, The New Industrial State (Boston, Houghton Mifflin, 1967) our analysis of Galbraith parallels that of Scott, "Stages of Corporate Development", pp. 45 - 47.

top management to the technostructure. The technostructure is the group of technically oriented personnel that engages in product planning, market planning, guides projects, and assures their final outcome.

5. The technostructure has no compulsion to maximize profits and, therefore, does not. Galbraith states that:

But with the rise of the technostructure, the notion, however tenuous, that a few managers might maximize their own return by maximizing that of the stockholders, dissolves entirely. Power passes down into the organization. Even the small stock interest of the top officers is no longer the rule. Salaries, whether modest or generous, are according to scale; they do not vary with profits.³

Because the technostructure does not seek to maximize profits, it naturally pursues the goal of perpetuating itself and increasing its power and influence. The important consequence of this, Galbraith claims, is that economic models, public policies, and attempts to exert macroeconomic control of the economy can no longer be based on the assumption of profit maximization on the part of corporations.

Whether or not forward planning and marketing techniques have the power to control the demand for products is a debatable point, but this study has not been addressed to that issue and is consequently not an appropriate forum for such a debate. Galbraith's propositions concerning the nature of planning within the firm and the locus of decision making power are, on the other hand, directly relevant to this study. We suggest that this view of corporate life ignores fifty years of evolution in the nature and structure of American industrial enterprise. That great size, economic stability, technology, and the increasing complexities of industry can create a phenomenon like the technostructure is hardly arguable. That it is ascendent and increasingly important is, however, quite a different matter. As early as 1920, managers at Du Pont discovered that size, technology and product diversity were producing administrative problems

³ Ibid., p. 116

that threatened management's ability to guide and control the company. Their solution was reorganization -- the birth of the multidivisional form of organization. Here is how Alfred Chandler described du Pont management's victory over the technostucture:

In September, 1921, the du Pont Company put into effect this new structure of autonomous, multidepartmental divisions and a general office with staff specialists and general executives

Unumbercumbered by operating duties, the senior executives at the general office now had the time, information, and more of a psychological commitment to carry on the entrepreneurial activities and make the strategic decisions necessary to keep the over-all enterprise alive and growing and to coordinate, appraise, and plan for the work of the divisions

(This structure) has served the du Pont Company effectively ever since.⁴

We now know that neither du Pont's problem nor its solution was unique; the dominant form of large business enterprise in the United States today is the diversified multidivisional corporation. Unlike the firm that Galbraith describes, these firms have split their operations into separate businesses, each controlled by a division manager who is responsible for current profit as well as planning. The division manager in such a firm is not far removed from the goal of profit maximization -- he is reviewed on his profit performance at least once a year and his compensation is typically dependent upon his performance. In a study of reward systems in product division companies, Salter found that systems of review and control were more formal than in functional organizations and that division manager's bonuses, stock options, and eligibility for advancement were tied to the formal systems of control.⁵ Thus, while a manager's salary

⁴ Alfred D. Chandler, Strategy and Structure (New York, Doubleday, 1966) p. 135

⁵ Malcolm S. Salter, Stages of Corporate Development : Implications for management control (Unpublished paper, Harvard Business School, 1967)

may not be tied to his ability to maximize profit, his compensation is and, what is more important to most managers, so are his chances for promotion.

The control and reward systems of product division organizations allow top management to create great pressure for profit maximization. Pressure so created can, in fact, be greater than that due to the competitive struggle among firms. There is nothing to prevent top management from setting goals for division managers that require performance above and beyond that normally expected from a single business firm in the industry. While such goals may not be met, their existence, or potential existence, can create a climate strongly conducive to profit maximization.

With regard to the superseding of markets, it is the capital markets rather than intermediate or consumer markets that have been absorbed into the infrastructure of the new type of corporation. While this may be no more preferable in a social-economic sense than control over product markets, it is an entirely different phenomenon with different effects. Both investment bankers and the market for equity securities now treat the firm as a whole and leave to top management the task of deciding which business, or even which industry, should receive new infusions of capital and which should be treated as a cash generator. Furthermore, recent history shows that most large firms have infrequently approached the equity markets for new capital, preferring to depend upon reinvested earnings for growth. This means that the diversified divisionalized firm is increasingly becoming the arbiter of inter-sectional shifts in funds.

If Galbraith's model does not accurately describe the modern corporation, what does it describe? Scott has observed that the type of firm that most closely matches Galbraith's model is the large integrated Dominant business corporation⁶ -- our Dominant Vertical firm. As we have noted, these firms have faced special

⁶ Scott, "Stages of Corporate Development", p. 48.

barriers to diversification and are increasingly becoming representatives of the "old industrial state."

The new type of firm has adapted to the imperatives of modern technology and modern markets — the need to meet demands for variety, to apply new skills and ideas to old products, to regularly rechannel resources into new product areas, to rapidly respond to new customer demands and to have the ability to bring talented men, money, and equipment together to work on a project for a time and then to disband the project and find efficient uses for these resources elsewhere. The new linkages are not vertical, from raw materials to processors to factories, etc., but horizontal and market related. A cheap efficient microwave oven is developed and firms engaged in food processing, packaging, oven equipment and cooking utensils must all respond. The high density photo-storage library becomes possible and firms who publish books, make paper and ink, sell microfilm equipment, as well as those involved in computers, educational technology, data transmission technology, copying equipment and eventually the Telephone Company must all respond in some way.

This view of the firm and the data we have found that pertains to it have important implications for businessmen and makers of public policy.

Implications for Management. The critical resource of the modern diversified divisionalized firm is general management skill, and the change from the old to the new industrial structure has multiplied several-fold the number of general management positions to be filled in U.S. Industry. The training and effective employment of generalists must become the prime concern of any firm that has goals that include growth by diversification or participation in the new technologies that quickly produce a proliferation of new products

Not only must top management train and make use of general managers, but it must learn how to fine tune the product division form of organization. At a time when most large companies

were merely experimenting with diversification and the product division structure was an atypical curiosity, the progressive firm could gain a competitive edge in many industries merely by reorganizing along product division lines. Such a move simplified the life of top management and provided for greater visibility of individual product performance. Now, however, most large firms have adopted the product division structure and competitive advantages that come from superior organization must be sought by fully exploiting the potentials of this structure. For example, we would expect that highly diversified firms would greatly benefit from improved systems of strategic review and formal systems of intervening in divisional operations during crisis situations.

With regard to the efficacy of various strategies of diversification, it would be pleasant to report that the Dominant Constrained and Related Constrained are the optimal strategies and should be followed by all prudent firms. Our data, however, only indicated that there is an association between these strategies and high levels of financial performance. While it may be that these strategies tend to produce good rates of growth and profit, it is more likely that firms that are already rapidly growing and profitable think it wise to restrict their scopes of activity to business that directly relate to their currently successful areas of competence.

We can, nevertheless, suggest that the intensive cultivation of a single field has proven, on the average, financially more successful than bold moves into uncharted areas. While many consider them to be the most interesting and technologically powerful firms, the Related Linked companies have experienced distinctly average levels of performance over the past twenty years.

Our finding that Related Linked and Unrelated Passive strategies have not proven particularly rewarding to most firms should be of interest to managements that are planning to embark

on programs of increased diversification. During the 1960s, many Related Constrained and Dominant Constrained firms moved to Related Linked and Unrelated Passive strategies only to discover, our data shows, that expansion into new tenuously related or unrelated areas, though it makes good reading in annual reports, does not always produce the desired results. In particular, the formation of large Unrelated Passive firms through the merger of two diversified but unrelated firms has been strikingly unsuccessful. Managements expecting to exploit the synergistic effects of bringing two such firms together should be warned that the organizational problems rising from this type of merger have, on the average, nullified any possible beneficial gains due to scale or synergy.

Managements who are responsible for the affairs of Dominant Vertical firms are probably already aware that their companies face special barriers to diversification. While it would be foolhardy to suggest that fully integrated firms abandon their commitments to their major businesses and seek diversification, we can point out that increased integration will only heighten the barriers to diversification. Firms whose growth patterns have placed them at the decision point between increased integration and product-market diversification should realize that, although integration may provide temporary improvements in costs and permit an increase in size, other firms making similar decisions have eventually found themselves heavily invested in stagnant low profit industries. While integration itself can make good economic sense, it implies a long term commitment to an industry, a pattern of operations, and an administrative system that preclude rapid redeployment of resources and adaptation to change.

Finally, our findings on the variability in earnings experienced by firms following various strategies of diversification should be of interest to managers who expect product-market diversification to protect their firms from financial instability.

We found that although the categories of diversification strategy did not differ significantly in earnings variability itself, there were significant differences in the ratio of earnings variability to growth rates in earnings -- the risk premium ratio (RPR). This means that variability is strongly affected by growth and the firm's strategy of expansion is the critical factor influencing the degree of variability. Firms that followed Dominant Constrained and Related Constrained strategies had the most favorable risk premium ratios, and all other categories had poorer, and approximately equivalent, risk premium ratios. The implication is that both very little and very great diversity produce equivalent variability in earnings, but that diversity that is carefully controlled is the best form of diversification for reducing fluctuations in earnings. We suggest that the factor responsible for this result is the ability of the diversified, but "Constrained", firm to employ the beneficial effects of negatively correlated returns by replacing products that are stagnant or declining with close functional substitutes that are profitable and growing for reasons related to the decline of the original products. In other words, the "Constrained" firm ties its fortunes to the satisfaction of a particular functional need (such as business information processing, permanent image making, or convenience foods). It then develops a variety of products that relate to this fairly constant need and continuously searches its product line for weak points, innovating to meet changing tastes and needs.

In contrast to this strategy, the firm that seeks stability through diversification into tenuously related or unrelated areas is merely averaging risks rather than exploiting negatively correlated returns. At the most, such a policy can only reduce uncertainty to the level experienced by the economy as a whole.

Public Policy Implications. If we assume that one of the important objectives of American economic public policy is to strengthen competition and foster market structures that are free

and in equilibrium, assumptions regarding the behavior of the corporate sector must play an important role in the policy process. If models of business behavior that are based on single business firms are applied to the new industrial structure, the conclusions thereby derived will be wrong and the policies adopted will be ineffective. Of particular importance is the recognition that most large enterprises now employ decentralized administrative structures and control systems that strongly induce division level managers to meet and exceed their profit targets. This means that policies designed to affect corporate behavior that are not translated into profit goals by the firm's control system will have little chance of influencing behavior. For example, corporate goals such as reducing pollution will probably never create more than rhetoric unless division level managers are provided with incentives to curb pollution that are as strong as the rewards tied to profit and growth. As another example, Bower states that:

If, , the goal of an anti-inflationary policy is a slowing of the rate of private investment, some means must be found to induce the corporate officers to take pressure for current earnings and growth off their divisions. As long as division managers will rise and fall on relative profit and growth in the short run, they will put great pressure on their top management to find investment capital. The pressure of the inside management on top management is then likely to be very strong to keep investing, no matter how severe outside pressure may become.⁷

Another area that should be of concern to economists and public policy makers is the behavior and fate of the Dominant Vertical firm. The issue is not their economic power, but their lack of it in modern terms. As noted in Chapter V, the one industry strategies of these companies have left them tied to stagnant businesses and the policy of vertical integration has deepened their commitments to particular materials and processes.

⁷ Joseph L. Bower, "Planning Within the Firm", The Journal of the American Economic Association, May 1970, p. 193.

Unless these firms can adapt to the new technologies, new types of competition, and new forms of organization, they will become the railroads of the future -- necessary but weak and inefficient. A proposal to make these firms even larger through diversification may seem radical (or perhaps reactionary) in the light of current concern over the misuse of power by large firms, but we suggest that it may no longer be valid to assume that all large firms follow oligopolistic patterns of competition and tend to maximize sales, market share, or growth at the expense of return. The multidivisional firm can exert great pressure on division managers for current profit and the diversified firm can rechannel resources out of stagnant areas into ones that promise greater return and, presumably, in which new capital would produce social and economic benefits. An additional point that deserves mention is that it also may no longer be assumed that the total economic power of a firm is translated into an ideological and political commitment to a product. The multidivisional diversified company can bear economic, social, and legal change in isolated areas at a much lower cost than can a single business enterprise.

The "Dominant Syndrome" should be of particular interest to European planners because many of their policies are designed to create just this type of firm. The all too frequent response of European businessmen and planners to American competition has been to assume that large and more financially powerful firms are needed. McArthur and Scott report that French planners are trying to extend ideas of economies of scale, ideas that apply most appropriately to industries like steel and oil, to the electrical equipment, chemical and consumer products industries. They state that:

The French prescription would probably result in domination of each industry by one or two companies. This would amount to the creation of a French "champion" in each industry, on the pattern of P  chiney in aluminium. This French champion would face little if any competition from other French firms; instead it would carry the French colours

against foreign competitors.⁸

While size may certainly be a help to certain firms, without the diversity that allows cross-fertilization among business activities, or more importantly, without the generalist skills necessary to manage a diversified firm, the results may be the creation of a set of Dominant Vertical or Unrelated Passive companies, ill equipped to face the realities of modern competition. More vital than size is the creation of firms that have organizational structures that encourage rapid innovation and economic performance at the product-market level; equally vital is the training and effective use of the generalists that must be responsible for the affairs of such enterprises.

⁸ John H. McArthur and Bruce R. Scott, Industrial Planning in France, (Boston, Division of Research, Harvard Business School, 1969) p. 525.

APPENDIX A

This appendix consists of Table A-1 which lists all of the firms in the sample. The firms are listed in alphabetical order and, unless the firm came into being after 1949 or was merged or acquired before 1969, information on product lines, rank in the 500, and the strategic and structural classifications that were assigned to them are given for 1949, 1959, and 1969.

The first list of businesses for each firm gives the largest single business (underlined) and other businesses. Businesses that were judged to be unrelated to the firm's largest group of somehow related businesses are contained in square brackets. Lists of businesses after the first again give the largest single business but do not repeat businesses in which the firm was still active. Instead, it lists businesses that were ADDED and EXITED from since the time of the last list.

Ranks among the top 500 firms refer to assets in 1949 and sales in 1959 and 1969. In addition, only ranks among the top 100 are given exactly for 1949; others are given as 101-200 or 200-500.

The strategic and structural categories assigned to each firm are indicated by symbols. Their meaning is as follows:

<u>Strategy Symbol</u>	<u>Strategic Class</u>
S	Single business
DV	Dominant Vertical
DC	Dominant Constrained
DL	Dominant Linked
DU	Dominant Unrelated
RC	Related Constrained
RL	Related Linked
UP	Unrelated Passive
AC	Acquisitive Conglomerate
<u>Structure Symbol</u>	<u>Structure Class</u>
F	Functional
FS	Functional with subsidiaries
PD	Product division
G	Geographic division
H	Holding company

TABLE A-1

THE SAMPLE: COMPANIES, PRODUCTS, AND CLASSIFICATIONS

		<u>Rank</u>	<u>Strat.</u>	<u>Struct.</u>
ABBOTT LABORATORIES				
1949	<u>Antibiotics</u> , hormones, bulk medicinal chemicals, proprietary drugs, animal health products.	201-500	RC	F
1959	<u>Antibiotics</u> , + as above. ADDED: artificial sweeteners, hospital solutions and diagnostic agents.	328	RC	F
1969	<u>Hospital supplies</u> : solutions and diagnostics; + as above. ADDED: <u>rubber and plastic hospital supplies</u> , eye care products, diet and infant foods, medical electronics instruments.	248	RC	PD
ABEX				
Prior to 1966 name was American Brake Shoe.				
1949	<u>Railroad brake shoes and related forgings</u> , other forged and cast parts for general industry, machined parts for industrial and farm equipment and automobiles, air compressors.	201-500	DC	H
1959	<u>Railroad brake shoes and related forgings</u> , + as above. EXITED: air compressors. ADDED: heavy hydraulic equipment, aerospace hydro-servo valves.	265	RC	PD
1968	Acquired by Illinois Central Industries.			
ACF INDUSTRIES				
Prior to 1954 name was American Car and Foundry.				

TABLE A-1 (continued)

ACF (continued)		Rank	Strat.	Struct.
1949	<u>Railroad cars</u> , mine hauling cars, storage tanks, wheels, other forgings and castings, [automobile fuel pumps and carburetors].	201-500	DL	FS
1959	<u>Railroad cars</u> , + as above. ADDED: heavy valves, pressure vessels, [aerospace electronics systems, airframe and missile components].	252	DU	PD
1969	<u>Railroad cars</u> , + as above. EXITED: electronics, aircraft and missile components. ADDED: hydraulic controls, [nylon hose and parts, nylon resins].	343	RL	PD
ADDRESSOGRAPH-MULTIGRAPH				
1949	<u>Addressograph machines, parts and accessories</u> ; offset office reproduction equipment.	--	DC	F
1959	<u>Addressograph machines, parts and accessories</u> , + as above; ADDED: type composition machine, office supplies.	306	DC	F
1969	As above. ADDED: <u>Office copiers</u> , microfilm printer, precision printed plastics.	244	RC	PD
AIR REDUCTION				
1949	<u>Gases</u> : medical, scientific and industrial; acetylene welding equipment.	201-500	DC	FS
1959	<u>Gases</u> , + as above. ADDED: speciality chemicals and plastic intermediates.	225	DC	PD

TABLE A-1 (continued)

AIR REDUCTION (continued)

	<u>Rank</u>	<u>Strat.</u>	<u>Struct.</u>
1969 [as above]. ADDED: Ferro-alloys, electron beam metallurgy, vacuum metals, special metals; [carbon and graphite products, electronic components, vinyl molded parts and hose].	221	UP	PD

ALLIS-CHALMERS MANUFACTURING

1949 <u>Farm equipment</u> , earthmoving equipment, electric power machinery, hydraulic machinery, marine engines, industrial process machinery of wide variety.	65	RL	PD
1959 <u>Farm equipment</u> , + as above. ADDED: materials handling equipment, diesel engines, atomic power plant equipment.	80	RL	PD
1969 <u>Farm equipment</u> , + as above. ADDED: electronic control systems, garden tractors, all terrain vehicles, fuel cells, household appliances, water pollution control equipment, [educational materials].	140	RL	PD

ALUMINIUM COMPANY OF AMERICA

1949 Integrated aluminium: <u>mill products</u> , primary aluminium, building products, cookware, cans, wire, architectural shapes, by-products and joint products: magnesium, bauxite, chemicals.	28	DV	FS
1959 Integrated aluminium: <u>architectural shapes and structures</u> , + as above ADDED: real estate development.	48	DV	FS

TABLE A-1 (continued)

ALCOA (continued)

	<u>Rank</u>	<u>Strat.</u>	<u>Struct.</u>
1969 Integrated aluminium: <u>architectural shapes and structures</u> , + as above.	67	DV	FS

AMERICAN BILTRITE RUBBER

Prior to 1951 name was Panco Rubber.

1949 <u>Rubber shoe heels and soles</u> , rubber floor tiles.	--	RC	F
1959 <u>Rubber shoe heels and soles</u> , + as above. ADDED: carpets, rubber chemicals, rubber industrial products.	--	RC	PD
1969 <u>Rubber shoe heels and soles</u> , + as above. ADDED: synthetic rubber.	481	RC	PD

AMERICAN CHAIN AND CABLE

1949 <u>Chain, cable, wire and wire products</u> , springs, machine tools, automobile brakes and steering mechanisms, materials handling equipment, pressure and hardness instruments, lawnmowers.	201-500	RC	FS
1959 <u>Chain, cable, wire and wire products</u> , + as above. ADDED: [Electronic instruments].	352	RC	FS
1969 <u>Chain, cable, wire and wire products</u> , + as above. ADDED: industrial tools, electro-mechanical relays, [blades and saws, pollution control systems, computer data loggers, gyros].	424	RL	PD

TABLE A-1 (continued)

		<u>Rank</u>	<u>Strat.</u>	<u>Struct.</u>
AMERICAN ENKA				
1949	<u>Rayon tire yarn.</u>	201-500	S	F
1959	<u>Rayon and nylon yarn and fiber.</u> ADDED: cable, wire, insulating materials.	366	DL	FS
1969	<u>Synthetic yarns and fibers,</u> + as above.	355	DL	PD
AMERICAN METAL CLIMAX				
Prior to 1957 name was American Metal.				
1949	Mining, smelting, refining: <u>copper</u> , lead, zinc, silver, gold.	201-500	DV	F
1959	Mining, smelting, refining: <u>copper</u> , lead, zinc, + as above. ADDED: Potash, molybdenum, vanadium, oil and gas production.	66	RC	PD
1969	As above. EXITED: oil and gas. ADDED: <u>aluminium</u> : primary and fabricated; iron ore, coal, [pre-cast concrete].	147	RC	PD
AMERICAN MOTORS				
Prior to 1954 name was Nash-Kelvinator.				
1949	<u>Automobiles</u> , [electrical appliances].	101-200	DU	F
1959	<u>Automobiles</u> , [electrical appliances].	47	DU	PD
1969	<u>Automobiles.</u> EXITED: Electrical appliances.	152	S	F

TABLE A-1 (continued)

		<u>Rank</u>	<u>Strat.</u>	<u>Struct.</u>	A-7
AMERICAN OPTICAL					
1949	<u>Eyeglasses and ophthalmic equipment</u> , microscopes, slide projectors, industrial safety equipment.	201-500	DC	F	
1959	<u>Eyeglasses and ophthalmic equipment</u> , + as above. ADDED: astronomical instruments, medical office furniture, aerospace optics, dentists' equipment.	418	RC	PD	
1967	Acquired by Warner-Lambert Pharmaceutical.				
AMERICAN STANDARD					
Prior to 1967 name was American Radiator and Standard Sanitary.					
1949	<u>Plumbing equipment</u> , fans and heat exchangers, bathroom accessories, heating equipment.	91	RC	F	
1959	<u>Plumbing equipment</u> , + as above. ADDED: heating and cooling systems, [aerospace guidance and industrial instruments].	85	RC	PD	
1969	<u>Plumbing equipment</u> , + as above. ADDED: water treatment and waste disposal equipment; [safes, business information systems, land development, air brakes].	82	UP	PD	
ARMCO STEEL					
1949	Integrated steel: steel ingots, <u>mill products</u> , building products, fabricated products.	101-200	DV	F	
1959	Integrated steel: <u>mill products</u> , + as above. ADDED: oil field equipment.	48	DV	PD	

TABLE A-1 (continued)

ARMCO (continued)		<u>Rank</u>	<u>Strat.</u>	<u>Struct.</u>
1969	Integrated steel: <u>mill products</u> , + as above. ADDED: non-metallic composites, [mobile homes, snowmobiles, equipment leasing].	63	DV	PD
ARMOUR				
1949	Integrated meat packer: <u>meat</u> , leather, soap, chemicals, dairy, poultry, vegetable oil.	30	DV	F
1959	Integrated meat packer: <u>meat</u> , + as above.	16	DV	PD
1969	Acquired by General Host.			
ARMSTRONG CORK				
1949	<u>Floor tiles</u> , insulation, corks and caps, bottles, rubber medical and shoe products, bricks, cement and tile building products, rugs.	101-200	DL	F
1959	<u>Floor tiles</u> , + as above.	157	DL	PD
1969	<u>Floor tiles</u> , + as above.	196	DL	PD
ASHLAND OIL				
Prior to 1970 name was Ashland Oil and Refining.				
1949	<u>Oil refining</u> .	201-500	S	F

TABLE A-1 (continued)

ASHLAND (continued)		<u>Rank</u>	<u>Strat.</u>	<u>Struct.</u>
1959	<u>Oil refining.</u> ADDED: petro-chemicals: ammonia, detergent, plasticizer.	150	S	F
1969	<u>Oil refining</u> , + as above. ADDED: synthetic rubber, [construction work, fiberglass boats, crushed stone].	101	RC	PD
ATLANTIC RICHFIELD				
Prior to 1966 name was Atlantic Refining.				
1949	<u>Integrated oil.</u>	34	S	F
1959	<u>Integrated oil.</u> ADDED: petro-chemicals, ammonia, plasticizers.	79	S	F
1969	<u>Integrated oil:</u> + as above. ADDED: nuclear fuel.	29	DV	F
AVCO				
Prior to 1959 name was AVCO Manufacturing.				
Prior to 1947 name was Aviation Corp.				
1949	<u>Refrigerators, freezers</u> , TV and radio sets, kitchen ranges and sinks, radar and fire control systems, [aircraft engines, boilers, buses, farm equipment, broadcasting].	101-200	UP	II
1959	As above. EXITED: almost all consumer appliances, buses, ship-yards. ADDED: <u>aerospace R & D</u> , electronic control and guidance systems, aerospace structures.	148	UP	PD

TABLE A-1 (continued)

AVCO (continued)

	Rank	Strat.	Struct.
1969 <u>Helicopter engines</u> , + as above. ADDED: [steel wire, abrasives, motion pictures, insurance, consumer finance, savings and loan companies, credit card finance].	129	AC	PD

BANGOR PUNTA

Prior to 1967 name was Bangor Punta Alegre Sugar.

Prior to 1964 name was Punte Alegre Sugar.

1949 <u>Sugar</u> .	--	S	H
1959 <u>Sugar</u> .	--	S	H
1969 EXITED: Sugar. ADDED: <u>light aircraft</u> [grain elevator, metals distribution, textile converting, railroad, foundry equipment, boats, jewelry, finance company, handguns, refrigeration equipment, tear gas, materials handling equipment, cottonseed oil, local power systems, recreational vehicles, knit fabrics].	288	AC	H

BECTION, DICKINSON

1949 <u>Thermometers</u> , syringes, bandages and other medical supplies.	--	RC	F
1959 <u>Thermometers</u> , + as above. ADDED: rubber gloves, surgical instruments, reagents, electronic components, laboratory equipment.	--	RC	FS
1969 As above. ADDED: laboratory animals, science education aids, bio-chemicals, medical electronic devices, laboratory environmental control, hospital computer systems, Sportade, navigational instruments, air-craft fabrics, packaging machinery.	417	RL	PD

TABLE A-1 (continued)

BELL INTERCONTINENTAL		Rank	Strat.	Struct.
Prior to 1960 name was Bell Aircraft.				
1949	<u>Helicopters</u> , rockets and missiles, aircraft subassemblies, R & D on advanced aircraft, [stove parts].	201-500	DC	F
1959	As above. ADDED: aviation electronic systems, [abrasive cleaning equipment, molded fiberglass, hydraulic and extrusion presses].	314	RC	PD
1969	<u>Abrasive cleaning equipment</u> , + as above. EXITED: helicopters, rockets, missiles, electronics, R & D, fiberglass, hydraulic presses. ADDED: [service station accessories, carbon paper, ink, air conditioning equipment, galvanized metal products].	552	UP	PD
BOISE CASCADE				
Prior to 1957 name was Boise Payette Lumber.				
1949	Integrated forest products: timber, <u>lumber</u> , building materials; non-wood building materials.	--	DV	F
1959	Integrated forest products: timber, <u>lumber</u> , building materials; + as above. ADDED: pulp, paper, paperbags and boxes [all extension of vertical chain].	321	DV	PD
1969	<u>Lumber and wood building products</u> , + as above. ADDED: pre-fab buildings and parts, plastic containers, [real estate development, recreational equipment, mobile homes, computer soft-ware].	55	UP	PD

TABLE A-1 (continued)

		<u>Rank</u>	<u>Strat.</u>	<u>Struct.</u>
BORDEN				
1949	<u>Dairy products</u> , chemical by-products, coffee.	67	DC	PD
1959	<u>Dairy products</u> , + as above. ADDED: other grocery products, [hose and tubing, cosmetics].	44	RC	PD
1969	<u>Dairy products</u> , + as above. ADDED: convenience and snack foods, pet foods, [perfumes and toiletries, adhesives, protective coatings, packaging films, ink, varnish, plastic housewares, diapers].	54	RL	PD
BORG-WARNER				
1949	<u>Auto parts</u> , household appliances, special steels and structural steel parts, electric motors, power-transmission devices, airplane engines.	101-200	RL	H
1959	<u>Auto parts</u> , + as above. ADDED: air-conditioning and heating systems, pumps, electronic components and instruments, rubber industrial products, materials handling equipment, [thermo-plastic resins, synthetic latex].	68	RL	PD
1969	<u>Auto parts</u> , + as above. EXITED: appliances. ADDED: electronics: aerospace, industrial controls, communications and data handling systems; [coin-op dry cleaners, plasticizers].	108	RL	PD

TABLE A-1 (continued)

		<u>Rank</u>	<u>Strat.</u>	<u>Struct.</u>
BRISTOL-MYERS				
1949	<u>Proprietary drugs</u> , toiletries, ethical drugs, [brushes and rollers, metal containers].	201-500	RC	PD
1959	<u>Proprietary drugs</u> , + as above. EXITED: brushes and rollers, metal containers.	307	RC	PD
1969	<u>Proprietary drugs</u> , + as above. ADDED: diet and infant foods, medical testing products, electric hairsetters, [household cleaners].	123	RC	PD
BROWN SHOE				
1949	<u>Shoes</u> .	201-500	S	F
1959	<u>Shoes</u> .	172	S	F
1969	<u>Shoes</u> . ADDED: handbags, luggage.	253	DC	PD
BRUNSWICK				
Prior to 1960 name was Brunswick-Balke-Collender.				
1949	<u>Bowling and billiards products</u> .	--	S	F
1959	<u>Bowling and billiards products</u> . ADDED: sports equipment [gym equipment, defense supplies and aircraft parts, plastic filament radomes].	175	DL	PD

TABLE A-1 (continued)

BRUNSWICK (continued)		<u>Rank</u>	<u>Strat.</u>	<u>Struct.</u>
1969	As above. EXITED: gym equipment, defense supplies. ADDED: [yachts, outboard motors, medical supplies and instruments, bulk chemicals].	230	AC	PD
BUCYRUS-ERIE				
1949	<u>Power cranes and excavators</u> , drilling machines and tools, tractor equipment.	201-500	RC	F
1959	<u>Power cranes and excavators</u> , + as above.	401	RC	PD
1969	<u>Power cranes and excavators</u> , + as above. ADDED: conveyors and conveyor systems, industrial packaging machinery.	588	RC	PD
BULOVA WATCH				
1949	<u>Watches</u> .	201-500	S	F
1959	<u>Watches</u> . ADDED: clock radios, radios, electronic timers, defense electronics, servo-control, electric razors, phonographs.	--	RL	FS
1969	<u>Watches</u> , + as above. EXITED: Razors, phonographs.	527	RC	FS
BUNKER-RAMCO				
Prior to 1968 name was Amphenol.				
Prior to 1965 name was Amphenol-Borg Electronics.				

TABLE A-1 (continued)

		<u>Rank</u>	<u>Strat.</u>	<u>Struct.</u>
BUNKER-RAMO				
1958	Formed by merger of Amphenol Electronics and Borg (Geo. W.) Co.			
1959	<u>Electronic components: cables and wire, connectors, switches, relays; electronic instruments, [auto clocks, pile fabrics].</u>	--	RC	PD
1969	<u>Electronic components</u> ; + as above. ADDED: electronic systems: information, guidance, communication; numerical controls, [electric advertising signs].	357	RL	PD
BUTLER MANUFACTURING				
1949	<u>Pre-fab building components</u> , farm equipment, storage and transport tanks, [dry-cleaning equipment, electric meter boxes].	--	RC	FS
1959	<u>Pre-fab building components</u> , + as above. EXITED: storage tanks, dry-cleaning equipment, electric meter boxes.	439	RC	FS
1969	<u>Pre-fab building components</u> , + as above. ADDED: <u>construction from pre-fab parts</u> , materials handling systems and electronic controls.	--	RL	PD
CALUMET AND HECLA				
Prior to 1952 name was Calumet and Hecla Consolidated Copper.				
1949	Integrated copper: <u>tubing</u> , refined, by-products [zinc].	201-500	S	F

TABLE A-1 (continued)

CALUMET AND HECLA (continued)

	<u>Rank</u>	<u>Strat.</u>	<u>Struct.</u>
1959 Integrated copper: <u>tubing</u> , + as above. ADDED: non-copper tubing, magnesium, uranium mining, heat-exchangers for power generators, [vulcanizing equipment, lumber].	455	DV	PD
1968 Acquired by Universal Oil Products.			

CAMPBELL SOUP

1949 <u>Canned soup</u> , other canned foods.	101-200	DC	F
1959 <u>Canned soup</u> , + as above. ADDED: frozen food.	88	DC	FS
1969 <u>Canned soup</u> , + as above. ADDED: baked goods, candy.	131	DC	FS

CAMPBELL TAGGERT ASSOCIATED BAKERIES

1949 Baked goods.	--	S	G
1959 Baked goods.	257	S	G
1969 Baked goods.	320	S	G

CANADA DRY

1949 <u>Soft drinks and syrups</u> , alcoholic beverages	--	DC	PD
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TABLE A-1 (continued)

CANADA DRY (continued)		<u>Rank</u>	<u>Strat.</u>	<u>Struct.</u>
1959	<u>Soft drinks and syrups</u> , + as above.	440	DC	PD
1968	Merged with Hunt foods to form Norton Simon.			
CARBORUNDUM				
1949	<u>Abrasives</u> , refractories, heating elements, resistors.	201-500	RC	F
1959	<u>Abrasives</u> , + as above. ADDED: refractory metals, ceramic electric products, abrasive machinery.	349	RC	PD
1969	<u>Abrasives</u> , + as above. EXITED: refractory metals. ADDED: industrial machinery, carbon and graphite products, china, liquid-metal pumps, waste incineration, filters.	323	RL	PD
CARRIER				
1949	<u>Air conditioners and refrigeration equipment</u> .	201-500	S	F
1959	<u>Air conditioners and refrigeration equipment</u> . ADDED: heating, steam-turbines, compressors and blowers, precision potentiometers, heavy motors and generators, electronic mechanisms.	181	DL	PD
1969	<u>Air conditioners</u> , + as above. EXITED: heavy motors and generators, refrigeration equipment, electronic mechanisms.	208	DL	PD
CATERPILLER TRACTOR				
1949	<u>Tractors and earthmoving equipment</u> , diesel-generator sets.	101-200	DC	F

TABLE A-1 (continued)

CATERPILLER (continued)

		<u>Rank</u>	<u>Strat.</u>	<u>Struct.</u>
1959	<u>Tractors and earthmoving equipment</u> , + as above.	53	DC	F
1969	<u>Tractors and earthmoving equipment</u> , + as above. ADDED: turbine engines.	42	DC	GW

CENTRAL SOYA

1949	<u>Soybean processing: oil, feed, meal, flour.</u>	201-500	S	F
1959	<u>Soybean processing</u> : + as above. ADDED: <u>chemical products of soy.</u>	162	S	F
1969	<u>Soybean processing</u> , + as above. ADDED: (part of vertical chain) <u>poultry</u> , agricultural chemicals.	194	S	FS

CHAMPION SPARK PLUG

1949	<u>Spark plugs.</u>	201-500	S	F
1959	<u>Spark plugs.</u>	375	S	F
1969	<u>Spark plugs.</u> ADDED: steel bars, aircraft safety equipment, [non-destructive testing equipment, medical diagnostic instruments, spraying and finishing equipment, x-ray tubes, magnetic tape player].	334	UP	FS

CHRYSLER

1949	<u>Cars</u> , trucks, air conditioners, powder metals, engines.	25	S	F
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TABLE A-1 (continued)

CHRYSLER (continued)				<u>Rank</u>	<u>Strat.</u>	<u>Struct.</u>
1959	<u>Cars</u> , trucks, + as above. ADDED: [ordinance, missiles].			9	DU	PD
1969	<u>Cars</u> , trucks, + as above. ADDED: [rockets].			6	DL	PD
CINCINNATI MILACRON						
Prior to 1970 name was Cincinnati Milling Machine.						
1949	<u>Machine tools</u> , cutting fluid, industrial chemicals.			--	DC	FS
1959	<u>Machine tools</u> , + as above. ADDED: grinding wheels, metal treating machines, plastic moldings.			370	DC	PD
1969	<u>Machine tools</u> , + as above. ADDED: numerical controls for machine tools, computers.			322	DL	PD
CLARK EQUIPMENT						
1949	<u>Fork trucks and towing tractors</u> , RR car parts, auto parts.			201-500	RC	PD
1959	<u>Fork trucks and tractors</u> , + as above. ADDED: shovels, cranes, trailers and vans.			218	RC	PD
1969	<u>Trucks and tractors</u> , + as above. ADDED: motor homes, [refrigeration equipment, food service equipment].			175	RC	PD

TABLE A-1 (continued)

CLEVITE			
Prior to 1952 name was Cleveland Graphite Bronze.			
	Rank	Strat.	Struct.
1949 <u>Auto parts:</u> precision engine parts.	--	S	F
1959 <u>Precision engine parts.</u> ADDED: [electronic components, micro-phones, instruments, phonograph needles, piezo-electric materials].	443	UP	PD
1969 Merged with Gould-National Batteries to form Gould.			
CLINTON FOODS			
1949 <u>Corn refining: starch, sugar, syrup, oils, pudding, preserves; frozen and canned foods.</u>	201-500	DC	PD
1956 Acquired by Standard Brands.			
CLUETT, PEABODY			
1949 <u>Men's clothing.</u>	201-500	S	F
1959 <u>Men's clothing.</u>	359	S	F
1969 <u>Men's clothing.</u> ADDED: <u>women's clothing</u> , retail clothing stores, knit mill, [luggage].	223	RC	PD
COCA-COLA			
1949 <u>Coke syrup.</u>	78	S	F

TABLE A-1 (continued)

COCA-COLA (continued)		<u>Rank</u>	<u>Strat.</u>	<u>Struct.</u>
1959	<u>Coke syrup.</u>	132	S	F
1969	<u>Soft drinks.</u> ADDED: orange juice, coffee.	79	RC	PD
COLLINS AND AIKMAN				
1949	<u>Upholstery fabrics,</u> other fabrics, yarn.	201-500	DC	F
1959	<u>Upholstery fabrics,</u> + as above. ADDED: car rugs, synthetic fur.	--	RC	PD
1969	<u>Apparel fabrics,</u> + as above. ADDED: knit fabrics, carpets, hosiery.	419	RC	PD
COLORADO MILLING AND ELEVATOR				
1949	<u>Flour milling:</u> flour, animal feed; merchants of grain, seed and coal.	201-500	S	F
1959	<u>Flour milling,</u> + as above.	318	S	F
1967	Merged with Great Western Sugar to form Great Western United.			
COLT INDUSTRIES				
Prior to 1964 name was Fairbanks Whitney.				
Prior to 1959 name was Penn-Texas.				
Prior to 1954 name was Pennsylvania Coal and Coke.				

TABLE A-1 (continued)

COLT INDUSTRIES (continued)		<u>Rank</u>	<u>Strat.</u>	<u>Struct.</u>
1949	<u>Coal.</u>	--	S	F
1959	<u>Coal.</u> ADDED: <u>Machine tools</u> , materials handling equipment, power shovels, pumps and compressors, engines, valves and controls, [oil and gas properties, fire-arms, caps and closures, electronic systems, home water systems, wire and cable].	284	AC	PD
1969	As above. EXITED: wire and cable, shovels, oil and gas. ADDED: carburetors, special steels, transformers.	155	AC	PD
COMMONWEALTH OIL REFINING				
1953	Incorporated.			
1959	<u>Oil refining.</u>	--	S	F
1969	<u>Oil refining.</u> ADDED: petro-chemicals, retail gas stations (both part of vertical chain).	452	DV	F
CONSOLIDATED CIGAR				
1949	<u>Cigars.</u>	201-500	S	F
1959	<u>Cigars.</u>	468	S	F
1968	Acquired by Gulf and Western.			

TABLE A-1 (continued)

CONTAINER CORPORATION OF AMERICA

		<u>Rank</u>	<u>Strat.</u>	<u>Struct.</u>
1949	<u>Paperboard, containers and fiber cans.</u>	101-200	S	PD
1959	<u>Paperboard containers and fiber cans.</u> ADDED: lumber, plastic packaging.	139	DC	PD
1968	Merged with Montgomery Ward to form Marco.			

CONTINENTAL OIL

1949	<u>Integrated oil company.</u>	61	S	F
1959	<u>Integrated oil company.</u> ADDED: petro-chemicals, detergent, fertilizer, ammonia.	61	DV	G
1969	<u>Integrated oil company,</u> + as above. ADDED: plastics, resin, coal, fiberglass.	35	DV	PD

CORNING GLASS WORKS

1949	<u>Electrical glass products,</u> glass housewares, fireglass, lab ware and optical glass, electronics glass.	101-200	RC	F
1959	<u>Electrical glass products,</u> + as above. ADDED: refractories, [nuclear fuel].	221	RC	PD

TABLE A-1 (continued)

CORNING (continued)

	<u>Rank</u>	<u>Strat.</u>	<u>Struct.</u>
1969 <u>Electrical glass products</u> , + as above. EXITED: nuclear fuel. ADDED: integrated circuits, medical equipment, home appliances, food service equipment, computer logic, [plastic packaging].	210	RC	PD

CPC INTERNATIONAL

Prior to 1959 name was Corn Products.

Prior to 1958 name was Corn **Products** Refining.

1949 Corn processing: <u>oil</u> , starch, syrup, sugar, feed, in bulk and for consumers.	201-500	DV	F
1959 Corn processing: <u>bulk corn mill products</u> , + as above. ADDED: grocery products, household products: dyes, rinses, wax.	65	RC	PD
1969 <u>Bulk corn mill products</u> , + as above. ADDED: seeds, industrial and agricultural chemicals, plastics and resins, food service.	92	RL	PD

CROWELL COLLIER AND MACMILLIAN

Prior to 1965 name was Crowell-Collier Publishing.

1949 Publishing: <u>magazines</u> , books.	201-500	DC	FS
1959 <u>Book publishing</u> . EXITED: magazines. ADDED: Radio station, record club.	--	DL	FS

TABLE A-1 (continued)

CROWELL (continued)		Rank	Strat.	Struct.
1969	Book <u>publishing</u> , + as above. ADDED: bookstores, educational supplies, adult education, laboratory equipment, information data banks, academic robes, school janitor services, magazines, film distribution, [musical instruments].	255	RL	PD
CROWN CORK AND SEAL				
1949	<u>Caps, cans, containers</u> , cork, bottling and dairy machines.	201-500	S	F
1959	<u>Caps, cans, containers</u> , + as above.	135	S	F
1969	<u>Caps, cans, containers</u> , + as above.	264	S	F
CRUCIBLE STEEL				
1949	Integrated specialty steel producer: <u>alloys</u> , high grade, specialty carbon steel; in semi-finished, mill and finished forms.	201-500	DV	F
1959	Integrated specialty steel producer: <u>alloy steels</u> , + as above. ADDED: titanium and titanium alloys.	209	DV	F
1968	Acquired by Colt Industries.			
CUDAHY				
Prior to 1966 name was Cudahy Packing.				
1949	Meat packing: <u>meat</u> , skins, soap, chemical by-products; poultry, dairy products.	201-500	DV	FS

TABLE A-1 (continued)

CUDAHY (continued)		<u>Rank</u>	<u>Strat.</u>	<u>Struct.</u>
1959	Meat packing: <u>meat</u> , + as above.	128	DV	FS
1969	Meat packing: <u>meat</u> , + as above.	274	DV	FS
CURTISS-WRIGHT				
1949	<u>Aircraft engines, frame parts and rockets</u> , metal parts for automobiles and industry, [motion picture equipment].	100	DC	PD
1959	<u>Aircraft engines, frame parts and rockets</u> , + as above. EXITED: motion picture equipment. ADDED: electronic instruments, military vehicles, rocket motors , missile R & D, hovercraft, nuclear reactor and instruments, [plastic textiles, plastic industrial parts, heavy earth-moving equipment].	135	UP	PD
1969	<u>Aircraft engines, frame parts and rockets, and aerospace electronics</u> , + as above. EXITED: plastic products, missiles, military vehicles, hovercraft. ADDED: [water management and pollution control, process equipment, shot peening, real estate development].	318	UP	PD
DANA				
1949	<u>Auto parts</u> .	101-200	S	F
1959	<u>Auto parts</u> . ADDED: industrial parts.	200	DC	F
1969	<u>Auto parts</u> , + as above. ADDED: boat trailers, car transporters.	173	DC	PD

TABLE A-1 (continued)

		<u>Rank</u>	<u>Strat.</u>	<u>Struct.</u>
DEERE				
1949	<u>Farm equipment.</u>	63	S	F
1959	<u>Farm equipment.</u> ADDED: fertilizer, industrial and construction machinery: tractors, bulldozers.	78	DC	F
1969	<u>Farm equipment</u> , + as above. ADDED: materials handling equipment.	112	DC	F
DI GIORGIO				
Prior to 1964 name was Di Giorgio Fruit.				
1949	Vertically integrated produce company: <u>growing and selling fresh produce</u> and wine, selling lumber and wooden boxes.	--	DV	FS
1959	As above. ADDED: <u>canned and packaged fruits, vegetables, some grocery specialties.</u>	393	RC	PD
1969	<u>Lumber</u> , + as above. EXITED: canned and packaged fruits and vegetables. ADDED: <u>house frames</u> , [retail and wholesale sale of food and drugs, soft drinks, snack food, furniture, campers, trailers, plastic cutlery, proprietary drugs, food accounting machinery].	268	UP	PD
DIAMOND INTERNATIONAL				
Prior to 1964 name was Diamond National.				
Prior to 1959 name was Diamond Gardner.				

TABLE A-1 (continued)

DIAMOND (continued)		<u>Rank</u>	<u>Strat.</u>	<u>Struct.</u>
Prior to 1957 name was Diamond Match.				
1949	Vertically integrated forest products: <u>lumber</u> , sanitary paper products, pulp products, matches, and small woodenware.	201-500	DV	FS
1959	As above. EXITED: sanitary paper products, pulp products. ADDED: <u>paper packaging</u> , advertising materials, charcoal briquets (all part of the vertical chain), packaging and papermaking machinery.	201	DV	PD
1969	Vertically integrated forest products: <u>lumber</u> , + as above. ADDED: commercial printing, cans and plastic packaging, fine paper, [poultry-raising equipment].	216	DV	PD
DOEHLER-JARVIS				
1949	<u>Die castings.</u>	201-500	S	F
1953	Acquired by National Lead.			
DONNELLY (R.R.) AND SONS				
1949	<u>Commercial printing and binding.</u>	201-500	S	F
1959	<u>Commercial printing and binding.</u>	312	S	F

TABLE A-1 (continued)

DONNELLY (continued)		Rank	Strat.	Struct.
1969	<u>Commercial printing and binding.</u>	304	S	PD
DOW CHEMICAL				
1949	<u>Industrial chemicals</u> , agricultural chemicals, plastic chemicals and film, pharmaceutical intermediaries.	52	RC	FS
1959	<u>Industrial chemicals</u> , + as above. ADDED: styrofoam, acrylic fiber, coated paper, [wrapping paper and ribbons].	58	RC	FS
1969	<u>Industrial chemicals</u> , + as above. ADDED: ethical drugs, animal health products, packaging machinery, medical diagnostic instruments and services, plastic building products, [aluminium extruded parts].	49	RL	GW
DRESSER INDUSTRIES				
1949	Equipment for oil and gas industries: <u>drills, rigs and bits</u> , pumps, engines, meters, etc.	201-500	RC	H
1959	<u>Drills, rigs and bits</u> , + as above. ADDED: electronic seismographs.	199	RC	PD
1969	<u>Equipment for oil and gas industry</u> , + as above. ADDED: cranes, hoists, refractories, lead mine, pneumatic tools, vacuum technology, glass, sand and clay, recording instruments, valves and fittings.	163	RL	PD

TABLE A-1 (continued)

DU MONT (ALLEN B.) LABORATORIES

		<u>Rank</u>	<u>Strat.</u>	<u>Struct.</u>
1949	<u>TV sets</u> , cathode ray tubes, TV broadcasting and broadcasting equipment and oscilloscopes.	201-500	DC	PD
1959	<u>Cathode ray tubes, and oscilloscopes</u> . EXITED: TV sets, broadcasting and broadcasting equipment, TV tubes. ADDED: mobile radio, automobile testing equipment, military electronics: TV, radar, telemetry.	--	DC	PD
1960	Acquired by Fairchild Camera.			

DUPONT, (E.I.) DE NEMOURS

1949	<u>Nylon and rayon fibers, yarn and fabrics</u> , explosives, dyestuffs, finishes, pigments, heavy industrial chemicals, tetraethyl lead, plastics, insecticides and other agricultural chemicals, refrigerants, fire arms (Remington), photographic film.	8	RL	PD
1959	<u>Synthetic fibers, yarns and fabrics</u> , + as above. ADDED: vinyl plastics, teflon, dacron, orlon, mylar polyester film, special rubber synthetics.	12	RL	PD
1969	<u>Synthetic fibers, yarns and fabrics</u> , + as above. ADDED: corfam shoe upper material, magnetic tape, synthetic elastomeric yarns, photo instruments, antifreeze, wide variety of new plastics.	15	RL	PD

TABLE A-1 (continued)

EASTMAN KODAK

	<u>Rank</u>	<u>Strat.</u>	<u>Struct.</u>
1949 <u>Photographic supplies</u> : films, cameras, projectors, plates and paper; motion picture films, chemicals, cellulose acetate products: yarn, plastic molding compounds; organic and commercial chemicals, military optical and other precision equipment, vacuum equipment, microfilm and equipment, and vitamins.	33	RC	F
1959 <u>Photographic supplies</u> , + as above. ADDED: light chemicals, office copiers, plastic parts and tubing, microfilm EDP.	45	RC	PD
1969 <u>Photographic supplies</u> , + as above.	27	RC	PD

EATON YALE AND TOWNE

Prior to 1963 name was Eaton Manufacturing.

1949 <u>Auto parts</u> .	101-200	S	F
1959 <u>Auto parts</u> . ADDED: industrial parts, heaters and air conditioners and dynamometers.	160	RC	F
1969 <u>Auto parts</u> , + as above. ADDED: boat engines, valves and controls for autos and industry, fork lift trucks, materials handling and construction equipment, [locks and builders hardware].	110	RL	PD

ELECTRIC AUTO-LITE

1949 Auto parts, <u>electrical parts and batteries</u> , castings and forgings,	101-200	RC	F
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TABLE A-1 (continued)

ELECTRIC AUTO-LITE (continued)

	<u>Rank</u>	<u>Strat.</u>	<u>Struct.</u>
1949 (continued) plastic parts, meters; industrial motors and generators, pressure and temperature gauges.			
1959 <u>Auto parts</u> , + as above	223	RC	PD
1963 Merged with Mergenthaler Linotype to form Eltra.			

ELTRA

1963 Formed by merger of Electric Auto-Lite and Mergenthaler Linotype.			
1969 Auto parts: <u>electric and batteries</u> , die castings, speedometers and tachometers; industrial batteries, die castings and measuring instruments, [refractories, plumbers' fittings, typesetting equipment, flour, and flour mixes].	238	AC	PD

ENDICOTT JOHNSON

1949 Shoes.	101-200	S	F
1959 Shoes.	287	S	F
1969 Shoes.	524	S	F

TABLE A-1 (continued)

ENGELHARD INDUSTRIES

Prior to 1958 name was Baker and Co.

		Rank	Strat.	Struct.
1949	<u>Precious metals: smelting, mining and fabricating</u> , quartz products, electrical wire and contacts.	101-200	DC	H
1959	<u>Precious metals: smelting, mining and fabricating</u> , + as above. ADDED: metal-based catalysts and chemicals.	320	DC	H
1967	Merged with Minerals and Chemicals Phillip to form Engelhard Minerals and Chemicals.			

EX-CELL-O

1949	<u>Precision parts and assemblies for aircraft</u> , machine tools, cutting tools, [dairy and packaging machinery].	--	RC	F
1959	<u>Precision parts and assemblies for aircraft</u> , + as above.	355	RC	F
1969	As above. ADDED: electronic products: computer memory systems, numerical controls for machine tools, tape readers, etc.	270	RC	PD

FAIRMONT FOODS

1949	<u>Dairy products</u> , poultry and eggs, animal feed, oils and fats, frozen food, store equipment and produce distribution.	201-500	RC	F
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TABLE A-1 (continued)

FAIRMONT FOODS (continued)

		<u>Rank</u>	<u>Strat.</u>	<u>Struct.</u>
1959	<u>Dairy products</u> , + as above. EXITED: oils and fats, animal feed, store equipment, produce distribution. ADDED: snack foods.	389	RC	F
1969	Dairy products, + as above. ADDED: soft drink syrups, retail convenience stores.	327	RC	PD

FALSTAFF BREWING

1949	Beer.	--	S	F
1959	Beer.	360	S	F
1969	Beer.	516	S	F

FIBREBOARD

Prior to 1966 name was Fibreboard Paper Products.

Prior to 1956 name was Pabco.

1949	<u>Paperboard containers</u> , glass bottles and jars, [electrotype, paint, roofing, flooring and insulating products].	101-200	DU	FS
1959	<u>Paperboard containers</u> , + as above. EXITED: glass containers. ADDED: <u>paperboard</u> .	334	DU	PD
1969	<u>Paperboard and paperboard containers</u> . EXITED: flooring, paint, and roofing products. ADDED: lumber, plywood, inks, real estate development.	450	DL	PD

TABLE A-1 (continued)

FIRESTONE TIRE AND RUBBER

		<u>Rank</u>	<u>Strat.</u>	<u>Struct.</u>
1949	Vertically integrated rubber company: <u>tires and tubes</u> , latex, flooring, insulation, etc., steel auto parts, plastic auto parts, and tire cord.	39	DV	FS
1959	Vertically integrated rubber company: <u>tires and tubes</u> , + as above. ADDED: synthetic fibers for tires, (part of vertical chain); plastics.	30	DV	FS
1969	Vertically integrated rubber company: <u>tires and tubes</u> , + as above. ADDED: seat belts.	37	DV	PD

FLINTKOTE

1949	<u>Building materials of asphalt, asbestos and paper</u> : roofing, insulation, wallboard, siding, building papers, floor coatings, tiles; industrial products from same materials: adhesives and cements, coatings, paperboard containers.	201-500	RC	FS
1959	<u>Building materials</u> , + as above. ADDED: lime and limestone products, building slabs, cement; gypsum building products, pipes.	131	RC	FS
1969	<u>Building products</u> , + as above. ADDED: spray machine for roofing.	77	RC	PD

FMC

Prior to 1961 name was Food Machinery and Chemical.

TABLE A-1 (continued)

FMC (continued)		<u>Rank</u>	<u>Strat.</u>	<u>Struct.</u>
1949	<u>Agricultural chemicals</u> , industrial: chemicals, phosphates, chlorides, food preparation and processing equipment; agricultural equipment: tractors, sprayers; industrial equipment: fire fighters.	201-500	RL	PD
1959	<u>Agricultural chemicals</u> ,+ as above. ADDED: plasticizers, packaging machinery, oil field equipment, pumps and water systems, [automotive service, waste disposal systems, industrial castings, pre-stressed concrete, defense materiel, rocket motors, electronic components].	131	RL	PD
1969	Agricultural chemicals, industrial chemicals, [industrial machinery, + as above; food and agricultural equipment, + as above; castings, defense materiel and rocket fuel, automotive services, pumps and water systems, waste disposal systems]. EXITED: Rocket motors. ADDED: <u>Synthetic fibers and films</u> ; fibers, tire cord, packaging film; [recreational vehicles, power transmission equipment, materials handling systems, railroad cars, construction machinery, auto parts, power tools].	77	AC	PD
FORD MOTOR				
1949	<u>Cars and trucks</u> , tractors, aircraft engines.	10	S	PD
1959	<u>Cars and trucks</u> , + as above. ADDED: [defense electronic systems].	3	DC	PD
1969	<u>Cars and trucks</u> , + as above. EXITED: aircraft engines. ADDED: [home electronic machines, and home appliances].	3	DU	PD

TABLE A-1 (continued)

FOREMOST DAIRIES		Rank	Strat.	Struct.
1949	<u>Dairy products</u> ; jams and jellies, frozen foods.	--	S	F
1959	<u>Dairy products</u> , + as above. EXITED: frozen foods. ADDED: light chemicals (fatty acids), canned fruits, vegetables and juices.	97	DC	G
1967	Merged with McKesson and Robbins to form Foremost-McKesson.			
FOSTER WHEELER				
1949	Power plant equipment; oil refinery equipment: chemical process equipment and the design and construction of entire plants in the above fields or other industries.	--	RC	F
1959	As above. ADDED: nuclear plant equipment.	268	RC	F
1969	As above. ADDED: pressure vessels, furnaces.	265	RC	PD
GAF				
Prior to 1968 name was General Aniline and Film.				
1949	<u>Photographic equipment</u> : cameras, film, paper, darkroom equipment, dyes and pigments, heavy and special chemicals, textile chemicals.	201-500	RL	PD
1959	<u>Photographic equipment</u> , + as above. ADDED: microfilm, data handling equipment.	273	RL	PD
1969	[As above]. ADDED: <u>asphalt, asbestos and gypsum building products: roofing, siding, flooring, insulation, coatings</u> ; industrial products from same materials: acoustical coverings, filters, etc.; [business forms, office copiers, micrographics, pharmaceutical, agricultural and polymer chemicals].	184	UP	PD

TABLE A-1 (continued)

GENERAL CABLE		<u>Rank</u>	<u>Strat.</u>	<u>Struct.</u>
1949	<u>Wire and cable.</u>	201-500	S	F
1959	<u>Wire and cable.</u> ADDED: poles and crossarms, insulated cord.	260	DC	F
1969	<u>Wire and cable</u> , + as above. ADDED: electrical switches and panels, pole line hardware, aerial lift vehicles, dryers for cables, cable pressure systems.	265	DC	PD
GENERAL ELECTRIC				
1949	<u>Electric power generation equipment</u> , motors, motor controls, lighting equipment, electric appliances, TV and radio, jet engines, electronic parts and systems.	9	RL	PD
1959	As above. ADDED: plastics, computers, semi-conductors, aerospace contracting.	4	RL	PD
1969	As above. ADDED: cable TV, TV-radio broadcasting, time-share computers, [rapid-transit cars, urban systems study].	4	RL	PD
GENERAL FOODS				
1949	<u>Coffee</u> , wide variety of grocery products.	73	RC	PD
1959	<u>Coffee</u> , + as above.	37	RC	PD
1969	<u>Coffee</u> , + as above.	46	RC	PD

TABLE A-1 (continued)

GENERAL HOST		Rank	Strat.	Struct.
Prior to 1967 name was General Baking.				
1949	<u>Baked goods.</u>	201-500	S	G
1959	<u>Baked goods.</u> ADDED: [restaurants, candy].	261	DU	G
1969	<u>Baked goods</u> , + as above. ADDED: frozen dinners, [tourist concessions, recreational area development, general retail stores].	425	AC	PD
GENERAL INSTRUMENT				
1949	<u>Components for electronic entertainment equipment:</u> tuners, other components; phonograph record changers, defense electronic components.	--	DC	F
1959	<u>Entertainment electronic components</u> , + as above. ADDED: semi-conductors, military electronic equipment and systems.	--	RC	PD
1969	As above. ADDED: data collecting, processing and display systems, registers and ticket issuers, cable TV, magnetic recording heads, thermo-electric products, electro-optical products.	369	RL	PD
GENERAL MILLS				
1949	Flour miller: <u>flour</u> , feeds, cereals, packaged grain-based foods, chemical and pharmaceutical milling by-products; contract machining for defense, small appliances, distribution of feeds and other farm supplies.	101-200	DV	PD

TABLE A-1 (continued)

GENERAL MILLS (continued)		<u>Rank</u>	<u>Strat.</u>	<u>Struct.</u>
1959	Flour miller: <u>flour</u> , + as above. EXITED: contract machining. ADDED: ready-to-bake foods, pet food, defense electronics, steroids, [sponges, plastic consumer products].	77	RC	PD
1969	Flour miller: flour, <u>cereals</u> , grain-based packaged foods, chemical and pharmaceutical by-products, + as above. EXITED: feed, home appliances, defense electronics, ready-to-bake foods. ADDED: <u>snack foods</u> , other grocery products, chemicals and resins, [restaurants, jewelry, clothes, mail-order merchandising, toys and games].	130	RC	PD
GENESCO				
Prior to 1959 name was General Shoe.				
1949	<u>Shoes, shoe stores.</u>	201-500	S	H
1959	<u>Shoes, shoe stores.</u> ADDED: clothing stores, jewelry stores, silverware, women's apparel.	173	DL	H
1969	<u>Shoes, shoe stores</u> , + as above. EXITED: jewelry stores, silverware. ADDED: men's apparel.	95	RL	PD
GILLETTE				
Prior to 1953 name was Gillette Safety Razor.				
1949	<u>Razors, blades, and shaving cream</u> ; hair care products.	201-500	DC	FS
1959	<u>Razors, blades, and shaving cream</u> , + as above. ADDED: cosmetics, proprietary drugs, contract R & D, [ball-point pens].	216	RC	PD

TABLE A-1 (continued)

GILLETTE (continued)		<u>Rank</u>	<u>Strat.</u>	<u>Struct.</u>
1969	<u>Razors, blades, and shaving cream.</u> EXITED: proprietary drugs. ADDED: men's toiletries, [disposable hospital supplies, electric appliances, photographic equipment, audio systems].	182	RC	PD
GLEN ALDEN				
Prior to 1959 name was Glen Alden Coal.				
1949	Coal mining.	--	S	F
1959	Coal mining. ADDED: air conditioners, fire trucks and equipment, movie theatres, textile finishing, auto parts, warehouses, gas and oil properties.	364	UP	PD
1969	[As above]. EXITED: coal, fire equipment, textile finishing, auto parts, gas and oil. ADDED: <u>alcoholic beverages</u> , [appliances, textiles, family consumer products: toothbrushes, sanitary products, etc.; men's and women's apparel, wholesale and retail; building materials, medical electronics; high technology products: Apollo space suits, air-supported structures].	--	AC	PD
1969	Acquired by Rapid-American.			
GLIDDEN				
1949	<u>Vegetable oils processor:</u> margarine, dressings and other foods, feed, 101-200 chemical by-products; spices, paint and pigments, pine gum processing: resin, turpentine.		RL	PD

TABLE A-1 (continued)

GLIDDEN (continued)		<u>Rank</u>	<u>Strat.</u>	<u>Struct.</u>
1959	<u>Paint and pigments</u> , + as above. EXITED: pine tar processing, animal feed, chemical by-products of soy processing. ADDED: titanium powder, organic chemicals.	230	RL	PD
1967	Acquired by SCM.			
GOODRICH (B.F.)				
1949	Rubber: <u>tires and tubes</u> , industrial goods, synthetic rubber, druggists' goods, shoes, latex, rubber chemicals; other chemicals; tire fabrics, organic colors, wheels and brakes for aircraft, vinyl plastic.	62	DV	FS
1959	Rubber: <u>tires and tubes</u> , + as above. ADDED: auto wheels and brakes, vinyl plastic: imitation leather, floor and wall coverings; petrochemicals, synthetic fiber, [aerospace and defense: fuel cells, rocket cases, fuel, motors, nose cones].	50	DV	PD
1969	Rubber: <u>tires and tubes</u> , + as above. ADDED: [materials handling system].	90	DV	PD
GOODYEAR TIRE AND RUBBER				
1949	Rubber: <u>tires and tubes</u> , consumer products, industrial products, aviation products, latex; chemicals, resins; synthetic rubber; vinyl goods, wheel rims.	32	DV	FS

TABLE A-1 (continued)

GOODYEAR (continued)		<u>Rank</u>	<u>Strat.</u>	<u>Struct.</u>
1959	Rubber: <u>tires and tubes</u> , + as above. ADDED: packaging film, plastics, [rockets and missiles, radar and military electronics].	21	DV	PD
1969	Rubber: <u>tires and tubes</u> , + as above. ADDED: auto parts, [heaters and air conditioners, railroad equipment, real estate development].	20	DV	PD
GRACE (W.R.)				
1949	<u>Import-export</u> ; airlines and shipping, agriculture, mining: tungsten and tin; banking and insurance; manufacturing: fertilizer, textiles, paints and varnishes, paper, vegetable oil, etc.	--	UP	H
1959	[As above]. ADDED: <u>chemicals</u> : industrial, agricultural, plasticizers, acids, petro-chemicals, chemical products such as packaging film, sealing compounds; atomic fuel.	93	AC	PD
1969	<u>Chemicals</u> , + as above. EXITED: airlines and shipping; textiles, vegetable oil, agriculture, paint, banking and insurance, atomic fuel. ADDED: <u>detergents and germicides</u> , <u>plastic products</u> : vinyl fabrics, imitation leather footwear, lubricants, [frozen foods, snack foods, mail-order gifts, medical diagnostic laboratory, restaurants, women's clothing, shoe stores, leather goods].	50	AC	PD
GREAT NORTHERN NEKOSSA				
Prior to 1970 name was Great Northern Paper.				
1949	Integrated <u>paper</u> : pulp.	201-500	S	F

TABLE A-1 (continued)

GREAT NORTHERN NEKOSSA (continued)		<u>Rank</u>	<u>Strat.</u>	<u>Struct.</u>
1959	<u>Integrated paper</u> , + as above. ADDED: Kraft container board.	--	S	F
1969	Integrated forest products: <u>newsprint</u> , specialty and coated paper, container board. ADDED: plywood.	461	DV	F
GULF OIL				
1949	<u>Integrated oil producer</u> .	7	S	F
1959	<u>Integrated oil producer</u> . ADDED: petro-chemicals, synthetic rubber.	7	S	F
1969	<u>Integrated oil producer</u> , + as above. ADDED: fertilizers, plastics, coal, atomic energy, [real estate development].	10	DV	FS
HANDY AND HARMON				
1949	<u>Refined precious metals</u> , for jewelry, dentistry, industry.	201-500	S	F
1959	<u>Refined precious metals</u> . ADDED: steel and nickel tubing.	437	DL	FS
1969	<u>Refined precious metals</u> , + as above. ADDED: steel bearings, belts and miscellaneous parts, heat treatment of metals.	416	RL	FS
HARSCO				
Prior to 1956 name was Harrisburg Steel.				
1949	<u>Steel forgings and castings</u> .	--	S	F

TABLE A-1 (continued)

HARSCO (continued)		<u>Rank</u>	<u>Strat.</u>	<u>Struct.</u>
1959	<u>Steel forgings and castings.</u> ADDED: non-ferrous castings, steel trackwork, auto tools, dies and frame parts, [drilling machines, plastic parts].	396	DC	PD
1969	<u>Steel forgings and castings,</u> + as above. EXITED: non-ferrous castings, auto tools, dies and parts, drilling machines, plastic parts. ADDED: steel structures, building materials: bricks, pipes; sewage treatment equipment, hoist towers, [farm, stone-crushing and materials handling machinery, defense vehicles, missile support equipment, power tools].	341	RL	PD
HART SCHAFFNER AND MARX				
1949	<u>Men's clothing and clothing stores.</u>	--	S	F
1959	<u>Men's clothing and clothing stores.</u>	451	S	F
1965	Dropped due to greater than 50% non-manufacturing.			
1969	Men's clothing and clothing stores. ADDED: women's clothing.	--	RC	PD
HERSHEY FOODS				
Prior to 1968 name was Hershey Chocolate.				
1949	<u>Chocolate and cocoa products.</u>	101-200	S	F
1959	<u>Chocolate and cocoa products.</u>	266	S	F

TABLE A-1 (continued)

Hershey (continued)		<u>Rank</u>	<u>Strat.</u>	<u>Struct.</u>
1969	<u>Chocolate and cocoa products.</u> ADDED: grocery products, prepared foods for food service, [appliances, cookware, vending services, business aids].	298	DL	PD
HOBART MANUFACTURING				
1949	Commercial food machinery, kitchen appliances, scales.	--	RC	F
1959	As above.	--	RC	FS
1969	As above. ADDED: furniture for food service.	427	RC	FS
HOOKER CHEMICAL				
Prior to 1958 name was Hooker Electrochemical.				
1949	<u>Industrial chemicals</u> , mainly chlorine-based, agricultural chemicals, pharmaceutical intermediates, dyestuffs, electrolytic cells.	--	RC	F
1959	<u>Industrial chemicals</u> , + as above. ADDED: plastic chemicals, plastic compounds and foam, detergents, feed, solid rocket fuel.	283	RC	PD
1968	Acquired by Occidental Petroleum.			
HORMEL (GEO. A.)				
1949	Meat packing: <u>meat</u> , gelatine, by-products, canned soup.	101-200	DV	F
1959	Meat packing: <u>meat</u> , + as above. ADDED: frozen food, baby food.	117	DV	F

TABLE A-1 (continued)

HORMEL (continued)		<u>Rank</u>	<u>Strat.</u>	<u>Struct.</u>
1969	Meat packing: <u>meat</u> , + as above. ADDED: prepared foods for mass feeding.	178	DV	PD
HYGRADE FOOD PRODUCTS				
1949	Meat packer: <u>meat</u> , by-products--dog food, soap, fertilizer; dairy foods, coffee, grocery products.	101-200	DV	F
1959	Meat packer: <u>meat</u> , + as above.	111	DV	F
1969	Meat packer: <u>meat</u> , + as above.	269	DV	F
INGERSOLL-RAND				
1949	Heavy pumps and compressors, condensers, internal combustion engines, hand pneumatic tools, air conditioners and refrigeration equipment, mining machinery.	201-500	RC	F
1959	As above.	269	RC	F
1969	As above. ADDED: consumer tools, plastic molding machinery, bearings, [industrial and surgical needles, precision parts].	160	RC	PD
INLAND CONTAINER				
1949	<u>Kraft paper containers and Kraft paper.</u>	201-500	S	F

TABLE A-1 (continued)

INLAND CONTAINER (continued)		Rank	Strat.	Struct.
1959	<u>Kraft paper containers and Kraft paper.</u>	400	S	F
1969	<u>Kraft paper containers and Kraft paper.</u> ADDED: printing inks, retailing of packaging machinery, [real estate development].	444	S	PD
INMONT				
Prior to 1969 name was Interchemical.				
1949	<u>Printing inks</u> , pigments and dyes, finishes: paint, varnish; coated fabrics, carbon paper and ribbon.	201-500	RC	PD
1959	<u>Printing inks</u> , + as above. EXITED: coated fabrics, paint and varnish. ADDED: clay, adhesives.	329	RC	PD
1969	<u>Printing inks</u> , + as above. ADDED: fine chemicals and pharmaceutical intermediates, foam rubber insulators, copier paper, auto finishes, [oyster farm, candy].	289	RC	PD
INTERNATIONAL BUSINESS MACHINES				
1949	<u>Tabulating machines</u> , radio and audio equipment, clocks, typewriters, timers and meters.	68	RC	F
1959	As above. EXITED: radio and audio equipment. ADDED: <u>electronic data processing systems</u> .	27	RC	PD
1969	<u>EDP systems</u> , + as above. ADDED: dictating machines.	5	DC	PD

118

TABLE A-1 (continued)

		<u>Rank</u>	<u>Strat.</u>	<u>Struct.</u>
INTERSTATE BRANDS				
Prior to 1969 name was Interstate Bakeries.				
1949	<u>Bakery products.</u>	201-500	S	F
1959	<u>Bakery products.</u>	324	S	F
1969	<u>Bakery products.</u> ADDED: frozen pies, institutional canned vegetables, dog food.	374	S	PD
IOWA BEEF PROCESSORS				
Prior to 1970 name was Iowa Beef Packers.				
1960	Incorporated.			
1969	<u>Beef packing.</u>	167	S	F
IPL				
Prior to 1968 name was International Packers Limited.				
1950	Incorporated.			
1959	<u>Meat packing</u> ; vegetable oil milling, poultry and grocery products distribution.	113	DC	F
1969	Merged with Deltec Panamerica SA to form Deltec International.			

TABLE A-1 (continued)

		<u>Rank</u>	<u>Strat.</u>	<u>Struct.</u>
JOHNSON AND JOHNSON				
1949	Bandages, band-aids and surgical supplies, baby care products, sanitary paper products, tooth and hair brushes, non-prescription pharmaceuticals, adhesives.	101-200	RC	F
1959	As above. ADDED: textiles, molded plastic medical equipment, ethical drugs, chemicals for textile manufacture, glue.	151	RC	PD
1969	Bandages and band-aids and surgical supplies, + as above. EXITED: textiles, glue. ADDED: <u>medical and surgical instruments</u> , elastic fabric.	170	RC	PD
JONES AND LAUGHLIN STEEL				
1949	Integrated steel producer: <u>steel mill products</u> , fabricated products, structures, coal chemicals; distribution of oil well supplies.	35	DV	F
1959	Integrated steel producer: <u>steel mill supplies</u> , + as above.	51	DV	F
1968	Acquired by Ling-Temco-Vought.			
JOSLYN MANUFACTURING AND SUPPLY				
1949	Construction materials for electric, telephone and telegraph lines: poles and cross arms, insulators and switches, wood preservatives, lightning rods; street lighting equipment, iron castings, steel bars, wires and shapes, [kitchen ware].	201-500	RC	F

TABLE A-1 (continued)

JOSLYN (continued)		<u>Rank</u>	<u>Strat.</u>	<u>Struct.</u>
1959	As above. EXITED: [kitchen ware].	405	RC	F
1969	As above. ADDED: surge arrestors, fiberglass insulators, wood floor blocks, marine and industrial hardware.	606	RC	F
KAISER ALUMINIUM AND CHEMICAL				
Prior to 1949 name was Permanente Metals.				
1949	Integrated aluminium producer: <u>sheet</u> , ingot, chemical by-products.	101-200	DV	F
1959	Integrated aluminium producer: <u>sheet</u> , + as above. ADDED: foil and packaging, rod and bar, screening and roofing, cans, electrical conductors (all part of vertical chain).	101	DV	PD
1969	Integrated aluminium producer: <u>sheet</u> , + as above. ADDED: urethane foams; agricultural chemicals; nickel, iron and uranium mining; modular buildings and building parts; [real estate development, cattle production].	125	DV	PD
KARAGHEUSIAN				
1949	<u>Carpets and rugs.</u>	201-500	S	F
1959	<u>Carpets and rugs.</u>	-	S	F
1964	Acquired by Stevens (J.P.).			

TABLE A-1 (continued)

KELSEY-HAYES

Prior to 1956 name was Kelsey-Hayes Wheel.

	<u>Rank</u>	<u>Strat.</u>	<u>Struct.</u>
1949 <u>Auto parts</u> ; wheels, brakes, small parts.	201-500	S	F
1959 <u>Auto parts</u> , + as above. ADDED: airplane parts: engine parts, transmissions, gear-boxes; hand tools, [vacuum-melted metals].	224	RL	F
1969 <u>Auto parts</u> , + as above. EXITED: hand tools, vacuum-melted metals. ADDED: heavy iron and steel castings, aluminium auto parts, contract making of missile parts.	259	RL	F

KEYSTONE CONSOLIDATED INDUSTRIES

Prior to 1969 name was Keystone Steel and Wire.

1949 Integrated steel company: <u>wire and wire products</u> , (fencing, barbed, builders' products, industrial), semi-finished steel, nails; locks, hardware.	201-500	DV	FS
1959 Integrated steel: <u>wire and wire products</u> , + as above. ADDED: wire tying machines and related tools; agricultural equipment: gates, feeders; [plastic extrusions].	331	DV	FS
1969 Integrated steel: <u>wire and wire products</u> . ADDED: precision metal parts.	449	DV	PD

KIMBERLY-CLARK

1949 Integrated paper producer: cellulose wadding, white paper.	101-200	DV	F
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TABLE A-1 (continued)

KIMBERLY-CLARK (continued)		<u>Rank</u>	<u>Strat.</u>	<u>Struct.</u>
1959	Integrated paper producer: <u>cellulose wadding</u> , + as above. ADDED: sanitary paper products, special purpose papers, stationery.	125	DV	F
1969	Integrated paper producer: + as above.	134	DV	PD
KOPPERS				
1949	<u>Engineering and construction</u> : of coke ovens, gas and chemical plants, etc, tar processor: creosote, tars, pitches, chemicals, gas and coke, wood preserving of treated wood products, plastics and plasticizers, industrial chemicals and resins, metal products: contract machining, shaft couplings and propellers, piston rings; coke and gas plant equipment.	101-200	RL	F
1959	<u>Tar processor</u> , + as above. ADDED: plastic film, dyestuffs, paper box machinery; air and gas filters, noise abatement equipment.	193	RL	PD
1969	<u>Wood preserving and treated wood products</u> , + as above. ADDED: <u>laminated wood products</u> ; modular homes; glass reinforced plastics, fiberglass extrusions, thermo-plastics, polyester floor tile; water and sewage treatment, metallurgy.	209	RL	PD
LEAR SIEGLER				
Prior to 1962 name was Siegler.				
Prior to 1954 name was Siegler Heater.				
1959	<u>Home electronics</u> : radio, TV, phonographs, tape recorders, cable,	--	UP	PD

TABLE A-1 (continued)

LEAR SIEGLER (continued)		Rank	Strat.	Struct.
1959	(continued) and closed circuit TV; aerospace: structures, large antennas, avionics, telemetry, hydraulic jacks; [materials handling equipment, electronic testing equipment and industrial controls, tools; heaters, furnaces, air conditioning].			
1969	<u>Home electronics</u> , + as above. EXITED: cable TV, aerospace structures. ADDED: aircraft electric power systems, flight control systems, pumps, actuators; lasers; computers; home lighting; [metal-cutting tools, auto parts; food display equipment; mattresses, bed frames; business schools].	186	AC	PD
LEHIGH PORTLAND CEMENT				
1949	<u>Cement</u> .	101-200	S	F
1959	<u>Cement</u> .	438	S	F
1969	<u>Cement</u> . ADDED: aggregate concrete; [carpets, furniture].	--	DU	PD
LIBBEY-OWENS-FORD				
Prior to 1968 name was Libbey-Owens-Ford Glass.				
1949	<u>Plate and auto glass</u> , insulating glass, mirrors; plastics and resins.	101-200	DC	F
1959	<u>Plate and auto glass</u> , + as above. EXITED: plastics and resins.	147	DC	FS

TABLE A-1 (continued)

LIBBEY-OWENS-FORD (continued)

	<u>Rank</u>	<u>Strat.</u>	<u>Struct.</u>
1969 <u>Plate and auto glass</u> , + as above. ADDED: ornamental bronze; auto seals, [hose and fluid couplings, cargo tiedown equipment; industrial rubber products; custom plastic and cellulose products].	232	RL	FS

LIEBMANN BREWERIES

1949 <u>Beer and ale</u> .	201-500	S	F
1959 <u>Beer and ale</u> .	--	S	F
1964 Acquired by Pub United (Rheingold > 1964).			

LING-TEMCO-VOUGHT

Prior to 1961 name was Ling-Temco Electronics.
 Prior to 1960 name was Ling-Altec Electronics.
 Prior to 1959 name was Ling Electronics.
 Prior to 1956 name was L.M. Electronics.

1953 Incorporated.

1959 Electronic components, information display systems, super-power electronics, broadcasting and sound reproduction: TV, stereos, recording equipment, sonar and radar, measurement devices.	--	RC	PD
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TABLE A-1 (continued)

LING-TEMCO-VOUGHT (continued)

	Rank	Strat.	Struct.
1969 [as above]. EXITED: electronic components. ADDED: <u>meat packing</u> : meat, fats, by-products, [athletic goods, pharmaceutical chemicals, industrial chemicals and resins; aircraft, missiles, weapon systems, aerospace electronic systems; rockets and launch vehicles; engineering and architecture, insulates wire; steel: mill products, structural products, modular housing; airline; banking and insurance, car rental; hotels; ski area; business colleges; floor coverings, industrial maintenance.	14	AC	H

LITTON INDUSTRIES

Prior to 1954 name was Electro-Dynamics.

1953 Incorporated.

1959 <u>Business machines</u> : computers, calculators, guidance systems and aircraft instruments; x-ray equipment.	322	RL	PD
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1969 <u>Guidance, navigational and control systems</u> , + as above. ADDED: tag punchers and readers, typewriters, business forms, [office furniture; materials handling systems; motors, machine tools; optical and medical instruments; dental supplies, special printing; maps; <u>textbooks</u> ; ship building, food display; prepared food; store fixtures].	39	AC	PD
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LOCKHEED AIRCRAFT

1949 <u>Aircraft and aircraft parts, ground equipment, air terminal and over-haul.</u>	101-200	S	F
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TABLE A-1 (continued)

LOCKHEED (continued)		Rank	Strat.	Struct.
1959	<u>Aircraft and aircraft parts</u> , + as above. ADDED: avionics, weapons systems, space satellites, missile work, [ship building and repair, heavy construction].	28	RC	PD
1969	<u>Aircraft and aircraft parts</u> , + as above. ADDED: computers, electronic components, oceanography, [plastic parts, steel fabricated products, materials handling equipment, meters and measuring devices].	41	RC	PD
LONGVIEW FIBRE				
1949	<u>Containerboard and Kraft paper</u> .	201-500	S	F
1959	<u>Containerboard and Kraft paper</u> .	--	S	F
1969	<u>Containerboard and Kraft paper</u> . ADDED: containers, bags.	--	S	FS
LONG-BELL LUMBER				
1949	Integrated lumber company: <u>lumber</u> , wood building materials, furniture; retail building materials outlets.	201-500	DV	F
1956	Acquired by International Paper.			
LORILLARD (P.)				
1949	<u>Cigarettes and tobacco</u> .	101-200	S	F

TABLE A-1 (continued)

LORILLARD (continued)		<u>Rank</u>	<u>Strat.</u>	<u>Struct.</u>
1959	<u>Cigarettes and tobacco.</u>	171	S	F
1968	Acquired by Lowe's Theatres.			
LUKENS STEEL				
1949	Integrated steel company: <u>steel plate shapes</u> ; steel plate, steel plate products, including machinery and steel machinery parts.	--	DV	F
1959	Integrated steel company: <u>steel plate shapes</u> , + as above. ADDED: alloys and clad materials, electrodes (both part of vertical chain).	453	DV	F
1969	Integrated steel company: <u>steel plate shapes</u> , + as above.	536	DV	F
MALLORY (P.R.)				
1949	<u>Batteries</u> ; capacitors, rectifiers; resistors, tuners, switches; special metals, welding tips.	--	DC	PD
1959	<u>Batteries</u> , + as above. ADDED: non-ferrous forgings and castings, steel rolling mill, [plastic molded household products].	435	RL	PD
1969	<u>Batteries</u> , + as above. ADDED: micro-modules.	--	RL	PD

TABLE A-1 (continued)

		<u>Rank</u>	<u>Strat.</u>	<u>Struct.</u>
MARATHON OIL				
Prior to 1962 name was Ohio Oil.				
1949	<u>Integrated oil company.</u>	201-500	S	F
1959	<u>Integrated oil company.</u>	130	S	F
1969	<u>Integrated oil company.</u> ADDED: petro-chemicals.	126	DV	F
MARTIN-MARIETTA				
Prior to 1961 name was Martin Company (The).				
Prior to 1956 name was Glen L. Martin.				
1949	<u>Aircraft and parts</u> , military rockets and missiles, weapons systems.	201-500	DC	F
1959	<u>Rockets and missiles</u> , + as above. ADDED: <u>electronic aero-space systems.</u>	84	DC	F
1969	<u>Missiles and electronic aerospace systems.</u> EXITED: aircraft, rockets. ADDED: [cement, lime, concrete and tile; inks and dyes; aluminium; engineering off shore oil installations; real estate development].	116	UP	PD
MAYTAG				
1949	<u>Washing machines.</u>	201-500	S	F
1959	<u>Washing machines.</u> ADDED: dryers, [missile products].	326	S	FS

TABLE A-1 (continued)

MAYTAG (continued)		Rank	Strat.	Struct.
1969	<u>Washing machines and dryers.</u> EXITED: missile products.	486	S	FS
MCCALL				
1949	<u>Women's magazine publishing;</u> clothes pattern printing, contract printing, publication of fashion books.	--	DC	PD
1959	<u>Women's magazine publishing,</u> + as above.	485	RC	PD
1968	Merged with Hunt Foods to form Norton Simon.			
MCGRAW EDISON				
Prior to 1957 name was McGraw Electric.				
1949	<u>Electrical appliances:</u> kitchen appliances, fans, heaters; street lighting; pipe and conduit, fuses, pole-line hardware, transformers, capacitors, switches.	201-500	RL	PD
1959	<u>Electrical appliances,</u> + as above. ADDED: industrial furnaces and dryers, batteries, dictating machines, small motors, electrical tools, [aircraft instruments, medical gases, juvenile furniture].	164	RL	PD
1969	<u>Electrical appliances,</u> + as above. EXITED: industrial furnaces, medical gases, juvenile furniture. ADDED: coin laundries and dry cleaners, erection of utility poles, decorative lighting, timing devices, [vacuum treatment of metal, textile machinery].	183	RL	PD

TABLE A-1 (continued)

		<u>Rank</u>	<u>Strat.</u>	<u>Struct.</u>
MCLOUTH STEEL				
1949	Steel mill products.	201-500	S	F
1959	Steel mill products.	254	S	F
1969	Steel mill products.	399	S	F
MEAD				
1949	<u>Paper, paperboard.</u>	101-200	S	F
1959	<u>Paperboard</u> , + as above. ADDED: <u>paperboard containers.</u>	138	RC	F
1969	<u>Paper</u> , + as above. ADDED: school supplies, modeling clay, crayons, [computer software, photographic testing and processing, iron, coke; cement, pipe, precision castings; effluent treatment, furniture, upholstery fabrics].	113	RC	PD
MENDEL				
1949	Integrated forest products company: <u>finished wood products</u> (furniture, doors, cabinets), lumber, paperboard containers.	201-500	DV	PD
1954	Acquired by Container Corporation of America.			

TABLE A-1 (continued)

MERCK AND COMPANY		Rank	Strat.	Struct.
1949	<u>Bulk vitamins</u> , antibiotics, hormones, and medicinal chemicals, agricultural chemicals, industrial chemicals.	201-500	RC	F
1959	<u>Vitamins in bulk</u> , + as above. ADDED: <u>prescription vitamins</u> , <u>ethical drugs</u> .	210	RC	PD
1969	<u>Ethical drugs</u> , + as above. ADDED: chemicals for water purification and water purification systems, commercial cleaning products.	174	RC	PD
MIDLAND-ROSS				
Prior to 1957 name was Midland Steel Products.				
1949	<u>Heavy steel auto parts and frames</u> , pneumatic controls, steel tubing.	201-500	DC	F
1959	<u>Heavy steel auto parts and frames</u> , + as above. ADDED: door openers, aircraft coupling and controls, [industrial dryers, plastic extruding machinery, furnaces, air conditioners; air purification equipment, glass plant; non-ferrous die casting].	422	UP	PD
1969	<u>Heavy steel auto parts and frames</u> , + as above. ADDED: iron and steel castings, [water and air pollution control equipment; winding and slitting machinery; electric plugs, connectors, fixtures and controls, grinding balls, jet engine and missile components, steel boxes, letter presses.	335	UP	PD
MILES LABORATORIES				
1949	<u>Proprietary medicines</u> , pharmaceutical chemicals.	--	RC	F

TABLE A-1 (continued)

MILES LABORATORIES (continued)

		<u>Rank</u>	<u>Strat.</u>	<u>Struct.</u>
1959	<u>Proprietary medicines</u> , + as above. ADDED: prescription medicines, diagnostic tests, enzymes.	497	RC	PD
1969	<u>Proprietary medicines</u> , + as above. ADDED: medical electronic instruments, laboratory supplies, cyclamates, dairy cultures, refined corn products, household cleaning products.	331	RL	PD

MINNESOTA MINING AND MANUFACTURING

1949	<u>Pressure sensitive electrical, magnetic tape</u> , adhesives, abrasives, gummed paper, heavy chemical by-products.	101-200	RL	PD
1959	<u>Tape</u> , + as above. ADDED: office copier, microfilm equipment, photo offset plates, electric ceramics, refractories, concrete, stone; coated paper, plastic insulator, ribbon, RFP sheet and pipe, varnish, resins, floor covering; synthetic rubber.	66	RL	PD
1969	<u>Tape</u> , + as above. ADDED: cameras, projectors, video-tape equipment; film processing, radio broadcasting, cable, EDP microfilm.	59	RL	PD

MOBIL OIL

Prior to 1966 name was Socony Mobil Oil.

Prior to 1955 name was Socony-Vacuum Oil.

Prior to 1934 name was Socony-Vacuum.

Prior to 1931 name was Standard Oil (N.Y.).

TABLE A-1 (continued)

MOBIL OIL (continued)		<u>Rank</u>	<u>Strat.</u>	<u>Struct.</u>
1949	<u>Integrated oil company.</u>	5	S	F
1959	<u>Integrated oil company.</u>	6	S	F
1969	<u>Integrated oil company.</u> ADDED: petro-chemicals, plastics, (both part of vertical chain), phosphates, paint and chemical coatings.	7	DV	FS
MOHASCO INDUSTRIES				
Prior to 1955 name was Alexander Smith.				
Prior to 1951 name was Alexander Smith and Sons, Carpet Co.				
1949	<u>Carpets</u> , floor tile, plastic wall coverings, plastic household utensils.	201-500	DC	FS
1959	<u>Carpets</u> . EXITED: floor tile, plastic wall coverings, plastic household utensils.	384	S	F
1969	<u>Carpets</u> . ADDED: furniture.	300	DC	PD
MONSANTO				
Prior to 1964 name was Monsanto Chemical.				
1949	<u>Plastics, synthetic resins, coatings</u> , phosphates, detergents, plasticizers, resins; pharmaceutical chemicals, rubber, petro-chemicals, heavy industrial chemicals, agricultural chemicals, wood preservatives, food additives.	86	RC	PD

TABLE A-1 (continued)

MONSANTO (continued)		<u>Rank</u>	<u>Strat.</u>	<u>Struct.</u>
1959	<u>Plastics, synthetic resins, coatings</u> , + as above. ADDED: integrated oil production, synthetic fibers, explosives.	70	RC	PD
1969	<u>Synthetic fibers</u> , + as above. ADDED: high protein food, water treatment processes, electronic testing instruments, automatic control valves.	43	RL	PD
NATIONAL BISCUIT				
1949	<u>Baked goods</u> , snack foods, cereals, dog food.	97	DC	F
1959	<u>Baked goods</u> , + as above. ADDED: cake mixes, other grocery products.	104	DC	F
1969	<u>Baked goods</u> , + as above. ADDED: candy, frozen baked goods, institutional food service.	157	DC	PD
NATIONAL CASH REGISTER				
1949	Business machines: <u>cash registers</u> , accounting and adding machines.	201-500	DC	F
1959	<u>Accounting machines</u> , + as above. ADDED: electronic data processing computers, commercial and aircraft; typewriter paper.	108	RC	F
1969	<u>Accounting machines</u> , + as above. ADDED: defense communications electronics, paper capsules for drugs.	87	RC	F

TABLE A-1 (continued)

NATIONAL CASTINGS		Rank	Strat.	Struct.
Prior to 1961 name was National Malleable and Steel Castings.				
1949	Iron and steel castings, for railroad and automobiles.	201-500	S	F
1959	Iron and steel castings. ADDED: grinding balls.	--	S	PD
1965	Acquired by Midland-Ross.			
NATIONAL CONTAINER				
1949	Paperboard and paperboard containers.	201-500	S	F
1956	Acquired by Owens-Illinois.			
NATIONAL LEAD				
1949	<u>Paints and pigments</u> , lead chemicals, paint oils; lead products: pipes, sheets, washers, wires, lead alloy products: solders, type; other metal products: bearings, castings; acid handling equipment, titanium alloy manufacturing; lubricating devices, [well drilling equipment].	84	RL	FS
1959	<u>Paints and pigments</u> , + as above. ADDED: refractories, non-ferrous die casting, nickel processing; screws; nuclear metals, R & D, processing aluminium tooling.	82	RL	FS

TABLE A-1 (continued)

NATIONAL LEAD (continued)		Rank	Strat.	Struct.
1969	<u>Paints and pigments</u> , + as above. ADDED: [chemicals for petroleum industry; aluminium architectural structures, plastics for industry; hardwood veneer; bank, precision parts for electronics].	122	RL	PD
NATIONAL STEEL				
1949	Integrated steel producer: <u>steel strips and sheet</u> , bars, shapes, structural shapes, tin plate, pig iron and ores, chemical by-products.	43	DV	F
1959	Integrated steel: <u>strips and sheet</u> , + as above. ADDED: coated steel (part of vertical chain).	54	DV	F
1969	Integrated steel: <u>strips and sheet</u> , + as above. ADDED: aluminium building products.	91	DV	F
NATIONAL SUPPLY				
1949	Oil well drilling equipment: tubes, pumps, motors, etc., and distribution of these and other oil well supplies.	201-500	RC	FS
1958	Acquired by Armco Steel.			
NEW YORK TIMES				
1949	<u>Newspaper publishing</u> , radio station, facsimile transmission.	--	DC	FS
1959	<u>Newspaper publishing</u> , + as above. EXITED: facsimile transmission.	377	DC	FS
1969	<u>Newspaper publishing</u> , + as above. ADDED: book publishing, magazine publishing, teaching materials.	381	DL	FS

TABLE A-1 (continued)

NEWPORT NEWS SHIPBUILDING

		<u>Rank</u>	<u>Strat.</u>	<u>Struct.</u>
1949	<u>Shipbuilding</u> , hydraulic turbines, heavy industrial machinery.	201-500	S	F
1959	<u>Shipbuilding</u> , + as above.	238	S	F
1968	Acquired by Tenneco.			

NORTH AMERICAN ROCKWELL

Prior to 1967 name was North American Aviation.

1949	<u>Airplanes and parts</u> , rocket engines, missiles, electronic and electro-mechanical equipment for aerospace, designing, building and operating nuclear reactors.	101-200	DC	F
1959	<u>Airplanes and parts</u> , + as above. ADDED: electronic systems for aerospace.	38	RC	PD
1969	[As above]. ADDED: <u>auto parts</u> , industrial machine parts, textile machinery, graphic arts equipment.	30	UP	PD

NORTHERN PAPER MILLS

1949	Sanitary paper products: toilet paper, paper towels, napkins.	201-500	S	F
1953	Acquired by Marathon.			

OLIN

Prior to 1969 name was Olin-Mathieson Chemical.

Prior to 1954 name was Mathieson Chemical.

TABLE A-1 (continued)

OLIN (continued)		Rank	Strat.	Struct.
1949	Industrial heavy chemicals: <u>bulk heavy alkalis</u> , chlorides, nitrogen products, etc, agricultural chemicals.	101-200	RC	F
1959	[As above]. ADDED: <u>primary aluminium, non-ferrous fabricated parts</u> , [arms, ammunition, explosives; cellophane, cigarette filter and paper, lumber, kraft paper, outdoor furniture, nuclear fuel, pharmaceuticals: drugs, vitamins, proprietary products.	60	UP	PD
1969	<u>Primary aluminium, non-ferrous parts</u> , + as above. EXITED: pharmaceuticals, nuclear fuel. ADDED: <u>aluminium: consumer products, architectural products</u> , [polyvinyl chloride and polyvinyl chloride pipe, outdoor recreation: equipment, area development, books; mobile homes, real estate development; wood furniture, floors].	99	UP	PD
OTIS ELEVATOR				
1949	<u>Elevators and escalators</u> .	--	S	F
1959	<u>Elevators and escalators</u> . ADDED: fork-lift trucks, [military electronics; snipping and maintaining automatic bowling pinsetters].	203	DL	FS
1969	<u>Elevators and escalators</u> . EXITED: military electronics.	192	DL	FS
OUTBOARD MARINE				
Prior to 1956 name was Outboard Marine and Manufacturing.				
1949	<u>Outboard motors</u> .	--	S	F
1959	<u>Outboard motors</u> . ADDED: power lawnmowers, chain saws, utility vehicles.	259	DC	F

4-160

TABLE A-1 (continued)

OUTBOARD MARINE (continued)		<u>Rank</u>	<u>Strat.</u>	<u>Struct.</u>
1969	<u>Outboard motors</u> , + as above. ADDED: snowmobiles, camping trailers, grass care equipment.	290	DC	PD
OWENS-ILLINOIS				
Prior to 1965 name was Owens-Illinois Glass.				
1949	<u>Glass containers, closures</u> , glass insulators, blocks, glass laboratory equipment, glass TV tubes, glass tableware, paper containers, plywood and veneer, plastic molding machines, [brush-making machines, brushes].	85	DC	PD
1959	<u>Glass containers, closures</u> , + as above. EXITED: plywood and veneer, brushes and brush machines. ADDED: <u>plastic containers</u> , paperboard, paper bags, plastic tableware, precision glass electronic components.	76	RC	PD
1969	<u>Glass and plastic containers, closures</u> , + as above. ADDED: fiber cans, plywood, glass resins, glass for fiber optics, optical systems, gyro test systems, engineering for aerospace, paper cups and plates.	85	RL	PD
PACIFIC CAR AND FOUNDRY				
1949	railroad cars; logging and construction equipment, trucks and buses, shipbuilding and repair, structural steel plates and shapes.	--	RC	F
1959	<u>Trucks</u> , + as above. ADDED: <u>off-highway construction vehicles</u> .	374	RC	F
1969	<u>Trucks and construction vehicles</u> , + as above.	227	DC	F

TABLE A-1 (continued)

		Rank	Strat.	Struct.
PACIFIC MILLS				
1949	Textiles: cotton, rayon, worsted, for apparel, furnishings, industry and other uses.	101-200	RC	PD
1959	Acquired by Burlington Industries.			
PENNWALT				
Prior to 1969 name was Pennsalt.				
Prior to 1957 name was Pennsylvania Salt Manufacturing.				
1949	Agricultural chemicals, heavy industrial chemicals, household chemical products: cleansers, water softeners; chemical builders' products: paint, rust inhibitor.	--	RC	F
1959	As above. ADDED: fertilizer.	428	RC	PD
1969	As above. ADDED: specialized equipment: tabletting, freeze-drying, plastic molding, separating; health products: ethical and proprietary drugs, dentists' equipment and supplies.	248	RL	PD
PET				
Prior to 1966 name was Pet Milk.				
1949	<u>Dairy products</u> : evaporated milk, milkpowder, ice cream mix, cottage cheese.	101-200	S	FS
1959	<u>Dairy products</u> , + as above. ADDED: frozen foods.	270	DC	FS

TABLE A-1 (continued)

PET (continued)		Rank	Strat.	Struct.
1969	<u>Dairy products</u> , + as above. ADDED: diet foods, candy, snacks, fruit products, food stores, restaurants, service stations, store equipment, institutional food service.	181	RC	PD
PFIZER (CHAS. A.)				
1949	Industrial chemicals, pharmaceutical chemicals, food supplements.	201-500	RC	F
1959	As above. ADDED: ethical drugs: <u>antibiotics</u> , and others, plasticizers, lubricating additives, proprietary drugs, syringes.	189	RC	F
1969	<u>Antibiotics</u> , + as above. ADDED: plastics, adhesives, [lime, limestone, calcium, magnesium, powder metals, paint pigments, refractories].	139	RC	PD
PHILCO				
1949	<u>Radios, TV's, phonographs</u> : cathode ray tubes, air conditioners and refrigerators, electric ranges, government electronics, TV broadcasting.	101-200	RL	PD
1959	<u>Radios, TV's, phonographs</u> , + as above. ADDED: washing machines, transistors, special purpose computers.	122	RL	PD
1961	Acquired by Ford Motor.			
PHILLIP MORRIS				
1949	<u>Cigarettes and tobacco</u> .	101-200	S	F

TABLE A-1 (continued)

PHILLIP MORRIS (continued)		<u>Rank</u>	<u>Strat.</u>	<u>Struct.</u>
1959	<u>Cigarettes and tobacco</u> . ADDED: [adhesives, glassine paper, textile processing chemicals: printing].	156	DU	FS
1969	<u>Cigarettes and tobacco</u> , + as above. ADDED: [beer, gum, razor blades, toiletries; hospital and surgical supplies; labels, parts for power tools].	145	DU	PD
PHILLIPS PETROLEUM				
1949	<u>Integrated oil</u> ; fertilizer, carbon black (both part of vertical chain).	22	DV	FS
1959	<u>Integrated oil</u> , + as above. ADDED: synthetic rubber, polyethelene (both part of chain); plastic containers.	31	DV	FS
1969	<u>Integrated oil</u> , + as above. ADDED: packaging film and machinery; polyethelene coated paper; plastic pipe, paper containers; synthetic fibers [jute, twine, carpet yarn].	38	DV	FS
PILLSBURY				
Prior to 1958 name was Pillsbury Mills.				
1949	<u>Bulk flour and flour-based mixes</u> , consumer flour and mixes, animal feed, soy and feed by-products.	101-200	DC	F
1959	<u>Bulk flour and flour-based mixes</u> , + as above. ADDED: frosting mix.	126	RC	PD

TABLE A-1 (continued)

PILLSBURY (continued)		<u>Rank</u>	<u>Strat.</u>	<u>Struct.</u>
1969	<u>Bulk flour and flour mixes</u> , + as above. ADDED: snack foods, artificial sweetener, canned food, household cleaner, [poultry processing, restaurants, time-shared computer].	109	RC	PD
PITTSBURGH STEEL				
1949	<u>Steel mill products</u> : wire, tubing, semi-finished products.	101-200	S	FS
1959	<u>Steel mill products</u> , + as above.	282	S	FS
1966	Merged with Wheeling Steel to form Wheeling-Pittsburgh Steel.			
PPG INDUSTRIES				
Prior to 1968 name was Pittsburgh Plate Glass.				
1949	<u>Flat glass</u> , mirrors, optical glass, related chemicals, paint, varnish.	70	RL	PD
1959	<u>Flat glass</u> , + as above. ADDED: fiberglass yarn, insulation, resins, titanium dioxide.	72	RL	PD
1969	<u>Flat glass</u> , + as above. ADDED: polyvinyl chloride, fertilizers.	102	RL	PD
PROCTOR AND GAMBLE				
1949	<u>Soaps, detergents</u> , oils and fats, chemical dissolving cottonseed pulp.	48	DC	F
1959	<u>Soaps, detergents</u> , + as above. ADDED: <u>cleansers</u> , peanut butter, toiletries, baking mixes, sanitary paper goods; chemical dissolving pulp, bleach, disinfectant.	25	DC	PD

TABLE A-1 (continued)

PROCTOR AND GAMBLE (continued)

	<u>Rank</u>	<u>Strat.</u>	<u>Struct.</u>
1969 <u>Soaps, detergents, cleansers</u> , + as above. EXITED: bleach, disinfectant. ADDED: coffee.	26	RC	PD

PULLMAN

1949 <u>Railroad and street cars, forgings and parts</u> , [design and construction of petroleum and chemical plants, also separate sale of equipment: pipes, heat exchangers, pressure vessels, R & D: metallurgy, atomic energy].	81	UP	H
1959 <u>Railroad and street cars, forgings and parts</u> . EXITED: separate sale of process plant equipment; metallurgical R & D. ADDED: <u>truck trailers</u> , [furnaces, kilns, construction of paper mills; engineering of power plants, water systems, highway construction].	114	UP	H
1969 <u>Railroad and street cars, truck trailers</u> , + as above.	151	UP	PD

QUESTOR

Prior to 1968 name was Dunhill International.

1949 <u>Cigarettes, tobacco, pipes</u> : [toiletries, cosmetics, leather gifts, luggage].	--	DU	H
1959 <u>Cigarettes, tobacco, pipes</u> , + as above. ADDED: [freight terminals].	--	DU	H
1969 EXITED: cigarettes, tobacco, pipes, toiletries, cosmetics, leather goods; freight terminals. ADDED: <u>auto parts</u> , children's furniture, bottles, clothes, books, toys, recreational equipment; building products: ventilators, shutters, metal grid ceiling system.	376	AC	PD

TABLE A-1 (continued)

		Rank	Strat.	Struct.
RALSTON PURINA				
1949	<u>Animal feeds, soy products</u> , animal health products, cereal, Ry-Krisp.	101-200	DC	F
1959	<u>Animal feeds, soy products</u> , + as above. ADDED: dog food.	81	DC	F
1969	<u>Animal feed, soy products</u> , + as above. ADDED: grocery products, tuna and fish meal; [restaurant food supplying, restaurants].	78	DC	PD
RATH PACKING				
1949	Meat packing: <u>meat</u> , by-products.	101-200	DV	F
1959	Meat packing: <u>meat</u> , + as above. ADDED: chemical by-products.	176	DV	F
1969	Meat packing: <u>meat</u> , + as above.	349	DV	F
RAYBESTOS-MANHATTEN				
1949	<u>Asbestos and rubber auto parts</u> ; asbestos and rubber industrial products; packings and gaskets; bowling balls; powdered metal parts; auto brake service equipment; abrasive and diamond wheels.	201-500	RC	F
1959	<u>Asbestos and rubber auto parts</u> , + as above. ADDED: hydraulic brake parts and fluid; teflon and nylon industrial products, resin-asbestos molding material and aerospace components.	423	RC	F
1969	<u>Asbestos and rubber auto parts</u> , + as above. ADDED: rivets, riveting machines.	525	RC	FS

TABLE A-1 (continued)

		<u>Rank</u>	<u>Strat.</u>	<u>Struct.</u>
RAYONIER				
1949	Integrated forest products: <u>cellulose</u> , pulp, paper.	201-500	S	F
1959	Integrated forest products: <u>cellulose</u> , + as above. ADDED: lumber, [tannin].	313	S	F
1968	Acquired by International Telephone and Telegraph.			
REMINGTON RAND				
1949	<u>Typewriters</u> , adding and accounting machines, supplies and forms; electric shavers.	101-200	RC	PD
1955	Merged with Sperry to form Sperry Rand.			
REPUBLIC AVIATION				
1949	<u>Highspeed aircraft</u> .	201-500	S	F
1959	<u>Highspeed aircraft</u> . ADDED: R & D missiles.	228	S	F
1965	Acquired by Fairchild Hiller.			
REPUBLIC STEEL				
1949	Integrated steel producer: <u>mill products</u> , fabricated products.	29	DV	F
1959	Integrated steel producer: <u>mill products</u> , + as above.	36	DV	F

TABLE A-1 (continued)

REPUBLIC STEEL (continued)		Rank	Strat.	Struct.
1969	Integrated steel producer: <u>mill products</u> , + as above.	69	DV	F
REVLON				
Prior to 1955 name was Revlon Products.				
1949	<u>Cosmetics and toiletries</u> , hair care preparations, perfumes, manicure equipment.	--	DC	F
1959	<u>Cosmetics and toiletries</u> , + as above. ADDED: proprietary drugs, [shoe polish].	323	DC	FS
1969	<u>Cosmetics and toiletries</u> , + as above. EXCLUDED: shoe polish. ADDED: beauty parlor fixtures, ethical drugs.	296	DC	PD
REYNOLDS METALS				
1949	Integrated aluminium: <u>mill products</u> , fabricated products, primary aluminium, pigments.	101-200	DV	F
1959	Integrated aluminium: <u>mill products</u> , + as above. ADDED: plastic wrap.	91	DV	F
1969	Integrated aluminium: <u>mill products</u> , + as above.	114	DV	PD
RICHFIELD OIL				
1949	<u>Integrated oil</u> .	101-200	S	F
1959	<u>Integrated oil</u> . ADDED: petro-chemicals.	178	DV	F

TABLE A-1 (continued)

RICHFIELD OIL (continued)		<u>Rank</u>	<u>Strat.</u>	<u>Struct.</u>
1966	Merged with Atlantic Refining to form Atlantic Richfield.			
ROCKWELL MANUFACTURING				
1949	<u>Gas, oil and water meters and regulators</u> , valves, pumps, hydraulic transmissions, [parking and fare meters, power tools].	201-500	UP	H
1959	<u>Fluid meters, regulators</u> , + as above. ADDED: <u>pneumatic and electronic process control instruments</u> , [gas and diesel engines, steel and iron foundry, automatic voting machines].	327	UP	PD
1969	<u>Power tools</u> , + as above. EXCLUDED: pneumatic and electronic process control instruments. ADDED: industrial machine tools, [brass plumbing fittings].	344	UP	PD
ROCKWELL-STANDARD				
Prior to 1958 name was Rockwell Spring and Axel.				
1953	Formed by merger of Standard Steel Spring and Timken Detroit Axel.			
1959	<u>Axles and transmissions for autos and other vehicles</u> , vehicle springs, seats, brakes and other parts, [lighting standards, filters, light aircraft].	163	RC	PD
1967	Merged with North American Aviation to form North American Rockwell.			

TABLE A-1 (continued)

		<u>Rank</u>	<u>Strat.</u>	<u>Struct.</u>
ROHM AND HASS				
1949	<u>Organic resins</u> , plastic powder and sheet, chemicals for leather and textile industries; agricultural chemicals.	201-500	RC	FS
1959	<u>Organic resins</u> , + as above. ADDED: detergent.	211	RC	PD
1969	<u>Organic resins</u> , + as above. ADDED: elastic fabric; silk and synthetic yarn; ethical drugs, animal health products; medical testing laboratory and instruments.	233	RC	PD
ROHR				
Prior to 1961 name was Rohr Aircraft.				
1949	Formed as spinoff of Newport Steel.			
1949	<u>Aircraft power packages</u> , air frame components.	--	DC	F
1959	<u>Aircraft power packages</u> , + as above.	237	DC	F
1969	<u>Aircraft power packages</u> , + as above. ADDED: rocket motor cases and nozzles, large antennas, power boats, [computer software and systems].	348	DC	PD
ROYAL MCBEE				
Prior to 1954 name was Royal Typewriter.				
1949	<u>Typewriters, supplies</u> .	201-500	S	F
1959	<u>Typewriters, supplies</u> . ADDED: keysort machines, business forms, office copying machines, general purpose computer.	376	DC	F
1965	Acquired by Litton Industries.			

TABLE A-1 (continued)

RUBEROID		Rank	Strat.	Struct.
1949	Asphalt and tar building and industrial materials: <u>roofing</u> , siding, felt, coatings; asbestos-cement building products: roofing, siding, building board; Kraft building and packaging paper.	201-500	RC	F
1959	<u>Asphalt and tar roofing</u> , + as above. ADDED: gypsum building materials; roofing granules; asbestos auto parts, floor tiles.	317	RC	F
1967	Acquired by GAF.			
RYAN AERONAUTICAL				
1949	<u>Airplanes and airplane parts</u> , missiles.	--	S	F
1959	<u>Airplanes and airplane parts</u> , + as above. ADDED: aerospace and military electronic systems.	442	RC	F
1967	Acquired by Teledyne.			
SANDERS ASSOCIATES				
1951	Incorporated.			
1959	<u>Defense electronic systems</u> , R & D for defense electronic systems; electronic aerospace instruments, microwave equipment; servo-valves, gyroscopes, blowers; oceanographic equipment.	--	RC	FS
1969	<u>R & D for defense electronic systems</u> , + as above. ADDED: lasers, long-range communication systems, commercial computer systems, information handling and display systems.	440	RC	PD

TABLE A-1 (continued)

		<u>Rank</u>	<u>Strat.</u>	<u>Struct.</u>
SCHERING				
1949	Ethical drugs: <u>hormones</u> , antihistamines, sulfas, etc., proprietary drugs; cosmetics; bulk pharmaceutical chemicals.	--	RC	F
1959	Ethical drugs: <u>hormones</u> , + as above. ADDED: animal health products.	463	RC	F
1969	Ethical drugs: <u>hormones</u> , + as above.	408	RC	PD
SCHLITZ (JOS. P.) BREWING				
1949	Beer.	101-200	S	F
1959	Beer.	274	S	F
1969	Beer.	243	S	F
SCOTT PAPER				
1949	Integrated paper company: <u>sanitary paper products</u> , wax paper, industrial paper wipers.	101-200	DV	F
1959	Integrated paper: <u>sanitary paper products</u> , + as above. ADDED: sulphite papers; urethane foam, plastic cups.	153	DV	F
1969	Integrated paper: <u>sanitary paper products</u> , + as above. ADDED: plastic wrap, plastic coating, printing paper; non-woven textiles, [projectors, copiers, film and microfilm, educational films and slides].	154	DV	PD

TABLE A-1 (continued)

SERVEL	Rank	Strat.	Struct.
1949 <u>Refrigerators</u> , central air conditioners, heaters, ice makers.	201-500	DC	F
1959 [Icemakers]. EXITED: refrigerators, air conditioners, heaters. ADDED: <u>batteries</u> .	---	S	F
1967 Acquired by Clevite.			
SPERRY			
1949 <u>Instruments and controls</u> : electronic instruments and controls, gyroscopes, navigational equipment, computing systems, precision instruments; motors and generators; photocells [farm and quarry equipment; hydraulic equipment: variable speed transmissions; packaging equipment, winches, aluminium roofing and siding.	101-200	UP	PD
1955 Merged with Remington Rand to form Sperry Rand.			
ST. REGIS PAPER			
1949 Integrated paper company: <u>paper bags</u> , ground wood paper, Kraft pulp, paper and paperboard, plastic/paper laminates; bag-making and filling machines.	99	DV	F
1959 Integrated paper company: + as above. ADDED: <u>paper containers</u> , waxed paper, printing paper, lumber, plywood (all part of integrated chain), polyethelene film and bags, injection molded plastics.	92	DV	F
1969 Integrated paper: <u>Kraft pulp, paper and paperboard</u> , + as above. ADDED: stationery, school supplies (both part of chain); [dairy products processing and handling equipment; special industrial equipment].	133	DV	PD

TABLE A-1 (continued)

STALEY (A.E.) MANUFACTURING		Rank	Strat.	Struct.
1949	<u>Soy bean products</u> : oil, meal, sauce; corn products: starch, syrup, oil; refining crude soy and corn oil, chemical by-products.	101-200	RC	F
1959	As above. ADDED: industrial chemical products, polymers, marsh-mallows.	203	RC	PD
1969	<u>Soy: oil, meal, corn: starch, syrup, oil</u> , (i.e., 'food ingredients'). ADDED: <u>spices, flavorings</u> , household cleaner, grocery products; adhesives.	315	RL	PD
STANDARD BRANDS				
1949	<u>Coffee</u> , tea, grocery products; bakers' ingredients, liquor.	101-200	RC	F
1959	<u>Coffee</u> , + as above. ADDED: dog food, refined corn products.	102	RC	FS
1969	<u>Coffee</u> , + as above. ADDED: candy, [industrial chemicals; latex, resin, adhesives].	120	RC	FS
STANDARD KOLLSMAN INDUSTRIES				
Prior to 1960 name was Standard Coil Products.				
1949	<u>TV and radio-tuners</u> .	--	S	F
1959	Tuners. ADDED: air navigation instruments, air guidance systems, precision motors.	489	RL	PD
1969	As above. ADDED: automobile cigarette lighters; electric blankets; semi-conductors, rectifiers; aerospace electronic systems, switches,	--	RL	PD

TABLE A-1 (continued)

STANDARD KOLLSMA, INDUSTRIES (continued)

1969 (continued) fire extinguishers, rear view mirrors.

STANDARD OIL (OHIO)

	<u>Rank</u>	<u>Strat.</u>	<u>Struct.</u>
1949 <u>Integrated oil company.</u>	69	S	F
1959 <u>Integrated oil company.</u> ADDED: ammonia, urea, nitric acid, plastic intermediates.	118	S	F
1969 <u>Integrated oil company</u> , + as above. ADDED: molded plastics, plastic housewares, plastic powder, plastic pipe, building materials; coal, water heaters, motels, restaurants; vending machines, [low density glass foam; mass food service].	97	DV	FS

STANLEY WORKS

1949 Power tools, hand tools; full line of hardware, coatings and finishes, 201-500 steel stripping.		RC	PD
1959 As above. ADDED: aluminium windows, automatic door opener, drapery hardware.	373	RC	PD
1969 <u>Building materials: hardware, windows, door openers</u> , + as above. ADDED: <u>doors, curtain walls</u> ; [electronic stampings].	354	RC	PD

STAUFFER CHEMICAL

1949 Basic industrial chemicals, agricultural chemicals, battery cases.	201-500	RC	FS
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TABLE A-1 (continued)

STAUFFER CHEMICAL (continued)		Rank	Strat.	Struct.
1959	As above. EXITED: battery cases. ADDED: petro-chemicals; ethical drugs; organo-metals; plastic foam; plastic pipes and tubes.	202	RC	PD
1969	As above. ADDED: refractory metal alloys, plasticizers; sanitary chemicals; coated fabrics; vinyl wall covering.	215	RC	PD
STERLING DRUG				
1949	<u>Proprietary drugs</u> ; ethical drugs; pharmaceutical chemicals; soaps and cleansers; shoe polish; ammonia, bleach, toiletries; dye; ink; industrial chemicals.	201-500	RC	PD
1959	<u>Proprietary drugs</u> , + as above. EXITED: ammonia, bleach. ADDED: animal health products; industrial waste treatment; insecticides, [pet accessories].	217	RC	PD
1969	<u>Ethical drugs</u> , + as above. EXITED: pet accessories. ADDED: cosmetics and perfumes; wax; starch.	200	RC	PD
STEVENS (J.P.)				
1949	Textiles: <u>rayon and cotton "grey goods"</u> ; wide variety of woolen and worsted finished textile products.	96	DC	F
1959	<u>"Grey goods"</u> , + as above. ADDED: non-woven fabrics, plastics.	94	DC	F
1969	As above. EXITED: plastics. ADDED: hosiery, carpets.	115	RC	F

TABLE A-1 (continued)

STEWART-WARNER		Rank	Strat.	Struct.
1949	Auto parts: fuel pumps, gears, instruments, auto servicing equipment; die castings; hardware, [TV, radio receivers; phonographs, military electronic devices; heaters, furnaces].	201-500	UP	PD
1959	As above. EXITED: TV, radio. ADDED: automobile lights, [heat exchangers, air conditioners; industrial electronics: automatic controls for machines, facsimilie transmission, meters, switches].	347	UP	PD
1969	As above. ADDED: lubricating equipment, vehicle servicing, [data display systems, integrated circuits, pneumatic and hydraulic equipment for aircraft and missiles, fluid materials handling equipment].	441	UP	PD
STOKELY-VAN CAMP				
1949	<u>Canned fruits, vegetables and cooked food</u> , frozen fruits and vegetables.	201-500	DC	FS
1959	<u>Canned food</u> , + as above. ADDED: frozen main dishes, grocery products.	275	DC	FS
1969	<u>Canned food</u> , + as above. ADDED: oils, margarine, gatorade.	307	RC	FS
STUDEBAKER-WORTHINGTON				
1967	Formed by merger of Studebaker and Worthington.			
1969	<u>Electrical industrial products and electronic components</u> : power	141	RL	PD

TABLE A-1 (continued)

STUDEBAKER-WORTHINGTON (continued)

	<u>Rank</u>	<u>Strat.</u>	<u>Struct.</u>
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1969 (continued) plants, motors and generators, transformers, silicon rectifiers, electron tubes, read-out devices; turbines, pumps, compressors; air conditioners, heaters, refrigerators, [springs, forgings; construction equipment: water meters; utility vehicles, automotive products: oil and gas treatments, brakes, brake-fluid, lamps, flashers; locomotives].

SUNDSTRAND

Prior to 1959 name was Sundstrand Machine Tool.

1949	<u>Machine tools</u> , aircraft hydraulic machinery, industrial hydraulic machinery, sanding and rubbing equipment, castings.	--	RC	F
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1959	<u>Aircraft: hydraulic machinery</u> , + as above. ADDED: <u>aircraft hydraulic drives and motors</u> ; power supplies for missiles.	457	RL	PD
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1969	<u>Aircraft hydraulic machinery, drives and motors</u> , + as above. ADDED: aviation: electronic controls and instruments, music and announcement systems, mechanical actuators, light-weight castings; numerical and computer controls for machine tools; materials handling equipment.	310	RL	PD
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SUNSHINE BISCUIT

1949	<u>Bakery products</u> , cereal, dog food.	201-500	DC	F
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1959	<u>Bakery products</u> , + as above. ADDED: snack foods.	242	DC	F
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1965	Acquired by American Tobacco.			
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TABLE A-1 (continued)

		<u>Rank</u>	<u>Strat.</u>	<u>Struct.</u>
SYBRON				
Prior to 1968 name was Ritter Pfaudler.				
1965	Formed by merger of Ritter and Pfaudler Permutit.			
1949	<u>Steel processing tanks.</u>	--	S	F
1959	<u>Steel processing tanks.</u> ADDED: water and waste treatment equipment; sanitation chemicals; [brewers' equipment].	--	DC	PD
1969	Steel processing tanks, + as above. ADDED: <u>electronic and pneumatic process control instruments</u> , [dental and hospital and laboratory equipment; textile chemicals; government R & D].	311	UP	PD
SYLVANIA ELECTRIC PRODUCTS				
1949	<u>Radio and TV tubes, special military tubes</u> , radio sets, TV chassis; light bulbs and light fixtures; electronic components; radar components.	101-200	RL	F
1959	Acquired by General Telephone and Electronics.			
TEXACO				
Prior to 1959 name was Texas Co. (The).				
1949	<u>Integrated oil company</u> ; petro-chemicals.	6	DV	F
1959	<u>Integrated oil</u> , + as above. ADDED: synthetic rubber, [uranium mining and processing].	8	DV	FS
1969	<u>Integrated oil</u> , + as above. ADDED: fertilizer.	8	DV	FS

TABLE A-1 (continued)

TEXAS INSTRUMENTS		<u>Rank</u>	<u>Strat.</u>	<u>Struct.</u>
Prior to 1951 name was Geophysical Service.				
1949	<u>Seismic exploration</u> ; military electronics; military electro-mechanical equipment.	--	DC	F
1959	As above. ADDED: <u>electronic components</u> : semi-conductors, transistors, etc., gravity meters; optical components; special clad metals, motor controls; nuclear fuel.	234	RL	PD
1969	<u>Electronic components</u> , + as above.	135	RL	PD
TIMES MIRROR				
1949	<u>Newspaper publishing</u> ; paper; job printing.	201-500	DC	F
1959	<u>Newspaper publishing</u> , + as above. ADDED: paper bags; TV station; TV programs.	394	DL	PD
1969	<u>Newspaper publishing</u> , + as above. EXITED: TV station, programs. ADDED: lumber, plywood; books publishing, binding; chart and map printing; magazine publishing; globes and teaching aids; [CATV].	256	RL	PD
TIME				
1949	<u>Magazine publishing</u> , news films.	101-200	DC	FS
1959	<u>Magazine publishing</u> . EXITED: news films. ADDED: book publishing; printing supplies and equipment; pulp, paper, radio, TV broadcasting.	177	DC	FS

TABLE A-1 (continued)

TIME (continued)	<u>Rank</u>	<u>Strat.</u>	<u>Struct.</u>
1969 <u>Magazine publishing.</u> ADDED: textbooks; educational materials and supplies; CATV; [business information service].	180	RL	PD
TRANS UNION			
Prior to 1968 name was Union Tank Car.			
1949 <u>Building and leasing railroad tank cars.</u>	--	S	F
1959 <u>Building and leasing railroad tank cars.</u> ADDED: tanks for cars, storage tanks; castings and forgings; oil and water treatment equipment, [atomic reactor].	342	RL	PD
1969 <u>Building and leasing railroad tank cars,</u> + as above. ADDED: [import-export; credit bureau, real estate development, fasteners].	459	RL	PD
TRW			
Prior to 1965 name was Thompson Ramo Wooldridge.			
Prior to 1958 name was Thompson Products.			
1949 <u>Aircraft engine parts,</u> aircraft fluid and fuel control parts; auto engines and parts; electronic switches; [pumps, valves, couplings].	201-500	RC	PD
1959 <u>Aircraft engine parts,</u> + as above. ADDED: radar; rocket nozzles; space systems, missile power; avionics; parts for entertainment electronics; TV cameras and related equipment; transformers, capacitors, semi-conductors; electronic systems R & D; [industrial electronic control systems; process control and general computers, records, tapes; nuclear power unit; special chemicals].	109	UP	PD

TABLE A-1 (continued)

TRW (continued)		Rank	Strat.	Struct.
1969	<u>Electronic components</u> , + as above. ADDED: RFP components, electronic educational systems; automated industrial plants; high-speed camera; [coatings, plastic chemicals, small arms, ball and roller bearings; oil and gas exploration and service].	61	UP	PD
UNIROYAL				
Prior to 1967 name was United States Rubber.				
1949	Rubber: <u>tires and tubes</u> , manufactured industrial and consumer, medical items, footwear, textile, latex, etc.; adhesives, coatings, seals; insulated wire and cable, bulk plastics, plastic film, tire yarn and textile, agricultural chemicals.	37	DV	PD
1959	Rubber: <u>tires and tubes</u> , + as above. EXITED: wire, cable. ADDED: rubber flooring, orlon textiles, plastic pipe; synthetic rubber; plastic consumer products; [fuel cells].	43	DV	PD
1969	Rubber: <u>tires and tubes</u> , + as above.	66	DV	PD
UNITED AIRCRAFT				
1949	<u>Airplane engines</u> , propellers, helicopters, military aircraft.	101-200	DC	PD
1959	<u>Airplane engines</u> , + as above. EXITED: military aircraft. ADDED: aircraft fuel systems, controls, gyros and other parts, military electronics systems; commercial and industrial control systems; meteorological system; aerospace R & D.	35	DC	PD

TABLE A-1 (continued)

UNITED AIRCRAFT (continued)		Rank	Strat.	Struct.
1969	<u>Airplane engines</u> , + as above. ADDED: turbines and compressors for industry and ships; fuel cells, rocket engines, nuclear power plant, electron beam welding, electronic components, telemetry devices, medical electronic equipment, turbo-trains, industrial valves, [glass fiber/plastic material].	36	DC	PD
UNITED STATES STEEL				
1949	Integrated steel: <u>sheet, strip</u> , other mill products, wire, pipe, tube, building materials, pre-fabricated buildings; coke oven chemicals; construction of buildings, bridges, ships, barges; railroad cars; oil well equipment, cement.	3	DV	FS
1959	Integrated steel: <u>pipes, tubes</u> , + as above. EXITED: ships, barges, railroad cars.	5	DV	FS
1969	Integrated steel: <u>pipes, tubes</u> , + as above. ADDED: titanium, titanium semi-fabricated products; aluminium building products; engineering consulting service; industrial, agricultural chemicals; plasticizers; chemical coatings; plastic parts; [lumber; real estate development].	12	DV	FS
USM				
Prior to 1968 name was United Shoe Machinery.				
1949	<u>Shoe machinery</u> , boxes, cartons, adhesives, resins; shoe hardware.	201-500	S	FS
1959	<u>Shoe machinery</u> , + as above. ADDED: special industrial equipment; [military electronics, aerospace electronics; nuclear R & D; equipment for electronics industry; data processing machinery; power transmission for aircraft].	411	RC	FS

TABLE A-1 (continued)

USM (continued)		<u>Rank</u>	<u>Strat.</u>	<u>Struct.</u>
1969	<u>Shoe machinery</u> , + as above. ADDED: [fasteners; bolts, motors, molds for plastics industry, and plastic parts; precision parts; stampings; electronic components].	271	UP	PD
WARD FOODS				
Prior to 1964 name was Ward Baking.				
1949	<u>Bread and cake</u> .	201-500	S	F
1959	<u>Bread and cake</u> . ADDED: <u>frozen pies</u> .	381	S	F
1969	<u>Bakery products</u> . ADDED: meat packing, coffee, tea; grocery products; dairy products; candy; seafood; institutional foods; restaurants; [food machinery].	252	RL	PD
WARNER AND SWASEY				
1949	<u>Machine tools</u> ; textile machinery; earth-moving equipment; foundry, [astronomical instruments and observatories].	--	DC	F
1959	<u>Machine tools</u> , + as above.	--	RL	F
1969	<u>Machine tools</u> , + as above. EXITED: astronomical instruments and observatories. ADDED: numerical controls for machine tools, diagnostic computers; spectrometers, circuit boards.	482	RL	PD
WARNER-LAMBERT PHARMACEUTICAL				
Prior to 1955 name was Warner-Hudnut.				
Prior to 1950 name was Warner (Wm. R.) and Co.				

TABLE A-1 (continued)

WARNER-LAMBERT (continued)		Rank	Strat.	Struct.
1949	<u>Proprietary medicines</u> : cosmetics and toiletries.	--	RC	PD
1959	<u>Proprietary medicines</u> , + as above. ADDED: ethical drugs; tooth-brushes; [molded plastics, glass containers].	248	RC	PD
1969	As above. EXITED: glass containers, toothbrushes. ADDED: eyeglasses and ophthalmic goods; optical instruments; medical instruments; medical electronics; gum, candies.	138	RL	PD
WESTINGHOUSE ELECTRIC				
1949	<u>Motors and controls</u> , turbines, condensers, generators; transformers; switchgear; appliances; radio and TV sets and stations; military electronics, missiles; elevators, escalators; air and marine engines, [industrial education, housing construction].	17	RL	PD
1959	<u>Electric power machinery</u> , + as above. ADDED: avionics, radar, industrial electronics systems; integrated circuits.	15	RL	PD
1969	<u>Electrical power machinery</u> , + as above. ADDED: rocket engines, computer controlled industrial systems, [learning systems].	17	RL	PD
WESTVACO				
Prior to 1969 name was West Virginia Pulp and Paper.				
1949	Integrated forest products: <u>paper</u> , paperboard, chemical by-products.	101-200	DV	F

TABLE A-1 (continued)

WESTVACO (continued)				Rank	Strat.	Struct.
1959	Forest products, + as above.	ADDED: <u>paper containers</u> .		198	DV	PD
1969	Forest products, + as above.	ADDED: [land development].		242	DV	PD
WEYERHAEUSER						
Prior to 1959 name was Weyerhaeuser Lumber.						
1949	Forest products: <u>lumber</u> , pulp, plywood, miscellaneous wood products.		101-200	DV	F	
1959	Forest products: <u>lumber</u> , + as above.	ADDED: paperboard, particle board; cartons; coated paper.	95	DV	PD	
1969	Forest products: <u>lumber</u> , + as above.	ADDED: wood building products, chemical by-products, fine paper, [ski area; banking, home building].	89	DV	PD	
WHITE MOTOR						
1949	<u>Trucks and buses</u> .		201-500	S	F	
1959	<u>Trucks</u> .	EXITED: buses. ADDED: diesel engines.	133	DC	F	
1969	<u>Trucks</u> , + as above.	ADDED: farm equipment, industrial tractors, rebuilt auto parts, [design, building, installation of heating and cooling systems].	118	RC	PD	
WHITTAKER						
Prior to 1964 name was Telecomputing.						

TABLE A-1 (continued)

WHITTAKER (continued)

		<u>Rank</u>	<u>Strat.</u>	<u>Struct.</u>
1949	<u>Digital electronics.</u>	--	S	F
1959	Digital electronics. ADDED: <u>valves, controls for aircraft and missiles</u> ; gyros, warhead test equipment; high energy batteries, precision electronic components, electronic and electro-mechanical parts for aircraft and missiles, aircraft and missile sub-assemblies; radar and communications systems; air traffic control systems.	--	RC	PD
1969	As above. ADDED: <u>industrial and commercial metals</u> : ferro-alloys, other alloys, pipes, casing, tubing, valves, fittings, architectural products: aluminium siding and trim, pipes, fittings; pipelines, fasteners, [railroad rolling stock, materials handling vehicles, fiberglass boats, sailboats; custom-made industrial polymers, also polymer coatings and pastes; resins; aerospace exotic materials].	148	AC	PD

WORTHINGTON

Prior to 1952 name was Worthington Pump and Machinery.

1949	Pumps; electric power generation and distribution equipment; compressors; engines, heat transfer equipment; heavy construction equipment; air conditioners, refrigerators; turbines; water and sewage treatment equipment.	201-500	RL	F
1959	As above. ADDED: home heaters and incinerators; electric control devices and relays; valves and controls for ships, missiles, industry; precision valves and actuators.	251	RL	PD
1967	Merged with Studebaker to form Studebaker-Worthington.			

TABLE A-1 (continued)

		<u>Rank</u>	<u>Strat.</u>	<u>Struct.</u>
WRIGLEY (WM. JR.)				
1949	<u>Chewing gum.</u>	201-500	S	F
1959	<u>Chewing gum.</u>	401	S	F
1969	<u>Chewing gum.</u>	480	S	F
XEROX				
Prior to 1961 name was Haloid Xerox.				
Prior to 1958 name was Haloid.				
1949	<u>Photocopying machines; microfilm projectors.</u>	--	S	F
1959	As above. ADDED: <u>xerographic equipment and supplies.</u>	--	DC	PD
1969	<u>Xerographic equipment and supplies</u> , + as above. ADDED: military and aerospace optical products; [digital computers and software, educational materials; book publishing].	71	DL	PD
YALE AND TOWNE MANUFACTURING				
1949	<u>Materials handling equipment</u> , [locks; builders hardware].	201-500	UP	PD
1959	<u>Materials handling equipment.</u>	289	UP	PD
1963	Merged with Eaton Manufacturing to form Eaton Yale and Towne.			

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